

Research Methodology for Commerce and Management

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Sean Gallagher

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PREFACE

Business research program is preferred by businesses to train and educate their managers and other state members in a vast array of different fields. Business research can relate to economics, business strategy, ethics, etc. Undertaking research in business management is important since it aids a business plan for the future, based on what may have occurred in earlier times. If performed actively it can help an organization to make plans on how to become more viable in its field.

Commerce is the activity of buying and selling of goods and services, especially on a large scale. The system includes legal, economic, political, social, cultural and technological systems that are in operation in any country or internationally. Thus, commerce is a system or an environment that affects the business prospects of economies. In fact, this endeavour is an attempt to synthesis the essential information in unique blend. It gives clear and explicit answers to the questions that arise when the students learn the subject and the scholars do the research.

Quantitative research refers to the systematic empirical investigation of quantitative properties and phenomena and their relationships. The objective of quantitative research is to develop and employ mathematical models, theories and/or hypotheses pertaining to phenomena. The process of measurement is central to quantitative research because it provides the fundamental connection between empirical observation and mathematical expression of quantitative relationships.

The purpose of research is to inform action. Thus, your study should seek to contextualize its findings within the larger body of research. Research must always be high quality in order to produce knowledge that is applicable outside of the research setting with implications that go beyond the group that has participated in the research. Furthermore, the results of your study should have implications for policy and project implementation.

Business and management research is a systematic inquiry that helps to solve business problems and contributes to management knowledge. Business research can include financial data, consumer feedback, product research and competitive analysis. Executives and managers who use business research methods are able to better understand their company, the position it holds in the market and how to improve that position.

As a species, human beings appear to be infinitely inventive and constantly searching for ways to improve their lives. Many important accomplishments and innovations resulting from this quest have changed the course of history along the way. Within the current lifetime, however, arguably none is more important than the progress that has been made in countless disciplines as well as everyday activities as a result of computer technology. The application of computers both at home and at work helps us to better organize our lives, more easily perform repetitive or logical tasks, and access our creativity. Like the Renaissance and the Industrial Revolution before it, the Information Age continues to transform our lives and the ways in which we perform many of our tasks. The trend does not appear to be coming to a quick end, however, and it is reasonable to assume that the proliferation of technological invention and innovation will not continue to change the way that we do things well into the future.

This book embodies a set of papers, conceptual as well as empirical, related to various specialized areas of research in commerce and management.

– *Sean Gallagher*

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Introduction to Research

INTRODUCTION

Research can be defined as the search for knowledge, or as any systematic investigation, to establish novel facts, solve new or existing problems, prove new ideas, or develop new theories, usually using a scientific method.

The primary purpose for basic research (as opposed to applied research) is discovering, interpreting, and the development of methods and systems for the advancement of human knowledge on a wide variety of scientific matters of our world and the universe. Scientific research relies on the application of the scientific method, a harnessing of curiosity. This research provides scientific information and theories for the explanation of the nature and the properties of the world around us. It makes practical applications possible. Scientific research is funded by public authorities, by charitable organizations and by private groups, including many companies. Scientific research can be subdivided into different classifications according to their academic and application disciplines.

Artistic research, also seen as ‘practice-based research’, can take form when creative works are considered both the research and the object of research itself. It is the debatable body of thought which offers an alternative to purely scientific methods in research in its search for knowledge and truth.

Historical research is embodied in the historical method. Historians use primary sources and other evidence to systematically investigate a topic, and then to write histories in the form of accounts of the past. The phrase *my research* is also used loosely to describe a person’s entire collection of information about a particular subject.

ETYMOLOGY

As per the Merriam-Webster Online Dictionary, the word *research* is derived from the Middle French “*recerche*”, which means “to go about seeking”, the term itself being derived from the Old French term “*recerchier*” a compound word from “re-” + “cerchier”, or “sercher”, meaning ‘search’. The earliest recorded use of the term was in 1577.

Definitions

Research has been defined in a number of different ways.

A broad definition of research is given by Martin Shuttleworth-“In the broadest sense of the word, the definition of research includes any gathering of data, information and facts for the advancement of knowledge.” The Merriam-Webster Online Dictionary defines research in more detail as “a studious inquiry or examination; especially: investigation or experimentation aimed at the discovery and interpretation of facts, revision of accepted theories or laws in the light of new facts, or practical application of such new or revised theories or laws”.

Meaning of Research

Research is defined as the scientific investigation of phenomena which includes collection, presentation, analysis and interpretation of facts that lines an individual’s speculation with reality.

Characteristics of Research

1. Empirical. Research is based on direct experience or observation by the researcher.
2. Logical. Research is based on valid procedures and principles.
3. Cyclical. Research is a cyclical process because it starts with a problem and ends with a problem.
4. Analytical. Research utilizes proven analytical procedures in gathering the data, whether historical, descriptive, experimental and case study.
5. Critical. Research exhibits careful and precise judgment.
6. Methodical. Research is conducted in a methodical manner without bias using systematic method and procedures.
7. Replicability. The research design and procedures are replicated or repeated to enable the researcher to arrive at valid and conclusive results.

Qualities of a Good Researcher

1. Research-oriented
2. Efficient
3. Scientific
4. Effective
5. Active
6. Resourceful
7. Creative

8. Honest
9. Economical
10. Religious.

Characteristics of the Researcher

1. Intellectual Curiosity. A researcher undertakes a deep thinking and inquiry of the things, and situations around him.
2. Prudence. The researcher is careful to conduct his research study at the right time and at the right place wisely, efficiently, and economically.
3. Healthy Criticism. The researcher is always doubtful as to the truthfulness of the results.
4. Intellectual Honesty. An intelligent researcher is honest to collect or gather data or facts in order to arrive at honest results.
5. Intellectual Creativity – A productive and resourceful investigator always creates new researches.

Values of Research to Man

1. Research improves the quality of life.
2. Research improves instruction.
3. Research improves students' achievement.
4. Research improves teacher's competence.
5. Research satisfies man's needs.
6. Research reduces the burden of work.
7. Research has deep-seated psychological aspects.
8. Research improves the exportation of food products.
9. Research responds to the economic recovery and austerity measure of the country.
10. Research trains graduates to become responsive to the economic development of the country and to compete globally.

Types of Research

1. Basic Research. This is also called as “fundamental research” or “pure research”. It seeks to discover basic truths or principles.

Examples:

- Boyle's Law
- Charles' Law
- Archimedes' Principle
- Hooke's Law
- Newton's Law

2. Applied Research. This type of research involves seeking new applications of scientific knowledge to the solution of a problem, such as a development of a new system or procedure, new device, or new method in order to solve the problem.

3. **Developmental Research.** This is a decision-oriented research involving the application of the steps of the scientific method in response to an immediate need to improve existing practices.

Classification of Research

1. **Library Research.** This is done in the library where answers to specific questions or problems of the study are available.
2. **Field Research.** Here, research is conducted in a natural setting.
3. **Laboratory Research.** The research is conducted in artificial or controlled conditions by isolating the study in a thoroughly specified and equipped area. The purposes are: (1) to test hypotheses derived from theory, (2) to control variance under research conditions, and (3) to discover the relations between the dependent and the independent.

The Variable

Meaning of Variable

A variable is defined as a quantity susceptible of fluctuation or change in value or magnitude under different conditions.

Types of Variable

1. **Independent Variable.** This is the stimulus variable which is chosen by the researcher to determine the relationship to an observed phenomenon.
2. **Dependent Variable.** This is the response variable which is observed and measured to determine the effect of the independent variable.
3. **Moderator Variable.** This is a secondary or special type of independent variable chosen by the researcher to determine if it changes or modifies the relationship between the independent and dependent variables.
4. **Control Variable.** This is the variable that is controlled by the investigator in which the effects can be neutralized by eliminating or removing the variable.
5. **Intervening Variable.** This is a variable which interferes with the independent and dependent variables, but its effects can either strengthen or weaken the independent and dependent variables.

Components of the research process

1. Problem/ Objectives
2. Hypotheses
3. Theoretical/ Conceptual Framework
4. Assumptions
5. Review of Related Literature
6. Research Design

7. Data Collection
8. Data Processing and Statistical Treatment
9. Analysis and Interpretation
10. Summary, Conclusions and Recommendations.

Research: Meaning and Characteristics

John W. Best and James V. Kahn in their book entitled “Research in Education” says “Research is a more systematic activity that is directed toward discovery and the development of an organized body of knowledge.

Research may be defined as the systematic and objective analysis and recording of controlled observations that may lead to the development of generalizations, principles, or theories, resulting in prediction and possibly ultimate control of events.”

The characteristics of research according to John W. Best and James V. Kahn are:

1. Research is directed toward the solution of a problem.
2. Research emphasizes the development of generalizations, principles, or theories that’ will be helpful in predicting future occurrences. Research usually goes beyond the specific objects, groups, or situations investigated and infers characteristics of a target population from the sample observed. Research is more than information retrieval, the simple gathering of information.
3. Research is based upon observable experience or empirical evidence. Certain interesting questions do not lend themselves to research procedures because they cannot be observed.
4. Research demands accurate observation and description. Researchers may choose to use quantitative measuring devices when possible. When this is not possible or appropriate to answer the’ researchers question, they may choose from a variety of qualitative, or non quantitative, descriptions of their observations. Good research utilizes valid and reliable data gathering procedures.
5. Research involves gathering new data from primary or first-hand sources or using existing data for a new purpose. The students are expected to read a number of encyclopedias, books, or periodical references and to synthesize the information in a written report. Merely reorganizing or restating what is already known and has already been written, valuable as it may be as a learning experience, is not research. It adds nothing to what is known.
6. Although research activity may at times be somewhat random and unsystematic, it is more often characterized by carefully designed procedures that apply rigorous analysis.
7. Research requires expertise. The researcher knows what is already known about the problem and how others have investigated it. He or she has searched the related literature carefully and is also thoroughly grounded in the terminology, concepts, and technical skills necessary to understand and analyze the data gathered.

8. Research strives to be objective and logical, applying every possible test to validate the procedures employed, the data collected, and the conclusions reached. The researcher attempts to eliminate personal bias. There is no attempt to persuade or to prove an emotionally held conviction. The emphasis is on testing rather than on proving the hypothesis.
9. Research involves the quest for answers to unsolved problems. However, previous important studies are deliberately repeated, 'using identical or similar procedures, with different subjects, -different settings, and at a different time. This process is replication, a fusion of the words repetition and duplication. Replication is always desirable to "confirm or to raise questions about the conclusions of a previous study.
10. Research is characterized by patient and unhurried activity. It is rarely spectacular, and researchers must expect disappointment and discouragement as they pursue the answers to difficult questions.
11. Research is carefully recorded and reported. Each important term is defined, limiting factors are recognized, procedures are described in detail, references are carefully documented, results are objectively recorded, and conclusions are presented with scholarly Caution and restraint. The written report and accompanying data are made available to the scrutiny of associates or other scholars. Any competent scholar will have the information necessary to analyze, evaluate, and even replicate the study.
12. Research sometimes requires courage. The history of science reveals that many important discoveries were made in spite of the opposition of political and religious authorities.

The Polish scientist Copernicus (1473- 1543) was condemned by church authorities when he announced his conclusion concerning the nature of the solar system-.

His theory, in direct conflict with the older Ptolemaic theory, held that the sun, not the earth, was the center of the solar system. Copernicus angered supporters of prevailing religious dogma, who viewed his theory as a denial of the story of creation as described in the book of Genesis. Modern researchers in such fields as genetics, sexual behavior, and even business practices have aroused violent criticism from those whose personal convictions, experiences, or observations were in conflict with some of the research conclusions.

The rigorous standards of scientific research are apparent from an examination of these characteristics.

The research worker should be a scholarly, imaginative person of the highest integrity, who is willing to spend long hours painstakingly seeking truth." The chief characteristics of educational research as described by Lulla, Murty and Taneja in their book 'Essentials of Educational Research' are presented below:

Research -

- is highly purposeful, dealing with the problems of immediate and remote concern to the teachers and educationists.
- follows a systematic process of investigation as precisely, objectively and scientifically as possible;
- involves determination of the problem to be studied, formulation of hypotheses, gathering of information and necessary data from the concerned sources and using different tools of investigation;
- employs scientific methods, objective procedures, logical arguments and inductive reasoning;
- attempts to organize the data in quantitative or qualitative terms to arrive at statistical inference;
- emphasizes the discovery of new facts or interpretation of known facts in a new perspective;
- has some underlying philosophic theory;
- depends on the ability, ingenuity and experience of the research for its conclusions and interpretations;
- demands interdisciplinary approach to solve many of its problem;
- demands subjective interpretation and deductive reasoning in some cases; and,
- uses class-rooms, schools and departments of education as the laboratories for conducting experiments studies and surveys.

Purposes of Research

John W. Best and James V. Kahn have mentioned three different types of purposes of research. These are:

Fundamental or Basic Research

If the goal of research is for the development of theories by the discovery of broad generalizations or principles, which often leads to knowledge for knowledge's sake, is the approach of basic or fundamental research. Fundamental research is usually carried on in a laboratory or other sterile environment, sometimes with animals. This type of research, which has no immediate or planned application, may later result in further research of an applied nature.

Applied Research

Applied research has most of the characteristics of fundamental research, including 'the use of sampling techniques and the subsequent inferences about the target population. However, its purpose is improving a product or a process—testing theoretical concepts in actual 'problem situations. Most educational research is applied research, for it attempts to develop generalizations about teaching-learning processes and instructional materials.

Action Research

Action research is focused on immediate application, not on the development of theory or on general application. It has placed its emphasis on a problem here and now in a local setting. Its findings are to be evaluated in terms of local applicability, not universal validity.

Types of Educational Research

John W. Best and James V. Kahn say “Any attempt to classify types of educational research poses a difficult problem”. They agree the fact that practically every textbook suggests-a different system of classification provides convincing evidence that there is no generally accepted scheme.

According to them “it should be noted that the system of classification is not important in itself but only has value in making the analysis of ‘research processes more comprehensible. Actually, all research activities involve the elements of observation, description, and the analysis of what happens under certain circumstances. Practically, all studies fall under one, or a combination, of the following types.

1. Historical research describes what was. The process involves investigating, recording, analyzing, and interpreting the events of the past for the purpose of discovering generalizations that are helpful in understanding the past and the present, and, to a limited extent, in anticipating the future.
2. Quantitative descriptive research uses quantitative methods to describe what is, describing, recording, analyzing, and interpreting conditions, that exist. It involves some type of comparison or contrast and attempts to discover relationships between existing non-manipulated variables. Some form of statistical analysis is used to describe the results of the study.
3. Qualitative descriptive research uses non-quantitative methods to describe what is. Qualitative descriptive research uses systematic procedures to discover non-quantifiable relationships between existing variables.
4. Experimental research describes what will be when certain variables are carefully controlled or manipulated. The focus is on variable relationships. As defined here, deliberate manipulation is always a part of the experimental method.

Definition of Research

Research can be defined as the search for knowledge, or as any systematic investigation, with an open mind, to establish novel facts, usually using a scientific method. The primary purpose for applied research is discovering, interpreting, and the development of methods and systems for the advancement of human knowledge on a wide variety of scientific matters of our world and the universe. Scientific research relies on the application of the scientific method, a harnessing of curiosity. This research provides scientific information and theories for the explanation of the nature and the

properties of the world around us. It makes practical applications possible. Scientific research is funded by public authorities, by charitable organizations and by private groups, including many companies. Scientific research can be subdivided into different classifications just as to their academic and application disciplines. Artistic research can take form when creative works are considered both the research and the object of research itself. It is the debatable body of thought which offers an alternative to purely scientific methods in research in its search for knowledge and truth. Historical research is embodied in the scientific method. The phrase my research is also used loosely to describe a person's entire collection of information about a particular subject.

Significance of Research

The purpose of research is to inform action. Thus, your study should seek to contextualize its findings within the larger body of research. Research must always be high quality in order to produce knowledge that is applicable outside of the research setting with implications that go beyond the group that has participated in the research. Furthermore, the results of your study should have implications for policy and project implementation.

One problem that often plagues progress in global health is the slow translation of research into practice. Oftentimes, a disconnect exists between those who create the evidence base and those who are positioned to implement the research findings. The underlying problem is in “the way in which the production of evidence is organized institutionally with highly centralized mechanisms, whereas the application of that science is highly decentralized.

This social distance prevails because scientists are more oriented to the international audiences of other scientists for which they publish than to the needs of practitioners, policy makers, or the local public.” Thus, as researchers, it is imperative to take steps to overcome this barrier. Publishing your study may be one initial step to make your research known to the global community. Other proactive measures can be taken to encourage the uptake of evidence-based interventions. For example, you can present your research findings at various venues such as the Unite for Sight sponsored Global Health and Innovation Conference. Furthermore, you can send the results of your study to local officials, policy-makers, and community leaders.

Goals of Research

There are relatively few published studies about eye care in developing countries, and Unite For Sight encourages all volunteers to consider developing a research study to contribute important knowledge to the eye care community on a global scale. Pursuing a research project will be a challenging and rewarding experience, and this opportunity enables you to pursue an in-depth original study about a topic of interest.

Well-conducted research is vital to the success of global health endeavours. Not only does research form the foundation of program development and policies all over the

world, but it can also be translated into effective global health programs. Research draws its power from the fact that it is empirical: rather than merely theorizing about what *might* be effective or what *could* work, researchers go out into the field and design studies that give policymakers hard data on which they can base their decisions. Furthermore, good research produces results that are examinable by peers, methodologies that can be replicated, and knowledge that can be applied to real-world situations. Researchers work as a team to enhance our knowledge of how to best address the world's problems.

The “Iterative” Process of Research

Ultimately, the key to a successful research project lies in iteration: the process of returning again and again to the research questions, methods, and data, which leads to new ideas, revisions and improvements. It is easy to think of research as a step-by-step “1,2,3” process, but it is important to be fluid and open to change. Oftentimes, by discussing the research project with advisers and peers one will find that new research questions need to be added, variables need to be omitted, and other changes made. As a proposed study is examined and re-examined from different perspectives, it may begin to transform and take a different shape. This is to be expected and is a component of a good research study. In addition, it is important to examine study methods and data from different viewpoints to ensure a comprehensive approach to the research question. In conclusion, there is no one formula for developing a successful research study, but it is important to realize that the research process is cyclical and iterative.

Research Limitations

Every study, no matter how well it is conducted, has some limitations. This is why it does not seem reasonable to use the words “prove” and “disprove” with respect to research findings. It is always possible that future research may cast doubt on the validity of any hypothesis or the conclusions from a study.

Limitations of Case Studies

Case studies may be viewed as having the most limitations. You cannot make causal conclusions from case studies. This is true because we cannot rule out alternative explanations. It is always unclear about the generality of the findings of a case study. A case study involves the behaviour of one person. The behaviour of one person may not reflect the behaviour of most people. Thus, we do not know how others may behave.

Limitations of Correlational Studies

Correlational research also has the same limitations as case studies. Correlational research merely demonstrates that we can predict a variable from another variable. It is demonstrating that two variables are associated. However, two variables can be associated without there being a causal relationship between the variables. We cannot

make causal conclusions from correlational findings because we cannot rule out all alternative explanations for correlational findings. Thus, making causal conclusions from correlational findings is a logical error. If we find that A is associated with B, it could mean that A caused B, B caused A, or some third variable caused both A and B without there being any causal relationship between A and B. Even if we could rule out one of the possible relationships, we cannot rule out all alternative explanations from correlational studies. For every correlational study, there is the possibility that some third variable caused the two variables without there being a causal relationship between the variables.

Correlational research may also have limitations with respect to the generality of the findings. Perhaps the study involved a specific group of people, or the relation between the variables were only investigated in some situations. Thus, it may be uncertain whether the correlational findings may generalize to other people or situations.

Limitations of Randomized Experiments

Experiments involving the random assignment of participants to conditions may allow us to make causal conclusions if the variables that are manipulated are not confounded with other variables. However, there still may be limitations with respect to the generality of the findings. The experiment may have involved a specific group of people, certain situations, and only some of the possible conceptualizations of variables. Thus, we may not know whether the findings will be generalized to other people, situations, or conceptualizations of the variables.

Research and Development

The phrase research and development (also R and D or, more often, R&D), according to the Organization for Economic Co-operation and Development, refers to “creative work undertaken on a systematic basis in order to increase the stock of knowledge, including knowledge of man, culture and society, and the use of this stock of knowledge to devise new applications”. Research and development is often scientific or towards developing particular technologies and is frequently carried out as corporate or governmental activity.

Overview

New product design and development is more often than not a crucial factor in the survival of a company. In an industry that is changing fast, firms must continually revise their design and range of products. This is necessary due to continuous technology change and development as well as other competitors and the changing preference of customers. Without an R&D program, a firm must rely on strategic alliances, acquisitions, and networks to tap into the innovations of others.

A system driven by marketing is one that puts the customer needs first, and only produces goods that are known to sell. Market research is carried out, which establishes

what is needed. If the development is technology driven then it is a matter of selling what it is possible to make. The product range is developed so that production processes are as efficient as possible and the products are technically superior, hence possessing a natural advantage in the market place.

R&D has a special economic significance apart from its conventional association with scientific and technological development. R&D investment generally reflects a government's or organization's willingness to forgo current operations or profit to improve future performance or returns, and its abilities to conduct research and development.

The top eight spenders in terms of percentage of GDP were Israel (4.53%), Sweden (3.73%), Finland (3.45%) Japan (3.39%), South Korea (3.23%), Switzerland (2.9%), Iceland (2.78%) and United States (2.62%). The Commitment to Development Index ranks these countries, rewarding them for research and development that support the creation and dissemination of innovations of value to developing countries.

In general, R&D activities are conducted by specialized units or centers belonging to companies, universities and state agencies. In the context of commerce, "research and development" normally refers to future-oriented, longer-term activities in science or technology, using similar techniques to scientific research without predetermined outcomes and with broad forecasts of commercial yield.

Statistics on organizations devoted to "R&D" may express the state of an industry, the degree of competition or the lure of progress. Some common measures include: budgets, numbers of patents or on rates of peer-reviewed publications. Bank ratios are one of the best measures, because they are continuously maintained, public and reflect risk.

In the U.S., a typical ratio of research and development for an industrial company is about 3.5% of revenues. A high technology company such as a computer manufacturer might spend 7%. Although Allergan (a biotech company) tops the spending table with 43.4% investment, anything over 15% is remarkable and usually gains a reputation for being a high technology company. Companies in this category include pharmaceutical companies such as Merck & Co. (14.1%) or Novartis (15.1%), and engineering companies like Ericsson (24.9%). Such companies are often seen as credit risks because their spending ratios are so unusual.

Generally such firms prosper only in markets whose customers have extreme needs, such as medicine, scientific instruments, safety-critical mechanisms (aircraft) or high technology military armaments. The extreme needs justify the high risk of failure and consequently high gross margins from 60% to 90% of revenues.

That is, gross profits will be as much as 90% of the sales cost, with manufacturing costing only 10% of the product price, because so many individual projects yield no exploitable product. Most industrial companies get only 40% revenues.

On a technical level, high tech organizations explore ways to re-purpose and repackage advanced technologies as a way of amortizing the high overhead. They often reuse

advanced manufacturing processes, expensive safety certifications, specialized embedded software, computer-aided design software, electronic designs and mechanical subsystems. Research has shown that firms with a persistent R&D strategy outperform those with an irregular or no R&D investment programme.

Pharmaceuticals

Research often refers to basic experimental research; development refers to the exploitation of discoveries. Research involves the identification of possible chemical compounds or theoretical mechanisms.

In the United States, universities are the main provider of research level products. In the United States, corporations buy licences from universities or hire scientists directly when economically solid research level products emerge and the development phase of drug delivery is almost entirely managed by private enterprise.

Development is concerned with proof of concept, safety testing, and determining ideal levels and delivery mechanisms. Development often occurs in phases that are defined by drug safety regulators in the country of interest. In the United States, the development phase can cost between \$10 to \$200 million and approximately one in ten compounds identified by basic research pass all development phases and reach market.

Pharmaceuticals market, however are extremely complex in many respects. Large public sector investment in biomedical R & D influence private companies' choices about what to work and how intensively to invest in research and development. The returns on private sector R & D are attractive, on average but they vary considerably from one drug to the next. Consumer demand for prescription drug is often indirect, mediated by doctors and health insurers. New drug must undergo costly and time consuming testing before they sold.

Business

Research and development is nowadays of great importance in business as the level of competition, production processes and methods are rapidly increasing. It is of special importance in the field of marketing where companies keep an eagle eye on competitors and customers in order to keep pace with modern trends and analyse the needs, demands and desires of their customers.

Unfortunately, research and development are very difficult to manage, since the defining feature of research is that the researchers do not know in advance exactly how to accomplish the desired result. As a result, higher R&D spending does not guarantee "more creativity, higher profit or a greater market share".

R&D Alliance

An R&D alliance is a mutually beneficial formal relationship between two or more parties to pursue a set of agreed goals while remaining independent organisations, where acquiring new knowledge is a goal by itself.

The different parties agree to combine their knowledge to create new innovative products. Thanks to funding from government organizations, like the European Union's Seventh Framework Programme (FP7), and modern advances in technology, R&D alliances have now become more efficient. Every company has their own research and develop department.

R&D Reorientation

Industrial sector is transforming from vertical integration to horizontal structure. So far this transformation has taken place in manufacturing, but now it's the turn for R&D to make a paradigm shift. The restructuring trends of manufacturing predict the future of R&D to be focused, financially independent, universally expanded and distributed.

In 2003 on July the 25th with their article titled Reorienting R&D for a Horizontal Future Downey, Greenberg and Kapur illustrated that a horizontal structure of development in all industrial disciplines is required because of emerging markets, cost cutting needs and globalization. Collaborative networks in a decentralized environment are producing better results because of flexibility and cost reduction. R&D is spreading across the globe for faster development and unique innovations. Building block structure is getting popular in R&D because it is more manageable and focused.

In a challenging market environment after manufacturing now companies are eyeing to reduce the development cost where R&D needs to justify its existence as an independent profit centre with increased productivity and reduced operational costs. This can only be achieved through collaboration and trade-offs. Formation of horizontal collaborative network is challenging because it incorporates the issues of adaptability, security, cultural differences, varying standards, trust and ownership. Globalization is pushing R&D to concentrate on global design approach with the design flexibility where products can be tailored as per different geographic markets.

Innovating products in response to market requirements will keep organizations in business whereas knowing their own productivity and capacity will enable them to operate more effectively. Firms can make better investment decisions if they can predict their development cost. This will directly add to the bottom line of the business. In a race to improve the productivity, lower the cost and save time, firms should not forget to retain the expertise of their core competencies. The proper synchronization of global resources must be addressed in order to save time and money. Companies must support the development of innovation management system for future growth.

Companies are decentralizing their R&D units across the globe to extract the time and diversity benefits. Customer centric orientation is most profitable approach for any firm to operate in a competitive environment. R&D is an area where 'The World is Flat' theory applies perfectly.

World is shrinking with the increasing pace of information technology, and the same time decentralization is making our days longer. In this fast pace environment managing intellectual property is more critical for businesses to keep themselves in the race.

Never ending attempts should be made to optimize the development with minimal resources. Flattening the innovation process will result in technological development with greater profits and efficiency.

Scientific Research

Generally, research is understood to follow a certain structural process. Though step order may vary depending on the subject matter and researcher, the following steps are usually part of most formal research, both basic and applied:

1. Observations and Formation of the topic
2. Hypothesis
3. Conceptual definitions
4. Operational definition
5. Gathering of data
6. Analysis of data
7. Test, revising of hypothesis
8. Conclusion, reiteration if necessary.

A common misunderstanding is that by this method a hypothesis could be proven or tested. Generally a hypothesis is used to make predictions that can be tested by observing the outcome of an experiment. If the outcome is inconsistent with the hypothesis, then the hypothesis is rejected. However, if the outcome is consistent with the hypothesis, the experiment is said to support the hypothesis. This careful language is used because researchers recognize that alternative hypotheses may also be consistent with the observations. In this sense, a hypothesis can never be proven, but rather only supported by surviving rounds of scientific testing and, eventually, becoming widely thought of as true. A useful hypothesis allows prediction and within the accuracy of observation of the time, the prediction will be verified. As the accuracy of observation improves with time, the hypothesis may no longer provide an accurate prediction. In this case a new hypothesis will arise to challenge the old, and to the extent that the new hypothesis makes more accurate predictions than the old, the new will supplant it.

Exploratory Research

Exploratory research is a type of research conducted for a problem that has not been clearly defined. Exploratory research helps determine the best research design, data collection method and selection of subjects. It should draw definitive conclusions only with extreme caution. Given its fundamental nature, exploratory research often concludes that a perceived problem does not actually exist.

Exploratory research often relies on secondary research such as reviewing available literature and/or data, or qualitative approaches such as informal discussions with consumers, employees, management or competitors, and more formal approaches through in-depth interviews, focus groups, projective methods, case studies or pilot studies.

The Internet allows for research methods that are more interactive in nature. For example, RSS feeds efficiently supply researchers with up-to-date information; major search engine search results may be sent by email to researchers by services such as Google Alerts; comprehensive search results are tracked over lengthy periods of time by services such as Google Trends; and websites may be created to attract worldwide feedback on any subject.

The results of exploratory research are not usually useful for decision-making by themselves, but they can provide significant insight into a given situation. Although the results of qualitative research can give some indication as to the “why”, “how” and “when” something occurs, it cannot tell us “how often” or “how many”.

Exploratory research is not typically generalizable to the population at large. Social exploratory research “seeks to find out how people get along in the setting under question, what meanings they give to their actions, and what issues concern them. The goal is to learn ‘what is going on here?’ and to investigate social phenomena without explicit expectations.”

This methodology is also at times referred to as a grounded theory approach to qualitative research or interpretive research, and is an attempt to unearth a theory from the data itself rather than from a predisposed hypothesis.

Earl Babbie identifies three purposes of social science research. The purposes are exploratory, descriptive and explanatory. Exploratory research is used when problems are in a preliminary stage. Exploratory research is used when the topic or issue is new and when data is difficult to collect. Exploratory research is flexible and can address research questions of all types (what, why, how). Exploratory research is often used to generate formal hypotheses. Shields and Tajalli link exploratory research with the conceptual framework working hypothesis.

Applied Research

Applied research in administration is often exploratory because there is need for flexibility in approaching the problem. In addition there are often data limitations and a need to make a decision within a short time period. Qualitative research methods such as case study or field research are often used in exploratory research.

There are three types of objectives in a marketing research project.

- Exploratory research or formulative research
- Descriptive research
- Causal research.

Exploratory research or formulative research: The objective of exploratory research is to gather preliminary information that will help define problems and suggest hypotheses.’

Descriptive Research: The objective of descriptive research is to describe things, such as the market potential for a product or the demographics and attitudes of consumers who buy the product.’

Causal research: The objective of causal research is to test hypotheses about cause-and-effect relationships.

Constructive Research

Constructive research is perhaps the most common computer science research method. This type of approach demands a form of validation that doesn't need to be quite as empirically based as in other types of research like exploratory research.

Nevertheless the conclusions have to be objectively argued and defined. This may involve evaluating the "construct" being developed analytically against some predefined criteria or performing some benchmark tests with the prototype.

The term "construct" is often used in this context to refer to the new contribution being developed. Construct can be a new theory, algorithm, model, software, or a framework.

The "fuzzy info from many sources" tab refers to different info sources like training materials, processes, literature, articles, working experience etc. In the "solution" tab, "theoretical framework" represents a tool to be used in the problem solving.

The "practical relevance" tab it refers to empirical knowledge creation that offers final benefits. The "theoretical relevance" tab it gives the new theoretical knowledge that needs scientific acceptance: the back arrow to "theoretical body of knowledge" tab.

Steps to be followed in "practical utility" tab (a):

- set objectives and tasks
- identify process model
- select case execution
- interview case organization
- prepare simulation
- run simulation
- interpret simulation results
- give feedback.

Steps to be followed in "epistemic utility" tab (b):

- constructive research
- case research
- surveys
- qualitative and quantitative methods
- theory creating
- theory testing.

Empirical Research

Empirical research is a way of gaining knowledge by means of direct and indirect observation or experience. Empirical evidence (the record of one's direct observations or experiences) can be analyzed quantitatively or qualitatively. Through quantifying

the evidence or making sense of it in qualitative form, a researcher can answer empirical questions, which should be clearly defined and answerable with the evidence collected (usually called data). Research design varies by field and by the question being investigated. Many researchers combine qualitative and quantitative forms of analysis to better answer questions which cannot be studied in laboratory settings, particularly in the social sciences and in education.

In some fields, quantitative research may begin with a research question (e.g., “Does listening to vocal music during learning a word list have an effect on later memory for these words?”) which is tested through experimentation in a lab. Usually, a researcher has a certain theory regarding the topic under investigation. Based on this theory some statements, or hypotheses, will be proposed (e.g., “Listening to vocal music has a negative effect on learning a word list.”). From these hypotheses predictions about specific events are derived (e.g., “People who study a word list while listening to vocal music will remember fewer words on a later memory test than people who study a word list in silence.”). These predictions can then be tested with a suitable experiment. Depending on the outcomes of the experiment, the theory on which the hypotheses and predictions were based will be supported or not.

Terminology

The term empirical was originally used to refer to certain ancient Greek practitioners of medicine who rejected adherence to the dogmatic doctrines of the day, preferring instead to rely on the observation of phenomena as perceived in experience. Later empiricism referred to a theory of knowledge in philosophy which adheres to the principle that knowledge arises from experience and evidence gathered specifically using the senses.

In scientific use the term empirical refers to the gathering of data using only evidence that is observable by the senses or in some cases using calibrated scientific instruments. What early philosophers described as empiricist and empirical research have in common is the dependence on observable data to formulate and test theories and come to conclusions.

Usage

The researchers attempts to describe accurately the interaction between the instrument (or the human senses) and the entity being observed. If instrumentation is involved, the researcher is expected to calibrate his/her instrument by applying it to known standard objects and documenting the results before applying it to unknown objects. In other words, it describes the research that has not been taken place before and their results.

In practice, the accumulation of evidence for or against any particular theory involves planned research designs for the collection of empirical data, and academic rigor plays a large part of judging the merits of research design. Several typographies for such designs have been suggested, one of the most popular of which comes from Campbell

and Stanley (1963). They are responsible for popularizing the widely cited distinction among pre-experimental, experimental, and quasi-experimental designs and are staunch advocates of the central role of randomized experiments in educational research.

Scientific Research

Accurate analysis of data using standardized statistical methods in scientific studies is critical to determining the validity of empirical research. Statistical formulas such as regression, uncertainty coefficient, t-test, chi square, and various types of ANOVA (analyses of variance) are fundamental to forming logical, valid conclusions. If empirical data reach significance under the appropriate statistical formula, the research hypothesis is supported. If not, the null hypothesis is supported (or, more correctly, not rejected), meaning no effect of the independent variable(s) was observed on the dependent variable(s).

It is important to understand that the outcome of empirical research using statistical hypothesis testing is never proof. It can only *support* a hypothesis, *reject* it, or do neither. These methods yield only probabilities. Among scientific researchers, empirical *evidence* (as distinct from empirical *research*) refers to objective evidence that appears the same regardless of the observer. For example, a thermometer will not display different temperatures for each individual who observes it. Temperature, as measured by an accurate, well calibrated thermometer, is empirical evidence. By contrast, non-empirical evidence is subjective, depending on the observer. Following the previous example, observer A might truthfully report that a room is warm, while observer B might truthfully report that the same room is cool, though both observe the same reading on the thermometer. The use of empirical evidence negates this effect of personal (i.e., subjective) experience. Ideally, empirical research yields empirical evidence, which can then be analyzed for statistical significance or reported in its raw form.

Philosophy of Research

You probably think of research as something very abstract and complicated. It can be, but you'll see that if you understand the different parts or phases of a research project and how these fit together, it's not nearly as complicated as it may seem at first glance. A research project has a well-known structure—a beginning, middle and end. We introduce the basic phases of a research project in *The Structure of Research*. We also introduce some important distinctions in research: the different *types of questions* you can ask in a research project; and, the major components or parts of a research project.

Before the modern idea of research emerged, we had a term for what philosophers used to call research—logical reasoning. So, it should come as no surprise that some of the basic distinctions in logic have carried over into contemporary research. In *Systems of Logic* we discuss how two major logical systems, the inductive and deductive methods of reasoning, are related to modern research.

OK, you knew that no introduction would be complete without considering something having to do with assumptions and philosophy. (I thought I very cleverly snuck in the stuff about logic in the last paragraph). All research is based on assumptions about how the world is perceived and how we can best come to understand it.

Of course, nobody really *knows* how we can best understand the world, and philosophers have been arguing about that very question for at least two millennia now, so all we're going to do is look at how most contemporary social scientists approach the question of how we know about the world around us. We consider two major philosophical schools of thought—Positivism and Post-Positivism—that are especially important perspectives for contemporary social research (OK, I'm only considering positivism and post-positivism here because these are the major schools of thought. Forgive me for not considering the hotly debated alternatives like relativism, subjectivism, hermeneutics, deconstructivism, constructivism, feminism, etc.

If you really want to cover that stuff, start your own Web site and send me your URL to stick in here). Quality is one of the most important issues in research. We introduce the idea of validity to refer to the quality of various conclusions you might reach based on a research project. Here's where I've got to give you the pitch about validity.

Most students roll their eyes, curl up into a fetal position or go to sleep. They think validity is just something abstract and philosophical (and I guess it is at some level). But I think if you can understand validity—the principles that we use to judge the quality of research—you'll be able to do much more than just complete a research project. You'll be able to be a virtuoso at research, because you'll have an understanding of *why* we need to do certain things in order to assure quality. You won't just be plugging in standard procedures you learned in school—sampling method X, measurement tool Y—you'll be able to help create the next generation of research technology. Enough for now—more on this later.

Structure of Research

Most research projects share the same general structure. You might think of this structure as following the shape of an hourglass. The research process usually starts with a broad area of interest, the initial problem that the researcher wishes to study. For instance, the researcher could be interested in how to use computers to improve the performance of students in mathematics. But this initial interest is far too broad to study in any single research project (it might not even be addressable in a lifetime of research). The researcher has to narrow the question down to one that can reasonably be studied in a research project. This might involve formulating a hypothesis or a focus question. For instance, the researcher might hypothesize that a particular method of computer instruction in math will improve the ability of elementary school students in a specific district.

At the narrowest point of the research hourglass, the researcher is engaged in direct measurement or observation of the question of interest. Once the basic data is collected,

the researcher begins to try to understand it, usually by analyzing it in a variety of ways. Even for a single hypothesis there are a number of analyses a researcher might typically conduct. At this point, the researcher begins to formulate some initial conclusions about what happened as a result of the computerized math programme. Finally, the researcher often will attempt to address the original broad question of interest by generalizing from the results of this specific study to other related situations. For instance, on the basis of strong results indicating that the math programme had a positive effect on student performance, the researcher might conclude that other school districts similar to the one in the study might expect similar results.

Components of a Study

What are the basic components or parts of a research study? Here, we'll describe the basic components involved in a causal study. Because causal studies presuppose descriptive and relational questions, many of the components of causal studies will also be found in those others. Most social research originates from some general problem or question.

You might, for instance, be interested in what programmes enable the unemployed to get jobs. Usually, the problem is broad enough that you could not hope to address it adequately in a single research study.

Consequently, we typically narrow the problem down to a more specific research question that we can hope to address. The research question is often stated in the context of some theory that has been advanced to address the problem. For instance, we might have the theory that ongoing support services are needed to assure that the newly employed remain employed. The research question is the central issue being addressed in the study and is often phrased in the language of theory.

For instance, a research question might be:

- Is a programme of supported employment more effective (than no programme at all) at keeping newly employed persons on the job?

The problem with such a question is that it is still too general to be studied directly. Consequently, in most research we develop an even more specific statement, called an hypothesis that describes in *operational* terms exactly what we think will happen in the study.

For instance, the hypothesis for our employment study might be something like:

- The Metropolitan Supported Employment Programme will significantly increase rates of employment after six months for persons who are newly employed (after being out of work for at least one year) compared with persons who receive no comparable programme.

Notice that this hypothesis is specific enough that a reader can understand quite well what the study is trying to assess. In causal studies, we have at least two major variables of interest, the cause and the effect. Usually the cause is some type of event, programme, or treatment. We make a distinction between causes that the researcher can control

(such as a programme) versus causes that occur naturally or outside the researcher's influence (such as a change in interest rates, or the occurrence of an earthquake). The effect is the outcome that you wish to study. For both the cause and effect we make a distinction between our idea of them (the construct) and how they are actually manifested in reality. For instance, when we think about what a programme of support services for the newly employed might be, we are thinking of the "construct."

On the other hand, the real world is not always what we think it is. In research, we remind ourselves of this by distinguishing our view of an entity (the construct) from the entity as it exists (the operationalization). Ideally, we would like the two to agree. Social research is always conducted in a social context. We ask people questions, or observe families interacting, or measure the opinions of people in a city. An important component of a research project is the units that participate in the project. Units are directly related to the question of sampling. In most projects we cannot involve all of the people we might like to involve. For instance, in studying a programme of support services for the newly employed we can't possibly include in our study everyone in the world, or even in the country, who is newly employed. Instead, we have to try to obtain a representative sample of such people.

When sampling, we make a distinction between the theoretical population of interest to our study and the final sample that we actually measure in our study. Usually the term "units" refers to the *people* that we sample and from whom we gather information. But for some projects the units are organizations, groups, or geographical entities like cities or towns. Sometimes our sampling strategy is multi-level: we sample a number of cities and within them sample families. In causal studies, we are interested in the effects of some cause on one or more outcomes.

The outcomes are directly related to the research problem—we are usually most interested in outcomes that are most reflective of the problem. In our hypothetical supported employment study, we would probably be most interested in measures of employment—is the person currently employed, or, what is their rate of absenteeism. Finally, in a causal study we usually are comparing the effects of our cause of interest (*e.g.*, the programme) relative to other conditions (*e.g.*, another programme or no programme at all).

Thus, a key component in a causal study concerns how we decide what units (*e.g.*, people) receive our programme and which are placed in an alternative condition. This issue is directly related to the research design that we use in the study. One of the central questions in research design is determining how people wind up in or are placed in various programmes or treatments that we are comparing.

LIMITATIONS OF RESEARCH

The limitations of the research are first addressed by discussing the purposes of the research, then discussing the limitations of the data management, data analysis, and the validity of the analysis.

PURPOSES OF RESEARCH

The “trustworthiness” of research depends on “What counts as knowledge?”. The general purposes of research can include knowledge production, understanding, and prediction. This study has focused on the first two purposes: production of applied knowledge and process knowledge, both of which are suitable for developmental research. Applied knowledge is context-specific, useful for the solution of practical problems, while process knowledge is usually specified in terms of models. A qualitative approach is called for in the study of context and contextual influences. A second purpose of research is promoting understanding of a process; in this case, the instructional approach and the underlying phenomena, including participation and dialogue. A secondary aspect of understanding is a better understanding of what it requires in using the development cycle as a form of inquiry.

DATA MANAGEMENT

Data management involves the procedures used for a systematic, coherent process of data collection, storage and retrieval for the purpose of high quality, accessible data, the documentation of analysis, and retention of data. The design decisions, implementation, and evaluation of the instructional design course were documented in eight data sources: working logs, electronic mail, syllabi, conferences, draft ID projects, completed ID projects, course evaluations, and self-evaluations. These data sources, which were based on observations, interviews, or documents, were event-driven, meaning that they served our instructional needs to watch, ask, and examine. These observations, interviews, and documents were in place prior to the conceptual framework of the study. As a result, the data sources were not as complete, tightly defined, or structured across the six cases if they had been researcher-driven. Some data sources, such as syllabi, course evaluations, and self-evaluations, evolved to suit the learning needs of the students.

However, because we had presented at research conferences, we had collected and stored data for each case, as well as conducted analysis with most of the data sources, although using different methodologies. These research efforts can be regarded as interim analyses in which we became familiar with procedures in recording observations and personal conferences, as well as retaining and analyzing documents. Over the six cases, we came to have a better understanding of the instructional setting, being sensitive to research opportunities and more systematic in our data collection and management efforts, but also retaining instruction and responsiveness to learners as our top priority.

DATA ANALYSIS

Within-Case Analysis

Data displays, structured summaries, and charts allowed a condensed view of the data sources and revealed that some further analysis was needed, such as coding of structured summaries to reveal themes as well as to identify exceptions and differences.

Across-Case Analysis

A means to report the changes in design decisions, implementation, and evaluation of the model across the six Cases. "Each case has a specific history-which we discard at our peril-but it is a history contained within the general principles that influence its development". This summary attempted to preserve the uniqueness of each case, yet also make comparisons along the developmental cycle. In an effort to extend external validity, what participants "did, said, or designed," were examined in multiple settings.

The processes of participation and dialogue within participation were examined in six different, developmental configurations and can be viewed as a replication of the focus of the study. A set of generalizations on how the model was implemented, as well as conditions necessary for its use. The danger to this generalization is that "multiple cases will be analysed at high levels of inference, aggregating out the local webs of causality and ending with a smoothed set of generalizations that may not apply to any single case". The goal was to better understand the overall processes at work across the cases; in this case, teacher and student thinking, participation and dialogue, and teacher responsiveness, represented within design decisions, class implementation, and model evaluation. We did not average, for example, course evaluation results, as one way to avoid misinterpretation and superficiality and to preserve case uniqueness.

Analytic Validity

In traditional instances of qualitative data collection and analysis, the research "shifts between cycles of inductive data collection and analysis to deductive cycles of testing and verification".

In this study, sources of data were already in place prior to conceptualizing a conceptual study framework. However, in this study the details of the conceptual framework and the subsequent data analysis of the six cases cycled back and forth to realise more appropriate matches of methodology and method to existing data sources and research objectives. The analytic cycle for this study could be better described as one which moved between conceptual framework, case analysis, and being clear as to the purpose of the study. Although being clear as to the purpose of the study is preferable before constructing a methodology, such purposes are not always clear based on the complexity of processes to be studied, amount of data, and personal involvement over time.

Potential shortcomings in this research that are sources for bias include the large amount of data which may have led to missing important information or overweighting some findings due to focusing on a particular and large set of data. Personal involvement with the course also increases the possibility that our recorded observations in working logs highlighted particular incidents while ignoring others. On the other hand, the working log recorded observations or design decisions that would have been lost to our collective memories over the five years of involvement with the course. Personal involvement as a co-instructor also implied a danger in being selective and overconfident

with some data. Another shortcoming was not checking descriptions with each case of students and additional peer review outside of the co-instructor.

To address these shortcomings, we used multiple data sources for triangulation to achieve an agreement of one data source with another. Multiple sources of data, such as working logs, email, and syllabi, also provided different strengths and complemented each other. Syllabi compactly recorded design decisions, while working logs and email documented our thinking that influenced these design decisions. The data sources were a mix of student-generated and instructor-generated data. A significant amount of time and effort was spent in carefully assembling the data reduction notebooks, which included data displays of structured summaries and charts.

The effort was made towards a goal of being able to track in the documents the reasoning behind descriptions and summary and the generalizations made about the model.

During the analysis of these data sources we looked for contrasts, comparisons, and exemplars and reported these during the data reduction so as not to filter out outliers and extreme instances. For example, in the ID project analysis we added a column to record any unique features of the project that might not be addressed by the completeness, consistency, and coherency criteria. Replication of the conceptual framework across multiple cases helped to provide evidence that what was described in one Case was based upon the details of the instructional approach and uniqueness of the setting and participants.

We attempted to remain "descriptive" and to report what decisions were made for the ID course, what occurred during the implementation of the reflexive approach, and systematically report the evaluation of the model in terms of student performance on ID projects, their perceptions of their learning and the course and data informing our responsivity to learner needs. Another means of addressing verification of findings and ceases was an "auditing" by the dissertation chair who was also co-instructor in the cases under study. Through periodic reviews of methodology and analysis, numerous inconsistencies in design decisions, for example, were identified and prompted for clarification. Feedback also prompted me to clarify procedures used to analyse the different data sources. This feedback, although one of the responsibilities of a doctoral chairperson, is another aspect of the reflexive stance that we had used on previous research: the need to assume regular, ongoing, and self-conscious documentation of our teaching. The working logs also served as a "reflexivity journal" to record these efforts.

TYPES OF RESEARCH

Research can be classified by purpose or by method. If we categorise it by purpose, it would fall into two major categories: Basic Research and Applied Research, while in case of method, it would be deductive research and inductive research.

BASIC RESEARCH

Also called Pure or fundamental Research, it is undertaken for increase in knowledge. There is no direct benefit as it is a research for the sake of research.

It is conducted to satisfy any curiosity such as:

- What makes things happen,
- Why society changes and
- Why social relations are in a certain way.

In fact, it is the source of most new theories, principles and ideas. Basic research rarely helps anyone directly. It only stimulates new ways of thinking. The main motivation is to *expand man's knowledge*. There is absolutely no commercial value to the discoveries resulting from such research.

However, in the long run, it forms the basis of applied research or development commercial products. If basic work is done first, then applied spin-offs often eventually result from this research.

As Dr. George Smoot of LBNL says, "People cannot foresee the future well enough to predict what's going to develop from basic research. If we only did "applied research", we would still be making better spears."

To sum up, basic research is purely theoretical to increase our understanding of certain phenomena or behaviour but does not seek to solve any existing problem.

APPLIED RESEARCH

It is use of basic research or past theories, knowledge and methods for solving an existing problem. It deals with practical problems. It is opposed to pure research which is not problem-oriented but for the increase in knowledge which may or may not be used in future.

In the present world situation, more emphasis is being given to applied research to solve problems arising out of overpopulation and scarcity of natural resources. Applied research should not be treated the same as Research and Development which is involved in developing products demanded by the existing clients.

Applied Research, on the other hand, focuses on uncovering what needs are not being met and use that information in designing products or services that would create their own demand. Thus, applied research brings in new customers and also provides better products and services to the existing customers. In old days, the mobile phone was expensive, bulky and had a short range. Applied Research foresaw that this product would have a limited market and stressed on cost-cutting, reduced weight and long-distance communication. Such measures caused a heavy demand.

METHODS OR APPROACHES

In research, there are based two methods known as the *deduction* and *induction*. Both are widely used in research projects. This helps the researchers to understand, explain, or predict business phenomena. Deduction follows an approach which is "top-

down” or “from general to specific.” On the other hand, induction is “bottom-up” in nature or from specific to general. In deduction, we start from a theory and try to prove it right with the help of available information. In induction, we observe some happening, deduct a pattern and draw a conclusion.

Example of Deduction

- All men are mortal.
- Socrates is a man
- Socrates is mortal

Example of Induction

- This ice is cold.
- All ice is cold.

Take another example: $3 + 5 = 8$ and eight is an even number. Therefore, an odd number added to another odd number will result in an even number.

**Example of Both Deduction
And Induction**

- You push the light switch and find no light
- You ask the question, Why no light?
- You infer a reason to answer the question and explain the fact that the bulb is burned out.
- You use this hypothesis to conclude that the light will not go on when we push the switch. We know from experience that burned-out bulb will not light.

DEDUCTIVE METHOD

Like Sherlock Holmes, the starting point is a given situation and moving to a specific conclusion. All relevant information is assembled, studied to solve the mystery. Aristotle, a Greek philosopher, described deduction as “drawing conclusions by applying rules or principles; logically moving from a general rule or principle to a specific solution”. It is the process of reaching a conclusion that is guaranteed to follow, if the evidence provided is true and the reasoning used to reach the conclusion is correct.

An investigator is distracted with the noise and thinks that “noise adversely affects mental problem solving.” It can be turned into a research question like “would it be more conducive to solve mental puzzle when there is no noise.”

In order to test it, two puzzles are given to the same group, one in noise-less conditions and other in a blaring noise. If the results of test conducted under noisy-environment are poor it can be deduced or concluded that controlling the aversive noise is helpful in solving mental puzzles. Someone may say “why research, it is apparent?” It is logical to say it but it had been previously proved and awareness has been created of the noise pollution. In fact, noise can be measured in terms of decibel (dB) and by research, it has

been established how much noise is tolerable under what conditions. Zero dB describes perfect silence while 140 dB would reflect a gunshot.

In deductive method, the premises provide a guarantee of the true cease.

Consider the following example:

- There are 32 books on the top-shelf of the book case,
- There are 12 books on the lower-shelf of the book case.
- There are no books anywhere on the bookcase

It can be concluded there are 44 books in the bookcase.

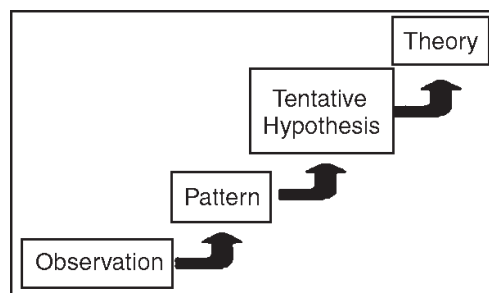


Fig. Inductive Approach

INDUCTIVE METHOD

Also known as Inductive reasoning or logic, this method involves moving from a specific situation to a general cease. This is a way of “theory-building”, whereby “specific facts are used to create a theory that explains relationships between the facts and allows prediction of future knowledge”. This method is more open-ended and exploratory. This does not give 100% guarantee of truth but probability of being true.

Here is an example:

- Socrates was Greek.
- Most Greeks eat fish.
- Socrates ate fish.

While the premises were true, it is possible that cease was false. Maybe Socrates was allergic to fish, for example. It can be said that inductive approach is weaker than deductive approach as there is possibility of arriving at false cease. However, it is not entirely true.

2

Research Design

INTRODUCTION

In our analogy between research and travel we compared the research design of a project to a journey planner or itinerary. If we consider what goes into the planning of a journey we get some idea of the functions of a research design. On having decided on my destination, I must, as a traveller, consider the best route by means of which to reach it.

‘Best’ implies taking into consideration factors such as time of the year, costs, mode of transportation and the route. In planning the journey I am constrained by what my travel agent and I know about the route from existing maps and guidebooks, and by resources like time and money. But I shall eventually produce an itinerary that meets my needs and which I will follow to reach my destination. A research design is like a route planner.

It is a set of guidelines and instructions on how to reach the goal that I have set for myself. The notions of ‘plan’ and ‘design’ are commonly used in the construction industry where drawing up building plans or architectural designs invariably precedes the actual construction of a building.

Similarly, the research design could be viewed as the ‘blueprint’ of the research project that precedes the actual research process. Building plans and blueprints are step-by-step outlines of what needs to be done. They specify the materials and the specifications just as to which they are to be used, the critical deadlines against which particular stages must be completed, and so forth. In the construction industry such

plans and designs are necessarily very detailed. But in other spheres of everyday life, designs differ in their degree of articulation and detail.

Someone who wishes to design a new dress might have only a very general 'picture' or 'idea' in her mind of how the design should look. An artist who composes a new picture might have only the faintest of ideas when starting out. Designs differ in terms of detail and finality. A major determining factor in this process is the degree of control that is deemed necessary for the project. Control and planning are paramount considerations where the risk of error is high and disastrous consequences could result if things are not well planned.

This is typically the case where lives are at risk, as in the case of the construction of a bridge or an apartment building. In other cases, where the question of risk is less important, designs are more flexible, open-ended and less fixed. Although we do not usually encounter similar degrees of risk in the social sciences the distinction remains valid.

Certain areas of pharmaceutical and medical research are obviously high-risk fields. In highly structured research such as experimental designs aimed at testing research hypotheses, the design is a framework of clearly formulated decision steps. In a semistructured and open-ended project such as qualitative exploratory research aimed at developing new hypotheses, the design is more flexible. We can once again illustrate this with our analogy of a journey.

If I embark on a business trip which will hopefully see the conclusion of months of extended negotiations and the closing of a multimillion rand contract, I will ensure that all my travel arrangements are taken care of well in advance, even to the extent of making contingency plans should something go wrong. In contrast, if I am embarking on a leisurely trip along the West coast and I intend to explore the area in my own time, questions of meeting deadlines, making bookings, and contingency plans may not even arise.

This discussion illustrates the importance of the research problem in a project. The structure and particular logic of a research design is determined by the formulation of the research problem. The degree of structure in our design will be a direct function of the research goals that we have set for ourselves.

RESEARCH DESIGN AS MAXIMISING VALIDITY

Our construction and travelling examples have shown that the need for design and planning is most urgent when errors and mistakes have to be eliminated. This even applies to research where the design is relatively open-ended. Although it is seldom possible to plan a project in such detail that all error will be eliminated, it is usually possible to identify certain typical threats to validity and to adjust one's design accordingly. The rationale for a research design is to plan and structure a research project in such a way that the eventual validity of the research findings is maximised through either minimising or, where possible, eliminating potential error.

The Notion of Validity

We argued that ‘validity’ should be viewed as a synonym for ‘best approximation to the truth’. Very briefly, our argument there was that, although scientists work under the epistemic imperative or search for truth, there are various ontological and sociological constraints that seriously curtail the attainment of this ideal, except in the simplest cases of singular descriptive statements.

This does not mean that we should abandon the search for truth, only that we have to accept that at best our research can only produce better or worse approximations of the truth. But merely setting our aims lower—from attaining truth to aiming at approximation of the truth above absolute truth—does not by itself clarify the notion of ‘validity’.

The key question is: how does one recognise valid research? How does one know when one study is more valid than another? These are essential questions, because unless we have a clear idea what the criteria for ‘validity’ are, there is no sense in defining the function of research designs as ‘maximising validity’.

Our approach is as follows: first, we have to identify the key dimensions of validity. Secondly we argue that the only feasible way to ‘maximise validity’ is by either minimising or eliminating all foreseeable threats to validity in the research process.

The discussion suggests that we should regard ‘validity’ as a criterion that is applicable to the whole research process. One way to do this is to look at the change in the meaning of ‘validity’ when applied to each of the main stages in the research process. In the following discussion of the stages in the research process, we shall identify, in each case,

- The stage in the research process;
- The major sources of error
- The particular ‘outcomes’ or ‘products’ of that stage in the research process and
- The appropriate criterion of validity as it applies to That outcome.

We need to emphasise that ‘validity’ is an epistemic criterion, which means that it is a quality of the elements of knowledge. These knowledge elements are the products of the various stages of decision making in research. The actual process of decision making is more or less objective. Objectivity is a criterion of the process, which means that it is a methodological criterion. We would therefore argue that research uses relatively objective methods when conceptualising, sampling, defining, analysing and collecting data. Let us elaborate. In each case in the discussion, we:

- Define what is understood by the particular stage in the research process;
- Identify the epistemological criterion; and
- Identify methodological criteria appropriate to the particular stage.

Conceptualisation

‘Conceptualisation’ refers to both the clarification and the analysis of the key concepts in a study and also to the way in which one’s research is integrated into the body of

existing theory and research. As far as the first meaning is concerned, 'conceptualisation' is synonymous with 'conceptual analysis' and involves the clear and unambiguous definition of central concepts.

'Conceptualisation' also refers to the underlying theoretical framework that guides and directs the research. When the research question or problem is formulated in the form of a research hypothesis, two of the important epistemic criteria are empirical testability and explanatory potential. The question surrounding empirical testability is whether one can foresee or even indicate how the hypothesis will be tested. The question of explanatory potential refers to the degree of theoretical support or embeddedness enjoyed by the hypothesis.

If the hypothesis is derived from an established theoretical framework that has been successfully applied to explain similar phenomena in the past, it strengthens the conclusions that could be drawn from the research. If not, the particular hypothesis has still to 'prove itself'. The 'outcome' of the conceptualisation phase is a research hypothesis which should, I suggest, meet the criterion of 'theoretical validity'.

Operationalisation

During the process of operationalisation a measuring instrument such as a questionnaire or scale is developed. Ideally, this instrument constitutes a valid measure of the key concepts in the research question. The outcome is a measuring instrument and the predominant epistemological criterion is measurement validity.

It has become customary to distinguish aspects or dimensions of measurement validity such as face validity, construct validity, criterion and predictive validity. What methodological criteria are applicable in the construction of valid measuring instruments? Firstly the population from which one selects items to construct the instrument must be exhaustive with regard to the phenomenon being investigated. Secondly the categories used in the scale or questionnaire must be unambiguous and mutually exclusive. Thirdly scales must meet the criterion of unidimensionality, which means that a single scale cannot be used to measure two or three different dimensions or aspects of a phenomenon.

Sampling

During the process of selecting or sampling the aim is to get a sample that is as representative as possible of the target population. Representativeness is the underlying epistemic criterion of a 'valid', that is, unbiased sample. The methodological criteria applied in the process of sampling are: clear definition of the population, systematic drawing of the sample, drawing probability rather than non-probability samples and observing the advantages of multi-stage versus simple random sampling.

Data Collection

During data collection, the researcher collects various kinds of empirical information or data, for instance historical, statistical or documentary data. This is accomplished

through various methods and techniques of observation such as document analysis, content analysis, interviewing and psychometric testing. There are a number of methodological criteria that ought to be followed during the process of data collection. These include suspension of personal prejudices and biases, systematic and accurate recording of the observations, establishment of trust and rapport with the interviewee and creating optimal conditions in terms of location or setting for the collection of the data. The outcome of the process is a set of data or empirical information and the epistemological criterion is that of reliability. We aim to produce reliable data. This means that if we were to use the same measures and hold the conditions under which the data are collected as constant as possible, we should get the same data from situation to situation. Reliability is hence synonymous with stability or consistency over time.

Conceptualisation (defining key concepts)

The term ‘conceptualisation’ is used here as a synonym for ‘conceptual analysis’ or ‘conceptual explication’. Assume that a researcher has decided to conduct a study to establish the relationship between political conservatism and racial prejudice. Even a person with no training in the social sciences would know that the concepts ‘conservatism’ and ‘racial prejudice’ have many connotations. In our everyday life these concepts form part of commonly held attitudes and value orientations.

In the language game of the social sciences, the concepts have become embedded in a variety of models and theories in sociology and political science. It is obvious that familiarity with the most important theories relating to the research problem is an essential precondition for an adequate conceptualisation. One of the most striking characteristics of theories in the social sciences is the incidence of highly abstract and multidimensional concepts. In the social sciences, concepts such as values, culture, solidarity, maturity, meaning, power, peace, revolution, alienation, anomie, structure, function, rite, religion, depression, social distance, anxiety, aggression, motivation, intelligence and success are unavoidable.

Many of these concepts have their roots in the world of social sciences research and are therefore usually linked exclusively to certain theories or models. However, even concepts such as power, freedom and revolution, which are part and parcel of everyday life and language, acquire new meaning when they become integrated in a theory in the social sciences such as, for example, that of Karl Marx. The fact that concepts acquire meaning, or even new meaning, within a conceptual framework such as a theory, a model or a typology, has led philosophers of science to refer to such concepts as ‘theoretical concepts’ or ‘constructs’.

We shall demonstrate that the aim in empirical research is to operationalise such constructs meaningfully by rendering them either measurable or observable. We shall discuss in some detail how a highly theoretical sociological concept such as ‘alienation’ can be explicated by means of theoretical definition, after which the notion of ‘theoretical validity’ will be defined more clearly.

Although Hegel was the first author to use the term ‘alienation’ in a theoretically interesting manner, Karl Marx is generally accepted as the first person to have developed a consistent and systematic theory about the concept. Marx endorsed Hegel’s view that alienation is a reality that arises when an individual feels that he or she has lost control. However, Marx differed from Hegel, Feuerbach and others in his view of the origin of alienation. He believed that it stemmed from economic factors, and more specifically that it was a consequence of capitalism:

- In what does alienation consist? First that the work is external to the worker, that it is not a part of his nature, that consequently he does not fulfil himself in his work but denies himself ... His work is not voluntary but imposed, forced labour. It is not the satisfaction of a need, but only a means of satisfying other needs. The object produced by labour, its product, now stands opposed to it as an alien being, as a power independent of the producer ... The performance of work is at the same time its objectification. This performance appears, in the sphere of political economy, as a vitiation of the worker, objectification as a loss and as servitude to the object, and appropriation as alienation.

One can only really understand this paragraph against the backdrop of Marx’s emphasis on the importance of human beings as labourers or makers. A human being attains self-realisation through labour or productivity. Marx, the capitalist system, as it existed at the time of his writing, resulted in human beings being alienated from the product of their labour by a system of unequal and unjust relations of production.

This system therefore separated people into two clearly identifiable classes: the owners and the workers. The fundamental inequity of the system stems from the structure of the production process. In relative terms, Marx, the worker contributes more to the actual production process, while the owner derives a far greater benefit. The worker’s productive ability is reduced to an object or thing. In other words, it is regarded as simply one more commodity on the market. Alienation therefore inevitably results when a quality which is intrinsic to the existence of man is reduced to a mere object or commodity.

The first clear definition of ‘alienation’ was therefore encountered in Marx’s economic theory. Despite the fact that it is a highly theoretical and abstract concept, we now have a clearer grasp of its meaning. The reason for this improved understanding is of the fact that the relationship between ‘alienation’ and better-known concepts such as ‘labour’, ‘production relationships’ and ‘inequality’ have been clarified within the framework of a theory.

These concepts are obviously still highly abstract terms. Nonetheless, the fact that the term ‘alienation’ has been embedded in a network of other related concepts, leads to a more precise definition of its meaning. In 1959 Melvin Seeman published an article entitled *On the meaning of alienation* in which he further elucidated the notion of ‘alienation’. His point of departure was that it was possible to define modern mass society more clearly by emphasising five essential structural elements, namely:

1. The development of impersonality and a reduction of relationships as a result of differences in status;
2. The development of a bureaucracy that leads to secularisation;
3. An increase in social differentiation and job specialisation;
4. Increasing social mobility; and
5. An increase in scale or size.

These five elements are fundamental to three factors that are relevant to alienation, namely loss of control over work and product, lack of integration within large organisational structures, and a low level of accessibility to reward values. Seeman maintained that the objective alienation in mass society eventually leads to five socio-psychological phenomena:

- Powerlessness;
- Normlessness;
- Isolation;
- Self-estrangement; and
- Meaninglessness.

Each of these five phenomena are subsequently defined in greater detail. Powerlessness refers to an individual's perception that he does not have complete control of his behaviour. Normlessness refers to the perception that socially unacceptable behaviour is necessary for the attainment of specific goals. Meaninglessness may be defined as a low expectation of being able to make meaningful predictions about future consequences of behaviour. Isolation is a tendency to attribute limited value to convictions or ideals that are typically highly valued. Self-estrangement indicates a degree of dependence on specific forms of behaviour for expected future consequences of behaviour. This is obviously a coherent theory. An explanation of the causes of alienation is provided.

As demonstrated by Marx, conceptual analysis by means of theoretical definition clearly involves explication of the concept by using other concepts that are sometimes more familiar. In the subject under discussion the concepts of powerlessness, normlessness, meaninglessness, isolation and self-alienation were used. In other definitions of alienation different dimensions of the concept are emphasised. Keniston emphasises the distinction between alienation from society and self-alienation:

- In societies in which the transition from childhood to adulthood is unusually painful, young people often form their own youth culture with a special set of anti-adult values and institutions, in which they can at least temporarily negate the feared life of the adult ... alienation of man from his own creative potentialities, embedded in his fantasy life.

In his typology of the dimensions of alienation, Stroup included indifference, isolation, self-estrangement, powerlessness, loneliness, meaninglessness, disenchantment and anonymity. The attempts of other scientists to define 'alienation' more precisely in different theories and typologies could also be referred to. However, these examples will suffice.

Theoretical Validity

What is involved in the theoretical definition of concepts? Concepts, or rather constructs, such as for example, alienation typically have many ‘shades’ of meaning—a variety of connotative elements. Theoretical concepts are rich in connotation. One could use the analogy of a field of meaning to illustrate this idea. Wittgenstein uses the term ‘family of resemblances’. Within a given field of meaning, certain dimensions or aspects of meaning are more closely associated than others.

Together, these dimensions within a field of meaning constitute the connotation ascribed to the concept. The relationships between these dimensions in meaning is not a matter of coincidence—it is not simply given. It is only within the framework of a theory or model that such relationships are systematically defined. And this is the function of a theoretical definition: to arrange or logically systematise the most important dimensions of the meanings of theoretical concepts. In this context, to arrange logically implies that the logical rules of correct classification, and the rules of mutual exclusion and exhaustiveness have to be adhered to.

This can be explained as follows: Assume that we needed to develop a classification of types of societies on the basis of their levels of development. The ‘principle of classification of’ or the ‘dimension along which’ societies are classified is ‘level of development’. The classification that we apply is the following: industrialised societies, agrarian societies and high-technology societies.

Obviously this simple typology would not be acceptable because there is a large degree of overlap between the first and third categories; they are not mutually exclusive. This is one way of saying that the principle of classification has not been adhered to. The distinction between ‘industrialised societies’ and ‘high-technology societies’ is not sufficiently clear because they both cover a similar part of the dimension of ‘level of development’.

Using the example of alienation, we have been able to demonstrate that a good theoretical definition implies that the essential dimensions of the meaning of a concept have been identified, and that, as far as possible, these dimensions are mutually exclusive. On face value, Stroup’s typology would appear to be lacking in terms of compliance with the second requirement in that the dimensions of isolation and loneliness, and also what he calls indifference and disenchantment, could be regarded as overlapping categories.

On the other hand, Seeman’s five dimensions appear to be valid, exhaustive and mutually exclusive categories, even on cursory inspection. But the notion of ‘theoretical validity’ should not be confined to conceptual clarity. Rose, following Phillips, introduced the term ‘internal theoretical validity’ and listed three characteristics of acceptable theoretical explication. These are ‘clarity’, ‘scope’ and ‘systematic import’.

Rose described each of these terms as follows:

- Clarity is the concept’s potential for leading to indicators, which depends on the degree to which it implies a chain of lower-level concepts; scope is the

breadth of the class of phenomena to which the concept applies; and systematic import is the extent to which the concept is used in propositions and theories.

We rarely, if ever, judge a theory solely in terms of conceptual clarity. We also ask whether it explains a lot of phenomena and also how integrated the various concepts and statements of the theory are. Because the connotative and denotative dimensions of concepts are so closely related, the ultimate test of a theory, model or typology is the extent to which it leads to valid information on the phenomena that it is supposed to describe or explain.

Critical Assignment

In reading 1, Giorgi defends a specific definition of the concept of 'religiosity' and also propounds a distinction between 'religiosity' and 'the religious'. Reconstruct and summarise her arguments. Why is it important to Giorgi to make this distinction?

Creating the Research Design

Defining a research problem provides a format for further investigation. A well-defined problem points to a method of investigation. There is no one best method of research for all situations. Rather, there are a wide variety of techniques for the researcher to choose from. Often, the selection of a technique involves a series of trade-offs. For example, there is often a trade-off between cost and the quality of information obtained. Time constraints sometimes force a trade-off with the overall research design. Budget and time constraints must always be considered as part of the design process.

Many authors have categorized research design as either *descriptive* or *causal*. Descriptive studies are meant to answer the questions of who, what, where, when and how. Causal studies are undertaken to determine how one variable affects another. McDaniel and Gates (1991) state that the two characteristics that define causality are *temporal sequence* and *concomitant variation*.

The word *causal* may be a misnomer. The mere existence of a temporal relationship between two variables does not prove or even imply that A causes B. It is never possible to prove causality. At best, we can theorize about causality based on the relationship between two or more variables, however, this is prone to misinterpretation. Personal bias can lead to totally erroneous statements. For example, Blacks often score lower on I.Q. scores than their White counterparts. It would be irresponsible to conclude that ethnicity causes high or low I.Q. scores. In social science research, making false assumptions about causality can delude the researcher into ignoring other (more important) variables. There are three basic methods of research: 1) survey, 2) observation, and 3) experiment (McDaniel and Gates, 1991). Each method has its advantages and disadvantages.

The *survey* is the most common method of gathering information in the social sciences. It can be a face-to-face interview, telephone, or mail survey. A personal interview is one of the best methods obtaining personal, detailed, or in-depth information.

It usually involves a lengthy questionnaire that the interviewer fills out while asking questions. It allows for extensive probing by the interviewer and gives respondents the ability to elaborate their answers. Telephone interviews are similar to face-to-face interviews. They are more efficient in terms of time and cost, however, they are limited in the amount of in-depth probing that can be accomplished, and the amount of time that can be allocated to the interview. A mail survey is generally the most cost effective interview method. The researcher can obtain opinions, but trying to meaningfully probe opinions is very difficult.

Observation research monitors respondents' actions without directly interacting with them. It has been used for many years by A.C. Nielsen to monitor television viewing habits. Psychologists often use one-way mirrors to study behavior. Social scientists often study societal and group behaviors by simply observing them. The fastest growing form of observation research has been made possible by the bar code scanners at cash registers, where purchasing habits of consumers can now be automatically monitored and summarized.

In an *experiment*, the investigator changes one or more variables over the course of the research. When all other variables are held constant (except the one being manipulated), changes in the dependent variable can be explained by the change in the independent variable. It is usually very difficult to control all the variables in the environment. Therefore, experiments are generally restricted to laboratory models where the investigator has more control over all the variables.

Sampling

It is incumbent on the researcher to clearly define the target population. There are no strict rules to follow, and the researcher must rely on logic and judgment. The population is defined in keeping with the objectives of the study. Sometimes, the entire population will be sufficiently small, and the researcher can include the entire population in the study. This type of research is called a *census* study because data is gathered on every member of the population. Usually, the population is too large for the researcher to attempt to survey all of its members. A small, but carefully chosen *sample* can be used to represent the population. The sample reflects the characteristics of the population from which it is drawn.

Sampling methods are classified as either *probability* or *nonprobability*. In probability samples, each member of the population has a known probability of being selected. Probability methods include random sampling, systematic sampling, and stratified sampling. In nonprobability sampling, members are selected from the population in some nonrandom manner. These include convenience sampling, judgment sampling, quota sampling, and snowball sampling. The other common form of nonprobability sampling occurs by accident when the researcher inadvertently introduces nonrandomness into the sample selection process. The advantage of probability sampling is that *sampling error* can be calculated. Sampling error is the degree to which a sample

might differ from the population. When inferring to the population, results are reported plus or minus the sampling error. In nonprobability sampling, the degree to which the sample differs from the population remains unknown.

Random sampling is the purest form of probability sampling. Each member of the population has an equal chance of being selected. When there are very large populations, it is often difficult or impossible to identify every member of the population, so the pool of available subjects becomes biased. Random sampling is frequently used to select a specified number of records from a computer file.

Systematic sampling is often used instead of random sampling. It is also called an *Nth name selection* technique. After the required sample size has been calculated, every Nth record is selected from a list of population members. As long as the list does not contain any hidden order, this sampling method is as good as the random sampling method. Its only advantage over the random sampling technique is simplicity.

Stratified sampling is commonly used probability method that is superior to random sampling because it reduces sampling error. A stratum is a subset of the population that share at least one common characteristic. The researcher first identifies the relevant strata and their actual representation in the population. Random sampling is then used to select subjects for each stratum until the number of subjects in that stratum is proportional to its frequency in the population.

Convenience sampling is used in exploratory research where the researcher is interested in getting an inexpensive approximation of the truth. As the name implies, the sample is selected because they are convenient. This nonprobability method is often used during preliminary research efforts to get a gross estimate of the results, without incurring the cost or time required to select a random sample.

Judgment sampling is a common nonprobability method. The researcher selects the sample based on judgment. This is usually an extension of convenience sampling. For example, a researcher may decide to draw the entire sample from one “representative” city, even though the population includes all cities. When using this method, the researcher must be confident that the chosen sample is truly representative of the entire population.

Quota sampling is the nonprobability equivalent of stratified sampling. Like stratified sampling, the researcher first identifies the strata and their proportions as they are represented in the population. Then convenience or judgment sampling is used to select the required number of subjects from each stratum. This differs from stratified sampling, where the strata are filled by random sampling.

Snowball sampling is a special nonprobability method used when the desired sample characteristic is rare. It may be extremely difficult or cost prohibitive to locate respondents in these situations. Snowball sampling relies on referrals from initial subjects to generate additional subjects. While this technique can dramatically lower search costs, it comes at the expense of introducing bias because the technique itself reduces the likelihood that the sample will represent a good cross section from the population.

Data Collection

There are very few hard and fast rules to define the task of data collection. Each research project uses a data collection technique appropriate to the particular research methodology. The two primary goals for both quantitative and qualitative studies are to maximize response and maximize accuracy.

When using an outside data collection service, researchers often *validate* the data collection process by contacting a percentage of the respondents to verify that they were actually interviewed. Data *editing* and *cleaning* involves the process of checking for inadvertent errors in the data. This usually entails using a computer to check for out-of-bounds data.

Quantitative studies employ deductive logic, where the researcher starts with a hypothesis, and then collects data to confirm or refute the hypothesis. *Qualitative* studies use inductive logic, where the researcher first designs a study and then develops a hypothesis or theory to explain the results of the analysis.

Quantitative analysis is generally fast and inexpensive. A wide assortment of statistical techniques are available to the researcher. Computer software is readily available to provide both basic and advanced multivariate analysis. The researcher simply follows the preplanned analysis process, without making subjective decisions about the data. For this reason, quantitative studies are usually easier to execute than qualitative studies.

Qualitative studies nearly always involve in-person interviews, and are therefore very labor intensive and costly. They rely heavily on a researcher's ability to exclude personal biases. The interpretation of qualitative data is often highly subjective, and different researchers can reach different conclusions from the same data. However, the goal of qualitative research is to develop a hypothesis—not to test one. Qualitative studies have merit in that they provide broad, general theories that can be examined in future research.

Data Analysis

Modern computer software has made the analysis of quantitative data a very easy task. It is no longer incumbent on the researcher to know the formulas needed to calculate the desired statistics. However, this does not obviate the need for the researcher to understand the theoretical and conceptual foundations of the statistical techniques.

Each statistical technique has its own assumptions and limitations. Considering the ease in which computers can calculate complex statistical problems, the danger is that the researcher might be unaware of the assumptions and limitations in the use and interpretation of a statistic.

Reporting the Results

The most important consideration in preparing any research report is the nature of the audience. The purpose is to communicate information, and therefore, the report should be prepared specifically for the readers of the report.

Sometimes the format for the report will be defined for the researcher (e.g., a dissertation), while other times, the researcher will have complete latitude regarding the structure of the report. At a minimum, the report should contain an abstract, problem statement, methods section, results section, discussion of the results, and a list of references.

Validity and Reliability

Validity refers to the accuracy or truthfulness of a measurement. Are we measuring what we think we are? “Validity itself is a simple concept, but the determination of the validity of a measure is elusive”.

Face validity is based solely on the judgment of the researcher. Each question is scrutinized and modified until the researcher is satisfied that it is an accurate measure of the desired construct. The determination of face validity is based on the subjective opinion of the researcher.

Content validity is similar to face validity in that it relies on the judgment of the researcher. However, where face validity only evaluates the individual items on an instrument, content validity goes further in that it attempts to determine if an instrument provides adequate coverage of a topic. Expert opinions, literature searches, and pretest open-ended questions help to establish content validity.

Criterion-related validity can be either predictive or concurrent. When a dependent/independent relationship has been established between two or more variables, criterion-related validity can be assessed. A mathematical model is developed to be able to predict the dependent variable from the independent variable(s). *Predictive validity* refers to the ability of an independent variable (or group of variables) to predict a future value of the dependent variable. *Concurrent validity* is concerned with the relationship between two or more variables at the same point in time.

Construct validity refers to the theoretical foundations underlying a particular scale or measurement. It looks at the underlying theories or constructs that explain a phenomena. This is also quite subjective and depends heavily on the understanding, opinions, and biases of the researcher.

Reliability is synonymous with repeatability. A measurement that yields consistent results over time is said to be reliable. When a measurement is prone to random error, it lacks reliability. The reliability of an instrument places an upper limit on its validity (Spector, 1981). A measurement that lacks reliability will necessarily be invalid. There are three basic methods to test reliability: test-retest, equivalent form, and internal consistency.

A *test-retest* measure of reliability can be obtained by administering the same instrument to the same group of people at two different points in time. The degree to which both administrations are in agreement is a measure of the reliability of the instrument. This technique for assessing reliability suffers two possible drawbacks. First, a person may have changed between the first and second measurement. Second,

the initial administration of an instrument might in itself induce a person to answer differently on the second administration. The second method of determining reliability is called the *equivalent-form* technique. The researcher creates two different instruments designed to measure identical constructs. The degree of correlation between the instruments is a measure of equivalent-form reliability. The difficulty in using this method is that it may be very difficult (and/or prohibitively expensive) to create a totally equivalent instrument.

The most popular methods of estimating reliability use measures of *internal consistency*. When an instrument includes a series of questions designed to examine the same construct, the questions can be arbitrarily split into two groups. The correlation between the two subsets of questions is called the *split-half* reliability. The problem is that this measure of reliability changes depending on how the questions are split. A better statistic, known as Chronbach's alpha (1951), is based on the mean (absolute value) interitem correlation for all possible variable pairs. It provides a conservative estimate of reliability, and generally represents "the lower bound to the reliability of an unweighted scale of items". For dichotomous nominal data, the KR-20 is used instead of Chronbach's alpha.

Variability and Error

Most research is an attempt to understand and explain *variability*. When a measurement lacks variability, no statistical tests can be (or need be) performed. Variability refers to the dispersion of scores. Ideally, when a researcher finds differences between respondents, they are due to true difference on the variable being measured. However, the combination of systematic and random errors can dilute the accuracy of a measurement. *Systematic error* is introduced through a constant bias in a measurement. It can usually be traced to a fault in the sampling procedure or in the design of a questionnaire. *Random error* does not occur in any consistent pattern, and it is not controllable by the researcher.

Conceptualisation (Formulating Research Hypotheses)

What do we mean when we refer to the 'concept-ualisation' of a study over and above the definition of our key concepts? As suggested, demonstrate that 'conceptu-alisation' also involves embedding or incorporating one's research into the body of knowledge that is pertinent to the research problem being addressed.

To do this, the researcher must first do a thorough literature search of previous theoretical and empirical work in this field and then relate her work to the existing literature. What does this mean and why are literature reviews important?

- A literature review serves as a 'map' or 'maps' of the terrain. With reference to our analogy of the journey, we have to realise that other researchers have 'travelled' this way before. In areas where there has been a concentrated focus on a specific phenomenon, a researcher has an obligation to acquaint herself

with any publications on major research already conducted in the field, the most widely accepted theoretical positions and the most recent debates.

- A review of research also provides guidelines, or at least suggestions, on the design of one's own project. By studying studies on a particular topic, one not only learns about the 'maps' and 'guidebooks', but also about the 'itineraries', that is the different ways that people have travelled this terrain.
- An intensive study of the existing body of knowledge yields various kinds of 'resources'. These include conceptual resources such as useful theoretical formulations or definitions of key concepts that are encountered in a specific field; methodological resources, such as a reliable and valid scale or questionnaire; and appropriate examples of qualitative and quantitative techniques.
- Literature searches are sometimes done by researchers who intend to replicate research. In such cases one is interested in both the methodology and the substantive results of research.
- Finally, anyone planning to research a field which has hitherto enjoyed limited attention, either worldwide or locally, can learn a great deal by studying related fields and from the designs and methods used.

When we refer to a literature review as a kind of 'research map', we must bear in mind that there are different kinds of research maps. When planning a journey we have a choice of any number of maps including largescale maps of countries, detailed town maps, guidebooks to countries and cities, and guidebooks that provide historical information as well. Similarly, there is a range of resources from which to choose when a literature review is undertaken. There are specific resources for specific research reviews. A simple classification of these resources and their main applications is given in table.

Table. Sources for Literature Searches

Kind of Literature Source	Application
Dictionaries, encyclopaedias, textbooks	1. Provide standard definitions of central concepts in a discipline. 2. Provide descriptions of main research areas. 3. Provide useful historical overviews of main figures and traditions in a discipline.
Annual reviews, state-of-the-art special editions of journals)	1. Usually include authoritative theart reviews (usually in reviews of the most prominent theories and research in specific problem areas. 2. Include succinct statements and discussions of key debates and issues.
Monographs/books	1. Comprise extended and well-contextualised in-depth studies on topics. 2. Normally include well-documented theoretical arguments on key issues in current debates.

Journal articles

1. Include topical discussions on the latest theoretical and empirical issues.
2. Include brief reports on key findings and new advances in methodology.
3. Include book reviews and discussions. Articles are useful 'second-order' introductions to primary sources.

In broad and somewhat simplified terms, the following guidelines apply: The more exploratory and open-ended the study, the more useful it will be to look at general sources such as encyclopaedias and review articles and also to use broad search strategies.

For more validation and structured studies, it is more useful to consult subject literature such as books and journal articles and to use more focussed search strategies. The level and context of the research project is an equally important consideration.

The fundamental differences involved in writing undergraduate assignments, master's theses, doctoral dissertations and journal articles. We shall discuss the differences in the kind of literature review required for each of these situations, and also the style of reporting. In summary then: a survey of the literature is an essential component of any study because it is the main access point or gateway to the relevant body of knowledge.

Through reading and studying the work of other academics, we learn how to improve our own research methods, ask the right questions and identify potentially useful answers to such questions. In fact, the body of literature is best viewed as a three-dimensional space that can be explored by a researcher to best locate and position her own work.

We focussed on the issue of conceptual analysis and the meaning of conceptual or theoretical validity. We showed how a study on alienation would benefit from a review of the theories on alienation and specifically what it means to arrive at a conceptually valid definition of highly abstract concepts. Another important reason for reviewing the literature is that it provides ideas, hypotheses and conjectures for one's own research. The remainder of this stage is devoted to a discussion of the distinction between different kinds of hypotheses and between hypotheses, assumptions and postulates.

Kinds of Hypotheses

Scientific statements differ with regard to the degree of evidentiary support that they enjoy. In this respect it is customary to distinguish between 'hypothetical statements' or 'hypotheses' and 'substantiated or validated statements', to which we usually refer as 'empirical statements'. When we first formulate a statement without knowing whether we have any empirical warrant to accept it as reasonably valid or even true, we call this a hypothesis. A hypothesis is a statement that makes a provisional or conjectural knowledge claim about the world. A 'good' hypothesis is empirically testable, which means that we must be able to specify clearly what data would provide support or rejection for it. Before discussing the criteria for a 'good hypothesis' we must distinguish between different kinds of hypotheses.

Existential and Relational Hypotheses

Hypotheses can be classified into two main groups, namely existential and relational hypotheses. An existential hypothesis is a provisional statement about a certain state of affairs, that is, it makes a claim that something is the case.

For example:

- Sixty per cent of rural South Africans are functionally illiterate.
- More than seventy per cent of all South Africans are opposed to legalized abortion.
- Durban has the highest crime rate in the country.

Statements such as these are claims that a certain entity has a certain property and what the value of that property is. Existential hypotheses are more common in exploratory research where the main purpose is to find out what the case is. Relational hypotheses postulate that a certain kind of relationship exists between two or more variables. It has become customary to distinguish further between correlational hypotheses and causal hypotheses, depending on the kind of relationship that is being postulated. A correlational hypothesis might claim that there is a positive or negative relationship between people's educational level and their tolerance of other people, that there is a relationship between stress and productivity. The first hypothesis postulates a positive relationship, namely that tolerance of other people or groups increases as level of education increases. The second hypothesis postulates a negative relationship, namely that, as stress in the workplace is increased, workers become less productive.

Research Design and Approach

The activity theory perspective suggests that a focus on teaching, unrelated to students' perceptions about their own learning and its context, is unlikely to be successful in changing outcomes. My approach focuses on the experience of learning Statistics at university from the point of view of the student. Consistent with my theoretical framework based on the ideas of Leont'ev and Vygotsky, I explore the network of relationships between learners, the subject matter and context. My investigation took place at a large and traditional metropolitan university, the oldest in Sydney. The participants were second year Psychology students, all of whom were required to study Statistics. Statistics is introduced in first year Psychology at The University of Sydney, but it is as a compulsory component of the second year course that it plays a major role, contributing one quarter of the final assessment mark. The instruments I used included participant observation, interviews and surveys. My project consists of two investigations.

Study One is entitled "Exploring Mature Students' Learning of Statistics". The context for this study was the Mathematics Learning Centre where I work. The participants in this study were of great interest to me as an educator concerned with ameliorating students' difficulties with Statistics. Study Two is called "Understanding University Students Learning Statistics as a Service Course to Psychology". In this study, I widened the context of my inquiry to include the general Statistics class. For my investigations,

I drew on a triad of research tools—naturalistic inquiry, procedures based on the phenomenographic method and quantitative methods for describing data and exploring patterns and relationships. Among the instruments I developed was the Approaches to Learning Statistics Questionnaire.

Psychology Students at the Mathematics Learning Centre

In Study One I utilise a naturalistic perspective to draw on the rich accounts of five students who were studying Psychology after many years away from the educational setting. My aim is to tell the story of these students in a way that is generalisable to others—that is, I try to uncover recognisable insights. I was in no way involved with the formal assessment of these students and worked very intensively and closely with them. Hence I was able to develop a close relationship with the participants of Study One and had more access to their perceptions and feelings than is usually the case for university educators in statistics. My analysis is based mainly on interview data, complemented by written expressions from various sources. The five participants were selected “purposively” for their range of experiences and attributes. As is often the case with qualitative research, the data for this study consisted of descriptions, direct quotations and excerpts, obtained by close psychological contact with those being studied.

The study is an example of action research in that my actions as a researcher were inseparable from those designed to assist the students in their learning of Statistics. As I have outlined in Gordon this means that the methodology for the study developed through the “action research spiral”.

They describe this spiral as recurring cycles of:

- Planning, acting, observing and reflecting, with each of these activities being systematically and self-critically implemented and interrelated.

View of Psychology Students Learning Statistics

Study Two was grounded in Study One and broadened the perspective on students learning Statistics. The participants of Study Two were Psychology II Students who took part in a survey during their Statistics lecture, near the end of the first semester. The survey was completed during 20 minutes of the lecture time and was the major source of data for this study. Other data for this study included assessment results and vignettes from interviews with students selected from those who agreed to contribute further to my research.

Background of Phenomenography as a Research Tool

One major focus of Study Two was to elicit students’ conceptions of the subject matter, in the context in which they were studying it. To do this I adopted a phenomenographic approach. The term phenomenography was first used by Ference Marton to describe programmes of research which had the common aim of describing peoples’ conceptions.

The results of phenomenographic analysis are categories of description of the qualitatively different ways students conceive of the phenomenon being investigated. This focus on describing conceptions is not the sole aim of the investigations in which phenomenography is used as a research tool. Rather it is learning and teaching that are being investigated based on the focusing of conceptions. Conceptions are the lens through which the phenomenographic researcher views learning.

Marton describes phenomenography as a method:

- For investigating the qualitatively different ways in which people experience, conceptualise, perceive and understand various aspects of, and phenomena in, the world around them.

Phenomenography was developed within the framework of education research at the University of Gothenburg, Sweden. It is concerned with the relations between people and the objects of their perceptions or content of their thoughts. Marton calls phenomenography a “research specialization”, that is, a combination of a research orientation and a research approach to describing and comparing conceptions and understandings.

The research orientation emphasises the importance of description and the use of categories as forms of expressing the conceptions. The research approach is characterised by the open explorative form of data collection and the interpretative analysis of data. Säljö, a forerunner in the development of the phenomenographic approach, comments that in more recent studies, the term “way of experiencing” is used in preference to the term “conception”.

This emphasises that:

- The prime interest of phenomenographic research is in finding and delimiting the variation in ways of experiencing reality.

Hence, in deriving categories of description for students’ conceptions of Statistics, I am attempting to find relationships between the students and the subject matter which express their ways of experiencing the subject, Statistics, as they were learning it. Svensson argues that underpinning the phenomenographic specialisation is an assumption that knowledge is fundamentally a “question of meaning in a social and cultural context”. Such an approach views phenomena systemically and avoids separating person and context.

As Säljö in an editorial of Learning and Instruction, concludes:

- Human experiences are inescapably cultural in nature, and learning and growth take place within cultural boundaries.

This view of the intertwining of culture, mind and action is consistent with a Vygotskian assumption that there is no duality between self and context; between thinking and acting. The phenomenographic view of knowledge is that it is relational, created through human thinking and activity which relates to the external world—a relation between thought and social and cultural life. Hence my choice of phenomenography as one of my research tools is consistent with the theoretical perspective with which I am framing this investigation.

The relational view of learning furnished by phenomenography has formed the basis of much of the research into student learning in higher education. Phenomenography has been fruitful in exploring students' conceptions and understanding in a number of topics, including physics; literature interpretation; essay writing and computer programming. In summary, this research indicates that learners' experiences should be considered as involving the ways students relate to the learning environment, their goals and intentions, as well as their learning strategies. My colleagues and I have found phenomenographic methods useful to explore the qualitatively different ways in which students experience learning mathematics. Two studies were conducted.

In the first study we investigated the conceptions of mathematics and approaches to learning it of students entering university. In the second investigation we explored these students' conceptions of a fundamental mathematical concept, namely that of a function. Traditionally, phenomenographic studies are built around interviews. However, in these investigations, my colleagues and I extended previous research by constructing categories from the written responses of a large number of students. This developed phenomenography as a research tool. In Study Two, I built on our previous phenomenographic research by looking at the variation in the way second year Psychology students conceived of the nature of the subject matter, Statistics. The phenomenographic perspective provided my basis for identifying categories of description of students' conceptions. These revealed qualitatively different ways of experiencing Statistics. In much research, the categories of description or variables are imposed on the data by an expert researcher.

In Study Two, in keeping with phenomenographic practice, the categories of description were constituted in relation to the data—from interpretations of the survey responses. This does not imply that I, as researcher, viewed the data free of ideas based on prior experience or theoretical orientation.

Indeed:

- The act of categorisation is by its very nature subjective.

Further, my research is deliberately set in the framework of activity theory. However, instead of characterising students' responses to the three open-ended questions in the survey by means of an established taxonomy, the categories emerged in the context of the study. By keeping the questions as general as possible, I tried not to shape students' responses. Rather I tried to explore the students' points of view. This enhances the ethnographic validity of this research, that is, the sense of the research in terms of the lives and awareness of the participants in it.

The phenomenographic method analyses meanings constituted within a particular context and content domain. My categories were derived in a particular setting. However, they and their structure as an organised set, may prove useful in contexts other than the one from which they emerged.

In accord with the phenomenographic approach:

- The categories of description are supposed to be replicable, but the way in which they were found is not.

Conceptions may be expressed in different forms of action but they are most accessible through language. Variation in conceptions, expressed by people, immediately brings in the cultural and social context in which these conceptions are reported. The students' reports on their ways of conceiving Statistics, although limited to the students' awareness and abilities in written verbal expression, provided me with a rich source of information for characterising these experiences. This experiential perspective is referred to as a "second order" perspective by Marton.

He contrasts this with phenomenology, based on Husserl's imperative to return to the essence of a phenomenon, through immediate experience and free from conceptual thought—a first order enterprise. Phenomenography is a perspective on how things appear to people—how phenomena are experienced, rather than what they are "in reality" or whether or not such a reality can be described.

Phenomenography was derived from empirical and pragmatic research in education. Its commitment to philosophical foundations is therefore not always clear. The procedures defining its use are neither prescriptive nor indisputable. Indeed, Hasselgren and Beach argue that "there is no genuine consensus method of phenomenography".

To them:

- Phenomenography is research which is simply concerned with how things are understood, the experiences of the process of formation of understandings at individual levels, and their distributions in specific collectivities. While my way of applying this research tool is individual, I have tried to maintain the essence of phenomenographic research as a way of understanding the multiple realities of students. That is, my appropriation of Marton's term signifies this approach. I view phenomenography as a research tool which bridges the gap between empirical, analytic methods, which are recognised by the research community, and students' subjective awareness of their own diverse conceptions and experiences.

Introduction to Analysis of Students' Approaches to Learning Statistics

I analysed students' approaches to learning Statistics by means of a questionnaire, the Approaches to Learning Statistics Questionnaire. My questionnaire was modified from the Approaches to learning Mathematics Questionnaire which, in turn, was derived by my colleagues and myself from the Study Process Questionnaire. Biggs refers to three approaches to learning: deep, surface and achieving. He differentiated each of these into two aspects—the motive, or intention, and the strategy. Hence, in the Study Process Questionnaire the items are divided into those referring to motive or intention and those referring to strategy.

Each motive-strategy combination defines a distinct approach to learning—deep, surface or achieving. Early work by Biggs, on samples of students from universities and Colleges of Advanced Education, indicated that the achieving approach was equally related to the surface and deep approaches in tertiary students.

The surface and deep approaches, however, are referred to by Biggs as:

- Independent ways in which students may become involved in learning. In this project I was concerned with deep and surface approaches to learning. Achieving approaches were of less theoretical interest to me and were not investigated. Considerable research has been done on the Study Process Questionnaire. For example, a literature review on approaches to studying indicates that mature students seem more likely than younger students to adopt deep approaches to learning, while surface approaches decline with age. However, completion rates and possible bias in these studies, for example, in the postal returns of surveys in Biggs' investigations, renders this finding problematic.

Richardson also reviewed the literature on cultural specificity of approaches to studying in higher education. Investigations using the Study Process Questionnaire in countries as diverse as Australia, Nepal, Britain and Nigeria yielded a similar two factor structure for this questionnaire—one indicating a deep approach and the other a surface approach.

The research also suggests that orientations towards understanding the meaning of learning materials, that is, deep approaches to learning, reflect a consistent agreement across different cultures with regard to the goals of higher education.

On the other hand, this research indicates that patterns of responses on the subscales measuring surface approaches to learning, are more distinctive to differing cultural contexts. Concern has been expressed about the factor structure underlying Biggs' differentiation of the Study Process Questionnaire into motive and strategy components. In my analysis of the Approaches to Learning Statistics Questionnaire I did not find evidence for a breakdown into intention and strategy factors. I therefore analysed the items of the ALSQ in terms of two subscales: one measuring a general surface approach and the other a deep approach to learning Statistics.

Biggs characterises a surface approach and a deep approach to an academic task by the following.

A surface approach is distinguished by:

- A view of the task as imposed, a demand to be met;
- A fragmented view of the task—parts are unrelated to each other and the task is unrelated to other tasks;
- Worries about the time taken by the task;
- A lack of recognition of any personal meaning in the task;
- Reliance on memorisation and ways of reproducing the details.

A deep approach reflects:

- Interest and enjoyment in the task;
- A search for underlying meaning;
- Efforts to make the learning personally meaningful—relating task to personal experience or to the real world;

- Holistic approach—relating parts of the task to each other and to other knowledge;
- Theorising about the task.

The surface approach and the deep approach, as defined by Biggs, correspond closely with the surface and deep levels of processing distinguished by Marton and Säljö. Later, Marton and Säljö referred to these as surface and deep approaches to learning. In contrast to Biggs, however, Marton and Säljö see surface and deep approaches as the opposite ends of a single continuum, rather than as independent ways of learning. Marton and Säljö showed that students would adopt one of two methods of processing information according to their intentions.

If their aim was merely to memorise and reproduce material, or, as Biggs puts it:

- To display the symptoms of having learned, They would adopt a surface level of processing; if they wanted to maximise their understanding of the underlying meaning, they would adopt a deep strategy.

Hence Marton and Säljö inferred a relationship between strategy and intention. What the student intends to gain from the learning determines the strategy used. In recent research, Marton, Watkins and Tang make the point that while a surface approach to learning is often associated with rote learning, this is not what characterises it.

- Rather, a surface approach is characterised by a focus on the learning material or task in itself and not, as would be the case for the deep approach, on the meaning or purpose underlying it.

Their ideas accord with Leont'ev's proposition that what distinguishes actions are the goals driving them.

Investigating Individuals in Socially Significant Practices

My project is an example of what Chaiklin describes as “investigating individuals in socially significant practices”.

These are studies in a “yet-to -be-embodied” research tradition that:

- Try and develop an account of the actions of individuals participating in a societally significant practice while it is occurring, by an analysis that locates the practice in a social, societal and/or historical perspective. Chaiklin proposes that many different theoretical perspectives and methods could contribute to such research. He summarises such projects as having five common characteristics, which I assert are fulfilled by my project.
- Firstly, these studies take concrete, meaningful, societal practices as the direct object of study. That is, the setting for the study is an actual example of human practice and would be so whether or not it was the object of study. In my research I have tried to get as close as possible to the participants and interpret situations which were as typical as possible for students learning Statistics.

- Secondly, the practice takes place in a recognisable societal institution, in this case an institution of higher learning.
- Thirdly, such studies have a definite theoretical interest. Insights into the practice of learning Statistics are important elements of my research but so are insights into and extensions of the activity theory framework in which I have interpreted this practice.
- Fourthly, there is the idea that knowledge and actions are social as well as individual.

I draw on Vygotsky's and Leont'ev's idea of "co-knowing" to illustrate this notion—statistical knowledge is socially created and re-created through the consensus of practitioners. This collective development of knowledge is facilitated through tools, for example, by the easy and quick exchange of ideas through electronic media. In my university, Statistics is continually being transformed by each lecturer and committee involved in organising the curriculum and setting the examinations. Students come to know Statistics through interacting with peers and teachers, as well as through their own reflections and actions. Socially organised constraints and resources surround individual actions.

- *Finally, the particular practice selected for such studies has:*
 - Significant consequences for the people participating in these practices.

Learning Statistics can change the lives of those participating in this practice. In some cases, the outcomes could be instrumental in determining students' academic and career paths.

For example, students' examination results in Statistics could determine whether they would attain a degree or be accepted into Psychology Honours. Learning Statistics can also powerfully affect students' personal development as was the case for some of the participants of Study One.

Design Research Methods

There are several different kinds of design research methods, tools and activities. The most common methods are:

Fieldwork

Fieldwork is about getting out into the real world and interacting with people - watching and talking to people where they work, live and play. Fieldwork is a key method for design research and should be considered first above all others. The understanding that you gain from being out in the real world is profoundly different from traditional surveys and focus groups.

Fieldwork is done in small teams, where one or two people visit with an individual or a small group like parents or a family.

Fieldwork focuses on observation and interviews. Because interviews are conducted closer to or where the activity happens, it's easier to ask the right questions and participants feel comfortable in the setting and can answer more accurately.

Fieldwork includes activities like:

Behavioural Interviews

What

Behavioural interviews are open-ended, non-directed interviews that focus on participants' behaviour. These interviews explore stories and explanations of how and why things happen and work best if held in context at home or work.

Why

Behavioural interviews help uncover values, motivations, priorities, activities, and relationships between elements of the larger system. They connect the research team with real people and they allow the team to investigate the underlying issues behind trends that may not have been revealed through analytics or surveys.

How

1. Create an interview guide that covers questions for all the topics you're interested in for this research. This isn't intended to be a script; instead use it as a guide to help keep you on track of all the topics you want to explore. Make sure you can fit your questions into the allotted time.
2. Interview at least eight individual participants by exploring each topic with them using open-ended questions. Pause and invite further explanation of interesting topics, anecdotes or other elements. Use your guide to direct conversation rather than as a script that you read from verbatim.
3. Record notes during the interview (you may want to work in a team, with one person taking notes and one interviewing).

Paired Interviews

What

Paired interviews are behavioural interviews with a pair of participants rather than individual participants.

Why

Paired participants will talk with each other as well as with the interviewer, offering alternative perspectives, additional detail, or a reality check on how things happen. While paired interviewing lets you include more participants in the same time slot, some participants will only talk about more personal things during an individual

interview. There are also fewer opportunities to investigate underlying issues or to understand specific activities because there is less time to devote to each individual.

How

1. Recruit pairs of participants for interviews. Participants may have an existing relationship, such as co-workers, neighbours, a domestic couple, or a parent and child. Pairs may also be created ad hoc by scheduling separately recruited people. If you create ad hoc pairs, match participants accordingly.
2. Carry out the interviews in the same way as behavioural interviews.

Remote Interviews

What

Remote interviews are behavioural interviews carried out over the telephone or through videoconferencing software.

Why

Holding remote interviews allows you to include people from across the province and to recruit participants from a larger population instead of just your local users. Phone interviews are also often easier to schedule for participants and research teams. However, phone interviews lack context from the environment and cues from body language, and it's often harder to build rapport with remote participants than with those in person.

How

1. Recruit participants in the same way as you would an in-person interview, but schedule a telephone call instead of a face-to-face interview.
2. Consider using a conference bridge that allows you to record the call.

Observation and Shadowing

What

Observation and shadowing is about spending time observing a participant during their daily routine, noting context, activities, floor plans, interactions and technology.

When it's appropriate and when you have consent, take photos and video. Observation and shadowing is often combined with behavioural interviewing.

Why

Watching someone in context can surface new insights, show research teams particular angles to explore, and give a better behavioural balance to interviews. Observation and shadowing can be followed up with a dedicated interview, or teams may only ask about situations as they come up.

How

1. Recruit participants for observation and shadowing.
2. Arrive at their work or home promptly and introduce yourself and the project. Confirm that they understand that you're going to spend some time shadowing their work, just like an intern or someone new on the job.
3. Answer any questions and have them sign the research consent form and provide any incentives at this time.
4. Ask them to show you what they do. You may ask someone to focus on specific kinds of activities if you're supporting those specific tasks with your project, or have them carry on a "typical day" if you have a broader focus.
5. Let the person explain their activity over the time that you spend together. Ask follow-up questions to clarify, especially points most relevant to your areas of interest.
6. Give your participant some language in advance to explain to others why you're with them, e.g., "Oh, Jim is from the Intranet project and is tagging along with me today to learn more about my work."
7. Wrap up by clarifying any outstanding points and thanking them for their time.

*Call Centre Shadowing**What*

Call centre shadowing is when you listen in on call centre calls related to your particular area(s) of interest.

Why

Call centres aggregate real user challenges so you can quickly and easily get a sense of the concerns that people deal with without going into the field.

How

1. Contact a relevant call centre. Finding an appropriate call centre may be a challenge, particularly if call support has been outsourced.
2. Arrange to visit and listen in on agent calls. Most call centres have equipment for new hires to listen in on calls as part of their training.
3. Follow the call centre's policies and procedures on disclosure of another listener on the call.
4. Take notes about the calls that you listen to.
5. If possible, discuss your observations with several agents while you are onsite at the call centre.
6. You may also be able to look at call centre logs for certain kinds of calls, depending on the call centre.

Tips

Words of caution regarding call centre shadowing:

- Call centre calls only give you a small sample of your overall users - you won't get any information about the people who don't need the call centre.
- Because you can't engage with the callers, you'll only get a fragment of their experience.

Digital Fieldwork*What*

Digital fieldwork uses social media, photo sharing, gaming, online communities and other digital artifacts to inform your field research. You may choose to conduct online interviews, look at artifacts like vacation photos or observe online behaviour in chat rooms, forums, online games or social sites.

Why

Digital fieldwork is often easier to conduct than in-person fieldwork. It can cost less, take less time, and require less permission. However, the quality of data is often lower than with in-person or remote fieldwork. Also, the kinds of questions you can explore are restricted to the kinds of digital data you can find online, and that data only reflects people with the resources and desire to participate online.

How

There are a variety of ways to look at people's stories and behaviour online. They fall into three main approaches with increasing effort and involvement with individuals: collecting artifacts and stories, observing behaviour, and interacting online.

1. Search photo sites such as Facebook or Flickr for relevant images posted by individuals.
2. Search blogs, social sites, and Tumblr for stories that people have shared that involve your topic of interest. You can use Google's "site:" modifier to restrict searches to specific sites like Facebook, Blogspot, or WordPress.
3. Find discussion forums, chat rooms or other online communities dedicated to relevant topics and simply observe and record people's interactions and the discussions they have. Twitter hashtags may also provide a source of ongoing conversation to observe.
4. After collecting and observing, you may engage with participants in discussion to ask questions, explore topics and otherwise gain more information. The guidelines for interviews are useful in these interactions. Remember to be respectful of individual community norms - you may want to use private messaging or otherwise keep a low profile as you engage online with individuals.

Co-Design and Gamestorming

Co-design is when you work with participants or partners on the design process instead of doing the work for them. This is a significant shift in how we do things but giving up some control and embracing co-design and co-creation is worth the benefits that it offers.

Participatory design methods involve a facilitated group of stakeholders making things to express their values, needs and ideas.

Design games - or *gamestorming* - help build common ground and better express needs. Some games involve making concrete things and some revolve around activities or words.

Participants may make metaphorical artifacts like collages to explore certain topics and values (like safety or family), or they may make literal artifacts like prototypes to explore different ideas for how to build something.

A literal artifact should be used to facilitate a deeper conversation about why someone suggests a certain feature or function instead of just a suggestion or feature request.

Ecosystem Research

Ecosystem research examines the greater context in which your design research is situated in an effort to better understand the overall ecosystem of a service. It focuses on relevant trends, relationships and interconnectivity within the system, possible alternatives, the consequences of not providing the service and assessing the best fit for the service within the greater ecosystem.

Focus Groups

Focus groups bring people together to discuss a specific topic guided by a skilled moderator. Topics can range from large-scale policy direction to specific service offerings.

Focus groups are useful for identifying public sentiment and preferences, but they're based on what people say and not what people actually do. Studies show that what people say they like or dislike can be quite different from what they demonstrate during observation.

Case in Point: Electronics giant Philips conducted a focus group to determine whether the new sport yellow boom box should be offered for sale to the public. During the focus groups, everyone preferred the sport yellow boom box over the black boom box because it was youthful, vibrant and exciting. On their way out, participants were offered a choice of the yellow or the black boom box. They all chose the black one.

Self-Reporting

Self-reporting occurs when you ask people to tell you something specific about a product or service without researcher involvement. It can be done in person, or through online forms, email, texting or other channels. Self-reporting tools include online feedback forms, structured surveys, questionnaires, card sorting and diary studies.

Self-reporting allows you to reach a large number of people easily and to work with remote participants. However, because it is difficult to expand on participants' responses as the researcher is not present, it works best when paired with other tools that allow for further understanding of what is being examined.

Inspection and Critique

Inspection uses a set of guidelines, heuristics or a structured approach to review a design. These design reviews may be done by an expert or by a mixed group. The guidelines may be extensive like the usability.gov evidence-based guidelines or may be heuristics based on industry best practices like Nielsen and Molich's Ten Heuristics for Web Usability. A set of web pages, mobile screens or service journeys are evaluated methodically against the set of guidelines.

The areas that perform well or poorly are highlighted. Inspection is often used as an evaluation tool to improve usability, especially to catch more obvious interface errors before committing to a larger investment in usability testing. Critique plays a similar role during the design process but is less tied to specific guidelines. A designer asks team members to review concepts, sketches or screens and each member shares what they like, what they would improve, what they would omit and if there is anything else they would like to see.

Usability Evaluation

Usability evaluation is having real users do real tasks with either a prototype or an existing, live interface. Typically, a test moderator introduced specific goals like "try to renew your driver's licence" and prompts the participant to describe out loud what they're expecting to see, what path they expect to take and why they choose certain actions.

This approach helps identify areas where the labeling, flow or functionality doesn't fit with people's mental models and expectations. It can also help evaluate performance on certain tasks (like time saved completing a task) and the overall usability of a system. Testing sessions can be done one-on-one with a participant and an observer, in pairs or in groups of participants. It can also be done remotely with a screen-sharing tool.

Analytics

Analytics examines the log files and other behavioural traces that real users leave when surfing the web. Web analytics can show traffic trends, search terms and the paths people take through a site. But it doesn't provide the full picture; it doesn't tell you why people are going where they're going and why they're searching for certain terms. It also doesn't capture an entire cross-channel service experience as people move between web, phone and in-person interactions, so analytics methods are more meaningful when partnered with other research tools.

Inventories and Audits

Two common inventories are content inventories and service inventories. A content inventory captures all the web pages for a given site and a service inventory itemizes services. A content inventory is a record of every single web page your organization is responsible for and is often created in a spreadsheet. Once this is done, the business owners or subject matter experts audit the content and look for redundant, outdated or trivial (ROT) content, identify what needs to be written or rewritten and ensure that owners are identified and a regular review schedule is established. Service inventories are similar but with metadata about service details, channels, availability, audience, prerequisites, outcomes and ownership captured.

ACTION RESEARCH DESIGN

Action Research (AR) represents a growing field of educational research whose chief identifying characteristic is the recognition of the pragmatic requirements of educational practitioners for organized reflective inquiry into classroom instruction. AR is an process designed to empower all participants in the educational process (students, instructors and other parties) with the means to improve the practices conducted within the educational experience. All participants were knowing, active members of the research process.

Action research has been described as an informal, qualitative, formative, subjective, interpretive, reflective and experiential model of inquiry in which all individuals involved in the study are knowing and contributing participants. Action research has the primary intent of providing a framework for qualitative investigations by teachers and researchers in complex working classroom situations. *Some of the most widely accepted definitions of Action Research include following:*

- *Action Research:* Aims to contribute both to the practical concerns of people in an immediate problematic situation and to the goals of social science by joint collaboration within a mutually acceptable ethical framework.
- Action Research is a form of self-reflective enquiry undertaken by participants in social (including educational) situations in order to improve the rationality and justice of
 - Their own social or educational practices,
 - Their understanding of these practices, and
 - The situations in which the practices are carried out. It is most rationally empowering when undertaken by participants collabor-atively... sometimes in cooperation with outsiders.
- *Action Research:* Is the systematic study of attempts to improve educational practise by groups of participants by means of their own practical actions and by means of their own reflection upon the effects of those actions.

The action research framework is most appropriate for participants who recognize the existence of shortcomings in their educational activities and who would like to

adopt some initial stance in regard to the problem, formulate a plan, carry out an intervention, evaluate the outcomes and develop further strategies in an iterative fashion. In short, action research is characterized by those constraints and strengths given a research methodology intended to be a workable technique for working classroom teachers.

ACTION RESEARCH DESIGN

The essentials of action research design are considered by Elliott as per the following characteristic cycle:

- Initially an exploratory stance is adopted, where an understanding of a problem is developed and plans are made for some form of interventionary strategy.
- Then the intervention is carried out.
- During and around the time of the intervention, pertinent observations are collected in various forms.
- The new interventional strategies are carried out, and the cyclic process repeats, continuing until a sufficient understanding of the problem is achieved.

The protocol is iterative or cyclical in nature and is intended to foster deeper understanding of a given situation, starting with conceptualizing and particularizing the problem and moving through several interventions and evaluations. A representation of an AR protocol by Kemmis is provided in Figure.

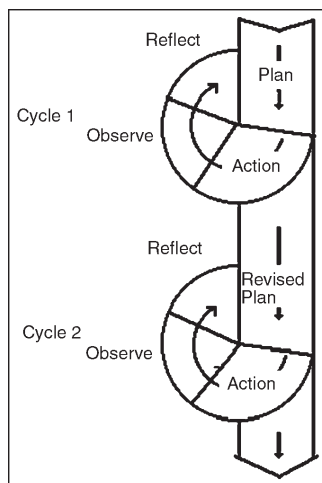


Fig. Action Research Protocol after Kemmis

Figure clearly displays the iterative nature of AR along with the major steps of planning, action, observation and reflection before revising the plan. This may be thought of as similar in nature to the numerical computing technique known as successive approximation - the idea is to close in upon a final goal or outcome by repeated iterations.

Later protocols reflect changes in the goal as determined via experience during the reflections of earlier iterations of AR. For instance, Figure reflects the evolution of the general idea or main topic of interest throughout the process.

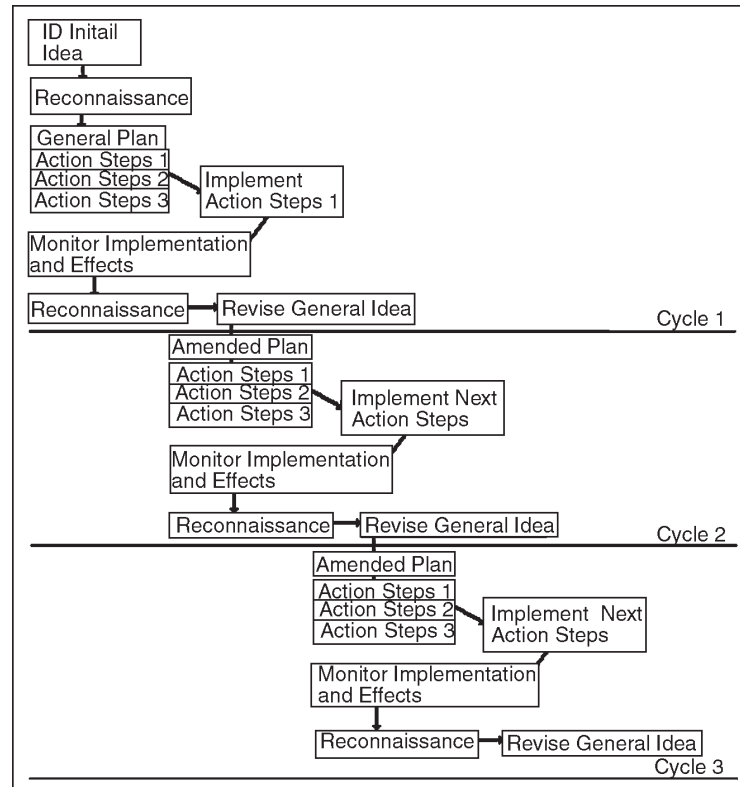


Fig. Action Research after Elliott

Elliott's model emphasizes constant evolution and redefinition of the original goal through a series of reconnaissances recurring every cycle. The reconnaissance necessarily includes some degree of analysis. This design permits much greater flexibility, and seeks to "...recapture some of the 'messiness' which the Kemmis version tends to gloss [over] ". Ebbutt further emphasizes the evolution of the overall plan through a spiral analogy.

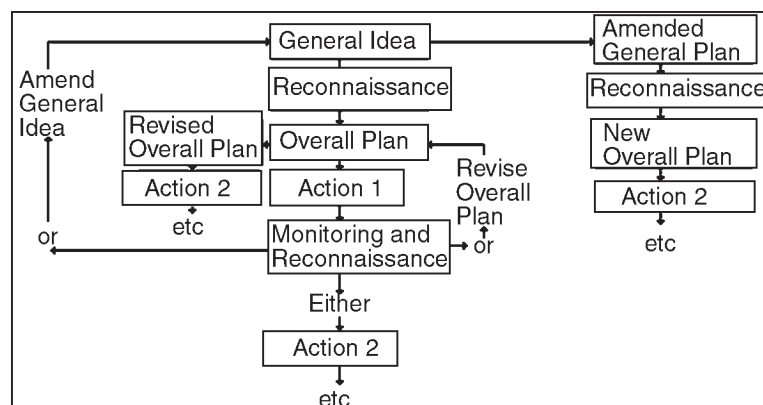


Fig. Action Research Protocol after Ebbutt

THE ROLE OF COMMUNICATION

Another distinguishing characteristic of Action Research is the degree of empowerment given to all participants. Involvement is of a knowing nature, with no hidden controls or preemption of direction by the researcher. All participants negotiate meaning from the data and contribute to the selection of interventionary strategies, including the university researchers, the teachers and the students.

THE ROLE OF REFLECTION

Another distinguishing characteristic of action research is the degree of empowerment given to all participants. Involvement is of a knowing nature, with no hidden controls or preemption of direction by the researcher. All participants including the university researchers, the teachers and the students negotiate meaning from the data and contribute to the selection of interventionary strategies.

Elliott considers the need for communication between all participants to be of paramount importance:

Since action research looks at a problem from the point of view of those involved it can only be validated in unconstrained dialogue with them....Since action research involves unconstrained dialogue between "researcher" (whether he be an outsider or teacher/researcher) and the participants, there must be free information flow between them.

Perhaps the key component involved in action research is the notion of praxis. Action research is intended to be the reflective counterpart of practical diagnosis. Schon describes the use of reflection to generate models from a body of previous knowledge. These models are used to re-frame a problem; then experiments are performed to bring about outcomes which are subjected to further analysis. This model (called reflection-in-action) frames means and ends interdependently and recognizes that there is little or no separation of research from practice, little or no separation of knowing and doing. Schon's model of reflection-in-action compliments the iterative and investigative natures of action research.

3

Research Policy and Research Management

There has been a clear increase in the role of the private sector in agricultural R&D, both profit and non-profit. This is due to two forces. On the demand side, it has resulted from democratization and decentralization of governance, increasing organization of civil society, and structural adjustment policies which have reduced the role and the budgetary capacity of the public sector.

This has led to the privatization of many agricultural services including variety development, seeds, animal health services, marketing boards, water delivery, and technology transfer.

On the supply side, advances in biotechnology, stronger intellectual property protection, improved information systems that facilitate access to scientific and market information, and more open market policies have expanded the range of technologies whose benefits can be appropriated by the private sector. For technologies which remain public goods, the supply side has also frequently been privatized under funding by foundations or national and international grants. Transfer of services to the private sector has been unevenly successful and the distribution of benefits from the change has been unequally distributed across the farm population, with some sectors benefiting while others have been left unattended. The CG focus on research management consequently needs to be broadened to understand the whole spectrum of public, non-profit, and private actors involved. Steps have been taken in this direction, with ISNAR working with NGOs, and looking at the complementary roles of the private and public sector, particularly in Latin America.

The panel consequently recommends that the CG mandate on research policy and research management be broadened to capture not just the traditional public management dimensions of R&D, irrigation, and forestry but also the public-private interface, the co-production of public goods by public and private sectors, and the role and management problems of NGOs and other non-profit organizations of civil society.

POLICY AND MANAGEMENT DECISIONS

We observed that most CG policy research had addressed the question of policy impact and policy advice, while little had been done on policy making and policy-making advice. Yet, the full payoff from policy impact analysis requires that it be imbedded within policy making analysis. This is because the political feasibility of policy recommendations requires assessing policy making.

For instance, recommendations of decentralized governance for participatory rural development may be vacuous, as shown by the experience with community development in India, without an understanding of the decision-making process at the local level. The policy-making game is fundamental for both positive analysis and for policy-making advice. Equally, important is to assess the credibility of policy in the sense of time consistency. If policy making institutions are not credible, for instance a Central Bank which is not autonomous from government and need to assume deficits created by soft budgets constraints, neither is monetary policy credible.

The need for political economy analysis also applies to management research. It is important in water management since much of the planned allocation of water is perverted by rent seeking. Patterns of rent seeking need to be understood to improve the efficiency of existing systems, to facilitate the transition to water markets, and to design effective water users' associations. It is also important in research management since rent seeking pressures distort the socially optimal or CG-optimal allocation of resources. In addition, not all resources that enter a research system contribute to research, with extensive deviations of funds in response to rent seeking forces in some countries.

This Panel is informed of previous recommendations made by internal and external reviewers as well as of past attempts to develop this important area of research and of modest ongoing initiatives. It recommends that every effort be made to continue to explore possibilities of making headways in research on the political economy of policy and management decisions.

RESEARCH ON INSTITUTIONS

An area of economic theory where there have been major advances in the last fifteen years is the "new" institutional economics. These theoretical advances have allowed to bring behaviour into traditional institutional economics and to thus understand the determinants of the choice of institutions for efficiency purposes.

This body of theory postulates that technological and institutional changes are brought about to reduce transactions and transformation costs (North and Wallis, 1994). In

particular, market failures due to transactions costs or to problems of imperfect information induce the quest for institutional alternatives that allow to achieve the objectives sought (production, consumption, or exchanges) more efficiently than through markets. This theory has been highly successful in the field of agrarian institutions, for instance to explain the choice of land tenure contracts (sharecropping versus fixed rent), optimum farm size (family versus commercial farms), labor relations (permanent versus seasonal labor; time versus piece rate wages), vertical integration, the Grafting of informal financial institutions (group lending, savings and loan cooperatives), the organization of cooperative users associations (water users' associations, service cooperatives, social forestry schemes), etc. It has also been important in the field of the environment, for instance for the definition of rules for the management of common property resources and the definition of incentive contracts for the conservation of resources. Finally, it has also been highly successful in the field of industrial organization and management to answer such questions as why firms are organized the way they are, how to design incentive compatible regulatory agencies, and how to design incentive systems in firms and organizations.

In general, use of this approach has been seriously lagging in CG research. Notable exceptions include work on intra-household food distribution and inequality (e.g., Haddad and Kanbur, 1990), and work on mutual insurance schemes and consumption smoothing using the ICRISAT panel data (but largely done by non-CG scientists). We are aware of the fact that some of the research on institutional economics has been highly academic, with little of relevance for normative policy purposes and risks of engaging in marginal issues for the CGIAR. Yet, this is not the rule. Very profitable uses have been made of this approach, ranging from socioeconomic research on household behaviour under imperfect markets, to the Grafting of agrarian institutions in response to the vacuum left by state contraction, and to the design of research and management institutions for technology and water among others. In management research, institutional issues arise in the design of the optimum interface between private and public roles. For germplasm conservation, questions of incentive structures for in-situ conservation require the design of implementing and regulatory agencies.

This Panel recommends that all Centres make a serious efforts to get better acquainted with the potential that the "new" institutional economics offers for their socioeconomic, management, and policy research. It suggests that scientists concerned in different centres collaborate in systematically exploring areas where worthwhile advances could be made using this approach.

Evolution of Classical Approach to Management

Traditional process of learning is either through observation and experiment. Nature or environment is considered uniform and when we observe certain phenomenon or events uniformly leading to the same result or results, we conclude a cause and effect relationship between the two.

This is learning by observation or in other words by experience. Earlier thinkers on management followed this approach in developing theories of management.

Learning principally is through empirical process and through analysis of the data collected through observation. Draw the principles of management by looking at and analysing the jobs that all managers commonly do.

This approach served as a starting point for pioneers on management science to verify the validity and improve the applicability of the principles and practices of management.

Analysis of observed data is what constitutes a case study. The observational method of case study helps arriving at logical conclusions about past experience and to test the same as standards for future events.

The German sociologist, Max Weber followed the classical approach and developed his theory of Bureaucracy, which portrays the structure and design of organisation characterised by a hierarchy of authority, formalised rules and regulations that serve to guide the coordinated functioning of an organization.

Basic Postulates of the Classical Approach by Max Weber

1. Management of an organization is considered as a chain of inter-related functions. The study of the scope and features of these functions, the sequence through which these are performed and their inter-relationship leads one to draw principles of management suitable for universal application
2. Learning principles of management is done through the past experiences of actual practicing managers
3. As business environment consists of uniform cycles exhibiting an underlying unity of realities, functions and principles of management derived through process of empirical reasoning are suitable for universal application
4. Emerging new managers through formal education and case study can develop skill and competency in management concepts and practices
5. The classical approach also recognised the importance of economic efficiency and formal organizational structure as guiding pillars of management effectiveness.
6. Business activity is based on economic benefit. Organizations should therefore control economic incentives

Neoclassical theory of management

There are 3 neoclassical theories:

Human Relations theory: Explains the modern advancement of Human Relations Management theory which takes into account human factors like the employer-employee relationship. Human relations theory is largely seen to have been born as a result of the Hawthorne experiments which Elton Mayo conducted at the Western Electrical Company.

The important strand in the development of modern management was the increase in attention to the human factors, which has become known as the ‘human relations school of management.’

The core aspect of Human Relations Theory is that, when workers were being observed and included in the research, they felt more important and valued by the company. As a result, their productivity levels went up significantly.

This represented a significant departure from many of the classical theories, particularly Fordism, as it went against the notion that management needed to control workers, and remove their autonomy at every step. Instead, it showed that by engaging with workers and considering their requirements and needs, company’s could benefit from increased productivity.

Behavioral theory: The behavioral management theory is often called the human relations movement because it addresses the human dimension of work. Behavioral theorists believed that a better understanding of human behavior at work, such as motivation, conflict, expectations, and group dynamics, improved productivity.

The theorists who contributed to this school viewed employees as individuals, resources, and assets to be developed and worked with — not as machines, as in the past. Several individuals and experiments contributed to this theory.

Social systems theory.: Developed by Niklas Luhmann is an option for the theoretical foundation of Human Resource Management (HRM). After clarifying the advantages of using a grand (social) theory as the basic theoretical perspective, the roots of this social systems theory - the deterministic view of systems as machines, the open systems approach and non-linear systems theory - are addressed.

Based on the view of social systems as autopoietically closed systems, five major contributions to a theoretical foundation of HRM are identified: (1) the conceptualisation of organising and managing human resources as social processes, thus overcoming an individualistic angle; (2) the new importance of individuals as essential element in the system’s environment; (3) the abstention from far reaching or highly unrealistic assumptions about the ‘nature’ of human beings; (4) the interaction between various levels and units of analysis built into the theory which is essential for comprehensive and in-depth analyses of HR phenomena and (5) the openness for additional theories for which social systems theory provides the overall framework. George Elton Mayo was in charge of certain experiments on human behavior carried out at the Hawthorne Works of the Western Electric company in Chicago between 1924 and 1927.

His research findings have contributed to organizational development in terms of human relations and motivation theory.

Elton Mayo’s contributions came as part of the Hawthorne studies, a series of experiments that rigorously applied classical management theory only to reveal its shortcomings. The Hawthorne experiments consisted of two studies conducted at the Hawthorne Works of the Western Electric Company in Chicago from 1924 to 1932. The first study was conducted by a group of engineers seeking to determine the

relationship of lighting levels to worker productivity. Surprisingly enough, they discovered that worker productivity increased as the lighting levels decreased — that is, until the employees were unable to see what they were doing, after which performance naturally declined. A few years later, a second group of experiments began.

Harvard researchers Mayo and F. J. Roethlisberger supervised a group of five women in a bank wiring room. They gave the women special privileges, such as the right to leave their workstations without permission, take rest periods, enjoy free lunches, and have variations in pay levels and workdays. This experiment also resulted in significantly increased rates of productivity.

Policy and Management Research in the System

The overall importance of social science research in the CGIAR and the relative importance of its three components—socioeconomic, policy, and management research. These data are derived from answers to the questionnaire sent to Centres for this study. The overall allocation of CG scientists to the social sciences is 17.4% and the corresponding budget allocation is 14.8%. This indicates that the social sciences have assumed a major role in the System, raising the question, whether this share seems to be too high or not. The smaller share of budgets compared to scientists presumably reflects the fact that social scientists are cheaper because they do not require extensive laboratory facilities. Social scientists are not only engaged in research but also importantly in outreach, including institutional development, and in training activities.

In addition to IFPRI, IIMI, and ISNAR which are primarily social science oriented, some other Centres are heavily vested in the social sciences. This includes CIFOR (34.8% of its scientists), ICRAF (17.3%), ICARDA (16.4%), ILRI (14%), CIP (13.5%), ICLARM (12.5%), and CIAT (10.2%). The larger crop improvement and primarily biophysical centres such as CIMMYT (8.1%), IRRI (8.3%), and ICRISAT (8.9%) have lower shares of their scientific staff in the social sciences. In the “Other Centres”, socioeconomic is vastly more important than policy and management research: overall the distribution of scientists is 8.3% to socioeconomic, 3.2% to policy, and 0.4% to management. In the three social science centres, policy is the dominant activity due to IFPRI’s weight (45.3% of all scientists), followed by management (9.8%), and socioeconomic (9.2%). These three social science centres account for 103 of the System’s 937 scientists, or 12.4% of the total.

Management Research

Centres specialized in management research are ISNAR for research on research policy and on the management of research, and IIMI for research on water management.

- ISNAR tackles several management issues in its programmes. Some of these are not “management” in a narrow sense, such as research programme design, which is concerned with both macro-level priority setting that belongs to policy, and with institute level scientific programme management. Other aspects that

have received considerable attention are monitoring and evaluation systems generally, as well as some specific thematic issues, such as organizing for natural resource management research. ISNAR's programme on "Management of Organizations and Resources" gave considerable attention to the development and installation in several countries of a NARO management information system (INFORM). Other aspects of ISNAR's research include the management of human (including gender), financial, physical, and information resources by NARSs.

- IIMI engages in a wide range of aspects of water management research that includes assessment and improvement of the performance of irrigated agriculture, sector-level management of irrigated agriculture, improvement of public irrigation organizations, local management of irrigation systems, and the sustainable management of water delivery and disposal. Cross-cutting themes include gender issues in the performance of irrigated agriculture, health and environmental problems related to irrigation, analysis of technological options, and development of a global data base on irrigated agriculture. An important feature of IIMI's management work is that, following extensive privatizations and devolution of management to users, it has expanded its research beyond the public management of irrigation systems to the role of water users associations and the determinants of cooperative behaviour in the management of common-property water resources.
- Among other Centres, most declare not to have a specific management research component, even though management aspects may enter indirectly through socioeconomic and policy research. Centres with no explicit management research are CIAT, CIMMYT, ICLARM, ICRAF, ICRISAT, IITA, ILRI, and WARDA.
- CIFOR devotes a small fraction (1.3%) of its budget to management research. This research includes a survey on research manpower in the South African development community, and identification of the policy and management barriers to application of forestry research results.
- CIP has minimal management research that mainly enters through projects with socioeconomic or policy objectives. This is for instance the case with research on characterization of constraints and opportunities for potato production in the Andes.
- ICARDA is conducting institutional research on property rights and public/local institutions that regulate use of rangelands (in cooperation with IFPRI). It is seeking support to expand its research on the public management of underground water for irrigation.
- IPGRI does research on the design, organization, and management of plant-genetic resource activities in NARSs.
- IRRI conducts management research on the setting of priorities in rice research. This includes analysis of yield gaps, identification of research priorities and

estimation of payoffs from research, and assessment of rice-research capacity in the Asian NARSs.

While principally engaged in policy research, IFPRI also conducts management research. This includes public management research which is an important component of public policy research. In addition, IFPRI has become increasingly involved in research on research management, water management, and forestry management, generally in collaboration with the corresponding specialized Centres, namely ISNAR, IIMI, and CIFOR.

Socioeconomic Research

This is the most important area of social science research in the Centres other than IFPRI (policy research) and ISNAR and IIMI (management research). The nature of this research has been analysed in another TAC special study and we only make general reference to this type of research principally because it is both essential for policy and management research and often hardly separable from these two. Even IFPRI, ISNAR, and IIMI do considerable socioeconomic research since analysis of the behaviour of consumers, producers, households, communities, associations, managers, rent seekers, and policy makers are fundamental cornerstones of policy and management research.

In principle, socioeconomic research conducted by Centres where biophysical research is dominant has as a purpose to guide research resource allocation toward priority areas, help in technology design, assess the determinants of adoption of new technologies, and determine the impact of diffusion of these technologies on welfare and the environment. For this type of research, the biophysical centres have a comparative advantage over IFPRI as they can most easily mobilize the necessary interdisciplinarity with biological scientists.

This research has an important service function in giving guidelines to researchers and the Centres. In fact, socioeconomic research covers a wide array of issues, showing that it has become an integral component of research programmes as research questions and the CO mandate have become increasingly more complex and holistic. No longer is socioeconomic research introduced as a prop to the definition and diffusion of technological change, but it has become an undissociable component of the search for solutions to complex interdisciplinary problems. This is, of course, the main reason why social science research has acquired growing importance in the CG's total research portfolio.

It is worth noticing that 17 out of 78 scientists devoted to socioeconomic research in the CGIAR, or 22%, are at ICRAF. This percentage raises to 25% when IFPRI, ISNAR, and IIMI are excluded. The Panel could not look into the reasons for this disproportionately high presence of socioeconomic research at this particular Center.

Some examples of the wide array of issues addressed by Centres' socioeconomic research is indicated by the following answers to the Centres' questionnaire:

- CIAT focuses mainly on the analysis of prototype production systems and participatory research methods.
- CIFOR conducts research on household strategies and community differentiation in the production of non-traditional forest products in Sri Lanka, household economics and forest use in Brazil, interactions between environment and population in Thailand, forests and food security, income generation and incentives for forest management, and the social dimensions of sustainability criteria.
- CIMMYT's socioeconomic research addresses questions of global maize research and the maize seed industries, the conservation of germplasm diversity, impact assessment of research in wheat and maize, priority setting and research resource allocation (with IFPRI), the determinants of adoption of long-term conservation investments, and the diagnosis of natural resource management issues.
- CIP's socioeconomic research is integrated in the same topics that are part of its policy research programme mentioned above.
- ICARDA's socioeconomic research focuses on the economic analysis of production systems and the ex ante and ex post analysis of new technology options. Projects address integrated crop/livestock production, water resource management, water harvesting, and farmers' participation in and use of local knowledge at breeding barley for specific adaptation.
- ICLARM conducts socioeconomic research on the dissemination and evaluation of genetically improved fish strains, the bioeconomic modelling of capture fisheries, and bioeconomic models of coastal aquaculture systems.
- ICRAF's socioeconomic research includes the development of typologies of land-use systems and farmers, the development of ecological-economic models which predict the adoption potential and sustainability of improved agroforestry practices, and the evaluation of the actual ecological, social, cultural, and economic impacts of adoption of agroforestry practices.
- IITA's socioeconomic research includes improving postharvest systems, the diversification of humid forest production systems, improvement of yam-based systems, and the determinants of cassava productivity under various ecological conditions.
- ILRI's research includes a wide range of themes focusing on the economic and environmental impacts of livestock disease control, the economics of mixed crop-livestock systems, the socioeconomic analysis of livestock production and natural resource management, the economics of peri-urban dairy systems, and impact assessment and priority setting studies.
- IPGRI's research addresses the role of gender in plant genetic resource conservation and use, documentation of indigenous knowledge for farmers and scientists, and the effect of conservation and use of germplasm on socioeconomic development through agricultural development.

- IRRI's socioeconomic research focuses on the determinants of adoption of knowledge intensive rice technology, understanding the risks in rainfed rice farming, and the impact of rice technology on rural income distribution and poverty.
- WARDA's research agenda includes farmers' perceptions and adoption of new rice varieties, the economics of rice-based cropping systems, agricultural intensification and women, economic constraints to rice based rotations in the Sahel, differentials in economic efficiency for small and large rice farmers, and the rate of return to rice breeding research.

Evolution of Strategic Management

Gluck, Kaufman, and Walleck propose a model of the evolution of strategic management within organizations whose managers have a propensity to plan. The model is based on a study undertaken by McKinsey and Company. Part of the reason for including a summary of their model here is to help students recognize the strategic management stage in which a particular organization might be. Such information may provide additional insight into the operation of the firm and help understand the degree of strategic management progress made by managers. More importantly, however, it may be useful in helping students understand strategic management as the highest level of a hierarchy of management activities.

The McKinsey analysis discovered four quite distinct phases of strategic management evolution. In Phase 1, financial planning, management focuses on the preparation of budgets with an emphasis on functional operations. Most organizations have a budgeting process, in at least rudimentary form, as a way of allocating resources among functional units, subsidiaries, or projects.

The second, forecast-based planning, follows naturally from the first as managers project budget requirements beyond the one-year cycle.

This phase represents an effort to extend managers' attention beyond the immediate future as scenarios are developed which describe their expectations about future time periods. Budgets are often constructed for several years at a time and are rolled over annually so that the appropriateness of a budgeted amount can be reviewed several times before it is operationalized. Phase 2 planning is very "now" oriented.

Current operations and characteristics are stressed in analyses of the firm and there is little attention to or patience for considering operational options or development of strategic changes. The business portfolio of a Phase 2 firm is often viewed as the final expression of strategy rather than as an input to the strategy formulation process. Current structure and business activities may be considered fixed, not as strategic variables.

Planners are required to adopt an external orientation and tools and procedures for environmental and internal assessment.

Concern centers on understanding the organization's environment and competitive position and generating ideas about how the company might better fit its environment.

Several choices, contingency plans, are often devised for how the company might fit its environment. Lower level planners and managers are often involved in the process of generating choices, an activity that soon puts top management in the position of choosing a plan in which it had little involvement in developing.

Strategic management, evolves as top management senses the need to more heavily invest in the planning process because of its lack of understanding of or involvement in the details of earlier plan development. Strategic management is the meshing of Phase 3 planning and operational management into one process.

It is analysis and conclusion that takes place year-round and ties performance evaluation and motivational programs to strategy.

Deliberateness of Strategy

Sometimes outsiders impute strategy to the behaviour of firms. Obviously, students analyzing case studies are placed in this position when they impute strategy from the data they are able to generate on the firm's operations. Similarly, journalists and the managers of competing firms may impute strategy to a firm's behaviour; and it may or may not accurately reflect the real strategy in place.

Outsiders may also imply intent to an imputed strategy. That is, they assume not only that the strategy they imputed from the firm's behaviour is the real strategy its employees are implementing, but they imply that this strategy is the one intended for the firm by its management. Seldom is this the case.

Mintzberg developed a taxonomy which is useful for discussing the realism and deliberateness of strategy. First, he distinguished between strategy that is the result of a plan, and of a pattern of behaviour.

He referred to them as "strategy as plan" and "strategy as pattern," respectively. Strategy as plan is a chosen course of action; it could be a real strategy (one intended for implementation) or a ploy (a tactical move whereby a competitor may be influenced into making a mistake). Some people think that Coca-Cola's rumored change in Coke's formula in the mid-1980s was such a ploy.

The implication is that Coca-Cola had not intended to really change the formula. Instead, it is thought, they publicized their plan to change formula, introduced a new product with a different formula that tasted a lot like a competitor's product, and finally graciously conceded to continue producing the old formula product when the public demonstrated a preference for it over the new—"similar to a competitor's—" formula. Incidentally, if this was in fact a ploy, it has to rank among the top marketing moves ever attempted by any business. Coca-Cola reaped an immediate increase in market share of about 15 per cent that thrust them once again into unquestioned dominance in the huge U.S. softdrink market.

Strategy as plan, when implemented, may or may not be what the firm ends up with. That is, the planned strategy could ultimately be either realized or unrealized. If it is realized, then the entire process would be a textbook case of strategy formulation and

implementation in the sense that the firm successfully implemented what was intended. But what happens if the planned strategy is implemented and, for some reason, the strategy that is realized is not the intended one? We might say that the planned strategy was unrealized, and the realized strategy (the one that seems to describe what the company is actually doing) arises out of some consistency in the behaviour of the company.

Mintzberg and Waters call this unintended realized strategy, “strategy as pattern,” or a pattern in a series of actions by the organization. Strategy as pattern is what you will end up with when you impute strategy to the behaviour of a company you are analyzing in a case study, or what journalists produce when they attribute a strategy to a company based only on its actions. Thus, a realized strategy could be either a deliberate strategy as plan, or an “undeliberate” strategy as pattern. If the realized strategy was planned and also accurately described the firm’s actions, then strategy as pattern and strategy as plan would be synonymous. However, when realized strategy is not intended strategy (that is, it was either not what was intended by management when they drafted a planned strategy, or they drafted no strategy at all), then it simply “grew” out of the activities of the company. In Mintzberg’s terms, it “emerged” as a pattern of behaviour in the absence of intension, or despite unrealized intentions.

A realized strategy is what a company is actually doing. If it is the one intended by management, then it is deliberate. If not, then the intended strategy was unrealized, and the realized strategy is emergent. An emergent strategy is, by definition, not deliberate.

However, a manager may choose not to consciously formulate strategy and, instead, “go with” the emergent one. But even here, the resultant emergent strategy would not have been deliberate in the same way an intended strategy would have been.

Often it is convenient to distinguish between intended and emergent strategies. When management performs no strategic management at all, they still will have a realized strategy that is emergent. This emergent strategy could be recognized by outsiders (and insiders for that matter) even though it may not have been intended by management.

Contingency Strategies

Managers may develop several versions of strategy, each of which is intended to fit a different environmental scenario. An environmental scenario consists of a set of environmental characteristics that describe a firm’s external circumstances. By structuring external variables into sets of related variables, different scenarios result. Thus several scenarios (usually three) can be arrayed on a probability distribution for which the horizontal axis is a measure of the extent to which they are viewed as “good” or “bad”—supportive or not supportive—to the firm.

In the scenarios that fall on the left side of the distribution are considered pessimistic (undesirable but only a small probability that they might have to be dealt with); those on the right as optimistic (desirable but probably will not materialize either); and those in the middle as most likely to occur and having some good and bad features.

Next a strategy set is established for each scenario. These could be entirely different strategies or variations of one. Evaluative criteria then are formulated for each scenario. These are variables to be monitored in order to figure out which of the scenarios is going to apply in the future.

Actually this is a matter of deciding whether the present scenario will continue to be the best description of the firm's environment or, instead, if one of the others seems to be emerging as most correct. By identifying critical values for evaluative criteria, the strategist knows beforehand under what set of conditions to switch from the present strategy set to the pessimistic or optimistic set.

Summary

This chapter serves to familiarize the reader with the underlying principles of strategic management and to outline the path through that field taken by the chapters. First, the history of strategic management as an academic discipline is reviewed to provide perspective on its current stage of development. Next, summarizes the results of some of the major studies conducted to answer the question "Does strategic management pay off?" Also discussed in this section are the primary methodological problems encountered in doing such research.

The mission statement is considered an input to the strategy formulation process, which is outlined in the next section. Strategy at four organizational levels comprise the focus of this section and the book's text. They are the enterprise, corporate, business, and functional levels.

Basic Methods of Market Research

While there are many ways to perform market research, most businesses use one or more of five basic methods: surveys, focus groups, personal interviews, observation, and field trials. The type of data you need and how much money you're willing to spend will determine which techniques you choose for your business.

Surveys: With concise and straightforward questionnaires, you can analyze a sample group that represents your target market. The larger the sample, the more reliable your results will be.

- In-person surveys are one-on-one interviews typically conducted in high-traffic locations such as shopping malls. They allow you to present people with samples of products, packaging, or advertising and gather immediate feedback. In-person surveys can generate response rates of more than 90 percent, but they are costly. With the time and labour involved, the tab for an in-person survey can run as high as \$100 per interview.
- Telephone surveys are less expensive than in-person surveys, but costlier than mail. However, due to consumer resistance to relentless telemarketing, convincing people to participate in phone surveys has grown increasingly difficult. Telephone surveys generally yield response rates of 50 to 60 percent.

- Mail surveys are a relatively inexpensive way to reach a broad audience. They're much cheaper than in-person and phone surveys, but they only generate response rates of 3 percent to 15 percent. Despite the low return, mail surveys remain a cost-effective choice for small businesses.
- Online surveys usually generate unpredictable response rates and unreliable data, because you have no control over the pool of respondents. But an online survey is a simple, inexpensive way to collect anecdotal evidence and gather customer opinions and preferences.

Focus groups: In focus groups, a moderator uses a scripted series of questions or topics to lead a discussion among a group of people. These sessions take place at neutral locations, usually at facilities with videotaping equipment and an observation room with one-way mirrors. A focus group usually lasts one to two hours, and it takes at least three groups to get balanced results.

Personal interviews: Like focus groups, personal interviews include unstructured, open-ended questions. They usually last for about an hour and are typically recorded. Focus groups and personal interviews provide more subjective data than surveys. The results are not statistically reliable, which means that they usually don't represent a large enough segment of the population. Nevertheless, focus groups and interviews yield valuable insights into customer attitudes and are excellent ways to uncover issues related to new products or service development.

Observation: Individual responses to surveys and focus groups are sometimes at odds with people's actual behaviour. When you observe consumers in action by videotaping them in stores, at work, or at home, you can observe how they buy or use a product. This gives you a more accurate picture of customers' usage habits and shopping patterns.

Field trials: Placing a new product in selected stores to test customer response under real-life selling conditions can help you make product modifications, adjust prices, or improve packaging. Small business owners should try to establish rapport with local store owners and Web sites that can help them test their products.

Framework for Choosing a Research Design

One reason why secondary research sources are often omitted is institutional. Most research is carried out by market research agencies which for various reasons find problems in offering desk research as a profit earning service. In this respect in-house research can have an edge over buying-in. Once desk research is completed (and assuming it does not yield all the information required for the project) primary research or fieldwork can be planned to fill in the gaps.

Think About

When did you last visit a commercial library? What do you think you could obtain from a library that you cannot obtain from the internet? Are you aware that some libraries

will carry out short desk research assignments for you for a small fee? If you had £5,000 to spend, how much could you find out by desk research?

How much could you find out by primary research?

Decisions in primary research design.

When contemplating the research method for primary data, the following should be considered:

- Whether the nature of the information sought is primarily quantitative or qualitative.

Sampling issues including:

- The universe to be covered e.g.: all adults, housewives, buyers of specific products in consumer markets and comparable aspects of industrial markets.
- Sectors or sub-groupings of the whole universe which are of specific interest and need to be considered in sampling design.
- Sampling method and size (number of interviews).
- Data collection methods – e.g. face to face or phone interviews, postal survey, observation etc.
- How the resulting data will be analysed – particularly relevant for more sophisticated techniques such as conjoint analysis.

The basis of which of the above should be covered in the design, depends on the research objectives, the information coverage, and the accuracy sought. Or at least this should be so in theory. In practice other factors have to be taken into account and especially the resources available including the budget.

For example, the ideal method may be face to face interviews with a sample of 1,000 people in their home. However, but for practical reasons (speed and cost) street interviews with 200 may be used instead. There is, of course, some point beyond which compromises are a danger to the accuracy of the result. What budget should be made available for the research project?

This is not a question of arguing that the budget should be whatever is needed to meet the research objectives at the required accuracy level. It is more a question of what funds are available or can be afforded for the project relative to other calls on business expenditure. Furthermore, even if cash is freely available, there are other considerations and especially the amount at risk in the decision which the research is to guide. For example, a research budget of £30,000 may be well worth spending if it is to guide a decision that entails capital expenditure of £5 million. However, if the decision has much lower cost implications, the value of doing the research will be less and obviously there is no point to spending £30,000 on research to decide whether to invest in a project entailing only this level of expenditure. For this reason, many businesses cannot afford to test the effectiveness of their advertising campaigns because the cost of the research is more than the campaigns themselves.

Apart from a cash budget, the major resource required for a research project is suitably qualified people. An in-house researcher may be able to carry out desk research but

would find it very difficult to carry out extensive fieldwork and for this would need the help of an agency. A research plan needs a timetable. The deadline for the project may well be driven by external events such as a board meeting or input for a marketing plan. The timeline available for the research often influences the research methods. Face to face interviews may be preferred but telephone interviews are quicker. Certainly good research can be carried out within a short timetable but beyond some point, quality will be compromised if the timetable is squeezed too much.

Need for and Meaning of Marketing Research

Many specific marketing problems can be solved on the basis of available information and existing knowledge. Past experience and previous research, coupled with sound judgment, have made possible the development of principles and rules of conduct that provide ready answers in a great many situations. New and unusual situations continually arise, however, and some old ones appear in new and even strange forms and under substantially different circumstances. In such cases, sufficient information or knowledge is not likely to be at hand as a basis for an intelligent solution. Instead, one of two general courses of action may be taken. One is to make a guess or perhaps an estimate of the situation and hope that it is the correct solution. The other involves the obtaining of necessary information in order that a reasonably accurate decision may be made on the basis of facts and sound application of principles.

The latter course of action entails the utilization of marketing research, because *the essential purpose of marketing research is to provide the necessary information or knowledge on which to base sound decisions in the solution of marketing problems when - readily available information is insufficient or completely lacking.*

Illustrations of situations where available knowledge and established principles are adequate for soundly conceived action are numerous and varied. The large-scale flour-miller, for example, receives regularly the crop reports of the U.S. Department of Agriculture. Such information, which relates to probable future supply, is an important factor influencing the price of grain. The interpretation of this information, in the light of accumulated experience in the trade, makes possible more intelligent decisions about inventory acquisitions than could be made otherwise. A second situation may be illustrated with respect to the case of a new and fairly small vegetable canning factory.

Because production is seasonal, the line of merchandise is small, and the potential market is widely scattered, knowledge of available principles indicates the use of brokers in the channel of distribution. Many additional examples of this kind could be cited. However, if general principles are not available, it becomes necessary to make a guess or to conduct a research investigation in search for the answer. The individual entrepreneur or manager who makes decisions on the basis of established principles owes his ability to make such decisions to the past research activities of marketing students or practitioners whose efforts have resulted in the formulation of principles or have provided accessible information as a guide in determining the course of action.

When principles are lacking or when available knowledge and information are less than adequate for sound decisions, it is the accepted function of marketing research to supply essential facts. Marketing facts comprise the information or knowledge requisite to the formulation of general principles, or more narrowly, to the sound solution of marketing problems within particular trades or institutions in a given situation. It is the task of those engaged in marketing research to determine what facts are needed, where and how they are to be assembled, how they are to be described and classified, what hypotheses are warranted by them, how those hypotheses can be verified by the assembly of additional facts, how those facts are to be finally interpreted, and what application may be made of the results.

From the foregoing discussion, the meaning of marketing research should be clear. It may be defined as any purposive investigation which has as its objective the obtaining of knowledge of any business activity necessary to ascertain the needs and wants of markets, plan product availability, effect transfers in ownership of products, provide for their physical distribution, and facilitate the entire marketing process. This covers research involving any market, marketing institution, marketing function, marketing cost, or marketing policy.

It applies equally to information obtained originally from the field, as in the case of a consumer questionnaire survey or an enumeration of business establishments, and to analyses of quantitative data and qualitative information available in published sources or in the files of the business enterprise, so long as the purpose of the investigation is that of adding to the fund of currently existing marketing knowledge.

Trends in Marketing Research

Marketing research, as a specialized field of activity, is relatively new. As recently as 1920, firms actively engaged in specific marketing research activities, carried on in a department created for such a purpose, were rare indeed. The Curtis Publishing Company and the General Motors Corporation are outstanding early examples. Even by 1940 it was still the rather exceptional company that recognized marketing research as a specific function. Early marketing research activity tended to be narrowly restricted, in line with common business attitudes about the then prevailing concept of the job of marketing—namely, to dispose of the output of field and factory through selling effort. Hence, pioneering marketing research activity, as carried on within business firms, tended to emphasize the collection of information about markets and about the selling function of marketing.

Since about 1950, the number of companies with specialized marketing research departments has greatly increased. It is now common to find persons with the title Director of Marketing Research among medium sized and large manufacturing companies, public utilities, publishing companies, advertising agencies, wholesaling corporations, retailing chains, financial institutions, transportation companies, and many other kinds of enterprises.

Not only has the number of marketing research practitioners increased greatly, but also the scope of their interests and activities has enlarged and become diversified. Whereas early marketing research tended to be confined to market characteristics and to analyses of selling effort, it is now commonplace for it to deal with buying policies, merchandise handling, location of facilities, transportation costs and methods, inventory levels and controls, order processing methods, risk management, credit and financial management, product development and testing, and general marketing administration.

The widening and enlarging scope of marketing research activities reflects an increasing acceptance of a broad and customer-oriented conception of marketing, compared with earlier and rather provincial attitudes that tended to equate marketing with selling effort. As the conception of marketing by business leaders broadens to coincide with that held by academicians, the scope of marketing research in business will correspondingly expand.

The nature of contemporary marketing research activity has also been influenced considerably by various trends relating to approaches, techniques, methods, and sources. First, there has been growing awareness that every business firm generates a great deal of internal information that can be utilized in solving marketing problems. Second, external sources of published and readily available marketing information have multiplied manifold and have been characterized by substantial improvement in quality of data. It is no small task for a research practitioner merely to keep informed about such information.

Third, many of the crude survey methods that were a common attribute of early marketing research have been supplanted by more advanced techniques borrowed from other areas in the social sciences. A fourth trend has been the greater use of mathematical models that involve sophisticated techniques of quantitative analysis and which require modern electronic data processing equipment to handle vast quantities of numerical information that could not practically be processed in any other way.

Social and Public Aspects

While marketing research is commonly and properly regarded as one of the most significant marketing functions within progressive business firms, certain of its broader implications and applications must not be overlooked. First, as will be conspicuously evident in the following major section, various governmental agencies are actively engaged in the marketing research function, as providers of a vast stream of highly valued marketing information which could never be collected and processed by any individual enterprise.

Second, marketing research often provides the factual basis for certain types of public policy. Illustrative is information collected for the purpose of anticipating effects of proposed antitrust or trade regulation legislation.

Other examples include data gathered by public agencies in regard to credit controls, price controls in periods of emergency, agricultural production control or price support

programs, public aids to business recovery in periods of recession, and so on. Such activities are not confined to the federal government but are also common at the state and local levels, especially in connection with programs to stimulate industrial development, modify taxation, regulate trade, or forecast revenues which may be dependent upon future market conditions. Third, activities of a marketing research nature are often carried on in connection with solving operating problems of public agencies. One example is extensive research conducted within the armed forces on the problem of logistics of supply, which is similar in many respects to the management of physical distribution functions in industry. Another example would be demand analysis for purposes of deciding upon the desirability of proposed price changes in certain state-owned liquor stores.

Marketing Research in Sampling

What is sampling? In market research, sampling means getting opinions from a number of people, chosen from a specific group, in order to find out about the whole group. Let's look at sampling in more detail and discuss the most popular types of sampling used in market research.

It would be expensive and time-consuming to collect data from the whole population of a market. Therefore, market researchers make extensive use of sampling from which, through careful design and analysis, marketers can draw information about their chosen market.

Sample Design

Sample design covers:

- Method of selection
- Sample structure
- Plans for analysing and interpreting the results.

Sample designs can vary from simple to complex. They depend on the type of information required and the way the sample is selected.

Sample design affects the size of the sample and the way in which analysis is carried out; in simple terms the more precision the market researcher requires, the more complex the design and larger the sample size will be.

The sample design may make use of the characteristics of the overall market population, but it does not have to be proportionally representative.

It may be necessary to draw a larger sample than would be expected from some parts of the population: for example, to select more from a minority grouping to ensure that sufficient data is obtained for analysis on such groups.

Many sample designs are built around the concept of random selection. This permits justifiable inference from the sample to the population, at quantified levels of precision. Random selection also helps guard against sample bias in a way that selecting by judgement or convenience cannot.

Defining the Population

The first step in good sample design is to ensure that the specification of the target population is as clear and complete as possible. This is to ensure that all elements within the population are represented.

The target population is sampled using a sampling frame.

Often, the units in the population can be identified by existing information such as pay-rolls, company lists, government registers etc.

A sampling frame could also be geographical. For example, postcodes have become a well-used means of selecting a sample.

Sample Size

For any sample design, deciding upon the appropriate sample size will depend on several key factors:

1. No estimate taken from a sample is expected to be exact: assumptions about the overall population based on the results of a sample will have an attached margin of error
2. To lower the margin of error usually requires a larger sample size: the amount of variability in the population, ie the range of values or opinions, will also affect accuracy and therefore size of the sample
3. The confidence level is the likelihood that the results obtained from the sample lie within a required precision: the higher the confidence level, the more certain you wish to be that the results are not atypical. Statisticians often use a 95% confidence level to provide strong conclusions
4. Population size does not normally affect sample size: in fact the larger the population size, the lower the proportion of that population needs to be sampled to be representative.

It's only when the proposed sample size is more than 5% of the population that the population size becomes part of the formulae to calculate the sample size

Types of Sampling

There are many different types of sampling methods, here's a summary of the most common:

Cluster sampling

Units in the population can often be found in certain geographic groups or "clusters" for example, primary school children in Derbyshire.

A random sample of clusters is taken, then all units within the cluster are examined.

Advantages

- Quick and easy

- Doesn't need complete population information
- Good for face-to-face surveys

Disadvantages

- Expensive if the clusters are large
- Greater risk of sampling error

Convenience sampling

Uses those who are willing to volunteer and easiest to involve in the study.

Advantages

- Subjects are readily available
- Large amounts of information can be gathered quickly

Disadvantages

- The sample is not representative of the entire population, so results can't speak for them - inferences are limited. future data
- Prone to volunteer bias

Judgement sampling

A deliberate choice of a sample - the opposite of random

Advantages

- Good for providing illustrative examples or case studies

Disadvantages

- Very prone to bias
- Samples often small
- Cannot extrapolate from sample

Quota sampling

The aim is to obtain a sample that is "representative" of the overall population.

The population is divided ("stratified") by the most important variables such as income, age and location. The required quota sample is then drawn from each stratum.

Advantages

- Quick and easy way of obtaining a sample

Disadvantages

- Not random, so some risk of bias
- Need to understand the population to be able to identify the basis of stratification

Simply random sampling

This makes sure that every member of the population has an equal chance of selection.

Advantages

- Simple to design and interpret
- Can calculate both estimate of the population and sampling error

Disadvantages

- Need a complete and accurate population listing
- May not be practical if the sample requires lots of small visits over the country

Systematic sampling

After randomly selecting a starting point from the population between 1 and $*n$, every n th unit is selected.

$*n$ equals the population size divided by the sample size.

Advantages

- Easier to extract the sample than via simple random
- Ensures sample is spread across the population

Disadvantages

- Can be costly and time-consuming if the sample is not conveniently located.

Research Strategies in Content Marketing

In the world of content marketing, strategy is king. The Content Marketing Institute's 2015 B2C Benchmark Report shows that 47 percent of the most effective content marketers have a documented strategy, while only 5 percent of the least effective do. This trend is mirrored in B2B content marketing, where 54 percent of the most effective practitioners have a documented strategy, compared with 11 percent of the least effective.

Having a documented strategy—and following it—seems to be the secret sauce behind content marketing success. But how exactly should you build that strategy?

A Modern Approach to Traditional Market Research

Content strategy should be determined by research and hard numbers, not hunches. With the proper tools and processes, you can use data to take the guess work out of marketing, thus creating the greatest likelihood of winning.

As with all marketing initiatives, it's important to conduct research to understand the four W's—who, what, where, and why—in order to maximize effectiveness. And when it comes to content marketing research, those four W's translate into four critical components that every strategy should have baked into it: audience, trends, media, and competition.

Audience assessment—the “who”

An audience assessment is, of course, the key to understanding the behavior of your prospects online. Specifically, this analysis will show you how your audience seeks information, how they evaluate brands and products, how they’re influenced throughout the buyer’s journey, and how they make purchases.

Research tactics for the audience assessment should include a combination of qualitative and quantitative research:

- Brand immersion
- Keyword research
- User-generated content analysis
- Customer interviews and surveys
- Expert interviews
- Analytics assessment

Altogether, these research tactics will allow you to put together comprehensive buyer personas, identify pain points, create a buyer’s journey map, and understand the purchasing considerations of your audience.

Trends assessment—the “what”

The goal of a trends assessment is to build the “what” component of your content strategy. By compiling and analyzing the most engaging content in your space, you can use hard data to identify trending content patterns and determine opportunities for contributions that are likely to resonate with your audience. The following research tactics can be used for a trends assessment:

- Sentiment analysis
- Angle identification
- Sub topic identification
- Compelling questions analysis
- Resources cited analysis
- Experts quoted analysis

These research practices will help you understand what’s being cited and shared by your audience, what questions are and aren’t being answered, and why people are interacting with content in the first place.

Media assessment—the “where”

A media assessment can be used to identify the top online media and influencers in your space. In conjunction with the trends assessment, the media assessment will help you understand how best to create content that will resonate with your audience and how to maximize the amplification of that content. This assessment should include the following:

- Industry publications analysis
- Industry social influencers analysis

- Industry experts analysis
- Paid media analysis
- Social engagement analysis

By identifying and analyzing the channels and outlets through which content performs well in your industry, you can set yourself up for success through earned and paid promotion efforts.

Competitive assessment—the “why”

In business, the drive to beat out competitors is the reason, or the “why,” behind content marketing. In particular, the competitive assessment is useful for identifying middle- and bottom-of-the-funnel opportunities that may or may not already be capitalized on by your competitors. A competitive content assessment may include the following:

- Content inventory
- Sentiment analysis
- Keyword gap analysis
- Site architecture and navigation audit
- Local SEO evaluation

Together, these research tactics will help you understand what your competitors are focusing on and where opportunities for content marketing growth lie.

Content Strategy Powered by Research

More than three-quarters of content marketers claim to have a strategy in place, but fewer than one-third have actually documented that strategy. Having a documented strategy helps increase effectiveness in all areas of content marketing, especially when that strategy is backed by quantitative and qualitative research.

Marketing Information Through Cooperation

Much marketing information reaches business establishments as a result of cooperative activity through which the members of some group share it. Two significant variations include trade associations and cooperative marketing organizations.

Trade Associations

Trade associations representing retailers, wholesalers, and manufacturers in the same line of business have long played a very important role in gathering and publishing information which is of value to their members. Through regular and special publications, reports upon various kinds of research activities are made available to members and, in some cases, to the general public. Surveys of typical operating ratios, studies of business techniques and procedures, the dissemination of current news of interest to the trade, and educational activities are among the numerous functions of the more important associations.

Cooperative Marketing Organizations

One of the important services of cooperative marketing organizations is that of collecting and disseminating marketing information to promote orderly marketing for the benefit of the membership. An example is provided by Florida Citrus Mutual, an organization with a membership of about 7,000 citrus fruit growers. Growers are divided into some 40 council groups. Officers are invited to headquarters in advance of each marketing season. At this time, officials present all available information relating to supply and demand conditions which will have a bearing upon the average price for the growers' fruit during the season. Such information is provided for each type of utilization, including fruit for fresh shipment, canned juice, and frozen concentrate. Throughout the season, a special weekly publication is sent to all members, advising them of all up-to-date information deemed to be of significance in the marketing of fruit available for disposition.

Private Sources

Public and cooperative sources of marketing information are supplemented by a wide variety of private sources. Some of the more important of these are indicated in the following paragraphs.

Trade and Professional Magazines

One of the most significant sources consists of trade or professional magazines or business papers. Most people have seen a few such publications as a matter of ordinary experience, but very few persons have any real knowledge of the vast number that is published regularly.

In one source, over 2,500 different trade or professional magazines are indexed and described. Some of these are of rather general interest and are read by executives and operating personnel in a variety of positions in all kinds of industries. Examples include *The Journal of Marketing*, *Business Week*, *Distribution Age*, *Fortune*, and the *Harvard Business Review*.

Certain magazines contain information of special significance to those charged with the responsibility of a particular function, which is the case with the periodicals *Credit and Financial Management*, *Sales Management*, and *Purchasing*. Many of them are primarily of interest to those in some particular line of trade or type of industry as suggested by the titles *Frozen Food Age*, *Men's Wear*, *Steel*, *Oil and Gas Journal*, *Hardware Age*, or *Building Supply News*. Some are of national significance, while others are of mere local or sectional interest as illustrated by the titles *Kansas City Grocer*, *New England Electrical News*, or *Michigan Drug Journal*. Trade magazines are of interest for several reasons.

First, they contain discussions of research activities and expert or editorial appraisals of current marketing developments. Second, many provide continuing feature columns or sections which contain reports of trade statistics, legislative developments or other

matters. Third, the trade advertising that they contain is informative in itself. Finally, many of the major publications have special annual issues, of which the following are of unusual significance:

Sales Management, “Survey of Buying Power Issue”, published annually on May 10. Contains current estimates of population, disposable family income, retail sales in total and by five store groups for cities over 10,000 population, counties, metropolitan areas, and states; data on industrial marketing potentials and wholesale trade for states and leading counties.

Printers’ Ink, “Advertisers’ Guide to Marketing”, published annually in October. Provides information summarized from marketing studies, advertising expenditures by major companies and by class of media, operations of advertising agencies, trends in various types of sales promotion as point of-purchase material, premiums, films, packaging; contains summaries of legal or regulatory developments in the field of marketing.

Industrial Marketing, “Market Data and Directory Number”, published annually in June. For each of some 90 industrial markets (e.g., apparel manufacturing, automotive, aviation, baking, etc.) provides information on current trends, what each industry buys, additional sources of information about the industry, current statistics on sales, production, number of plants, geographical distribution of the industry, trade publications and other magazines in each industrial classification. *Advertising Age*, “Annual Market Data Section”.

Contains an annotated description of many items of marketing information or statistics available from media, associations, and other groups. Compilation is classified under the following headings: national markets, farm markets, regional and local markets, distribution markets, industrial markets, professional markets, Canadian markets, and international markets.

Advertising Media

Many national magazines, some city newspapers, and radio and television stations publish research studies containing much valuable marketing analysis material. Such studies often contain quantitative information about the size of the market served by the media, qualitative characteristics of the consumer units in the market, their buying power, and the kind of appeals that can be used to reach them.

Special studies are often made to determine buying habits, preferences, or attitudes of consumers. Illustrative are the “Consumer Analysis” studies, published annually by the *Milwaukee Journal* and a number of other newspapers. Information is tabulated on brand preferences of consumers for hundreds of advertised products, the kind of stores in which the consumers purchase each type of product, and the extent of distribution for each product in stores in the community. Some newspapers also publish lists of local retail stores which are arranged in route order, so that they can be used by salesmen in making personal calls. Some national magazines publish their circulation and other

market figures by local areas in a manner that is helpful to manufacturers in establishing sales potentials, in delineating salesmen's territories, or in allocating local advertising expenditures.

University Bureaus of Business Research

Bureaus of business research have been established in some 45 universities for the purpose of obtaining and publishing information of value to the business community. Among the better known are those of Harvard, Ohio State, Michigan, Minnesota, Illinois, Indiana, Washington, and Texas. The bureau at Harvard is well known for its studies on the operation of certain types of wholesale and retail institutions, especially its annual reports on the operating results of department stores and limited price variety stores. The bureau at Ohio State has published a number of volumes dealing with various kinds of management problems in several wholesale and retail lines of business. Most of the bureaus have some regular publication which deals with business conditions in general and with sales trends and marketing problems, either nationally or in the state or region principally served by the university.

Marketing Research Companies

Increasing attention to the importance of the marketing research function has led to the development of a large number of institutions that specialize in this activity and do work for regular or occasional clients, usually on a fee basis. Many of these organizations are primarily engaged in conducting surveys and polls of consumer or trade opinion or attitudes. Some of them have a number of branch offices and employ a substantial staff equipped to handle large-scale interviewing surveys in all parts of the country. Certain companies provide a continuing kind of information service available on a subscription basis.

Among the kinds of marketing information which can be purchased in this manner are reports of sales by brands of products as ascertained by audits in retail stores; readership of newspaper advertisements in metropolitan papers; and extent of the audience reached by broadcast advertising media. Some agencies are highly specialized in terms of the service rendered, confining their activities to problems of product package design, to market testing, or to the testing of personnel for placement in marketing or selling positions. Other research organizations offer a very wide range of services and have expert marketing consultants capable of setting up a plan for doing research on any kind of a marketing problem and provide the machinery for doing the research work and reporting on it to their clients.

Other Private Sources

A variety of business institutions whose primary business is something other than the supplying of marketing information are nevertheless actively engaged in the performance of this function. As has been explained in earlier parts of this text, marketing information is one of the important services rendered for customers by brokers, selling

agents, wholesalers, commodity exchanges, advertising agencies, resident buying offices, and mercantile credit organizations. Other kinds of companies that often make studies pertaining to the market areas they serve are banks, public utilities, railroads, and large motor freight carriers.

EXAMPLES OF MARKETING RESEARCH IN PRACTICE

Most businesses find that marketing information from published sources is very helpful but that it does not completely satisfy their requirements. Even when published information is relevant to a particular marketing problem, considerable effort is often required to combine the data from various sources, apply statistical procedures, and make pertinent analyses. Often, however, a search of available published source reveals that the kind of data required for an intelligent marketing decision are not available and the individual company must go to considerable trouble or expense in developing suitable information. In either circumstance, the marketing research function is performed by the individual enterprise. In many large companies, this is primarily the responsibility of special marketing research personnel, but whether the company is large or small, various marketing executives or operating employees are engaged in this research function in one way or another.

FORECASTING SALES

One of the most significant uses of marketing information is that of estimating future sales. Short-range estimates, involving periods of several months to a year, are essential to retailing and wholesaling companies for determining their requirements for merchandise to be purchased; they are required by manufacturing companies in order to establish production schedules and plans for procuring needed raw materials and other purchased supplies and services. Forecasts covering longer periods, from five to twenty-five years, are utilized to resolve questions relating to plans for expanding physical facilities and for future personnel and capital needs. Different companies develop either type of forecast in a variety of ways, but in most cases the procedures involve use of some combination of data from published external sources and from internal business records.

Determining Optimum Size or Location of Physical Facilities

A problem of special importance to multiunit organizations is to determine how many establishments should be operated, how large they should be, and how they should be located to serve the market adequately and at minimum cost. Department stores face such problems in decisions relating to horizontal expansion through branch store units. Chain retailing organizations must resolve similar questions when adding new retail store units or in establishing wholesale distribution warehouses. Chain and branch house wholesalers encounter comparable problems when deciding to add new units or to discontinue operation of existing establishments. Manufacturing companies must resolve

similar questions when it comes to deciding upon new factory locations, the opening of district sales offices, or the operation of branch wholesale warehouses.

An approach to the problem may be illustrated with reference to a New England firm which served some of its customers by deliveries from its factory and others through a system of branch wholesale warehouses which it operated. A number of wholesale warehouses had been added to the distribution structure one at a time as the volume of business in various market areas increased. This had been done on the basis of common-sense judgments which, it was realized, often proved advantageous from the standpoint of one warehouse but sometimes had adverse effects on the operations of other warehouses in adjoining territories. Consulting analysts were employed to develop an answer to the question, "How large a territory should be served by a warehouse to result in a minimum total cost for warehousing, trucking between plant and warehouses, and delivery from warehouses to customers?" Detailed analyses were made of all items of operating expense in the plant and in the warehouses, and it was found that the cost of handling merchandise in each of the warehouses was related to two opposing factors:

- The volume of business handled in the warehouse and
- The size of the area served by it. A mathematical model of optimum cost operations for branch warehouses was then constructed. While these procedures are relatively complicated, the purpose is simply to classify variable costs according to their behaviour, with respect to volume of business and size of area served, and to determine whether the amount of business gained or lost by increasing or decreasing the size of the warehouse territory by small increments would or would not offset conflicting tendencies in the cost of serving a larger or smaller territory. As a result of such an analysis applied to all of the company's warehouse territories, it was learned that most of the existing warehouses were too large and that costs could be reduced by serving customers through a larger number of smaller warehouses.

A different type of approach to a similar problem is provided by a chain supermarket organization.

Through an elaborate study of the shopping habits of urban-area housewives, it was learned, among other things, that distance is not an important factor in discouraging patronage by its customers within the range of about 3 miles from a store location; beyond this range, however, it becomes an important deterrent.

As a consequence, the company attempts to blanket a metropolitan area with supermarkets located approximately 6 miles apart, so that no customers are more than about 3 miles from the nearest unit and its stores are not so close together that they compete excessively with each other.

This is not to be interpreted as a general principle of supermarket location, as different companies have varying experience, depending upon the average size of store, population density, conditions of accessibility, intensity of competition, general merchandising strategy, and other factors.

Evaluating Promotional Effort

A considerable amount of marketing research is devoted to the problem of evaluating objectively the possible different combinations of promotional activity in the marketing mix. The task of isolating the contribution of several different promotional variables when each one is ordinarily used together with others requires an experimental approach, with controls established to hold constant (to the maximum practical degree) other factors which could also contribute to measurable results. This may be illustrated by a major research programme undertaken within the Industrial Marketing Division of the Scott Paper Co. for the purpose of measuring the effectiveness of three different types of promotional activity, namely, magazine advertising, direct mail advertising, and special incentive contests for distributor salesmen.

The test involved the company's washroom product line—towels and toilet tissue sold through wholesale distributors to plants, hotels, stores, public buildings, and many other kinds of business users. For this experiment, 22 market areas were selected—11 test markets and 11 control markets. In the test markets the company conducted experimental promotions over a period of time involving the use of one or more of the three different types of promotional activities in question. The 11 combinations tested were as follows:

- Space advertising, high level
- Space advertising, low level
- Direct mail, high level
- Direct mail, low level
- Space advertising, high level, *plus* direct mail, low level
- Space advertising, low level, *plus* direct mail, high level
- Space advertising, low level, *plus* direct mail, low level
- Space advertising, low level, *plus* distributor sales incentive contest
- Direct mail, low level, *plus* distributor sales incentive contest
- Space advertising, low level, *plus* direct mail, low level, *plus* distributor sales incentive contest
- Distributor sales incentive contest only.

Each of the foregoing promotional approaches was tested in one market, and each such market was matched as closely as possible with a control market with respect to factors such as quality of distributors, company manpower, sales potential, and other matters besides those being tested that might have a bearing upon results. Each control market was, however, geographically removed from its corresponding test market, to avoid problems of overlapping promotional influence, especially in the case of regional or state-wide inserts in magazines used for space advertising. During and following the test, elaborate statistical analyses were made of company sales, inquiries from direct mail campaigns, user recognition of company advertising, familiarity of company name, and other matters related to the objectives of the research experiment. Comparisons of data could be made in the following ways:

- Results during the test period for each test market area compared with experience in that market during the two years preceding the test
- Results for each test market compared with those experienced in each other test market during the test period
- Results for each test market compared with those experienced in its corresponding control market, in which no special promotional activity was undertaken

Through mathematical analysis of such results, carried on by an operations research group within the company, a factual basis for decisions about the desirable relative importance of promotional ingredients in the marketing mix is believed to have been provided; however, the purpose here is merely to illustrate a complex type of experimental research design, not to discuss the relative merits of different approaches to promotional strategy.

Determining Current Purchases by Ultimate Consumers

Manufacturers of consumer goods, especially those who use a relatively long channel of distribution, have an urgent need for current marketing information showing the volume of purchases by ultimate consumers. When such a manufacturer sells through wholesalers or chain store warehouses that distribute his goods to retail stores, factory shipments during a given period of time are often a poor indication of consumer purchases.

Wholesale establishments and retail stores in the channel may carry inventories of some items varying from a one- or two-week supply to as much as a six- to twelve-month supply. Inventories on one or both levels may fluctuate widely from one month to another; as a consequence, factory sales may be at substantial variance from consumer purchases. Still, information on consumer purchases is vital if the manufacturer is to appraise intelligently the effect of his own or competitive promotional activity, the influence of seasonal factors upon consumer buying, or the presence or lack of speculative buying activity by institutions in the channel. Several methods are used to measure sales at the retail level, including

- Retail store audits of purchases and inventories;
- Purchase diary records maintained by samples of typical consumers; and
- Analysis of warranty cards mailed in by consumers making purchases. Two of these methods are illustrated in the following paragraphs.

Store Audit Procedure to Evaluate Promotional Strategy

Marketing information from store audits is a service usually purchased from marketing research organizations. Several companies supply this type of service, the best known being the A. C. Nielsen Co. This company serves over 400 companies by employing several hundred field representatives who make audits of merchandise movements in a scientifically designed sample of 19,000 food and drug stores in all parts of the United States. Through the cooperation of retailers, field auditors check inventories of designated

items every two months. On each store visit, they record the beginning inventory for the period, add all purchases made the store, and subtract the ending inventory, to derive sales to consumers during the period. As a result, the company is able to supply each client with a report that reveals

- Unit sales of its own brand by type of outlet and by region of the country,
- Unit sales for each major competitive brand, and
- The client's sales expressed as a per cent of total market sales for the type of item.

An extremely important application is that of evaluating the effect of consumer advertising activity. An illustration is provided by a large company that made a drastic modification in its advertising strategy which involved important changes in the quantity and type of advertising as well as in advertising media. During the four months following the change, factory sales of this product declined 19 per cent from the preceding period, which led the advertiser to conclude that the change in strategy was ineffective or had been damaging. Through the store audit reports, a quite different conclusion was reached.

This research information revealed that retailers' stocks of the item were high at the beginning of the promotional campaign and had been undergoing substantial liquidation during the four-month period. Moreover, during this period total market sales of all brands declined by 12 per cent. A comparison of the company's sales in relation to total market sales revealed that the company had actually increased *its share* of total consumer sales from 37 per cent at the beginning of the period to 43 per cent at the end of the four month period. Through research, it was revealed that a move which would have been judged a failure from the standpoint of factory sales had actually been very successful.

Continuous Consumer Panels

The process of obtaining information about the behaviour of consumers throughout a prolonged time period involves a research approach known as the continuous consumer panel. This is different from making a number of surveys at intervals in the sense that a relatively static group of consumers provides information about their individual behaviour on a continuing, day-to-day basis. Consumer panels are used for a variety of purposes, including the testing of new products, measurement of broadcast media audiences, and determining purchase behaviour. The method of obtaining information is by a self-administering form which is usually referred to as a diary. Consumers daily record all purchases made by family members, often supplemented by information about price, place of purchase, and so on. Reports are sent periodically to the research organization, where specific items of information are transferred to tabulating cards for sorting and analysis. Consumers are usually compensated in cash or premium merchandise for their cooperation.

As compared with store audits, purchase panels provide a wider range of data, including characteristics of consumers making or not making purchases of certain types or specific brands of products. One of the main advantages is the ability to measure

both gross and net changes in consumer purchases of a particular product during a time period. This can be illustrated as follows: During one six-month period, consumers purchased 1 million units of product X. Purchases in the next period dropped to 800,000 units, indicating a *net* change of 200,000 units. If 200,000 of the 800,000 units were purchased by people who did not make any purchases during the first period, the *gross* change would be 400,000 units. Since the research organization has complete information on the socioeconomic characteristics of all households in the sample, elaborate analyses can be made of purchase diary information in relation to such characteristics.

Warranty Card Analysis

Store audit and consumer panel procedures are widely used in the case of convenience goods but have not been widely applied to shopping or specialty goods where sales are much more infrequent at the retail level or involve significant seasonal variations. To determine the movement of such commodities to consumers, manufacturers of many forms of appliances make detailed statistical analyses of warranty cards. Such cards are usually given to the consumer at the time the product is delivered or installed, and must be filled out and mailed to the manufacturer in order to register the beginning of the warranty period. Customers must furnish their name and address, model and serial number of the unit, place of purchase, date of delivery or installation, and, often, other information of marketing significance.

In the Automatic Blanket and Fan Department of the General Electric Company, for example, about 1 million returned warranty cards are processed annually. A postal-size card is used, which is designed with holes around the edges that can be notched for machine sorting. The card is so designed that the consumer can provide a variety of information merely by placing check-marks in boxes and by filling in certain data, such as name, address, and date and place of purchase. From the flow of these cards, the department can accurately estimate its sales at the retail level for current periods, determine the average store-shelf life of products, and ascertain other consumer information, such as sex of buyer, reason for purchase (own use, gift, etc.), and advertising media that prompted the sale.

By periodically checking the total production of a given model during a long time period, and later comparing this with the total number of warranty cards eventually returned, and how many at different intervals after purchase, the department establishes formulas which enable it to estimate sales on a current basis, depending upon the number of warranty cards returned during a given week or month. Through this type of analysis, the manufacturer knows the rate of consumer sales from week to week and can judge whether consumer sales are up to expectations and, consequently, whether to maintain production at expected levels or to increase or curtail manufacturing activity, revise prices, or enter into different types of promotional activity.

4

Methods and Techniques of Data Collection

DATA COLLECTION

Data Collection helps your team to assess the health of your process. To do so, you must identify the key quality characteristics you will measure, how you will measure them, and what you will do with the data you collect.

What exactly is a key quality characteristic? It is a characteristic of the product or service produced by a process that customers have determined is important to them.

Key quality characteristics are such things as the speed of delivery of a service, the finish on a set of stainless steel shelves, the precision with which an electronic component is calibrated, or the effectiveness of an administrative response to a tasking by higher authority. Every product or service has multiple key quality characteristics.

When you are selecting processes to improve, you need to find out the processes, or process steps, that produce the characteristics your customers perceive as important to product quality.

Data Collection is nothing more than planning for and obtaining useful information on key quality characteristics produced by your process. However, simply collecting data does not ensure that you will obtain relevant or specific enough data to tell you what is occurring in your process. The key issue is not: How do we collect data? Rather, it is: How do we obtain useful data?

WELL-DEFINED DATA COLLECTION PROCESS

For your team to collect data uniformly, you will need to develop a Data Collection plan. The elements of the plan must be clearly and unambiguously defined- operationally defined. You may want to pause here and review the Operational Definitions module before you go on. Why does a team need Operational Definitions in order to collect useful data? Let's say three people are collecting data on the time it takes to perform a certain process step. Unless the exact moment when each action begins and the exact moment when it ends are operationally defined, each data collector will observe and record data based on his or her own understanding of the situation.

The Data Collection process will not be standardized or consistent. You will have collected data, but it probably won't be much good to you. Worse yet, you may make changes to your process based on flawed information.

Data Collection can involve a multitude of decisions by data collectors. When you prepare your Data Collection plan, you should try to eliminate as many subjective choices as possible by operationally defining the parameters needed to do the job correctly. It may be as simple as establishing separate criteria and a specific way to judge when a step begins and when it ends. Your data collectors will then have a standard operating procedure to use during their Data Collection activities.

Develop a Data Collection Plan

You should develop your Data Collection plan during the Plan Phase of the Plan-Do- Check-Act (PDCA) cycle. The PDCA cycle provides a framework for you to build an understanding of your process and how to obtain and interpret data that will lead to real process improvement. Although they can be time-consuming, planning sessions are extremely important because this is when you establish the guidance that helps you obtain the right data.

The Data Collection Plan Answer

Your team needs to develop the answers to the following questions as the basis for a sound Data Collection plan:

- Why do we want the data? What will we do with the data after we have collected them? The team must decide on a purpose for collecting the data.
- In the Plan Phase, your team should develop a working hypothesis which will serve as a guide to future investigation of the process. This hypothesis is an assumption based on already existing data and observations, such as your process Flowcharts or a Cause-and-Effect Diagram the team has prepared. You develop working assumptions and collect data to determine the process changes that will improve the key quality characteristics of your product or service. Your proposed change should be stated as an "If... then" statement.
- IF we change Step X in our process by doing..., we believe as suggested, THEN improve Y, which is a key quality characteristic of our product or service

This action focuses your team on the specific quality characteristic you want to improve, and sets the stage for where you will collect the data.

- Where will we collect the data? The location where data are collected must be identified clearly. This is not an easy step unless you tackle it from the following perspective:
 - Refer to the Flowcharts which depict both the current (“as is”) state of the process and the proposed (“should be”) state of the process after it has been modified. Focus on the process steps where the key quality characteristic you are trying to improve is produced.
 - Collect data from these process steps. You must collect data twice. First, you collect baseline data before you make any changes in your process. These baseline data will serve as a yardstick against which to compare the results of the process after changes have been made. Then, you must collect data after the change has been imposed on the process. To compare the before and after process, you will probably want to translate your data into graphic form using a Pareto Chart, Run Chart, or Histogram. The use of these tools is explained in separate modules.
 - Collect data on the key quality characteristic of the product or service at the end of your process. Again, before and after data must be collected. The comparison of before and after data validates whether the change actually improved the output of the process.
- What type of data will we collect? In general, data can be classified into two major types: attribute data and variables data.
 - Attribute data give you counts representing the presence or absence of a characteristic or defect. These counts are based on the occurrence of discrete events. As an example, if you are concerned with timely delivery of parts by your store keepers, you could develop a procedure that would give you a count of the number of supply parts they deliver on time and the number they deliver late (defects). This would give you attribute data, but it would not tell you how late a late delivery actually was. Two factors help determine whether attribute data will be useful:
 - Operational Definitions. You need to operationally define exactly what constitutes a defect. For the data collected in the example above to be useful, you would have to operationally define late. This may be a good time to review the module on Operational Definitions.
 - Area of Opportunity. For counts to be useful, they must come from a well-defined area of opportunity. You obtain a single count, or value, from each sample, or area of opportunity. For example, if you are collecting data on the number of defective bayonets received in each shipment of 200, the area of opportunity is the 200-bayonet shipment. The number of defective bayonets in the shipment gives you one count, or data point.

- Variables data are based on measurement of a key quality characteristic produced by the process. Such measurements might include length, width, time, weight, or temperature, to name a few. Continuing with the parts delivery example, you could collect variables data by tabulating the time it took to process an incoming supply request from receipt to validation of the National Stock Number (NSN); or the time from validation of the NSN to identification of the stock bin where the part is located; or the time required to post the obligation in the OPTAR Log; or the total time from receipt of the request to delivery of the part. This measurement, time, could be used to determine how timely or late the deliveries were.
- Who will collect the data? Many teams struggle with this question, but the answer is simple: Those closest to the data—the process workers—should collect the data. These people have the best opportunity to record the results. They also know the process best and can easily detect when problems occur. But remember, the people who are going to collect the data need training on how to do it, and the resources necessary to obtain the information, such as time, paper, pencils, and measurement tools.
- How do we collect the right data? You need to remember that you are collecting data for the purpose of improving the process, not the product it produces. Clearly, you want to collect the data that best describe the situation at hand. If you are going to use the data to make predictions about the performance of the process, you should collect small samples at regular intervals—let's say 4 or 5 units every other hour or each day. Since it is important to collect those 4 or 5 units in a short interval of time, you may want to use consecutive units or every other unit.
- But remember, the cost of obtaining the data, the availability of data, and the consequences of decisions made on the basis of the data should be taken into consideration when determining how much data should be obtained and how frequently it should be collected.

Data Collection Problems

Remembering that data form the basis for the effective, unemotional communication without which no process improvement effort can succeed, you need to avoid two significant problems associated with Data Collection.

Problem 1: Failure to establish Operational Definitions. You need to define, not simply identify, the following:

- When and how often you will collect the data
- How you will collect the data
- Units of measurement you will use in collecting the data
- The criteria for defects
- How you will handle multiple defects on single products

If you haven't thought about these issues, your Data Collection process may be doomed from the start. This is especially true when more than one person is collecting data. What is meaningful to one worker might not be to another. You have to take the time to develop adequate, clear-cut definitions, and train each collector to use those definitions.

Problem 2: Adding bias to the Data Collection process. You can never eliminate bias, but it is important to minimize it. Here are some ways your data can be biased:

- The process of collecting the data may affect the process being studied. If you are trying to make a process faster, taking data may either speed it up or slow it down.
 - On the one hand, the workers may speed up the way they work in the process, thus skewing the data in their favour. This may occur if they have a perception that the variables data they are collecting will show that they could be more efficient, productive, or effective. Once the Data Collection effort ceases, they may return to their old pace of operations.
 - On the other hand, the burden of Data Collection may cause a slowdown in the natural flow of the process.

If such events are affecting your improvement efforts, you need to alter your Data Collection plan.

- The attitudes and perceptions of the data collectors can affect what they see and how they record data. If there is a sense that the data will be used against them, workers may use the data collection process to cast a favourable light on the process being studied. You have to get past this fear in order to collect accurate data. You might want to consider an amnesty programme.
- Data collectors need to be assured that their leaders realise that the data gathered in the past may have been tainted by fear. This requires a commitment by your leadership that the new information—possibly less glowing or flattering—will not be compared against old data or their perception of how your process operated in the past.
- Failure to follow the established Data Collection procedures can add errors to the data. This bias occurs when the Data Collection instructions, training, or checksheets are not adequately prepared and tested in an operational environment. You need to conduct initial training on Data Collection and then perform a small-scale Data Collection trial to see if it all works smoothly. The small-scale trial may uncover some problems which need to be ironed out before you can actively pursue a larger scale Data Collection effort. The trial may reveal that you need to make a minor change in the checksheet to make it clearer or easier to use, or that additional training on Operational Definitions is required to calibrate the eyes of the data collectors.
- Data may be missing. Don't assume that missing data will show the same results as the data you collected. The fact that the data are missing is a clue

that they may be different from the rest. It is best to number the checksheets sequentially to make it easier to verify that you have them all and that all the required samples have been taken.

We use to Collect Data

Data are frequently collected using checksheets—structured forms that enable you to collect and organize data systematically. Because each checksheet is used for collecting and recording data unique to a specific process, it can be constructed in whatever shape, size, and format are appropriate for the Data Collection task at hand. Checksheets have three important uses:

- Record information on the key quality characteristics of your process for analysis using tools such as a Pareto Charts, Histograms, and Run Charts.
- Provide a historical record of the process over time.
- Introduce Data Collection methods to workers and supervisors who may not be familiar or comfortable with collecting data as a regular part of their jobs.

Types of checksheets are there

The most common types of checksheets collect data either in tabular form or in a location-style format. Occasionally you may encounter a graphic-style checksheet. No matter which type you are using, make sure that it is clear, complete, and userfriendly.

The three types of checksheets are described below:

1. *Tabular format:* A tabular checksheet—also known as a “tally sheet”—is easy for you and your team to use when you simply want to count how often something happens or to record a measurement. Depending on the type of data required, the data collector simply makes a checkmark in a column to indicate the presence of a characteristic, or records a measurement, such as temperature in degrees centigrade, weight in pounds, diameter in inches, or time in seconds.
2. *Location format:* A location checksheet allows you to mark a diagram showing the exact physical location of a defect or characteristic. An insurance adjuster’s pictorial claim form detailing your latest bumper bruise is an example of a location checksheet.
3. *Graphic format:* Another way of collecting data is by using a graphic form of checksheet. It is specifically designed so that the data can be recorded and displayed at the same time. Using this checksheet format, you can record raw data by plotting them directly onto a graph-like chart. An example of a checksheet which also produces a Run Chart as the individual data points are plotted and the adjacent points are joined with a straight line.

We develop useful Checksheets

There is no standardized format that you can apply to all checksheets. Instead, each checksheet is a form tailored to collect the information needed to improve a specific process.

Remember, a well-designed checksheet is the launching pad for an effective analysis in which data become meaningful information.

With that in mind, here are some guidelines to help you develop useful forms:

- Involve the process workers in developing the checksheet for their process.
- Label all columns clearly. Organize your form so that the data are recorded in the sequence seen by a person viewing the process. This reduces the possibility of data being recorded in the wrong column or not being recorded.
- Make the form user-friendly. Make sure the checksheet can be easily understood and used by all of the workers who are recording data.
 - Include brief instructions on the back of the form.
 - Create a format that gives you the most information with the least amount of effort. For example, design your checksheet so that data can be recorded using only a check mark, slant mark, number, or letter.
 - Provide enough space for the collectors to record all of the data.
 - Designate a place for recording the date and time the data were collected. These elements are required when the data are used with Run Charts or other tools which require the date and time of each observation.
 - Provide a place to enter the name of the individual collecting the data.
 - Allow enough space so data collectors can write in comments on unusual events. This information could be entered on the back of the form.

We review what we've learned

Now let's work through some examples that illustrate how to determine where to collect baseline data and how to use checksheets to capture them.

Example 1: Pharmacy Waiting Time: A team of hospital corpsmen working in a Naval Medical Clinic is attempting to improve service in the pharmacy by decreasing the time patients wait for their prescriptions (*i.e.*, the key quality characteristic).

The desired end state is reduced waiting time for the patients.

The team proceeds as follows:

- They develop a Flowchart of the process. After examining the Flowchart and discussing the steps, they conclude that a possible bottleneck in their process occurs at the point where a prescription is filled by the pharmacist. This is supported by general observations of team members.
- They develop a hypothesis along with their "should be" Flowchart.
- They opt to take baseline data using a checksheet to assess both the time required for the pharmacist's actions and the overall time required to process the prescription.

Example 2: Communications Gateway Site Setup Time: A team of Marines is investigating methods for improving the process to set up a communications gateway site for a tactical data link interface with Navy ships and aircraft. They want to reduce the time it takes to set up the site.

- They develop a Flowchart of their process and review it, identifying possible bottlenecks and slow operations.
- They hypothesize that they can improve their set-up time by focusing on the steps of the process dealing with unpacking the equipment.
- They opt to use a checklist to take baseline data on that portion of the process as well as on the overall time to complete the installation and checkout.

These two examples illustrate the point that you should collect data from the key process step or steps where the outcome can be most affected. But beware of making assumptions based only on a Flowchart. You should collect some initial data on all critical steps in the process. Then you should collect detailed data about steps that initially showed great variability, or those that took the longest to perform. Only when these data have been analysed can the cause of pain in the process be identified and acted upon.

Some examples of checklists that work

Remember, there are no standard Data Collection forms. Every checklist is unique to the process you are investigating. Through 22 at the end of this module are examples of checklists that are well-designed for their purposes. Careful design of your checklist may allow you to collect two or more factors on a single form.

DATA COLLECTION: PRIMARY RESEARCH METHODS

In this tutorial we continue our discussion of how marketers collect research data by examining the methods used for collecting primary data. Unlike secondary research, where data is initially obtained by someone other than the marketer, the responsibility for collecting data under primary research falls to the marketer. In general marketers can select from two basic approaches to data collections using primary methods:

- Qualitative Data Collection
- Quantitative Data Collection

Both methods offer advantages and disadvantages which are discussed in detail in this tutorial.

PRIMARY RESEARCH - ADVANTAGES

- *Addresses Specific Research Issues:* Carrying out their own research allows the marketing organization to address issues specific to their own situation. Primary research is designed to collect the information the marketer wants to know and report it in ways that benefit the marketer. For example, while information reported with secondary research may not fit the marketer's needs (e.g., different age groupings) no such problem exists with primary research since the marketer controls the research design.
- *Greater Control:* Not only does primary research enable the marketer to focus on specific issues, it also enables the marketer to have a higher level of control

over how the information is collected. In this way the marketer can decide on such issues as size of project (e.g., how many responses), location of research (e.g., geographic area) and time frame for completing the project.

- *Efficient Spending for Information:* Unlike secondary research where the marketer may spend for information that is not needed, primary data collections' focus on issues specific to the researcher improves the chances that research funds will be spent efficiently.
- *Proprietary Information:* Information collected by the marketer using primary research is their own and is generally not shared with others. Thus, information can be kept hidden from competitors and potentially offer an "information advantage" to the company that undertook the primary research.

PRIMARY RESEARCH - DISADVANTAGES

- *Cost:* Compared to secondary research, primary data may be very expensive since there is a great deal of marketer involvement and the expense in preparing and carrying out research can be high.
- *Time Consuming:* To be done correctly primary data collection requires the development and execution of a research plan. Going from the start-point of deciding to undertake a research project to the end-point to having results is often much longer than the time it takes to acquire secondary data.
- *Not Always Feasible:* Some research projects, while potentially offering information that could prove quite valuable, are not within the reach of a marketer. Many are just too large to be carried out by all but the largest companies and some are not feasible at all. For instance, it would not be practical for McDonalds to attempt to interview every customer who visits their stores on a certain day since doing so would require hiring a huge number of researchers, an unrealistic expense. Fortunately, as suggested, see in a later tutorial there are ways for McDonalds to use other methods (e.g., sampling) to meet their needs without the need to talk with all customers.

QUALITATIVE DATA COLLECTION

Sometimes referred to as "touchy-feely" research, qualitative data collection requires researchers to interpret the information gathered, most often without the benefit of statistical support. If the researcher is well trained in interpreting respondents' comments and activities, this form of research can offer very good information. However, it may not hold the same level of relevancy as quantitative research due to the lack of scientific controls with this data collection method. For example, a researcher may want to know more about how customers make purchase decisions. One way to do this is to sit and talk with customers using one-on-one interviews. However, if the interview process allows the researcher to vary what questions are asked (i.e., not all respondents are asked the same questions), then this type of research may lack controls needed to follow

a scientific approach. An additional drawback of qualitative research is that it can be time consuming and expensive and, consequently, only a very small portion of the marketer's desired market can participate in qualitative research. Due to the lack of strong controls in the research design (i.e., not as well structured, fewer participants), using results to estimate characteristics of a larger group is more difficult. Thus, qualitative data collection is generally not used for hypothesis testing. This is not to say qualitative research is not useful, it is very useful if its limitations are understood. It is widely employed for marketing research especially for research for the purpose of discovery, and to a lesser extent, explanation.

Qualitative data collection options include personal interviews, focus groups and observational research which are discussed next.

Qualitative Data: Personal Interviews

Talking to someone one-on-one allows a researcher to cover more ground than may be covered if a respondent was completing a survey. The reason lies with the researcher's ability to dig deeper into a respondent's comments to find out additional that might not emerge from initial responses.

Unfortunately, individual interviewing can be quite expensive and may be intimidating to some who are not comfortable sharing with a researcher.

Qualitative Data: Focus Groups

To overcome the drawbacks associated with personal interviews, marketers can turn to focus groups. Under this research format, a group of respondents (generally numbering 8-12) are guided through discussion by a moderator. The power of focus groups as a research tool rests with the environment created by the interaction of the participants. In well-run sessions, members of the group are stimulated to respond by the comments and the support of others in the group. In this way, the depth of information offered by a respondent may be much greater than that obtained through individual interviews.

However, focus groups can be costly to conduct especially if participants must be paid. To help reduce costs, online options for focus groups have emerged. While there are many positive aspects to online focus groups, the fact that respondents are not physically present diminishes the benefits gained by group dynamics. However, as technology improves, in particular video conferencing, the online focus group could become a major research option.

Qualitative Data: Observational Research

Watching customers as they perform activities can be a very useful research method, especially when customers are observed in a natural setting (e.g., shopping in a retail store, using products at home). In fact, an emerging research technique called ethnographic research has researchers following customers as they shop, work, and relax at home in order to see how they make decisions, use products and more.

Quantitative Data Collection

In general there are two basic types of primary research - quantitative data collection and qualitative data collection. Quantitative data collection involves the use of numbers to assess information. This information can then be evaluated using statistical analysis which offers researchers the opportunity to dig deeper into the data and look for greater meaning.

Certain information is by nature numerical. For example, asking a person their actual age or yearly income will result in a number. But under the right circumstances numbers can also be used to represent certain characteristics which are not on the surface considered numerical. This most often occurs with data collected within a structured and well-controlled scientific research design. For instance, research of customers' attitude toward a company's products may include the following:

Place an "X" on the line that best indicates your impression of the overall quality of our company's products:

Poor _ _ _ _ _ Excellent

In this example each line, which represents a potential customer response, could be assigned a number. For example, checking the left-most line could result in the researcher entering a "1", the next line a "2", the next line a "3" and so on. Once research is completed this question can undergo statistical analysis.

While quantitative analysis is potentially used for all types of research purposes it is most critical for hypothesis testing. In Step 6: Analyse Data, such analysis may prove very relevant by allowing the researcher to draw ceases.

Quantitative data collection comes in many forms but the most popular forms are surveys.

Quantitative Data: Surveys

This method captures information through the input of responses to a research instrument containing questions (i.e., such as a questionnaire).. Information can be input either by the respondents themselves (e.g., complete online survey) or the researcher can input the data (e.g. phone survey, mall intercept). The main methods for distributing surveys are via postal mail, phone, website or in person. However, newer technologies are creating additional delivery options including through wireless devices, such as smart phones.

Quantitative Data: Tracking

With tracking research marketers are able to monitor the behaviour of customers as they engage in regular purchase or information gathering activities. Possibly the most well-known example of tracking research is used by websites as they track customer visits. But tracking research also has offline applications, especially when point-of-purchase scanners are employed, such as tracking product purchases at grocery stores and automated collections on toll roads.

This method of research is expected to grow significantly as more devices are introduced that provide means for tracking. However, as we discussed in the Marketing Research Tutorial, some customers may see tracking devices as intrusive and many privacy advocates have raised concerns about certain tracking methods especially if these are not disclosed to customers.

Quantitative Data: Experiments

Marketers often undertake experiments to gauge how the manipulation of one marketing variable affects another (i.e., causal research).

The use of experiments has applications for many marketing decision areas including product testing, advertising design, setting price points and creating packaging. For example, a market researcher for a retail chain may want to study the effect on sales if a product display is moved to different locations in a store.

Unfortunately, performing highly controlled experiments can be quite costly. Some researchers have found the use of computer simulations can work nearly as well as experiments and may be less expensive, though the number of simulation applications for marketing decisions is still fairly limited.

Questionnaires

Questionnaires are a popular means of collecting data, but are difficult to design and often require many rewrites before an acceptable questionnaire is produced.

Advantages:

- Can be used as a method in its own right or as a basis for interviewing or a telephone survey.
- Can be posted, e-mailed or faxed.
- Can cover a large number of people or organisations.
- Wide geographic coverage.
- Relatively cheap.
- No prior arrangements are needed.
- Avoids embarrassment on the part of the respondent.
- Respondent can consider responses.
- Possible anonymity of respondent.
- No interviewer bias.

Disadvantages:

- Design problems.
- Questions have to be relatively simple.
- Historically low response rate (although inducements may help).
- Time delay whilst waiting for responses to be returned.
- Require a return deadline.
- Several reminders may be required.
- Assumes no literacy problems.

- No control over who completes it.
- Not possible to give assistance if required.
- Problems with incomplete questionnaires.
- Replies not spontaneous and independent of each other.
- Respondent can read all questions beforehand and then decide whether to complete or not. For example, perhaps because it is too long, too complex, uninteresting, or too personal.

Design of Postal Questionnaires

Theme and Covering Letter

The general theme of the questionnaire should be made explicit in a covering letter. You should state who you are; why the data is required; give, if necessary, an assurance of confidentiality and/or anonymity; and contact number and address or telephone number. This ensures that the respondents know what they are committing themselves to, and also that they understand the context of their replies. If possible, you should offer an estimate of the completion time. Instructions for return should be included with the return date made obvious. For example: 'It would be appreciated if you could return the completed questionnaire by... if at all possible'.

Instructions for Completion

You need to provide clear and unambiguous instructions for completion. Within most questionnaires these are general instructions and specific instructions for particular question structures.

It is usually best to separate these, supplying the general instructions as a preamble to the questionnaire, but leaving the specific instructions until the questions to which they apply. The response method should be indicated (circle, tick, cross, etc.). Wherever possible, and certainly if a slightly unfamiliar response system is employed, you should give an example.

Appearance

Appearance is usually the first feature of the questionnaire to which the recipient reacts. A neat and professional look will encourage further consideration of your request, increasing your response rate. In addition, careful thought to layout should help your analysis.

There are a number of simple rules to help improve questionnaire appearance:

- Liberal spacing makes the reading easier.
- Photo-reduction can produce more space without reducing content.
- Consistent positioning of response boxes, usually to the right, speeds up completion and also avoids inadvertent omission of responses.
- Choose the font style to maximise legibility.

- Differentiate between instructions and questions. Either lower case and capitals can be used, or responses can be boxed.

Length

There may be a strong temptation to include any vaguely interesting questions, but you should resist this at all costs. Excessive size can only reduce response rates. If a long questionnaire is necessary, then you must give even more thought to appearance.

Order

Probably the most crucial stage in questionnaire response is the beginning. Once the respondents have started to complete the questions they will normally finish the task, unless it is very long or difficult. Consequently, you need to select the opening questions with care. Usually the best approach is to ask for biographical first, as the respondents should know all the answers without much thought. Another benefit is that an easy start provides practice in answering questions.

Once the introduction has been achieved the subsequent order will depend on many considerations. You should be aware of the varying importance of different questions. Essential information should appear early, just in case the questionnaire is not completed. For the same reasons, relatively unimportant questions can be placed towards the end. If questions are likely to provoke the respondent and remain unanswered, these too are best left until the end, in the hope of obtaining answers to everything else.

Coding

If analysis of the results is to be carried out using a statistical package or spreadsheet it is advisable to code non-numerical responses when designing the questionnaire, rather than trying to code the responses when they are returned.

An example of coding is:

Male []	Female []
1	2

The coded responses are then used for the analysis.

Respondents to questionnaires rarely benefit personally from their efforts and the least the researcher can do is to thank them. Even though the covering letter will express appreciation for the help given, it is also a nice gesture to finish the questionnaire with a further thank you.

Questions

- Keep the questions short, simple and to the point; avoid all unnecessary words.
- Use words and phrases that are unambiguous and familiar to the respondent. For example, 'dinner' has a number of different interpretations; use an alternative expression such as 'evening meal'.

- Only ask questions that the respondent can answer. Hypothetical questions should be avoided. Avoid calculations and questions that require a lot of memory work, for example, 'How many people stayed in your hotel last year?'
- Avoid loaded or leading questions that imply a certain answer. For example, by mentioning one particular item in the question, 'Do you agree that Colgate toothpaste is the best toothpaste?'
- Vague words or phrases should be avoided. 'Generally', 'usually', or 'normally' are imprecise terms with various meanings. They should be replaced with quantitative statements, for example, 'at least once a week'.
- Questions should only address a single issue. For example, questions like: 'Do you take annual holidays to Spain?' should be broken down into two discreet stages, firstly find out if the respondent takes an annual holiday, and then secondly find out if they go to Spain.
- Do not ask two questions in one by using 'and'. For example, 'Did you watch television last night and read a newspaper?'
- Avoid double negatives. For example, 'Is it not true that you did not read a newspaper yesterday?' Respondents may tackle a double negative by switching both negatives and then assuming that the same answer applies. This is not necessarily valid.
- State units required but do not aim for too high a degree of accuracy. For instance, use an interval rather than an exact figure:

'How much did you earn last year?'

Less than £10,000 ☐

£10,000 but less than £20,000 ☐

Avoid emotive or embarrassing words - usually connected with race, religion, politics, sex, money.

Types of questions

Closed questions: A question is asked and then a number of possible answers are provided for the respondent. The respondent selects the answer which is appropriate. Closed questions are particularly useful in obtaining factual information:

Sex: Male ☐ Female ☐

Did you watch television last night? Yes ☐ No ☐

Some 'Yes/No' questions have a third category 'Do not know'. Experience shows that as long as this alternative is not mentioned people will make a choice. Also the phrase 'Do not know' is ambiguous:

Do you agree with the introduction of the EMU?

Yes ☐ No ☐ Do not know ☐

What was your main way of travelling to the hotel? Tick one box only. With such lists you should always include an 'other' category, because not all possible responses might have been included in the list of answers.

Sometimes the respondent can select more than one from the list. However, this makes analysis difficult:

Why have you visited the historic house? Tick the relevant answer(s). You may tick as many as you like.

Attitude questions: Frequently questions are asked to find out the respondents' opinions or attitudes to a given situation. A Likert scale provides a battery of attitude statements. The respondent then says how much they agree or disagree with each one:

Read the following statements and then indicate by a tick whether you strongly agree, agree, disagree or strongly disagree with the statement.

	Strongly agree	Agree	Disagree	Strongly disagree
My visit has been good value for money				

There are many variations on this type of question. One variation is to have a 'middle statement', for example, 'Neither agree nor disagree'. However, many respondents take this as the easy option.

Only having four statements forces the respondent into making a positive or negative choice. Another variation is to rank the various attitude statements, however, this can cause analysis problems:

Which of these characteristics do you like about your job? Indicate the best three in order, with the best being number

Varied work	☐
Good salary	☐
Opportunities for promotion	☐
Good working conditions	☐
High amount of responsibility	☐
Friendly colleagues	☐

A semantic differential scale attempts to see how strongly an attitude is held by the respondent. With these scales double-ended terms are given to the respondents who are asked to indicate where their attitude lies on the scale between the terms. The response can be indicated by putting a cross in a particular position or circling a number:

Work is: (circle the appropriate number)

Difficult	1 2 3 4 5 6 7	Easy
Useless	1 2 3 4 5 6 7	Useful
Interesting	1 2 3 4 5 6 7	Boring

For summary and analysis purposes, a 'score' of 1 to 7 may be allocated to the seven points of the scale, thus quantifying the various degrees of opinion expressed. This procedure has some disadvantages. It is implicitly assumed that two people with the same strength of feeling will mark the same point on the scale.

This almost certainly will not be the case. When faced with a semantic differential scale, some people will never, as a matter of principle, use the two end indicators of 1 and 7. Effectively, therefore, they are using a five-point scale. Also scoring the scale 1 to 7 assumes that they represent equidistant points on the continuous spectrum of opinion. This again is probably not true. Nevertheless, within its limitations, the semantic differential can provide a useful way of measuring and summarising subjective opinions.

Other types of questions to determine peoples' opinions or attitudes are:

- Which one/two words best describes...?
- Which of the following statements best describes...?
- How much do you agree with the following statement...?

Open Questions

An open question such as 'What are the essential skills a manager should possess?' should be used as an adjunct to the main theme of the questionnaire and could allow the respondent to elaborate upon an earlier more specific question. Open questions inserted at the end of major sections, or at the end of the questionnaire, can act as safety valves, and possibly offer additional information. However, they should not be used to introduce a section since there is a high risk of influencing later responses. The main problem of open questions is that many different answers have to be summarised and possibly coded.

Testing - pilot Survey

Questionnaire design is fraught with difficulties and problems. A number of rewrites will be necessary, together with refinement and rethinks on a regular basis. Do not assume that you will write the questionnaire accurately and perfectly at the first attempt. If poorly designed, you will collect inappropriate or inaccurate data and good analysis cannot then rectify the situation.

To refine the questionnaire, you need to conduct a pilot survey. This is a small-scale trial prior to the main survey that tests all your question planning. Amendments to questions can be made. After making some amendments, the new version would be re-tested. If this re-test produces more changes, another pilot would be undertaken and so on. For example, perhaps responses to open-ended questions become closed; questions which are all answered the same way can be omitted; difficult words replaced, etc.

It is usual to pilot the questionnaires personally so that the respondent can be observed and questioned if necessary. By timing each question, you can identify any questions that appear too difficult, and you can also obtain a reliable estimate of the anticipated completion time for inclusion in the covering letter. The result can also be used to test the coding and analytical procedures to be performed later.

Distribution and Return

It is usual to supply a prepaid addressed envelope for the return of the questionnaire. You need to explain this in the covering letter and reinforce it at the end of the

questionnaire, after the 'Thank you'. Finally, many organisations are approached continually for information. Many, as a matter of course, will not respond in a positive way.

Interviews: Interviewing is a technique that is primarily used to gain an understanding of the underlying reasons and motivations for people's attitudes, preferences or behaviour. Interviews can be undertaken on a personal one-to-one basis or in a group. They can be conducted at work, at home, in the street or in a shopping centre, or some other agreed location.

Personal interview Advantages:

- Serious approach by respondent resulting in accurate information.
- Good response rate.
- Completed and immediate.
- Possible in-depth questions.
- Interviewer in control and can give help if there is a problem.
- Can investigate motives and feelings.
- Can use recording equipment.
- Characteristics of respondent assessed - tone of voice, facial expression, hesitation, etc.
- Can use props.
- If one interviewer used, uniformity of approach.
- Used to pilot other methods.

Disadvantages Need to set up interviews:

- Time consuming.
- Geographic limitations.
- Can be expensive.
- Normally need a set of questions.
- Respondent bias - tendency to please or impress, create false personal image, or end interview quickly.
- Embarrassment possible if personal questions.
- Transcription and analysis can present problems - subjectivity.
- If many interviewers, training required.

Types of interview Structured:

- Based on a carefully worded interview schedule.
- Frequently require short answers with the answers being ticked off.
- Useful when there are a lot of questions which are not particularly contentious or thought provoking.
- Respondent may become irritated by having to give over-simplified answers.

Semi-structured

The interview is focused by asking certain questions but with scope for the respondent to express him or herself at length.

Unstructured

This also called an in-depth interview. The interviewer begins by asking a general question. The interviewer then encourages the respondent to talk freely.

The interviewer uses an unstructured format, the subsequent direction of the interview being determined by the respondent's initial reply. The interviewer then probes for elaboration - 'Why do you say that?' or, 'That's interesting, tell me more' or, 'Would you like to add anything else?' being typical probes.

This is a step-by-step guide to conducting an interview. You should remember that all situations are different and therefore you may need refinements to the approach.

Planning an Interview

- List the areas in which you require information.
- Decide on type of interview.
- Transform areas into actual questions.
- Try them out on a friend or relative.
- Make an appointment with respondent(s)—discussing details of why and how long.

Conducting an Interview

- *Personally:* Arrive on time be smart smile employ good manners find a balance between friendliness and objectivity.
- *At the start:* Introduce yourself re-confirm the purpose assure confidentiality - if relevant specify what will happen to the data.
- *The questions:* Speak slowly in a soft, yet audible tone of voice control your body language know the questions and topic ask all the questions.
- *Responses:* Recorded as you go on questionnaire written verbatim, but slow and time-consuming summarised by you taped - agree beforehand - have alternative method if not acceptable consider effect on respondent's answers proper equipment in good working order sufficient tapes and batteries minimum of background noise.

Telephone Interview

This is an alternative form of interview to the personal, face-to-face interview.

Advantages:

- Relatively cheap.
- Quick.
- Can cover reasonably large numbers of people or organisations.
- Wide geographic coverage.
- High response rate - keep going till the required number.
- No waiting.
- Spontaneous response.

- Help can be given to the respondent.
- Can tape answers.

Disadvantages:

- Often connected with selling.
- Questionnaire required.
- Not everyone has a telephone.
- Repeat calls are inevitable - average 2.5 calls to get someone.
- Time is wasted.
- Straightforward questions are required.
- Respondent has little time to think.
- Cannot use visual aids.
- Can cause irritation.
- Good telephone manner is required.
- Question of authority.

Getting Started

- Locate the respondent:
 - Repeat calls may be necessary especially if you are trying to contact people in organisations where you may have to go through secretaries.
 - You may not know an individual's name or title - so there is the possibility of interviewing the wrong person.
 - You can send an advance letter informing the respondent that you will be telephoning. This can explain the purpose of the research.
- Getting them to agree to take part:
 - You need to state concisely the purpose of the call - scripted and similar to the introductory letter of a postal questionnaire.
 - Respondents will normally listen to this introduction before they decide to co-operate or refuse.
 - When contact is made respondents may have questions or raise objections about why they could not participate. You should be prepared for these.

Ensuring Quality

- *Quality of questionnaire:* Follows the principles of questionnaire design. However, it must be easy to move through as you cannot have long silences on the telephone.
- *Ability of interviewer:* Follows the principles of face-to-face interviewing.

Smooth implementation

- *Interview schedule:* Each interview schedule should have a cover page with number, name and address. The cover sheet should make provision to record which call it is, the date and time, the interviewer, the outcome of the call and space to note down specific times at which a call-back has been arranged.

Space should be provided to record the final outcome of the call - was an interview refused, contact never made, number disconnected, etc.

- *Procedure for call-backs:* A system for call-backs needs to be implemented. Interview schedules should be sorted just as to their status: weekday call-back, evening call-back, weekend call-back, specific time call-back.

COMPARISON OF POSTAL, TELEPHONE AND PERSONAL INTERVIEW SURVEYS

Focus Group Interviews

A focus group is an interview conducted by a trained moderator in a non-structured and natural manner with a small group of respondents. The moderator leads the discussion. The main purpose of focus groups is to gain insights by listening to a group of people from the appropriate target market talk about specific issues of interest.

Observation

Observation involves recording the behavioural patterns of people, objects and events in a systematic manner.

Observational methods may be:

- Structured or unstructured
- Disguised or undisguised
- Natural or contrived
- Personal
- Mechanical
- Non-participant
- Participant, with the participant taking a number of different roles.

Structured or Unstructured

In structured observation, the researcher specifies in detail what is to be observed and how the measurements are to be recorded. It is appropriate when the problem is clearly defined and the information needed is specified.

In unstructured observation, the researcher monitors all aspects of the phenomenon that seem relevant. It is appropriate when the problem has yet to be formulated precisely and flexibility is needed in observation to identify key components of the problem and to develop hypotheses. The potential for bias is high. Observation findings should be treated as hypotheses to be tested rather than as conclusive findings.

Disguised or Undisguised

In disguised observation, respondents are unaware they are being observed and thus behave naturally. Disguise is achieved, for example, by hiding, or using hidden equipment or people disguised as shoppers.

In undisguised observation, respondents are aware they are being observed. There is a danger of the Hawthorne effect - people behave differently when being observed.

Natural or Contrived

Natural observation involves observing behaviour as it takes place in the environment, for example, eating hamburgers in a fast food outlet.

In contrived observation, the respondents' behaviour is observed in an artificial environment, for example, a food tasting session.

Personal

In personal observation, a researcher observes actual behaviour as it occurs. The observer may or may not normally attempt to control or manipulate the phenomenon being observed. The observer merely records what takes place.

Mechanical

Mechanical devices (video, closed circuit television) record what is being observed. These devices may or may not require the respondent's direct participation. They are used for continuously recording on-going behaviour.

Non-participant

The observer does not normally question or communicate with the people being observed. He or she does not participate.

Participant

In participant observation, the researcher becomes, or is, part of the group that is being investigated. Participant observation has its roots in ethnographic studies (study of man and races) where researchers would live in tribal villages, attempting to understand the customs and practices of that culture.

It has a very extensive literature, particularly in sociology (development, nature and laws of human society) and anthropology (physiological and psychological study of man). Organisations can be viewed as 'tribes' with their own customs and practices.

The role of the participant observer is not simple.

There are different ways of classifying the role:

- Researcher as employee.
- Researcher as an explicit role.
- Interrupted involvement.
- Observation alone.

Researcher as Employee

The researcher works within the organisation alongside other employees, effectively as one of them. The role of the researcher may or may not be explicit and this will have

implications for the extent to which he or she will be able to move around and gather information and perspectives from other sources. This role is appropriate when the researcher needs to become totally immersed and experience the work or situation at first hand.

There are a number of dilemmas. Do you tell management and the unions? Friendships may compromise the research. What are the ethics of the process? Can anonymity be maintained? Skill and competence to undertake the work may be required. The research may be over a long period of time.

Researcher as an Explicit Role

The researcher is present every day over a period of time, but entry is negotiated in advance with management and preferably with employees as well. The individual is quite clearly in the role of a researcher who can move around, observe, interview and participate in the work as appropriate. This type of role is the most favoured, as it provides many of the insights that the complete observer would gain, whilst offering much greater flexibility without the ethical problems that deception entails.

Interrupted involvement: The researcher is present sporadically over a period of time, for example, moving in and out of the organisation to deal with other work or to conduct interviews with, or observations of, different people across a number of different organisations. It rarely involves much participation in the work.

Observation alone: The observer role is often disliked by employees since it appears to be 'eavesdropping'. The inevitable detachment prevents the degree of trust and friendship forming between the researcher and respondent, which is an important component in other methods.

Choice of Roles

The role adopted depends on the following:

- *Purpose of the research:* Does the research require continued longitudinal involvement (long period of time), or will in-depth interviews, for example, conducted over time give the type of insights required?
- *Cost of the research:* To what extent can the researcher afford to be committed for extended periods of time? Are there additional costs such as training?
- *The extent to which access can be gained:* Gaining access where the role of the researcher is either explicit or covert can be difficult, and may take time.
- *The extent to which the researcher would be comfortable in the role:* If the researcher intends to keep his identity concealed, will he or she also feel able to develop the type of trusting relationships that are important? What are the ethical issues?
- *The amount of time the researcher has at his disposal:* Some methods involve a considerable amount of time. If time is a problem alternate approaches will have to be sought.

Case-studies

The term case-study usually refers to a fairly intensive examination of a single unit such as a person, a small group of people, or a single company. Case-studies involve measuring what is there and how it got there. In this sense, it is historical. It can enable the researcher to explore, unravel and understand problems, issues and relationships. It cannot, however, allow the researcher to generalise, that is, to argue that from one case-study the results, findings or theory developed apply to other similar case-studies. The case looked at may be unique and, therefore not representative of other instances. It is, of course, possible to look at several case-studies to represent certain features of management that we are interested in studying. The case-study approach is often done to make practical improvements. Contributions to general knowledge are incidental.

The case-study method has four steps:

- Determine the present situation.
- Gather background information about the past and key variables.
- *Test hypotheses:* The background information collected will have been analysed for possible hypotheses. In this step, specific evidence about each hypothesis can be gathered. This step aims to eliminate possibilities which conflict with the evidence collected and to gain confidence for the important hypotheses. The culmination of this step might be the development of an experimental design to test out more rigorously the hypotheses developed, or it might be to take action to remedy the problem.
- *Take remedial action:* The aim is to check that the hypotheses tested actually work out in practice. Some action, correction or improvement is made and a re-check carried out on the situation to see what effect the change has brought about.

The case-study enables rich information to be gathered from which potentially useful hypotheses can be generated. It can be a time-consuming process. It is also inefficient in researching situations which are already well structured and where the important variables have been identified. They lack utility when attempting to reach rigorous ceases or determining precise relationships between variables.

Diaries

A diary is a way of gathering information about the way individuals spend their time on professional activities. They are not about records of engagements or personal journals of thought! Diaries can record either quantitative or qualitative data, and in management research can provide information about work patterns and activities.

Advantages:

- Useful for collecting information from employees.
- Different writers compared and contrasted simultaneously.
- Allows the researcher freedom to move from one organisation to another.
- Researcher not personally involved.

- Diaries can be used as a preliminary or basis for intensive interviewing.
- Used as an alternative to direct observation or where resources are limited.

Disadvantages:

- Subjects need to be clear about what they are being asked to do, why and what you plan to do with the data.
- Diarists need to be of a certain educational level.
- Some structure is necessary to give the diarist focus, for example, a list of headings.
- Encouragement and reassurance are needed as completing a diary is time-consuming and can be irritating after a while.
- Progress needs checking from time-to-time.
- Confidentiality is required as content may be critical.
- Analyses problems, so you need to consider how responses will be coded before the subjects start filling in diaries.

Critical Incidents

The critical incident technique is an attempt to identify the more 'noteworthy' aspects of job behaviour and is based on the assumption that jobs are composed of critical and non-critical tasks. For example, a critical task might be defined as one that makes the difference between success and failure in carrying out important parts of the job. The idea is to collect reports about what people do that is particularly effective in contributing to good performance. The incidents are scaled in order of difficulty, frequency and importance to the job as a whole.

The technique scores over the use of diaries as it is centred on specific happenings and on what is judged as effective behaviour. However, it is laborious and does not lend itself to objective quantification.

Portfolios

A measure of a manager's ability may be expressed in terms of the number and duration of 'issues' or problems being tackled at any one time. The compilation of problem portfolios is recording information about how each problem arose, methods used to solve it, difficulties encountered, etc. This analysis also raises questions about the person's use of time. What proportion of time is occupied in checking; in handling problems given by others; on self-generated problems; on 'top-priority' problems; on minor issues, etc? The main problem with this method and the use of diaries is getting people to agree to record everything in sufficient detail for you to analyse. It is very time-consuming!

SECONDARY DATA COLLECTION

All methods of data collection can supply quantitative data (numbers, statistics or financial) or qualitative data (usually words or text). Quantitative data may often be

presented in tabular or graphical form. Secondary data is data that has already been collected by someone else for a different purpose to yours.

For example, this could mean using:

- Data collected by a hotel on its customers through its guest history system
- Data supplied by a marketing organisation
- Annual company reports
- Government statistics.

Secondary data can be used in different ways:

- You can simply report the data in its original format. If so, then it is most likely that the place for this data will be in your main introduction or literature review as support or evidence for your argument.
- You can do something with the data. If you use it (analyse it or re-interpret it) for a different purpose to the original then the most likely place would be in the 'Analysis of findings' section of your dissertation. A good example of this usage was the work on suicide carried out by Durkheim. He took the official suicide statistics of different countries (recorded by coroners or their equivalent) and analysed them to see if he could identify variables that would mean that some people are more likely to commit suicide than others. He found, for example, that Catholics were less likely to commit suicide than Protestants. In this way, he took data that had been collected for quite a different purpose and used it in his own study - but he had to do a lot of comparisons and statistical correlations himself in order to analyse the data.

Most research requires the collection of primary data (data that you collect at first hand), and this is what students concentrate on. Unfortunately, many dissertations do not include secondary data in their findings section although it is perfectly acceptable to do so, providing you have analysed it. It is always a good idea to use data collected by someone else if it exists - it may be on a much larger scale than you could hope to collect and could contribute to your findings considerably.

As secondary data has been collected for a different purpose to yours, you should treat it with care.

The basic questions you should ask are:

- Where has the data come from?
- Does it cover the correct geographical location?
- Is it current (not too out of date)?
- If you are going to combine with other data are the data the same (for example, units, time, etc.)?
- If you are going to compare with other data are you comparing like with like?

Thus you should make a detailed examination of the following:

- Title (for example, the time period that the data refers to and the geographical coverage).
- Units of the data.

- Source (some secondary data is already secondary data).
- Column and row headings, if presented in tabular form.
- Definitions and abbreviations, for example, what does SIC stand for? For example, how is 'small' defined in the phrase 'small hotel'? Is 'small' based on the number of rooms, value of sales, number of employees, profit, turnover, square metres of space, etc., and do different sources use the word 'small' in different ways? Even if the same unit of measurement is used, there still could be problems. For example, in Norway, firms with 200-499 employees are defined as 'medium', whereas in the USA firms with less than 500 employees are defined as 'small'.

There are many sources of data and most people tend to underestimate the number of sources and the amount of data within each of these sources.

Sources can be classified as:

- *Paper-based sources:* Journals, periodicals, abstracts, indexes, directories, research reports, conference papers, market reports, annual reports, internal records of organisations, newspapers and magazines
- *electronic sources:* CD-ROMs, on-line databases, Internet, videos and broadcasts.

The main sources of qualitative and quantitative secondary data include the following:

- Official or government sources.
- Unofficial or general business sources.

The output of all publishers of non-official sources is included in the most comprehensive directory available:

Mort D. (1997) Sources of Unofficial UK Statistics 3rd Edition Aldershot: Gower

The guide lists 1,059 statistical titles and series published by 635 different organisations. It excludes one-off surveys or market reports.

It lists references to the following types of sources:

- Trade associations
- Trade and other journals
- Private research publishers
- Stockbroking firms
- Large company market reports
- Local authorities
- Professional bodies
- Academic institutions.
- European Union (Community) sources.
- International sources.
 - Organisation for Economic Co-operation and Development (OECD)
 - United Nations and related organisations.

Sources for the last two categories are many and varied. If your dissertation requires these sources you need to conduct a more thorough search of your library and perhaps seek the assistance of the librarian.

Summary of Descriptive Statistics: So far you have collated the data and made some counts and determined percentages.

The next step is to summarise the data, if possible, with one or two summary statistics. Summary or descriptive statistics describe the original data set (the set of responses for each question) by using just one or two numbers - typically an average and a measure of dispersion.

- Mode is the most frequently occurring value, although it is not often used.
- Median is the middle value.
- Mean is found by adding the values and dividing by the number of values.
- Quartiles (Q1 and Q3) are the 25% and 75% values respectively and are measures of dispersion about the median.
- Standard deviation is a measure of the dispersion about the mean; a small standard deviation implies the data are tightly bunched about the mean, whereas a large standard deviation implies the data are widely scattered about the mean.

Different types of data (scales of measurement) can be summarised by different summary statistics. The table shows the types of averages and measures of dispersion for each type.

Data	Average	Dispersion
Nominal	Mode	
Ordinal	Mode	
	Median	
	Mean	
Numerical	Mode	
	Median	Quartiles
	Mean	Standard deviation

Note that as the level gets higher more statistics can be determined.

With nominal data it is only possible to show the number or percentage of people or items falling within each category. It is also possible to state which category includes the highest number of counts - the most popular category or modal category.

Care is needed when comparing such a mean with another mean, because the data are ordinal and not numerical. One question may give a weighted mean score of 2.1 and another question a weighted mean score of 4.2. It would be incorrect to say the result for the first question is twice as good as the result for the second question. Remember the weights are purely arbitrary. The data are not numerical and therefore the principle of ratios cannot be applied.

For numerical data, all three averages and their associated measures of dispersion can be determined. However, the mode tends to be of little interest with this type of data and may be ignored in most situations. It may be necessary to use a graphical technique such as a cumulative frequency curve (ogive) to determine the median and quartiles.

We can determine the mean and standard deviation using the formula:

$$\text{St Dev} = \sqrt{\frac{\sum fx^2}{\sum f} - \left[\frac{\sum fx}{\sum f} \right]^2}$$

The term in square brackets is the arithmetic mean. The approach to the calculation is to find the variance and then take the square root to obtain the standard deviation.

For the data in question 4 in the summary questionnaire we have:

Annual salary	Number of respondents
Less than £8,000	6
£8,000 but less than £12,000	12
£12,000 but less than £16,000	10
£16,000 but less than £20,000	6
£20,000 but less than £24,000	4
£24,000 and above	2

This can be re-written with closed classes and the units in £000s to eliminate the zeroes. Three columns of calculations are required as shown.

	Freq.	Mid Pt.		
	f	x	fx	fx^2
4 but less than 8	6	6	36	216
8 but less than 12	12	10	20	200
12 but less than 16	10	14	140	1,960
16 but less than 20	6	18	108	1,944
20 but less than 24	4	22	88	1,936
24 but less than 28	2	26	52	1,352
Total	40		544	8,608

$$\begin{aligned} \text{Variance} &= \frac{8,608}{40} - \left[\frac{544}{40} \right]^2 \\ &= 215.2 - (13.6)^2 \\ &= 215.2 - 184.96 \\ &= 30.24 \\ \text{St. dev.} &= \sqrt{30.24} \\ &= 5.5 \end{aligned}$$

The mean salary of respondents is £13,6000 per annum with a standard deviation of £5,500. (The results have been multiplied by 1,000 to get the correct units for the data.)

To find the median and quartiles first determine the percentage cumulative frequencies and then draw an ogive (cumulative frequency curve) and read off:

- 25% value to get the lower quartile Q1
- 50% value to get the median
- 75% value to get the upper quartile Q3.

	f	cf	%cf
4 but less than 8	6	6	15
8 but less than 12	12	18	45

12 but less than 16	10	28	70
16 but less than 20	6	34	85
20 but less than 24	4	38	95
24 but less than 28	2	40	100
Total	40		

The percentage cumulative frequencies are then plotted to correspond with the upper class limits. From the graph (not drawn here) the following approximate results are obtained:

Q1	=	£8,700
Median	=	£12,800
Q3	=	£17,300

All the summary statistics determined so far have used the results as presented on the summary sheet. An alternative approach is to use the raw data as presented on the data sheet. However, if you are going to use this approach the data ideally should be on a spreadsheet where the in-built statistical functions can be used.

These functions are:

Summary statistic	Excel function
Arithmetic mean	=AVERAGE(cell range of data)
Median	=MEDIAN(cell range of data)
Mode	=MODE(cell range of data)
Standard deviation	=STDEVP(cell range of data)
Q1	=QUARTILE(cell range of data,1)
Q3	=QUARTILE(cell range of data,3)

So far the analysis has been concerned with determining some summary statistics. Analysis, however, consists of more than this. In particular, analysis is concerned with establishing relationships between variables. There are two common approaches to this, namely, cross-tabulations and correlation analysis.

Cross-tabulations

A cross tabulation is a matrix in which all categories representing one variable are presented in rows, and all categories representing another variable are presented in columns. Although cross-tabulations can be constructed for any type of data they are particularly useful for analysing nominal and ordinal data.

For instance, consider the questionnaire on the attitudes of restaurant staff, you may believe that a worker's opinion of their manager is dependent upon the gender of the respondent. This hypothesis may be investigated by constructing a cross-tabulation for the two variables gender and opinion of manager. The blank table would look something like the following:

Gender (Q1)	Opinion of manager (Q2)			Total
	Good	Average	Poor	
	(1)	(2)	(3)	
Male	(1)			

Female (2)
Total

You would then count how many responses fall into each cell - this is called the cell frequency. Row totals, column totals and a grand total are inserted as well. This is a time-consuming process and requires care, but invariably leads to valuable information about the relationship between the two variables.

For this example and by referring to the data sheet, we can obtain the following completed cross-tabulation:

Gender	Opinion of manager (Q2)			Total (Q1)
	Good (1)	Average (2)	Poor (3)	
Male (1)	8	8	3	19
Female (2)	3	6	12	21
Total	11	14	15	40

Cross-tabulations can be analysed on two levels:

- Inspect the table to see if there are any patterns or cells with small and/or large cell frequencies. If there is, make a statement to reflect this pattern or possible relationship. If there is no obvious pattern, with frequencies being fairly even spread across the cells, then there is probably no relationship between the two variables.
- Test the independence of the two variables using a chi-square test. This is a sophisticated method and we do not cover it here because of the required theoretical underpinning and complexity of the technique.

If we consider the table, it would appear that most males (16 out of 19) have a reasonable opinion of their manager, whilst most females (12 out of 21) appear to have a poor opinion of their manager. This seems to indicate that the opinion of manager is dependent on the gender of respondent (for this sample!), or, there is a relationship between the two variables - gender and opinion of manager.

A possible pattern is less obvious but it would appear that males have a high rating of their work (13 out of 19 rating 1, 2 or 3), whereas females do not (16 out of 21 rating 3, 4 or 5). It would appear that the respondents' rating of their work is dependent on the gender of the respondent (for this sample).

The analysis could be continued by investigating the following hypotheses:

- Is 'rating of the manager' dependent on 'salary'?
- Is 'rating of the manager' dependent on 'age'?
- Is 'opinion of work' dependent on 'salary'?
- Is 'opinion of work' dependent on 'age'?
- Is 'salary' dependent on 'age'?

It is possible to cross tabulate every question with every other question. This produces so much information that the result is 'information overload' and you simply get very confused! You have to be selective about what you cross-tabulate.

With cross-tabulations do not use percentages, only use the actual frequencies. This is because the calculation of a percentage can be based on either row totals, or column totals, or the grand total. In other words, three percentages are possible for each cell.

Correlation Coefficient

This is another sophisticated technique that is commonly used to see if there is a linear relationship between two variables.

Diagrammatic Representation of data

Secondary data and the response counts or percentages associated with a question can be displayed in diagrammatic forms such as a line graph, bar chart or pie chart. These can greatly enhance your findings and subsequent discussion. If at all possible you should include some in your dissertation.

Whatever diagram you use, an associated commentary is essential. Do not leave it to the reader to work out what the diagram shows.

The commentary may:

- State the obvious, such as the largest and/or the smallest, or the trend
- Highlight something that is not so obvious. This is preferable.

Whenever a diagram is used you should position the diagram as close as possible to the associated commentary. Remember, the objective of using a diagram is to present something that is fairly complicated in an easy-to-understand manner as the dissertation is read. Putting it in the back of the dissertation disrupts the flow of reading and understanding.

Two common and popular mistakes made by students when presenting data or findings are as follows:

- The belief that every question or piece of data needs to have a diagram. This is not so. Be selective in the type and number of diagrams used.
- The use of a chart for simple data. For instance, a pie chart is often drawn for a question when there are just two responses, such as a gender breakdown of respondents. This is simplistic and unnecessary.
- Graphs and charts can be produced either within Word directly or by using ChartWizard in Excel and then importing the chart into the document. These require substantial, but important, technical skills. Give yourself time to re-learn them if you have forgotten them

Why we Analysis Data

The purpose of analysing data is to obtain usable and useful information.

The analysis, irrespective of whether the data is qualitative or quantitative, may:

- Describe and summarise the data
- Identify relationships between variables
- Compare variables

- Identify the difference between variables
- Forecast outcomes.

Before we look at the various ways of analysing, presenting and discussing data, we need to clarify the differences between qualitative research, quantitative research, qualitative data and quantitative data. Earlier, we distinguished between qualitative research and quantitative research. It is highly unlikely that your research will be purely one or the other - it will probably be a mixture of the two approaches. For instance, you may have taken a small sample (normally associated with qualitative research) but then conducted a structured interview or used a questionnaire (normally associated with quantitative research) to determine people's attitudes to a particular phenomenon (qualitative research). It is therefore likely that your 'mixed' approach will take a qualitative approach some of the time and a quantitative approach at others. It depends on where you are in the research process.

A misconception, and source of confusion for many people, is the belief that qualitative research generates just qualitative data (text, words, opinions, etc) and that quantitative research generates just quantitative data (numbers). Sometimes this is the case, but both types of data can be generated by each approach. For instance, a postal questionnaire or structured interview (quantitative research) will often gather factual information, for example, age, salary, length of service (quantitative data) - but may also seek opinions and attitudes (qualitative data). A second misconception is that statistical techniques are only applicable for quantitative data. Once again, this is not so. There are many statistical techniques that can be applied to qualitative data, such as ratings scales, that has been generated by a quantitative research approach.

Unfortunately, many people are worried about numbers, and in particular about statistics, and everything that word implies. Quantitative research and the analysis of quantitative data is consequently something to be avoided. This is rarely possible because qualitative data can also be analysed using statistics. An understanding of basic statistical terms and ideas and the ability to carry out some statistical analysis (elementary or otherwise) is essential for most researchers. Also competence in these techniques, even at a basic level, is a useful skill in its own right.

A third misconception is that qualitative data analysis is easy. There are many ways of conducting qualitative research and thus many ways of analysing the resulting (qualitative) data. For example, having conducted an interview, transcription and organisation of data are the first stages of analysis. This would then be continued by systematically analysing the transcripts, grouping together comments on similar themes and attempting to interpret them and draw ceases.

We deal with data that can be analysed statistically (quantitative data and some types of qualitative data) in the section called quantitative data analysis. We cover data that cannot, or is very difficult, to analyse statistically in the section called qualitative data analysis.

Purpose of analysing data is to obtain usable and useful information.

The analysis, irrespective of whether the data is qualitative or quantitative, may:

- Describe and summarise the data
- Identify relationships between variables
- Compare variables
- Identify the difference between variables
- Forecast outcomes.

Qualitative data is subjective, rich, and in-depth information normally presented in the form of words. In undergraduate dissertations, the most common form of qualitative data is derived from semi-structured or unstructured interviews, although other sources can include observations, life histories and journals and documents of all kinds including newspapers.

Qualitative data from interviews can be analysed for content (content analysis) or for the language used (discourse analysis). Qualitative data is difficult to analyse and often opportunities to achieve high marks are lost because the data is treated casually and without rigour. Here we concentrate on the content analysis of data from interviews.

Theory

When using a quantitative methodology, you are normally testing theory through the testing of a hypothesis. In qualitative research, you are either exploring the application of a theory or model in a different context or are hoping for a theory or a model to emerge from the data. In other words, although you may have some ideas about your topic, you are also looking for ideas, concepts and attitudes often from experts or practitioners in the field.

Collecting and Organising Data

The means of collecting and recording data through interviews and the possible pitfalls are well documented elsewhere but in terms of subsequent analysis, it is essential that you have a complete and accurate record of what was said. Do not rely on your memory (it can be very selective!) and either tape record the conversation (preferably) or take copious notes. If you are taking notes, write them up straight after the interview so that you can elaborate and clarify. If you are using a tape recorder, transcribe the exact words onto paper.

However you record the data, you should end up with a hard copy of either exactly what was said (transcript of tape recording) or nearly exactly what was said (comprehensive notes). It may be that parts of the interview are irrelevant or are more in the nature of background material, in which case you need not put these into your transcript but do make sure that they are indeed unnecessary. You should indicate omissions in the text with short statements.

You should transcribe exactly what is said, with grammatical errors and so on. It does not look very authentic if all your respondents speak with perfect grammar and BBC English! You may also want to indicate other things that happen such as laughter.

Each transcript or set of notes should be clearly marked with the name of the interviewee, the date and place and any other relevant details and, where appropriate, cross-referenced to clearly labelled tapes. These transcripts and notes are not normally required to be included in your dissertation but they should be available to show your supervisor and the second marker if required.

You may wonder why you should go to all the bother of transcribing your audiotapes. It is certainly a time-consuming business, although much easier if you can get access to a transcription machine that enables you to start and stop the tape with your feet while carrying on typing. It is even easier if you have access to an audio-typist who will do this labour intensive part for you.

The advantage of having the interviews etc in hard copy is that you can refer to them very quickly, make notes in the margins, re-organise them for analysis, make coding notations in the margins and so on. It is much slower in the long run to have to continually listen to the tapes. You can read much faster than the tape will play! It also has the advantage, especially if you do the transcription yourself, of ensuring that you are very familiar with the material.

Content Analysis

Analysis of qualitative data is not simple, and although it does not require complicated statistical techniques of quantitative analysis, it is nonetheless difficult to handle the usually large amounts of data in a thorough, systematic and relevant manner. Marshall and Rossman offer this graphic description:

"Data analysis is the process of bringing order, structure and meaning to the mass of collected data. It is a messy, ambiguous, time-consuming, creative, and fascinating process. It does not proceed in a linear fashion; it is not neat. Qualitative data analysis is a search for general statements about relationships among categories of data."

Marshall and Rossman, 1990:111: Hitchcock and Hughes take this one step further: "...the ways in which the researcher moves from a description of what is the case to an explanation of why what is the case is the case."

Hitchcock and Hughes 1995:295: Content analysis consists of reading and re-reading the transcripts looking for similarities and differences in order to find themes and to develop categories. Having the full transcript is essential to make sure that you do not leave out anything of importance by only selecting material that fits your own ideas. There are various ways that you can mark the text:

Coding paragraphs: This is where you mark each paragraph with a topic/theme/category with an appropriate word in the margin.

Highlighting paragraphs/sentences/phrases: This is where you use highlighter pens of different colours or different coloured pens to mark bits about the different themes. You could mark the bits relating to childcare and those relating to pay in a different colour, and so on. The use of coloured pens will help you find the relevant bits you need when you are writing up.

With both the methods you may find that your categories change and develop as you do the analysis. What is important is that you can see that by analysing the text in such a way, you pick up all the references to a given topic and don't leave anything out. This increases the objectivity and reduces the risk of you only selecting bits that conform to your own preconceptions.

You then need to arrange the data so that all the pieces on one theme are together.

There are several ways of doing this:

- *Cut and put in folders approach:* Make several copies of each transcript (keeping the master safe) and cut up each one just as to what is being discussed (your themes or categories). Then sort them into folders, one for each category, so that you have all together what each interviewee said about a given theme. You can then compare and look for similarities/differences/changes etc. Do not forget to mark each slip of paper with the respondent's name, initials or some sort of code or you won't be able to remember who said what. Several copies may be needed in case one paragraph contains more than one theme or category. This is time consuming and messy at first, but easier in the long run especially if you have a lot of data and categories.
- *Card index system:* Each transcript must be marked with line numbers for cross-referencing purposes. You have a card for each theme or category and cross-reference each card with each transcript so that you can find what everyone has said about a certain topic. This is quicker initially but involves a lot of referring back to the original transcripts when you write up your results and is usually only suitable for small amounts of data.
- *Computer analysis:* If you have access to a computer package that analyses qualitative data (e.g. NUDIST) then you can use this. These vary in the way they work but these are some of the basic common principles. You can upload your transcripts created in a compatible word-processing package and then the software allows you to mark different sections with various headings/themes. It will then sort all those sections marked with a particular heading and print them off together. This is the electronic version of the folders approach! It is also possible to use a word-processing package to cut and paste comments and to search for particular words.

There is a great danger of subjective interpretation. You must accurately reflect the views of the interviewees and be thorough and methodical. You need to become familiar with your data. You may find this a daunting and stressful task or you may really enjoy it - sometimes so much that you can delay getting down to the next stage which is interpreting and writing up!

Presenting Qualitative Data in your Dissertation

This would normally follow the topics, themes and categories that you have developed in the analysis and these, in turn, are likely to have been themes that came out in the

literature and may have formed the basis for your interview questions. It is usually a mistake to go through each interviewee in turn and what they said on each topic. This is cumbersome and does not give the scope to compare and contrast their ideas with the ideas of others.

Do not analyse the data on a question-by-question basis. You should summarise the key themes that emerge from the data and may give selected quotes if these are particularly appropriate.

Note how a point is made and then emphasized with an appropriate quote. The quotes make the whole text much more interesting and enjoyable to read but be wary of including too many. You should evaluate your own findings in this way and refer to the literature where appropriate. Remember the two concepts of presenting and discussing your findings. By presenting we mean a factual description/summary of what you found. The discussion element is your interpretation of what these findings mean and how they confirm or contradict what you wrote about in your literature section.

If you are trying to test a model then this will have been explored in your literature review and your methodology section will explain how you intend to test it. Your methodology should include who was interviewed with a clear rationale for your choices to explain how this fits into your research questions, how you ensured that the data was unbiased and as accurate as possible, and how the data was analysed. If you have been able to present an adapted model appropriate to your particular context then this should come towards the end of your findings section.

It may be desirable to put a small number of transcripts in the appendices but discuss this with your supervisor. Remember you have to present accurately what was said and what you think it means.

Analyzing survey data is an important and exciting step in the survey process. It is the time that you may reveal important facts about your customers, uncover trends that you might not otherwise have known existed, or provide irrefutable facts to support your plans. By doing in-depth data comparisons, you can begin to identify relationships between various data that will help you understand more about your respondents, and guide you towards better decisions.

It does not discuss specific usage of eSurveysPro for conducting analysis as it is intended to provide a foundation upon which you can confidently conduct your own survey analysis no matter what tool you use.

Three Common Mistakes

Before you dive into analyzing your survey results, take a look back at the big picture. What objectives were you trying to accomplish when you created your survey? Did your survey instrument meet those objectives? Is the data you collected the right data? Do you have sufficient data to properly reach a cease?

Although data analysis is the wrong time to try and rewrite your survey instrument, it is important to remember the scope of your project and stick to it. Many first time

surveyors attempt to read "between the lines" while analyzing data. They attempt to answer questions that were not asked by making inferences and assumptions from those that were asked. Doing so amounts to nothing more than guesswork. To avoid this temptation, remember this simple rule:

Rule 1: If you did not ask you do not know.

Another common mistake that many first time surveyors make is to attempt to change data to compensate for poor question design. For example, if a question asked a respondent to indicate his total household income using a scale of values, a mean and median cannot be calculated. Many people try to get around this by assigning each response a value representing the range. Even if the adjustment is made consistently across all responses, the resulting calculations will be wrong. Similarly, trying to analyse a multiple-choice question as if it was a single-select question will often provide erroneous information. In order to avoid this pitfall, remember this simple rule:

Rule 2: Do not alter data to compensate for bad survey design.

A second mistake inexperienced surveyors make is to project the findings to an audience that was not either part of the survey population or not adequately represented. For example, if an HR manager conducts a benefits survey and invites all employees to participate, most people would assume that the results represent all employees since everyone had an opportunity to participate. Provided that enough employees participate, the data might be statistically valid, but is it really representative of all employees? The answer is, it depends. If the survey collected data about employee demographics that could be compared to what is known about the company, then the results do reflect the company as a whole.

However, if 80% of the respondents are married and 50% of the total employee base is married, the results of the survey are skewed toward married people. If married people have different benefits needs than single people, using the survey results to make ceases about the entire employee pool would be less accurate than those ceases about the married employees or single employees independently. To avoid this temptation, remember this simple rule:

Rule 3: Do not project your data to people that did not respond.

The earlier you recognize flaws in your survey design and data collection, the more time you will save during analysis. If your questions do not provided the data you need to meet your survey objectives, you'll have to start over. If your questions are vague or ambiguous, you'll have to throw them out. If you do not have an adequate number of responses, you'll have to get more.

Survey Analysis

Analyzing any survey, web or traditional, consists of a number of interrelated processes that are intended to summarize, arrange, and transform data into information. If your survey objective was simply to collect data for your database or data warehouse, you do not have to do any analysis of the data. On the other hand, if your objective was

to understand the characteristics of typical customers, then you must transform your raw results into information that will enable you to paint a clear picture of your customers.

Assuming you need to analyse the data collected from your survey, the process begins with a quick review of the results, followed by editing, analysis, and reporting. To ensure you have accurate data before investing significant time in analysis, it is important that you do not begin analyzing results until you have completed the review and editing process.

Quick Review

Read all your results. Although, this seems like an obvious thing to do, many surveyors think that they can skip this step and dive right in to data analysis. A quick review can tell you lots about your project, including any flaws in questionnaire design or response population, before you spend hours of time in analyzing the data.

During the quick review, you should look at every question and see if the results "make sense". This "gut feel" check of the data will often uncover any issues with your survey project. Most surveyors already have an idea of how they expect their data to look.

A quick review of the data can help you quickly understand that tell you if the people that respond are the right people. For example, if you were conducting a survey of all the employees in a company and you knew that 10% were in the marketing department, 20% in sales, 45% in manufacturing, 5% in management, and 5% finance, and 15% research and development, you could reasonably expect your responses to be similarly distributed.

If your quick review disclosed 80% of your respondents were from the sales department, you know that your survey did not adequately capture a representative sample of all departments within the company.

The quick review can also highlight any problems with the survey instrument. Are most respondents answering all questions? If not, your questionnaire could be flawed in such a way that a person cannot complete the survey. A low response rate could mean your survey invitation was not compelling enough to encourage participation, or your timing was off and a follow-up reminder is needed.

Lastly, the quick review of the survey can show you what areas to focus on for detailed analysis. As stated earlier, most surveyors already know what they expect to get, so your quick review can show you the unexpected.

Editing and Cleaning

Editing and cleaning data is an important step in the survey process. Special care must be taken when editing survey data so that you do not alter or throw out responses in such a way as to bias your results. Although you can begin editing and cleaning your data as soon as results are received, caution should be used since any edits can be lost if the database is rebuilt. To be safe, wait until all data is received before you begin the editing and cleaning process.

To start, find and delete incomplete and duplicate responses. A response should be discarded if the respondent did not complete enough of the survey to be meaningful. For example, if a your survey was intended to determine future buying intentions across various demographic groups and the respondent did not answer any of the demographic questions, you should delete the response. On the other hand, if the respondent answered all the demographic questions but omitted their name or email address, then you should keep the response.

Duplicate responses are a unique issue for electronic surveys. Many tools, such as eSurveysPro, provide built in features to help minimize the risk of duplicate responses. Others, like the popular "infotainment" polls featured on many websites do nothing to eliminate duplicates. Without removing duplicates, your data will be skewed in favour of the duplicate response. Both the count and percentage of the whole will be affected by duplicate responses, and computed means and medians will also be thrown off. To find duplicate responses, carefully examine the answers to any open-ended questions. When two open-ended questions have the exact same answer, a duplicate response is likely to exist. Make sure the response is indeed a duplicate by comparing the answers to all the other questions, and then delete one of the responses if a match is found.

Data cleaning of web surveys usually involves categorizing answers to open-ended questions and multiple-choice questions that include an "other, please specify" response. Because of their nature, open-ended text response questions can provide significant value but they are nearly impossible to process without some form of summarization or tabulation. One of the easiest ways to summarize these questions is to build a list of themes and select the themes that apply as you read each response. Tools such as eSurveysPro allow you to add questions after a survey is run to do just this sort of thing.

A common problem in any survey that needs attention during the editing and cleaning process is when a respondent answers an "other, please specify" question by selecting "other" and then writing in an answer that was one of the listed response options. Without cleaning these answers, the "other" response will be overstated and the correct response will be understated. For example, a demographics question that asks for the respondent's role within the organization may have a response like "faculty, teacher, or student" and a respondent selects "other" and types "professor," you would want to clean the response by switching the other choice to the one for "faculty, teacher, or student".

Once the data preparation is complete, it is time to start analyzing the data and turning it into actionable information.

Detailed Analysis

Analysis is the most important aspect of your survey research project. At this point, you have collected a set of data that must now be turned into actionable information. The process of analysis can lead to a variety of alternative courses of action. Mistakes during analysis can lead to costly decisions down the road, so extreme caution and careful review must be followed throughout the process. Carelessness during analysis

can lead to disaster. What you do during analysis will ultimately determine if your survey project is a successful or not.

Depending on what type of information you are trying to know about your audience, you will have to decide what analysis makes sense. It can be as simple as reviewing the graphs that eSurveysPro automatically creates, or conducting in-depth comparisons between questions sets to identify trends or relationships. For most surveyors, a basic analysis using charts, cross tabulations, and filters is sufficient. On the other hand, more sophisticated users may wish to do a more complex statistical analysis using high powered analytical tools such as SPSS, Excel, or any number of number crunching applications.

Graphical Analysis

Graphical analysis simply means displaying the data in a variety of visual formats that make it easy to see patterns and identify differences among the results set. There are many different graphing options available to display data, the most common are Bar, Pie, and Line charts.

Bar charts use solid bars on an X and Y-axis that extend to meet a specific data value indicated on the chart and can be shown either vertically or horizontally. These charts are flexible and are most commonly used to display data from multiple-select, rank order, single-select matrix and numerical questions. Each response option is shown as an independent bar on the chart, and the length of the bar represents the frequency the response was chosen relative to all choices.

Pie charts, or circle graphs, have colourful "slices" representing segments of your data. These charts measure values as compared to a "whole", and the total percentages of the segments always add up to 100%. Pie charts are most useful with single-select questions because the each response is represented visually as a portion of the entire pie. It is easy to interpret which answer received the most responses in a pie chart by selecting the largest portion of the pie. When comparing two sets of data using a pie chart, it is important to make sure the colours used for each response option remain consistent in each chart. If represent the same response options in each chart, this way, a side-by-side visual comparison can quickly be made. Pie charts are not appropriate for multiple-select questions because each respondent can answer choose more than one option, and the sum of the option percentages will exceed 100%.

There are other graphing options such as line charts, area charts and scatter graphs, which are useful when displaying the same data over a period of time. However these formats are not as easy to interpret for casual users, so they should be used sparingly.

Frequency Tables

Frequency tables are another form of basic analysis. These tables show the possible responses, the total number of respondents for each part, and the percentages of respondents who selected each answer. Frequency tables are useful when a large number

of response options are available, or the differences between the percentages of each option are small. In most cases, pie or bar charts are easier to work with than frequency tables.

Response	Count	Per cent
Market Analysis	76	13.7%
Quantitative Analysis	150	27.0%
Strategic Planning	56	10.1%
Product Planning	33	5.9%
Promotional Communication	243	43.8%
Creating sales tools	152	27.4%
Providing channel support	157	28.3%

Cross Tabulation

Cross tabulations, or cross tabs, are a good way to compare two subgroups of information. Cross tabs allow you to compare data from two questions to determine if there is a relationship between them. Like frequency tables, cross tabs appear as a table of data showing answers to one question as a series of rows and answers to another question as a series of columns.

Base Question	Female	Male
Product Manager	57.2%	53.4%
Director	12.6%	14.2%
Product Marketing Manager	24.7%	23.1%
Programmmme Manager	2.8%	1.5%
Technical Product Manager	2.8%	7.7%
Total Counts	215	337

Cross tabs are used most frequently to look at answers to a question among various demographic groups. The intersections of the various columns and rows, commonly called cells, are the percentages of people who answered each of the responses. Females and males had relatively similar distribution among various job titles, with the exception of the tile of "Technical Product Manager", where 2.5 times as many males had the title as compared to females. For analysis purposes, cross tabs are a great way to do comparisons.

Filtering

Filtering is the most under-utilized tool used in analysis. Filters allow you select specific subsets of data to view. Unlike a cross tab, that compares two questions, a filter will allow you to examine all questions for a particular subset of the responses. By viewing only the data from the people who responded negatively, look at how they answered other questions. Find patterns or trends that help define why a person answered the way they did. You can even filter on multiple questions and criteria to do a more detailed search if necessary. For example, if you wanted to know the buying intentions

of men, over the age of 40, with income of about \$50,000, you would set a filter that would remove all those respondents that do not meet your criteria from the results set, thus enabling you to concentrate on the target population.

By applying filters to the date survey responses were received, you can see how the answers change from one time frame to the next. For instance, by continually running a customer satisfaction survey, you can assess changes in customer attitudes over time by filtering on the date the survey was received. You can also use a filter on date received to assess the impact of sales incentive programmes or new product offerings by comparing survey responses before and after the change.

Filters do not permanently remove the responses of those people that do not match the specified criteria; they simply eliminate them from the current view of the data, making it much easier to perform analysis. By looking at the same question with different filters applied, differences between the various respondents represented by the filter can be quickly seen. Because filters remain in effect until cleared, don't forget to clear them before attempting to analyse your survey responses as a whole, otherwise your observations will be inaccurate, and your recommendations flawed.

Simple Regression Analysis

Determining what factors have lead to a particular outcome is called regression analysis. The regression means you're working backwards from the result to find out why a person answered the way that they did. This can be based on how they answered other questions as well.

For example, you might believe that website visitors who had trouble navigating within your website are likely not return again. If 30% of the respondents said they had trouble navigating through the website and 40% said they would not return, you could look at only those that would not return to determine if poor navigation might be the case. After filtering to only those who would not return, if 30% or less said they had trouble navigating, then this is clearly not the "reason" visitors will not return. By filtering out those that would return, we expect the percentage to increase dramatically. If it does, we still cannot conclude that navigation is "the" reason, only that it might contribute to the respondents not returning. In order to know if it is "the" reason, we would need to ask a direct question.

Reporting

After analyzing your survey data, it is time to create a report of your findings. The complexity and detail need to support your ceases, along with your intended audience, will dictate the format of your report. CEO's require a different level of detail than line managers, so for maximum results consider who is going to receive your report and tailor it to meet their unique needs.

Visual reports, such as an HTML document or Microsoft PowerPoint presentation, are best suited for simple findings. These graphical reports are best when they are light

on text and heavy on graphs and charts. They are reviewed quickly rather than studied at length, and most ceases are obvious, so detailed explanations are seldom required. For more complex topics, a detailed report created in Microsoft Word or Adobe Acrobat is often required. Reports created using Word often include much more detailed information, report findings that require significant explanation, are extremely text heavy, and are often studied at great length and in significant detail.

No matter which type of report you use, always remember that information can be more powerfully displayed in a graphic format verses a text or tabular representation. Often, trends and patterns are more obvious and recommendations more effective when presented visually. Ideally, when making comparisons one or more groups of respondents, it is best to show a chart of each group's responses side-by-side. This side-by-side comparison allows your audience to quickly see the differences you are highlighting and will lead to more support for your ceases.

At the beginning of your report, you should review your survey objective and sampling method. This will help your audience understand what the survey was about, and enable you to avoid many questions that are outside of your original objectives. Your report should have a description of your sampling method, including who was invited to participate, over what time frame results were collected, and any issues that might exist relative to your respondent pool. Next, you should include your analysis and ceases in adequate detail to meet the needs of your audience. Include a table or graph for each area of interest and explain why it is noteworthy. You should make recommendations that relate back to your survey objectives. Recommendations can be as simple as conduct further studies to a major shift in company direction. In either case, your recommendation must be within the scope of your survey objective and supported by the data collected.

Administrative collections

Besides surveys and censuses, administrative data collection (general purpose administrative systems or administrative registries) intended to serve persons with disabilities can be an important method of gathering information about these people and their characteristics. In this method, any information collected is organized and becomes part of normal service administration procedure such as the information gathered using standard client intake forms for health, rehabilitation, or social work services.

Administrative data collections can take several forms, depending on the nature of the service, the format used, the type of information collected, and the method and frequency of collection. Some examples are population registers, vital registration systems, social security systems, registries of occupational injuries, rehabilitation programmes, assistive device services, and other services specifically designed to serve the needs of persons with disabilities.

Administrative registries are databases of records of people with particular characteristics and set up as part of the administration of services to these individuals.

Registers are either established during a registration survey or other point in time or they can be on-going and regularly updated.

Administrative records and registers, moreover, often provide unique information about persons with disabilities. The information is usually collected for reasons concerning the administration of the programme or service, but with care, can prove to be useful. For example, data about children and adolescents in special education programmes is an indication of participation rates in education; general invalid pension registries are often organized by disease or impairment; and domestic care allowances help to track rates of children and adults with severe disabilities.

Often these data are collected annually, which provides a source for trend data on the prevalence of impairments or disabilities.

Ministries, government departments, advocacy groups, and service providers often maintain administrative records which they use to monitor and evaluate programs and services. Given confidentiality concerns, only aggregated data may be available, but this can still be useful. It must be kept in mind, however, that data from registration systems cannot provide information about persons with disabilities who need a service or programme but do not receive it. Data about unmet need has to come from other sources.

Since the information collected in administrative data collections is limited to people receiving services, or otherwise known to the service agency (as in the case of people on a waiting list), these collections have limited coverage. Therefore administrative collections, despite all their benefits, are not good sources for estimating overall disability prevalence.

What are the benefits? As compared with censuses and sample surveys, administrative data collections have some advantages for disability data users:

Data from administrative collections are generally available on a regular or on-going basis, as they are part of regularly updated information systems linked to a service.

As administrative data are collected as part of the day-to-day running of a service, they use fewer resources than special purpose surveys.

If the government or other fund provider requires data collection as a condition of on-going funding to a service, the data collected by that service will tend to be more reliable, complete and of higher quality.

Information from administrative data collections is likely to be useful to a wide range of stakeholders such as service providers, higher-level bureaucrats, disability consumer groups, and researchers.

How can administrative data be used? Administrative data collections provide information on numbers and characteristics of service users, and the type, quantity and cost of services provided. Depending on the detail of the dataset and the complexity of the data collection format, a substantial amount of information about services, service users and service providers can be collected in this manner. Administrative data of this sort can then be adopted by service agencies, planners, consumer advocacy groups, and funding department for a variety of purposes:

- to provide an evidence base to support planning for future service delivery (e.g., by monitoring trends in client age or disability type);
- to indicate whether identified groups of people (e.g., particular ethnic groups) are accessing services as much as expected;
- to monitor the cost-effectiveness of service provision;
- to support a budget submission for increased funding.

Moreover, with only minimal analysis, these data can answer basic administrative questions such as:

- How many people were supported by the service in a given time period, and what were their characteristics and support needs?
- What type of support was provided and received?
- What quantity of support-e.g., in terms of staff hours-was provided and received?
- From whom was the support received (in terms of details of the service agency, such as size, staff profile, and hours)?
- What was the cost (total, per service type, per client) of providing these services?
- What were the outcomes for clients? (Examples of outcome questions include: Were clients satisfied with the services received? Were there increased levels of client participation in key life areas such as employment?)

Statistics/Methods of Data Collection

The main portion of Statistics is the display of summarized data. Data is initially collected from a given source, whether they are experiments, surveys, or observation, and is presented in one of four methods:

Textular Method

The reader acquires information through reading the gathered data.

Tabular Method

Provides a more precise, systematic and orderly presentation of data in rows or columns.

Semi-tabular Method

Uses both textual and tabular methods.

Graphical Method

The utilization of graphs is most effective method of visually presenting statistical results or findings.

Experiments

It's true that scientists utilize two methods to gather information about the world: correlations (a.k.a. observational research) and experiments.

Experiments, no matter the scientific field, all have two distinct variables. Firstly, an independent variable (IV) is manipulated by an experimenter to exist in at least two levels (usually “none” and “some”). Then the experimenter measures the second variable, the dependent variable (DV).

A simple example—Suppose the experimental hypothesis that concerns the scientist is that reading a Wiki will enhance knowledge. Notice that the hypothesis is really an attempt to state a causal relationship like, “if you read a Wiki, then you will have enhanced knowledge.” The antecedent condition (reading a Wiki) causes the consequent condition (enhanced knowledge). Antecedent conditions are always IVs and consequent conditions are always DVs in experiments. So the experimenter would produce two levels of Wiki reading (none and some, for example) and record knowledge. If the subjects who got no Wiki exposure had less knowledge than those who were exposed to Wikis, it follows that the difference is caused by the IV. So, the reason scientists utilize experiments is that it is the only way to determine causal relationships between variables. Experiments tend to be artificial because they try to make both groups identical with the single exception of the levels of the independent variable.

Sample surveys involve the selection and study of a sample of items from a population. A sample is just a set of members chosen from a population, but not the whole population. A survey of a whole population is called a census. A sample from a population may not give accurate results but it helps in decision making.

Examples

Examples of sample surveys:

- Phoning the fifth person on every page of the local phonebook and asking them how long they have lived in the area. (Systematic Sample).
- Dropping a quad. in five different places on a field and counting the number of wild flowers inside the quad. (Cluster Sample).
- Selecting sub-populations in proportion to their incidence in the overall population. For instance, a researcher may have reason to select a sample consisting 30% females and 70% males in a population with those same gender proportions. (Stratified Sample).
- Selecting several cities in a country, several neighbourhoods in those cities and several streets in those neighbourhoods to recruit participants for a survey (Multi-stage sample).

The term random sample is used for a sample in which every item in the population is equally likely to be selected.

Bias

While sampling is a more cost effective method of determining a result, small samples or samples that depend on a certain selection method will result in a bias within the results.

The following are common sources of bias:

- Sampling bias or statistical bias, where some individuals are more likely to be selected than others (such as if you give equal chance of cities being selected rather than weighting them by size).
- Systemic bias, where external influences try to affect the outcome (e.g. funding organizations wanting to have a specific result).

Survey Data and Questionnaires

Survey data can be gathered in many ways, such as with the use of questionnaires administered in person, interviews, and telephone surveys. Information attained through surveys benefits researchers by recording personal behaviours, attitudes, identities, and definitions from a high volume of respondents. Surveys are useful for gaining information that is difficult to discuss or cannot otherwise be calculated by observation, such as risk, fear, religious belief, and racial attitude. Ulmer and Wilson (2003) believe that the strength of using surveys comes from the fact that “the researcher can rely on outside observations. As a result, many scholars believe that surveys, under certain circumstances, provide more valid measures of phenomena than do official records”. Moreover, surveys provide a systematic and structured method for acquiring information on the same topic from a large group of people in a relatively short amount of time.

However, survey data does have its limits. Surveys require that respondents answer questions honestly and accurately, and researchers must be aware that questions are subject to interpretation by the respondents. Therefore, it is recommended that researchers check with respondents on their answers to make sure there is as much uniformity in their interpretation as possible, employ surveys with a published record of reliability, and determine that questions asked in the survey accurately represent the study’s intentions.

Esposito and Murphy (1999) believe the quantitative data typically acquired through survey responses do not give a very deep interpretation of occurrences, such as open-ended qualitative interviews would provide. These “snapshot”, point-in-time measures give a shallow view that fails to give a rich or true assessment of the aspect of life being studied.

In order to study life for research purposes, access must be obtained to the information. The nature of the research and questions asked help establish the types of access required. Remenyi, Williams, Money, and Swartz (1998) believe that good research sites should have an entrance, trust can be established, and possibilities for gathering credible/quality and suitable evidence are available.

Finally, questions concerning data collection should analyse whether the strategy for collection has been clearly stated or not. Without a clearly articulated method for data collection, replications are obstructed and validity and reliability might come into question. These questions should address whether or not the best approach has been chosen for answering the primary research question.

Mixed Methods

Although mixed methods is not within the scope of the question addressed in the beginning of this chapter, it is being briefly described here, because of its apparent utility in survey research which is addressed as the focus of this chapter. Also, it has been included in this piece out of hindsight and necessity for possible usage in a proposed-future dissertation by the author.

Determining whether to use qualitative or quantitative research should become more apparent as the research process unfolds. Hines (2000) believes that multiple methods provide construct validity, as well as internal and external validity, while allowing complex issues to be examined using the respondents' language. Creswell (2003) also indicates that using mixed-methods research will assist in making better interpretations because informants will be providing information that is both measurable and analysed through rich description. A mixed-methods approach will also aid in reducing researcher bias and allowing documentation to be measured and analysed more effectively. Likewise, Malterud (2001) states that, "qualitative studies can also be added to quantitative ones, to gain a better understanding of the meaning and implications of the findings".

Qualitative data is what often shapes the body of the research. Furthermore, it depicts the world as having several realities rather than one truth. On the other hand, quantitative research allows the use of data that identifies partial regularities, turnover in numbers, and statistical operations. Quantification shows a mixing and churning in numeric or sequential patterns and also emulates natural science. The practice of gathering and processing qualitative data should include the supplemental use of statistical hypothesis testing or comparing data sets with other samples and tests in order to confirm patterns.

Determining whether or not to use qualitative or quantitative research is not always so cut and dry. When great amounts of data are sought, quantitative research is often the most efficient and cost-effective research method. However, when data analysis and research questions require large amounts of description dealing with human experiences and perceptions, quantitative research could restrict the findings, making a qualitative approach ideal. The study itself should determine the method.

Ulmer and Wilson (2003) argue that "researchers can use qualitative data and statistical analysis without violating pragmatist ontological and epistemological positions" by meeting the following five conditions that serve to incorporate methods: (a) Using quantification and statistical analysis of outcomes of social processes to sensitize or lay the groundwork for qualitative study of the social processes in question themselves; (b) understanding that quantification should not be seen as synonymous with empiricism; (c) specifying that individual and joint actors, not variables, exercise agency (i.e., variables do not think or do things, people do, and we create variables to represent those things); (d) remembering that social causality lies not in variables or statistical models but in interpretive processes as people individually and jointly define situations and act within them; and (e) striving to keep the conceptual distance between

the quantified measures and the phenomena measured as small as possible (i.e., maximizing validity). As Hines (2000) asserts, there is no one best method for conducting survey research. Qualitative research is valuable for describing events, circumstances, and other areas of human understanding, including people's experiences and meanings in a normal social setting, as well as processes and structures. Similarly, Bryman (1993) believes qualitative research is best defined in studies of the social world; as it analyses and describes human behaviour from the point of view of those being studied. However, quantitative research too has value due to its ability to measure data, generalize findings, and make predictions. These research strategies should not be thought of as incompatible, but complementary. Their procedures for interpreting data and other types of information are interpreted differently, and they answer different types of questions. Their underlying principle for increasing understanding of a subject, however, is the same, and they can be used to interpret or enhance one another. Mixed-methods research utilizes both qualitative and quantitative methods in order to explore a topic in breadth and depth. Mixed methods lead to a "truer analysis," as Davies (2003) indicates, and quantitative discoveries should lead to a reflection on qualitative decision making in order to get a better understanding of the ideas in the "heads of economic actors and the models they use".

Research should occur through multiple methods of gathering data for the purpose of drawing a robust conclusion. As Davies (2003) explains, incorporation of both qualitative and quantitative designs in research could yield results that would not have been identified through use of one method or the other. Mixed method clearly has their advantages in research and is often the answer to the debate over which type of research is best.

Although not every tool and method in either quantitative or qualitative research was discussed in this chapter, it is clear that each method represents a different paradigm, making no possible answer as to which is conclusively best. One primary conclusion that can be drawn from this chapter is that for any study proposed, one will not know which research method will produce the most accurate conclusions without examining the kind of knowledge sought. Once a survey is initiated and the data is collected, clues to new avenues that researchers should pursue and new questions to be asked will be revealed. During this process, researchers need to be aware of personal biases in collecting and analysing data. They should find ways to keep their own and their respondents' biases in check in order to produce results that accomplish what scholars have yearned to do for centuries—help society grow in knowledge through realized truths.

Data Preparation

Data Preparation involves checking or logging the data in; checking the data for accuracy; entering the data into the computer; transforming the data; and developing and documenting a database structure that integrates the various measures.

Logging the Data

In any research project you may have data coming from a number of different sources at different times:

- mail surveys returns
- coded interview data
- pretest or posttest data
- observational data.

In all but the simplest of studies, you need to set up a procedure for logging the information and keeping track of it until you are ready to do a comprehensive data analysis. Different researchers differ in how they prefer to keep track of incoming data. In most cases, you will want to set up a database that enables you to assess at any time what data is already in and what is still outstanding. You could do this with any standard computerized database program (e.g., Microsoft Access, Claris Filemaker), although this requires familiarity with such programs. or, you can accomplish this using standard statistical programs (e.g., SPSS, SAS, Minitab, Datadesk) and running simple descriptive analyses to get reports on data status. It is also critical that the data analyst retain the original data records for a reasonable period of time — returned surveys, field notes, test protocols, and so on. Most professional researchers will retain such records for at least 5-7 years. For important or expensive studies, the original data might be stored in a data archive. The data analyst should always be able to trace a result from a data analysis back to the original forms on which the data was collected. A database for logging incoming data is a critical component in good research record-keeping.

Checking the Data For Accuracy

As soon as data is received you should screen it for accuracy. In some circumstances doing this right away will allow you to go back to the sample to clarify any problems or errors. There are several questions you should ask as part of this initial data screening:

- Are the responses legible/readable?
- Are all important questions answered?
- Are the responses complete?
- Is all relevant contextual information included (e.g., data, time, place, researcher)?

In most social research, quality of measurement is a major issue. Assuring that the data collection process does not contribute inaccuracies will help assure the overall quality of subsequent analyses.

Developing a Database Structure

The database structure is the manner in which you intend to store the data for the study so that it can be accessed in subsequent data analyses. You might use the same structure you used for logging in the data or, in large complex studies, you might have one structure for logging data and another for storing it. As mentioned above, there are

generally two options for storing data on computer — database programs and statistical programs. Usually database programs are the more complex of the two to learn and operate, but they allow the analyst greater flexibility in manipulating the data. In every research project, you should generate a printed codebook that describes the data and indicates where and how it can be accessed.

Minimally the codebook should include the following items for each variable:

- variable name
- variable description
- variable format (number, data, text)
- instrument/method of collection
- date collected
- respondent or group
- variable location (in database)
- notes.

The codebook is an indispensable tool for the analysis team. Together with the database, it should provide comprehensive documentation that enables other researchers who might subsequently want to analyse the data to do so without any additional information.

Entering the Data into the Computer

There are a wide variety of ways to enter the data into the computer for analysis. Probably the easiest is to just type the data in directly. In order to assure a high level of data accuracy, the analyst should use a procedure called double entry. In this procedure you enter the data once.

Then, you use a special program that allows you to enter the data a second time and checks each second entry against the first. If there is a discrepancy, the program notifies the user and allows the user to determine the correct entry. This double entry procedure significantly reduces entry errors.

However, these double entry programs are not widely available and require some training. An alternative is to enter the data once and set up a procedure for checking the data for accuracy.

For instance, you might spot check records on a random basis. Once the data have been entered, you will use various programs to summarize the data that allow you to check that all the data are within acceptable limits and boundaries. For instance, such summaries will enable you to easily spot whether there are persons whose age is 601 or who have a 7 entered where you expect a 1-to-5 response.

Data Transformations

Once the data have been entered it is almost always necessary to transform the raw data into variables that are usable in the analyses. There are a wide variety of transformations that you might perform. Some of the more common are:

- missing values: Many analysis programs automatically treat blank values as missing. In others, you need to designate specific values to represent missing values. For instance, you might use a value of -99 to indicate that the item is missing. You need to check the specific program you are using to determine how to handle missing values.
- item reversals: On scales and surveys, we sometimes use reversal items to help reduce the possibility of a response set. When you analyse the data, you want all scores for scale items to be in the same direction where high scores mean the same thing and low scores mean the same thing. In these cases, you have to reverse the ratings for some of the scale items. For instance, let's say you had a five point response scale for a self esteem measure where 1 meant strongly disagree and 5 meant strongly agree. One item is "I generally feel good about myself." If the respondent strongly agrees with this item they will put a 5 and this value would be indicative of higher self esteem. Alternatively, consider an item like "Sometimes I feel like I'm not worth much as a person." Here, if a respondent strongly agrees by rating this a 5 it would indicate low self esteem. To compare these two items, we would reverse the scores of one of them (probably we'd reverse the latter item so that high values will always indicate higher self esteem). We want a transformation where if the original value was 1 it's changed to 5, 2 is changed to 4, 3 remains the same, 4 is changed to 2 and 5 is changed to 1. While you could program these changes as separate statements in most program, it's easier to do this with a simple formula like:

New Value = (High Value + 1) - Original Value.

In our example, the High Value for the scale is 5, so to get the new (transformed) scale value, we simply subtract each Original Value from 6 (i.e., 5 + 1).

- scale totals: Once you've transformed any individual scale items you will often want to add or average across individual items to get a total score for the scale.
- categories: For many variables you will want to collapse them into categories. For instance, you may want to collapse income estimates (in dollar amounts) into income ranges.

Techniques of Data Collection

Basic requirements for scientific data are that it should be reliable and impartial. In Sociology these conditions are hard to meet. Yet numerous methods are used to minimize errors in data. Some of the commonly used sources in collecting data are:

- Existing materials including the official statistical record and historical and contemporary documents.
- Social surveys through questionnaire and schedules
- Interviewing
- Observation-Participants and non-participant.

Existing Material

Statistical Sources: Government statistics particularly census or statistics produced by large industrial or commercial firms, trade unions or other organizations provide one important account of data which sociologists can use in their analysis. An outstanding example of the imaginative use of official statistics in the positivist tradition is the study of suicide made by the famous French sociologist Emile Durkheim in the 19th century. However official statistics are the kind of data that are not collected by sociologists themselves and so there are problems while analysing the data.

Historical documents: Records and accounts of qualitative kind for example relating to belief, values, social relationship or social behaviour may also be contemporary or may refer to earlier periods.

There are several difficulties immediately present themselves in the use of records from the past. Few chroniclers of social relation and social action record observations in the systematic way in which the sociologists are interested. There are often intriguing and sympathetic records but the information that is vital to the sociologist is often missing.

Contemporary Records: Contemporary records relating to social relationship and social behaviour are seldom used as the sole source of information and sociological research. They are usually one source of a particular account or achievement.

Social Survey

The basic procedure in survey is that people are asked a number of questions on that aspect of behaviour which the sociologist is interested in. A number of people carefully selected so that their representation of their population being studied are asked to answer exactly the same question so that the replies to different categories of respondents may be examined for differences. One type of survey relies on contacting the respondents by letter and asking them to complete the questionnaire themselves before returning it. These are called Mail questionnaires. Sometimes questionnaires are not completed by individuals separately but by people in a group under the direct supervision of the research worker. A variation of the procedure can be that a trained interviewer asks the questions and records the responses on a schedule from each respondent.

These alternate procedures have different advantages and disadvantages. Mail questionnaires are relatively cheap and can be used to contact respondents who are scattered over a wide area. But at the same time the proportion of people who return questionnaires sent through post is usually rather small. The questions asked in mail questionnaires have also to be very carefully worded in order to avoid ambiguity since the respondents cannot ask to have questions clarified for them. Using groups to complete questionnaires means that the return rate is good and that information is assembled quickly and fairly. Administering the interview schedules to the respondents individually is probably the most reliable method. Several trained interviewers may be employed to contact specific individuals. The questionnaires and schedules can consist of both close-

ended and open-ended questions. Also a special attention needs to be paid to ensure that the questionnaires are filled in logical order.

Where aptitude questions are included great care must be exercised to ensure the proper words are used. In case of schedules emphasis and interactions may also be standardized between different individuals and from respondents to respondents. Finally proper sampling techniques must be used to ensure that the sample under study represents the universe of study. In order to enhance the reliability of data collected through questionnaires and schedules, these questionnaires and schedules must be pretested through pilot studies.

Interviewing

Social surveys may depend either on questionnaires that are self-administered or on schedules completed by trained research workers personally interviewing then is not a method of data collection distinct from social surveying but rather a technique which may vary from the brief formal contact as when the interviewer is working for the firms public opinion consultants or a market research organization and simply asks a housewife a few highly specific questions on limited range of topics to a long interview in which the research worker allows the respondents to develop points at leisure and take up others as he chooses.

The brief formal interview in which the working of the questions and the order in which they are asked is fixed is called structured interview while the freer discursive interview is called unstructured interview.

The object of using structured interview is to standardize the interview as much as possible and thus to reduce the effect that the interviewer's personal approach or biases may have upon the result and even when structured interviews are used, proper training can do a lot to ensure further the reliability and validity of research. The personality of the interviewer and the social characteristics that the respondents attribute to him can be having influence on the result.

The effort of interviewer's bias can be estimated by comparing one interviewer's result with other. The problem of interviewer's bias in an unstructured interview is much greater. Here the interviewer is left to his common devices as far as the way he approaches a respondent is concerned.

There is no fixed list of questions to work through. Instead the interviewer may work from a guide that will remind him of the topics he wishes to cover. The training of the interviewer is crucial here not simply training in the social skills of keeping the conversation going on a topic that the respondent may not be very interested in but also in acquiring sensitivity to those things his respondents tells him which are specially relevant to the theoretical topics he is pursuing. This means that unstructured interviews can be carried out by people trained in sociological theory. They are then able to size upon stray comments made by the respondents which can be developed and lead on to important theoretical insight.

Observation: Participant and non Participant

The rationale behind the use of observation in sociological research is that the sociologist should become party to a set of social actions sufficiently able to be able to assess directly the social relationship involved. The degree of involvement may vary considerable from being merely a watcher on the sidelines to be deeply involved in and being a part of what is going on. The former type of observation techniques are called non-participant while the latter is called participant observation.

Sometimes one way observations screen have been used to watch groups in actions that they are unaware that they are being watched and the observer cannot affect their actions by his presence. The sociologist is visibly present and is a part of the situation either as a sociologist or in another guise. Where the sociologist is merely an observer it is usually assumed that he knows enough about what the actors are doing to be able to understand their behaviour. Any sociological observer has then to some extent be a participant observer he must at least share sufficient cultural background with the actors to be able to construe their behaviour meaningfully but the degree of participation and of sharing of meaning may vary considerably. Examples of such studies are Nel Anderson's study of Hobo-Indians and William White study of Street Corner Society.

Sampling

For practical and cost reasons, it is often impossible to collect information about the entire population of people or things in which social researchers are interested. In these cases, a sample of the total is selected for study. Most statistical studies are based on samples and not on complete enumerations of all the relevant data. The main criteria when sampling are to ensure that a sample provides a faithful representation of the totality from which it is selected, and to know as precisely as possible the probability that a sample is reliable in this way. Randomization meets these criteria, because it protects against bias in the selection process and also provides a basis on which to apply statistical distribution theory that allows an estimate to be made of the probability that conclusions drawn from the sample are correct. A statistical sample is a miniature picture or cross-section of the entire group or aggregate from which the sample is taken. The entire group from which a sample is chosen is known as the population, universe or supply.

Simple Random Sampling

The basic type of random sample is known as a simple random sample, one in which each person or item has an equal chance of being chosen. Often a population contains various distinct groups or strata that differ on the attribute that is being researched.

Stratified Random Sampling

Stratified random sampling involves sampling of each stratum separately. This increases precision, or reduces time, effort and cost of allowing smaller sample sizes

for a given level of precision. For example, poverty is known to be most common among the elderly, the unemployed and single parent families, so research on the effect of poverty might will sample separately each of these three strata as part of a survey of poverty in the population as a whole which would permit the total sample size to be reduced because the investigator would know that the groups most affected by poverty were guaranteed inclusion.

Cluster Sampling

Cluster sampling is sometimes used when the population naturally congregates into clusters. For example, managers are clustered in organizations, so a sample of managers could be obtained by taking a random sample of organizations and investigating the managers in each of these. Interviewing or observing managers on this basis would be cheaper and easier than using a simple random sample of managers scattered across all organizations in the country. This is usually less precise than a simple random sample of the same size, but in practice the reduction in cost per element more than compensated for the decrease in precision.

Multi-stage Sampling

Sampling may be done as one process or in stages, known as multi-stage sampling. Multi-stage designs are common when populations are widely dispersed. Thus a survey of business managers might proceed by selecting a sample of corporations as first stage units, perhaps choosing these corporations with a probability proportionate to their size, and then selecting a sample of managers within these corporations at the second stage. Alternatively, a sample of individual factories or office buildings within each corporation could be chosen as the second stage units, followed by sample of managers in each of these as a third stage. Stratification can also be used in the design, if for example occupational sub-groups are known to differ from each other, by selecting state such as personnel, production, and finance management and sampling within each of these. For sampling to be representative, one needs a complete and accurate list of the first stage units that make up the relevant population, a basic requirement that is not always easily met. This forms the sampling frame. Selection from the frame is best done by numbering the items and using a table of random numbers to identify which items form the sample, though a quasi-random method of simply taking every item from the list is often appropriate. The reliability of a sample taken from a population can be assessed by the spread of the sampling distribution, measured by the standard deviation of this distribution, called the standard error. As a general rule, the larger is the size of the sample the smaller the standard error.

Area Sampling

In sampling of this kind small areas are designated as sampling units and the households interviewed include all or a specified fraction of those found in a canvass of

these designated small areas. The basic sampling units or segments chosen may be relatively large or relatively small depending on such factors as the type of area being studied, population distribution, the availability of suitable maps and other information and the nature and desired accuracy of the data being collected.

Data Collection Methods and Instruments

Measurement of Disability

Collecting data about persons with disabilities and their lives is difficult. It poses unique problems that data collection developers need to address in the design phase of the collection process. Two issues need to be addressed at the outset: what kind of data collection instrument should be used, and what unit of measurement should be employed.

Collection Instruments

The main types of instruments for collecting data about persons with disabilities are:

- Population censuses
- Sample surveys (either general social surveys or specific health and disability surveys)
- Administrative collections and registries.

Each of these tools can be used to measure aspects of disability in a population and each has its strengths and weaknesses.

The Population Census certainly has the advantage of providing complete population coverage. Unfortunately, it is difficult to collect accurate information about disability in a census since time constraints make it unlikely that more than 4 to 6 general disability questions can be asked.

Censuses, in some instances, also undercount children with disabilities and people with mild or moderate disabilities (in cases where the response categories are limited to 'Yes or No' options only). Still, for a crude measure of disability, and in the absence of other collection instruments, the census is useful.

Sample surveys are shorter surveys designed to be administered to a sub-population selected by some other instrument (often a census) that focus on specific issues. They are often put into the field to answer specific questions about a population.

As such, they provide the opportunity to ask more detailed questions about disability. More detailed information is useful in itself, of course, but it also helps to reduce the number of false positive and negative responses, thereby offering a more accurate prevalence measure. A sample survey may be an independent survey focusing entirely on disability, or a disability module added to an existing survey.

Administrative collections and registers are composed of data that is collected as part of the normal operation of some service or programme. An example is the information found on a client intake form.

These collections provide useful information on the characteristics of people accessing disability services as well as details about the services provided. They cannot give an accurate measure of disability prevalence since there is no guarantee of coverage and they tend to incorporate double counting.

The quality of administrative register data is closely related to the quality of the administrative system, in particular, how well it has been maintained and how closely the concepts align with the disability concepts of interest.

Choice of Selection and Measurement Unit

The second preliminary issue that needs to be addressed, whatever data collection instrument is chosen, is how to select the unit for which disability is to be measured. If the selection unit is the individual, then the individual will also be the measurement unit; if the selection unit is a collection of people-invariably a household-then, a decision has to be made as to whether the measurement unit is the household itself (that is, all individuals in the household) or some individual in the household that meets specified criteria (age, gender, unemployed, and/or others). These decisions depend in part on what kinds of data are needed. Is data required for the number of persons with disabilities and their characteristics, or for an estimate on the number of households that include individuals with disabilities? Data about individuals is important, but estimates at the household level are also useful for detailed analysis of living arrangements and access to help and assistance. There are also issues of practicality and efficiency. Having the household as the selection unit means that the survey or interview can be conducted in a setting in which information about other people can be efficiently collected, even if only one member of the household is given the full interview. Questions asked of a single household member may reveal another individual with a disability, thereby increasing the sample of person with disability, without adding to the number of households screened.

A note of caution applies, however, if the private household is chosen as the measuring unit. In such a case, thought needs to be given to the fact that this choice excludes residential care units, retirement homes, and other institutions such as prisons. Excluding people living in institutions underestimates the levels of disability for older people and for those with particular types of disability such as psychiatric disabilities. For a complete picture, if feasible, household surveys should be supplemented with institutional surveys of disability.

Population Censuses

Many countries have collected information about disability in their national population censuses. The United Nations' 2001 Guidelines and Principles for the Development of Disability Statistics suggests the inclusion of a 'general' disability question in censuses in order to get a rough idea of prevalence. In countries without a good household survey program, the census may be the only possible source for estimating disability prevalence and gaining an approximate estimate of types of disability in the country.

The amount of information on disability that can be collected in the population census is very limited, and is often confined to a single question. With only one question, false positive and false negative responses are more common and no complete measure of the number of persons with disabilities is possible, especially among children and the elderly. Still, census data should be readily used, where available, to develop more detailed follow-up surveys.

Many countries use both short and long forms of census questionnaires. The short form is for complete coverage of core topics, such as age, sex and location, and may also have a question on disability. A probability sample is then selected for the long form to be administered. The long form includes all questions in the short form plus supplementary questions for more detailed coverage of selected topics. Questions more detailed than a single general question on disability may be included in the long form.

Sample Surveys

Sample surveys are good methods of collecting data about persons with disabilities. They use sample selection procedures to identify a sub-population that is representative of the total population, unlike censuses that are designed to enumerate every household or individual in the country. While sample surveys are considerably cheaper than censuses, the size of the sample affects the quality of the data, which is particularly subject to sampling error. This can frustrate analyses that rely on cross-tabulation.

A small sample size also means that sample surveys cannot provide detailed information for very small geographical areas. Where such data is needed, census or administrative data collections or sophisticated data modelling techniques are more useful options.

Determining sample sizes required to produce statistics with an acceptable level of sampling error is the job of experienced survey methodologists. Unfortunately, the technical details of sample designs are beyond the scope of this manual. The UN Guidelines and Principles for the Development of Disability Statistics has some helpful comments on sample size for disability surveys. (As a rough guide, surveys that have sufficient size to yield valid unemployment estimates for a given geographic area are usually large enough to produce disability estimates for the same area for as long as the prevalence rates are similar.)

As mentioned, a sample survey for disability can either be an independent or dedicated survey, or a module to other surveys, such as a national health survey, a general social survey or labour force survey. Where these general social surveys are cyclical, it is very useful to have a disability module or a small set of disability questions added on.

As dedicated disability surveys: Surveys specifically dedicated to collect disability data are good sources of information about prevalence rates, the causes and types of disability, underlying health conditions, severity and duration of disability, and the use of and need for assistive devices, changes in environment, policies and public awareness on disability.

Dedicated disability surveys maximise the amount of information that can be collected to meet users' needs. Many countries have not attempted to conduct such disability surveys because they are more costly than simply adding a few questions into a census or already existing sample surveys.

As a module to other sample surveys: In many countries, national household surveys-covering topics such as health, education, living conditions, family income and expenditure, labour force participation, nutrition, time use, crime, and so on-are conducted on a regular basis. Whether every five years, annually, quarterly or monthly, these surveys try to identify short and long-term social trends.

A cost-efficient and effective way of collecting disability data is to add disability questions-or a 'disability module'-to one or several of these cyclical surveys. This approach has been used in many countries. Most commonly, disability modules have been added to labour force or health surveys, and sometimes to living conditions surveys.

There are advantages to on-going sample surveys rather than ad hoc or one-off sample surveys for disability. On-going surveys can make maximum use of the resources initially expended, including the time and other resources used in preparing for the data collection, as well as the trained personnel and other resources dedicated to collecting, processing, and analysing the data on a regular basis.

On-going survey programs also offer opportunities to learn from previous experiences so that the quality and usefulness of the information produced can be improved. They allow for measurement of change over time in key indicators such as frequency of types of disability, severity of disability, quality of life, opportunities and participation of persons with disabilities, and rehabilitation needs. These data can be exceedingly valuable for policy development and evaluation.

METHODS OF DATA COLLECTION

QUESTIONNAIRES

Questionnaires are one of the most popular methods of conducting scholarly research. They provide a convenient way of gathering information from a target population.

Advantages and Disadvantages of Written Questionnaires

Questionnaires are easy to analyse, and most statistical analysis software can easily process them. They are cost effective when compared to face-to-face interviews, mostly because of the costs associated with travel time. This is especially true for studies involving large sample sizes and large geographic areas. Written questionnaires become even more cost effective as the number of research questions increases.

Questionnaires are familiar to most people. Nearly everyone has had some experience completing questionnaires and they generally do not make people apprehensive. They are less intrusive than telephone or face-to-face surveys. When respondents receive a questionnaire in the mail, they are free to complete it on their own time-table. Unlike

other research methods, the respondent is not interrupted by the research instrument. On the other hand, questionnaires are simply not suited for some people. For example, a written survey to a group of poorly educated people might not work because of reading skill problems. More frequently, some people are turned off by written questionnaires because of misuse.

Written questionnaires reduce interviewer bias because there is uniform question presentation. Unlike in-person interviewing, there are no verbal or visual clues to influence a respondent to answer in a particular way. Many investigators have reported that interviewer voice inflections and mannerisms can bias responses. Written surveys are not subject to this bias because there is no interviewer. On the other hand, the lack of an interviewer limits the researcher's ability to probe responses. Structured questionnaires often lose the "flavour of the response", because respondents often want to qualify their answers. By allowing frequent space for comments, the researcher can partially overcome this disadvantage.

A common criticism of mail surveys is that they often have low response rates. Low response is the curse of statistical analysis, and it can dramatically lower confidence in the results. While response rates vary widely from one questionnaire to another, well-designed studies consistently produce high response rates.

When returned questionnaires arrive in the mail, it's natural to assume that the respondent is the same person you sent the questionnaire to. A number of researchers have reported that this may not actually be the case. Many times business questionnaires get handed to other employees for completion. Housewives sometimes respond for their husbands. Kids respond as a prank. For a variety of reasons, the respondent may not be who you think it is. In a summary of five studies sponsored by the British Government, Scott reports that up to ten per cent of the returned questionnaires had been completed by someone other than the intended person.

Response Rate

Response rate is the single most important indicator of how much confidence can be placed in the results of a mail survey. A low response rate can be devastating to the reliability of a study. Fortunately, "low response rates are not an inherent shortcoming of mail surveys", and the researcher must do everything possible to maximize response. Much of the research in questionnaire methodology has centred around techniques to maximize response. However, Jones and Lang point out that increasing the response rate does not necessarily improve the precision of survey results.

Following up on Nonrespondents

One of the most powerful tool for increasing response is to use follow-ups or reminders. "Traditionally, between 5 and 65 per cent of those sent questionnaires respond without follow-up reminders. These rates are too low to yield confident results". The need to follow up on nonrespondents is clear.

Researchers can increase the response from follow-up attempts by including another copy of the questionnaire. The most important consideration is that the investigator “designs the follow-up procedure by taking into consideration the unique characteristics of the people in the sample.” The most successful follow-ups have been achieved by phone calls.

Many researchers have examined whether postcard follow-ups are effective in increasing response. The vast majority of these studies show that a follow-up postcard increases response rate, and a meta-analysis by Fox, Crask, and Kim reveals an aggregate gain of 3.5 per cent. The postcard serves as an effective reminder for subjects who have forgotten to complete the survey.

Nonresponse Bias

Many studies have attempted to determine if there is a difference between respondents and nonrespondents. Some researchers have reported that people who respond to surveys answer questions differently than those who do not. Others have found that late responders answer differently than early responders, and that the differences may be due to the different levels of interest in the subject matter. One researcher, who examined a volunteer organization, reported that those more actively involved in the organization were more likely to respond.

Demographic characteristics of nonrespondents have been investigated by many researchers. Most studies have found that nonresponse is associated with low education. However, one researcher reported that demographic characteristics such as age, education, and employment status were the same for respondents and nonrespondents. Another study found that nonrespondents were more often single males.

Most researchers view nonresponse bias as a continuum, ranging from fast responders to slow responders. In fact, one study used extrapolation to estimate the magnitude of bias created by nonresponse. Another group of researchers argue that nonresponse should not be viewed as a continuum, and “that late respondents do not provide a suitable basis for estimating the characteristics of nonrespondents”.

Goal Definition

Most problems with questionnaires can be traced back to the design phase of the project. Well-defined goals are the best way to assure a good questionnaire design. One of the best ways to clarify the study goals is to decide how the information will be used.. Study goals should be committed to writing. When the goals of a study can be expressed in a few clear and concise sentences, the design of the questionnaire becomes considerably easier. The questionnaire is developed to directly address the goals of the study.

One important way to assure a successful survey is to include other experts and relevant decision-makers in the questionnaire design process. Their suggestions will improve the questionnaire and they will subsequently have more confidence in the results.

General Layout and Format Considerations

The physical appearance of a written survey may largely determine if the respondent will return it. Therefore, it is important to use professional production methods for the questionnaire—either desktop publishing or typesetting and keylining. Every questionnaire should have a title that is short and meaningful to the respondent. The rationale is that a questionnaire with a title will be perceived as more credible than one without.

Well-designed questionnaires include clear and concise instructions on how they should be completed. These must be very easy to understand, so use short sentences and basic vocabulary. The questionnaire itself should have the return address printed on it since questionnaires often get separated from the reply envelopes.

Questionnaires should use simple and direct language. The questions must be clearly understood by the respondent, and have the same meaning that the researcher intended. The wording of a question should be simple, to the point, and familiar to the target population. Surprisingly, several researchers have found that longer questions elicit more information than shorter ones, and that the information tends to be more accurate. However, it is generally accepted that questionnaire items should be simply stated and as brief as possible. The rationale is that this will reduce misunderstandings and make the questionnaire appear easier to complete. One way to eliminate misunderstandings is to emphasize crucial words in each item by using bold, italics or underlining.

Uncommon words, jargon, and abbreviations may be included in a questionnaire provided that they are familiar to the population being investigated. Slang is often ambiguous, and should be excluded from all questionnaires.

Questionnaires should leave adequate space for respondents to make comments. One criticism of questionnaires is their inability to retain the “flavour” of a response. Leaving space for comments will provide valuable information not captured by the response categories. Leaving white space also makes the questionnaire look easier and this might increase response.

Researchers should design the questionnaire so it holds the respondent’s interest. The goal is to make the respondent want to complete the questionnaire. One way to keep a questionnaire interesting is to provide variety in the type of items used. Varying the questioning format will also prevent respondents from falling into “response sets”.

The Order of the Questions

Questionnaires should begin with a few non-threatening and easy to answer items. If the first items are too difficult or threatening, there is little chance that the person will complete the questionnaire. People generally look at the first few questions before deciding whether or not to complete the questionnaire. The researcher can encourage response by starting with a few interesting and nonthreatening questions.

Likewise, the most important items should appear in the first half of the questionnaire. Respondents often send back partially completed questionnaires. By putting the most

important items near the beginning, the partially completed questionnaires will still contain important information.

Items on a questionnaire should be grouped into logically coherent parts. Grouping questions that are similar will make the questionnaire easier to complete, and the respondent will feel more comfortable. Questions that use the same response formats, or those that cover a specific topic, should appear together.

Each question should follow comfortably from the previous question. Writing a questionnaire is similar to writing anything else. Transitions between questions should be smooth. Questionnaires that jump from one unrelated topic to another feel disjointed and are not likely to produce high response rates. Most investigators have found that the order in which questions are presented can affect the way that people respond. One study reported that questions in the latter half a questionnaire were more likely to be omitted, and contained fewer extreme responses. Carp suggested that it may be necessary to prevent general questions before specific ones in order to avoid response contamination. McFarland reported that when specific questions were asked before general questions, respondents tended to exhibit greater interest in the general questions. A few researchers, however, have found that question-order does not effect responses. Bradburn and Mason reported that interviews involving self-reports and self-evaluations were unaffected by question order. Clancey and Wachslar found that responses to questions were similar regardless of where the questions appeared in a questionnaire. Bishop et al. reported that question-order effects existed in interviews, but not in written surveys. Ayidiya and McClendon reported mixed results, where some questions were subject to order effects, and other similar questions were not.

Anonymity and Confidentiality

As a general rule, long questionnaires get less response than short questionnaires. However, some studies have shown that the length of a questionnaire does not necessarily affect response. “Seemingly more important than length is question content.” A subject is more likely to respond if they are involved and interested in the research topic. Questions should be meaningful and interesting to the respondent.

Colour of the Paper

One study found that the colour of the paper. Nevertheless, Berdie, Anderson and Neibuhr suggest that colour might make the survey more appealing. Another early study examined the ink and paper colour combinations that provide the greatest legibility. The authors suggest three different ink colours for white paper: black, grass green, and lustre blue. The only other recommended combination is black ink on yellow paper. Some investigators have examined the effect of using a green paper compared to white paper. Two studies reported no significant differences in response rates, while another reported a 9.1 per cent difference. A meta-analysis of these studies calculated an average aggregate increase of 2.0 per cent when using a green questionnaire.

Incentives

Many researchers have examined the effect of providing a variety of nonmonetary incentives to subjects. These include token gifts such as small packages of coffee, ball-point pens, postage stamps, or key rings, trading stamps, participation in a raffle or lottery, or a donation to a charity in the respondent's name. Generally, nonmonetary incentives have resulted in an increased response.

There can be little doubt that monetary incentives increase response. Only a few investigators have reported no increase in response. The overwhelming majority have reported increased response by including monetary incentives.

Church conducted a meta-analysis of 38 prior studies that used some form of an incentive. Monetary and nonmonetary incentives were effective only when enclosed with the survey.

The promise of an incentive for a returned questionnaire was not effective in increasing response. The average increase in response rate for monetary and nonmonetary incentives was 19.1 per cent and 7.9 per cent, respectively.

Many researchers have found that higher monetary incentives generally work better than smaller ones. Armstrong proposed a diminishing return model, where increasing the amount of the incentive would have a decreasing effect on response rate. A meta-analysis performed by Fox, Crask, and Kim applied Armstrong's diminishing return model to fifteen studies. An incentive of 25¢ increased the response rate by an average of 16 per cent, and \$1 increased the response by 31 per cent.

It is not known whether the effects of incentives disappear after follow-up mailings. Kephart and Bressler found that a 25¢ incentive significantly increased response, however, the effect disappeared after one follow-up mailing. Another study using a 25¢ incentive reported a significant difference, and the difference continued to be significant even after three follow-up mailings. James and Bolstein reported that four mailings without an incentive produced a better response than one mailing with an incentive. However, incentives of \$1 and \$2 with follow-ups produced a significantly better response than the same number of follow-ups with no incentive. Nederhof used a ball-point pen as an incentive and reported that "the effect of nonmonetary incentives on response rates disappears after the first mailing."

It is not clear whether offering to share the results of the research provides sufficient incentive to affect response. Mullner, Levy, Byre, and Matthews report that mentioning the offer in a cover letter did not increase response, while two older studies found that it did.

Notification of a Cutoff Date

Several researchers have examined the effect of giving subjects a deadline for responding. While a deadline will usually reduce the time from the mailing until the returns begin arriving, it appears that it does not increase response, and may even reduce the response. One possible explanation is that a cutoff date might dissuade procrastinators

from completing the questionnaire after the deadline has past. A meta-analysis by Fox, Crask and Kim revealed an aggregate increase in response rate of 1.7 per cent, which was not significant.

Reply Envelopes and Postage

A good questionnaire makes it convenient for the respondent to reply. Mail surveys that include a self-addressed stamped reply envelope get better response than business reply envelopes, although they are more expensive since you also pay for the non-respondents.

Some investigators have suggested that people might feel obligated to complete the questionnaire because of the guilt associated with throwing away money--that is, the postage stamp. Others have pointed out that using a business reply permit might suggest advertising to some people. Another possibility is that a business reply envelope might be perceived as less personal.

Armstrong and Lusk performed a meta-analysis on 34 studies comparing stamped versus business reply postage. They calculated that stamped reply envelopes had a 9 per cent greater aggregate effect than business reply envelopes. In a subsequent meta-analysis on nine studies, Fox, Crask, and Kim reported an aggregate effect of 6.2 per cent.

The Outgoing Envelope and Postage

There have been several researchers that examined whether there is a difference in response between first class postage versus bulk rate.. A meta-analysis of these studies revealed a small, but significant, aggregate difference of 1.8 per cent. Envelopes with bulk mail permits might be perceived as "junk mail", unimportant, or less personal and thus will be reflected in a lower response rates.

A few researchers have also examined whether metreed mail or stamps work better on the outgoing envelope. The results of these studies suggest a small increase in response favouring a stamped envelope. A meta-analysis of these studies revealed that the aggregate difference was slightly less than one per cent. Many researchers have reported increased response rates by using registered, certified, or special delivery mail to send the questionnaire. The wisdom of using these techniques must be weighed against the consequences of angering respondents that make a special trip to the post office, only to find a questionnaire.

It is not clear whether a typed or hand-addressed envelope would affect response. One study, conducted at the University of Minnesota, reported that students responded better to hand-addressed postcards, while professors responded better to typed addresses. This researcher could find no studies that examined whether gummed labels would have a deleterious effect on response rate, although we might predict that response rate would be less for gummed labels because they have the appearance of less personalization.

This researcher could also find no studies that examined whether the colour of the envelope affects response rate. First impressions are important, and the respondent's first impression of the study usually comes from the envelope containing the survey. Therefore, we might predict that colour would have a positive impact on response because of its uniqueness.

The “Don’t Know”, “Undecided”, and “Neutral” Response Options

Response categories are developed for questions in order to facilitate the process of coding and analysis. Many studies have looked at the effects of presenting a “don’t know” option in attitudinal questions. The “don’t know” option allows respondents to state that they have no opinion or have not thought about a particular issue.

Holdaway found that the physical placement of the “undecided” category could change response patterns. Respondents were more likely to choose the “undecided” category when it was off to the side of the scale. This study also reported different response patterns depending on whether the midpoint was labeled “undecided” or “neutral”.

Bishop also found that the physical location of the middle alternative can make a difference in responses, and that placing the middle option at the last position in the question increased the percentage of respondents who selected it by over 9 per cent. Bishop states that “offering respondents a middle alternative in a survey question will generally make a significant difference in the ceases that would be drawn.” The middle option of an attitudinal scale attracts a substantial number of respondents who might be unsure of their opinion.

Poe et al. studied the “don’t know” option for factual questions. Unlike attitude questions, respondents might legitimately not know the answer to a factual question. Their findings suggest that the “don’t know” option should not be included in factual questions. Questions that excluded the “don’t know” option produced a greater volume of accurate data. They found no difference in response rate depending on the inclusion or exclusion of the “don’t know” option. Poe’s finding directly contradict several previous authors who advocate including a “don’t know” response category when there is any possibility that the respondent may not know the answer to a question.

Question Wording

The wording of a question is extremely important. Researchers strive for objectivity in surveys and, therefore, must be careful not to lead the respondent into giving the answer a desired answer. Unfortunately, the effects of question wording are one of the least understood areas of questionnaire research.

Many investigators have confirmed that slight changes in the way questions are worded can have a significant impact on how people respond. Several authors have reported that minor changes in question wording can produce more than a 25 per cent difference in people’s opinions.

One important area of question wording is the effect of the interrogation and assertion question formats. The interrogation format asks a question directly, where the assertion format asks subjects to indicate their level of agreement or disagreement with a statement. Schuman and Presser reported no significant differences between the two formats, however, other researchers hypothesized that the interrogation format is more likely to encourage subjects to think about their answers. Petty, Rennie and Cacioppo found that the interrogation format caused greater polarization in subjects' responses, suggesting that there was greater cognition than the assertion format.

Other investigators have looked at the effects of modifying adjectives and adverbs. Words like usually, often, sometimes, occasionally, seldom, and rarely are "commonly" used in questionnaires, although it is clear that they do not mean the same thing to all people. Simpson and a replication by Hakel looked at twenty modifying adjectives and adverbs. These researchers found that the precise meanings of these words varied widely between subjects, and between the two studies. However, the correlation between the two studies with respect to the relative ranking of the words was.99.

John Hoyt conducted a study on how people interpret quantifying adjectives. The results show that some of the adjectives had high variability and others had low variability. The following adjectives were found to have highly variable meanings: a clear mandate, most, numerous, a substantial majority, a minority of, a large proportion of, a significant number of, many, a considerable number of, and several. Other adjectives produced less variability and generally had shared meaning. These were: lots, almost all, virtually all, nearly all, a majority of, a consensus of, a small number of, not very many of, almost none, a damn few, hardly any, a couple, and a few.

Sponsorship

There have been several studies to determine if the sponsor of a survey might affect response rate. The overwhelming majority of these studies have clearly demonstrated that university sponsorship is the most effective. A meta-analysis of these studies revealed an aggregate increase in response rate of 8.9 per cent. Dillman suggested that this may be due to the past benefits that the respondent has received from the university. Another possibility is that a business sponsor implies advertising or sales to potential respondents.

Prenotification Letters

Many researchers have studied prenotification letters to determine if they increase response rate. A meta-analysis of these studies revealed an aggregate increase in response rate of 7.7 per cent. Dillman proposed that prenotification letters might help to establish the legitimacy of a survey, thereby contributing to a respondent's trust. Another possibility is that a prenotification letter builds expectation, and reduces the possibility that a potential respondent might disregard the survey when it arrives.

The prenotification letter should address five items:

1. Briefly describe why the study is being done.
2. Identify the sponsors.
3. Explain why the person receiving the pre-letter was chosen.
4. Justify why the respondent should complete the questionnaire.
5. Explain how the results will be used.

Cover Letters

The cover letter is an essential part of the survey. To a large degree, the cover letter will affect whether or not the respondent completes the questionnaire. It is important to maintain a friendly tone and keep it as short as possible.

The importance of the cover letter should not be underestimated. It provides an opportunity to persuade the respondent to complete the survey. If the questionnaire can be completed in less than fifteen minutes, the response rate can be increased by mentioning this in the cover letter.

Flattering the respondent in the cover letter does not seem to affect response. Altruism or an appeal to the social utility of a study has occasionally been found to increase response, but more often, it is not an effective motivator.

The cover letter should address seven items:

1. Briefly describe why the study is being done.
2. Identify the sponsors.
3. Mention the incentive.
4. Mention inclusion of a stamped, self-addressed return envelope.
5. Encourage prompt response without using deadlines.
6. Describe the confidentiality/anonymity policy.
7. Give the name and phone number of someone they can call.

Personalization in a Cover Letter

There are no definitive answers whether or not to personalize cover letters. Some researchers have found that personalized cover letters can be detrimental to response when anonymity or confidentiality are important to the respondent.

The literature regarding personalization are mixed. Some researchers have found that personalized cover letters with hand-written signatures helped response rates. Other investigators, however, have reported that personalization has no effect on response.

Signature on the Cover Letter

The signature of the person signing the cover letter has been investigated by several researchers. Ethnic sounding names and the status of the researcher do not affect response. One investigator found that a cover letter signed by the owner of a marina produced better response than one signed by the sales manager. The literature is mixed regarding whether a hand-written signature works better than one that is mimeographed.

Two researchers reported that mimeographed signatures worked as well as a hand-written one, while another reported that hand-written signatures produced better response. Another investigator found that cover letters signed with green ink increased response by over 10 per cent.

Postscript on the Cover Letter

It is commonly believed that a handwritten postscript in the cover letter might increase response. One older study did find an increase in response, however, two more recent studies found no significant difference.

Questionnaire Construction

A questionnaire is a series of questions asked to individuals to obtain statistically useful information about a given topic. When properly constructed and responsibly administered, questionnaires become a vital instrument by which statements can be made about specific groups or people or entire populations.

Questionnaires are frequently used in quantitative marketing research and social research. They are a valuable method of collecting a wide range of information from a large number of individuals, often referred to as respondents. Adequate questionnaire construction is critical to the success of a survey.

Inappropriate questions, incorrect ordering of questions, incorrect scaling, or bad questionnaire format can make the survey valueless, as it may not accurately reflect the views and opinions of the participants. A useful method for checking a questionnaire and making sure it is accurately capturing the intended information is to pretest among a smaller subset of target respondents.

Questionnaire Construction Issues

- Know how you will use the results of your research before you start. If, for example, the results won't influence your decision or you can't afford to implement the findings or the cost of the research outweighs its usefulness, then save your time and money; don't bother doing the research.
- The research objectives and frame of reference should be defined beforehand, including the questionnaire's context of time, budget, manpower, intrusion and privacy.
- How and from where you select the respondents will determine whether you will be able to generalize your findings to the larger population.
- The nature of the expected responses should be defined and retained for interpretation of the responses, be it preferences, facts, beliefs, feelings, descriptions of past behaviour, or standards of action.
- Unneeded questions are an expense to the researcher and an unwelcome imposition on the respondents. All questions should contribute to the objective(s) of the research.

- If you "research backwards" and determine what you want to say in the report then even though you don't know the exact answers yet, you will be certain to ask all the questions you need - and only the ones you need - in such a way to write your report.
- The topics should fit the respondents' frame of reference. Their background may affect their interpretation of the questions. Respondents should have enough information or expertise to answer the questions truthfully.
- The type of scale, index, or typology to be used shall be determined.
- The level of measurement you use will determine what you can do with and conclude from the data. If the response option is yes/no then you will only know how many or what per cent of your sample answered yes/no. You cannot, however, conclude what the average respondent answered.
- The types of questions should fit the statistical data analysis techniques available and your goals.
- Questions and prepared responses to choose from should be neutral as to intended outcome. A biased question or questionnaire encourages respondents to answer one way rather than another. Even questions without bias may leave respondents with expectations.
- The order or "natural" grouping of questions is often relevant. Prior previous questions may bias later questions.
- The wording should be kept simple: no technical or specialized words.
- The meaning should be clear. Ambiguous words, equivocal sentence structures and negatives may cause misunderstanding, possibly invalidating questionnaire results. Double negatives should be reworded as positives.
- If a survey question actually contains more than one issue, the researcher will not know which one the respondent is answering. Care should be taken to ask one question at a time.
- The list of possible responses should be collectively exhaustive. Respondents should not find themselves with no category that fits their situation. One solution is to use a final category for "other _____".
- The possible responses should also be mutually exclusive. Categories should not overlap. Respondents should not find themselves in more than one category, for example in both the "married" category and the "single" category - there may be need for a "not living with spouse" category.
- Writing style should be conversational, yet concise and accurate and appropriate to the target audience.
- Many people will not answer personal or intimate questions. For this reason, questions about age, income, marital status, etc. are generally placed at the end of the survey. This way, even if the respondent refuses to answer these "personal" questions, he/she will have already answered the research questions.
- "Loaded" questions evoke emotional responses and may skew results.

- Presentation of the questions on the page and use of white space, colours, pictures, charts, or other graphics may affect respondent's interest or distract from the questions.
- Numbering of questions may be helpful.
- Questionnaires can be administered by research staff, by volunteers or self-administered by the respondents. Clear, detailed instructions are needed in either case, matching the needs of each audience.

Types of Questions

- *Contingency questions*: A question that is answered only if the respondent gives a particular response to a previous question. This avoids asking questions of people that do not apply to them.
- *Matrix questions*: Identical response categories are assigned to multiple questions. The questions are placed one under the other, forming a matrix with response categories along the top and a list of questions down the side. This is an efficient use of page space and respondents' time.
- *Closed ended questions*: Respondents' answers are limited to a fixed set of responses. Most scales are closed ended. Other types of closed ended questions include:
 - *Yes/no questions*: The respondent answers with a "yes" or a "no".
 - *Multiple choice*: The respondent has several option from which to choose.
 - *Scaled questions*: Responses are graded on a continuum. Examples of types of scales include the Likert scale, semantic differential scale, and rank-order scale.
- *Open ended questions*: No options or predefined categories are suggested. The respondent supplies their own answer without being constrained by a fixed set of possible responses.

Examples of types of open ended questions include:

- *Completely unstructured*: For example, "What is your opinion of questionnaires?"
- *Word association*: Words are presented and the respondent mentions the first word that comes to mind.
- *Sentence completion*: Respondents complete an incomplete sentence. For example, "The most important consideration in our decision to buy a new house is..."
- *Story completion*: Respondents complete an incomplete story.
- *Picture completion*: Respondents fill in an empty conversation balloon.
- *Thematic apperception test*: Respondents explain a picture or make up a story about what they think is happening in the picture

Question Sequence

- Questions should flow logically from one to the next.

- The researcher must ensure that the answer to a question is not influenced by previous questions.
- Questions should flow from the more general to the more specific.
- Questions should flow from the least sensitive to the most sensitive.
- Questions should flow from factual and behavioural questions to attitudinal and opinion questions.
- Questions should flow from unaided to aided questions.
- The three stage theory, initial questions should be screening and rapport questions. Then in the second stage you ask all the product specific questions. In the last stage you ask demographic questions.

INTERVIEWS

An interview is a specialised type of communication, usually verbal, between two or more people and is carried out for a specific purpose. It is different from an ordinary conversation in that its form and purpose is structured.

The interview may be carried out for a number of reasons, for example:

- *To select someone for a job:* Personnel selection
- *To collect information:* Market or social research
- *To assist someone,:* A clinical or counselling interview

For an interview to have the best chance of success, the interviewer should usually show warmth and responsiveness.

This can be seen as a genuine interest in the person being interviewed. They should show understanding, and not stop the person being interviewed from expressing their thoughts and feelings.

Finally, the interviewer should not allow their own wishes, reactions or opinions to influence the outcome of the interview. From this you can see that interviewing is a skilled activity that takes experience and practice.

Planning an Interview

If you are going to carry out an interview, you need to think about the steps involved. Some preparation must be done before the interview to make sure that you get what you need from it, and some thought given afterwards to the information gathered:

- Decide on the purpose of the interview. For example, are you interviewing someone for a job? Are you trying to find out someone's opinion about something? Are you trying to help someone with a problem?
- Decide what kind of information you need to get from the interview to achieve the purpose. You should have a list of specific goals. If you were interviewing someone for a job, you would want to find out whether they were suited to the type of work and would fit in with the rest of the staff.
- Decide what questions are you going to ask. You should draw up a list of questions so that the answers help you to achieve your goals. For example, if

you were interviewing someone for the job of a postal worker, you might want to find out whether they were fit enough to carry a heavy sack and walk the streets.

- Carry out the interview. If you were interviewing someone for a job, you should be able to form a fair and honest assessment of the person being interviewed.
- Study the answers to your questions. Make sure that your goals have been achieved. If not, find out more information.
- Make a decision about the purpose of the interview. For example, the person interviewed has been successful and will be offered the job.

Personnel Selection Interviews

Purpose

A personnel selection interview has three purposes:

1. To get information from candidates about their abilities and ambitions;
2. To assess the likelihood of the candidates fitting in well with future colleagues;
3. To give information to the candidates to ensure that they are sufficiently informed about the job to decide whether they wish to carry on with their application.

Questions

All questions asked at interview must be relevant to the post, and should not stray into areas that employment law does not allow, such as those that might lead to discrimination on the grounds of gender, race or disability. If you draw up a job description of the activities involved in doing the job, and a 'person specification' that covers what experience, skills and abilities, training, education and qualifications a suitable candidate should have, you will be able to look for matches in a fair and objective way. When short-listing candidates for interview, you should do it on the basis of this matching process. Candidates have the right to ask why they were not short-listed or offered a job and therefore you must keep proper records of the questions asked and the selection criteria for this purpose. These must be kept for 6 months by law.

Format

Most interviews, including selection interviews, fall into four phases of welcome, collecting information, supplying information and parting. After this, you decide who to appoint, take up references, and offer the chosen candidate the job.

Welcome - The interviewer, or a panel of interviewers, usually greets the candidate, chats informally and explains briefly the procedure of the interview. The aim is to make the candidate relax. Collecting information - This is where you collect information about how the candidate matches the person specification.

All candidates should be asked more or less the same questions, and these should cover:

1. Work history, duties, likes, dislikes, achievements, working conditions, level of earnings, reasons for changing job, type of job desired;
2. Education and training, best and poorest subjects, grades, extra-curricular activities, training beyond undergraduate level;
3. Personal factors such as interests, hobbies, family status and commitments, geographical mobility, flexibility.

Supplying information - You give information to the candidate. The candidate may have questions about the demands of the job, the work environment, etc. and should be given the opportunity to ask them. Parting - The candidate leaves, usually with the promise of being informed of the results of the interview as soon as possible. It is good practice to offer expenses to candidates, especially those who are on low incomes or who are just starting on their career.

Conventions

- The interview will typically last between 10 and 60 minutes; the more important the job, the longer the interview. For some very senior posts, or for those with very specialist requirements, the interview and selection process can last several days.
- The interview will consist mainly of the interviewer, or the panel of interviewers, asking carefully selected and relevant questions, and then listening to the answers. Somebody should be responsible for recording the candidate's answers.
- The interview is usually a formal occasion in which both parties are formally dressed, behave politely and are not interrupted.
- The interview can take place across a desk or table, which makes note taking easy but can be intimidating, especially if there is a panel of interviewers on one side and the candidate on the other. It is more relaxing to sit at right angles to a candidate, perhaps in easy chairs or next to a low coffee table.

Interviewing in Market, Product or Social Research

Purpose

Interviews may be used in market research, consumer-product evaluation, social research on attitudes, or before any change is made in equipment or in working methods. It is also possible to reveal design faults in a product by talking to the users.

Methods of Data Collection

The Personal Interview Method

The personal interview is the most frequently used method. It is a direct form of investigation where trained interviewers get information from selected individuals. A

formal questionnaire is usually used for the interview. The place in which the interview is conducted depends on the subject and on the method of sampling. It may take place in the street or it may be a 'Hall Test', that takes place in a town hall or other public building.

If you want to speak to elderly or disabled people, this is best done in their homes or some place that they visit such as a lunch club. It should be by appointment only; never just knock on a door and expect to be let in. Children should always be accompanied if they are carrying out questionnaires in homes.

The Focused Interview Method

The focused interview differs from the personal interview in that respondents are encouraged to talk freely, while the interviewer reports what they say. They may even record it on a tape for later analysis. To keep the interview to the point, guiding questions are sometimes put to the respondent. Often, more information can be obtained this way, than when detailed, fixed questions are asked.

Disadvantages of personal and focused interviews:

1. There is great danger that the interviewer, when recording the answers, will interpret them in their own way.
2. The interpretation of the interviews by the researcher is also very difficult and gives an opportunity for personal bias to distort the findings.
3. Statistical analysis of the findings is either very difficult or impossible since the information may not be easy to quantify or code.
4. The interview takes a long time, which increases the cost per interview.
5. There is a high proportion of irrelevant information in the data.

The Telephone Interview Method

This is similar to the personal interview but obviously takes place over the telephone. Again the respondents are asked a set list of questions relevant to the study. This method is becoming very common for market research, as well as being used by service companies, such as banks, who want to make contact with their customers, either to improve services or to sell additional ones.

Advantages:

- It is a quick method for conducting a number of interviews within a short time.
- The interviewers can be supervised easily.
- The cost per interview is low.
- The sampling can be spread over the country and travelling is eliminated.
- People who might otherwise be inaccessible, can be interviewed, since the interview can be very brief and arranged for a time that is convenient to the respondent.

Disadvantages:

- Telephone subscribers may not be representative of the general population, Mobile phone numbers may be used which do not give information about the location of the respondent.
- Only a short questionnaire can be used.
- Observation is not possible, so that the interviewers have to rely totally on what the respondent tells them.
- The times during the day when respondents can be called are limited, for example to office hours, or early evening, and it is difficult to predict such times. Calling during meals or when people are watching TV can lead to negative responses.
- 'Call-backs' have to be made when the number is engaged.

The Panel Method

The panel method is used to study changes over time. A panel of the same respondents are used and they are questioned about the same sort of thing over a period of time. There are several types of panel including the consumer-purchasing panel, the consumer-product testing panel, or the television audience panel. The panel is often recruited by means of personal interviews of a cross-section of the population. The respondents are given diaries in which they enter, for example, every time they buy one of the products being surveyed. They return the diaries at regular intervals. Alternatively, the panel may be interviewed about the relevant facts periodically. This method has been successfully adopted for radio, television and opinion research.

Advantages:

- The method is especially useful for research into trends, as the same individuals are questioned over a period of time.
- Data collected over a period of time can be accumulated and the factors underlying the change can be analysed.
- The case histories of panel members can be established to give relevant background material to responses.
- The panel can sometimes be used for enquiries on other subjects, provided that these are not likely to affect the reliability of the panel for its main purpose.

Disadvantages:

- Individuals may leave the panel.
- If people refuse to join the panel it may no longer represent a target population.
- Being a member of the panel over a long period of time may affect the opinions and behaviour of panelists, so that their responses may no longer be representative of the population at large.
- The original recruiting drive is usually expensive so investment takes a long time to yield returns.
- Panel members are usually rewarded in some small way, which adds to the overall cost.

The Group Interview Method

A small and carefully selected group of people is invited to a discussion on whatever topic is under review. A questionnaire may be provided, but more usually, the group is encouraged to discuss the matter freely, following a basic agenda. The discussion may be recorded on tape, if the respondents agree, or an observer may be present to take notes.

The interviewer's task is to remain in the background and to intervene only to bring the discussion back to the point.

Advantages:

- Group interviews are appropriate in research concerned with motives and opinions where such factors as social status and acceptance are involved. Such factors are brought out through the dynamic group situation.
- The group interview is relatively inexpensive as one interviewer can listen to up to ten people at a time.
- The free discussion of the group on a topic can provide valuable information for pilot studies
- The spontaneity of the discussion may produce information and attitudes that cannot be obtained by other methods.

Disadvantages:

- It is usually very doubtful whether such a group can be regarded as really representative of the population at large.
- Statistical analysis of the material is usually difficult, if not impossible.
- The influence of the more vocal group members on group opinion is hard to estimate.
- In the group situation, people may assume roles and behaviour that are not characteristic of their usual behaviour.
- Some groups are difficult to assemble, for example, managing directors.

Self-administered and Group-administered

Questionnaires

The interviewer is often a person in authority such as a teacher. They briefly explain the purpose and requirements of the survey to the respondent, or group of respondents, and then leaves them alone to complete the questions. The method usually gives a high response rate and a minimum of interviewer bias. It has the benefit of a degree of interviewer assessment of the respondents, a better explanation to the respondents and personal contact.

For groups of respondents assembled together, the interviewer distributes the questionnaire to the assembly or reads aloud each question for immediate completion. All respondents then answer each question in the same order. Groups of forty can be handled in this way, but they can affect answers by copying, talking and asking questions.

Format of the Interview

The following stages usually occur:

- Decide on the aims of the study and what you are trying to investigate - usually called the 'hypothesis' or 'null hypothesis'. For example, 'men are more interested in sport than women'.
- Review the relevant literature and discuss with other people. This includes deciding how the data will be analysed, and making sure that the information is collected in a suitable format.
- Design the study, making the hypothesis specific to the situation. For example, 'men are more interested in playing golf than women'.
- Compose the interview questions that will bring out the information that you need for your study.
- Carry out a small test to make sure that you have phrased the questions adequately and modify them if you need to.
- Select a sample of subjects to be approached.
- Collect your data.
- Sort out and process your data. This can include giving codes to the answers so that they can be put on a computer spreadsheet for analysis by a statistical package.
- Do the statistical analysis.
- Put all of the results together and see what they are telling you. This is the part where the hypothesis is tested to see, for example, if men statistically do play more golf than women.
- Write up the results, relating the findings to other research, drawing ceases and interpretations.

Anonymity

Data obtained by means of questionnaires is confidential, in the sense that no responses or findings should ever be published which could be traced back to particular individuals. This is a requirement of the Data Protection Act. In enlisting co-operation for any survey, respondents are usually given this assurance and a guarantee of anonymity. This is often crucial in obtaining frank and revealing responses. Where possible, therefore, you should not ask respondents to put their names on, or to sign, their questionnaires. If you need to maintain a check on non-respondents, the questionnaires can be numbered and cross-referenced back to individuals.

5

Research Method in Business and Management

Business and management research is a systematic inquiry that helps to solve business problems and contributes to management knowledge. It is an applied research.

Four factors (Easterby-Smith, 2008) combine to make business and management a distinctive focus for research:

- Transdiscipline approach
- Information access is difficult since managers see information as competitive advantage on the market
- Managers are educated and want some information produced by the classical research method
- Finding must resolve practical management problems

Managers often need information of high quality to help them to make the right decision.

WHAT IS BUSINESS RESEARCH?

Business research is a field of practical study in which a company obtains data and analyzes it in order to better manage the company.

Business research can include financial data, consumer feedback, product research and competitive analysis. Executives and managers who use business research methods are able to better understand their company, the position it holds in the market and how to improve that position.

FINANCIAL DATA

Financial data takes qualitative information—such as sales reports, revenues and cost reports—to see what areas make money and what costs money. By reviewing data, managers can find the products, staff and departments that are most efficient and determine areas of unnecessary costs.

Consumer Feedback

Understanding what the public says about the products and services a company provides is essential to making sure the company is meeting consumer needs. Customer feedback includes case studies, focus groups, customer surveys and questionnaires.

Product Research

Product research seeks to improve the product to meet the needs of consumers. This may include technological advancements, improved customer service or access to the product through a variety of distribution channels.

Competitive Analysis

Competitive analysis is when one company compares its products and services to those of another company. This can be done to improve the product, create a niche or determine a more attractive price point to lure customers.

Industry Data

Using research tools such as the information compiled by Dun & Bradstreet can help a company to understand how the industry as a whole is doing.

This can help executives make decisions based on economic factors affecting their industry that are not limited to their own products.

Principals of Business Management

The Principles of Management are the essential, underlying factors that form the foundations of successful management. According to Henri Fayol (1841-1925) in his book *General and Industrial Management* (1916), there are fourteen ‘principles of management’. Management principles are statements of fundamental truth. These principles serve as guidelines for decisions and actions of managers. They are derived through observation and analysis of events which managers have to face in practice.

The specialization of the workforce according to the skills a person, creating specific personal and professional development within the labour force and therefore increasing productivity; leads to specialization which increases the efficiency of labour. By separating a small part of work, the workers speed and accuracy in its performance increases. This principle is applicable to both technical as well as managerial work.

The issue of commands followed by responsibility for their consequences. Authority means the right of a superior to give order to his subordinates; responsibility means

obligation for performance. This principle suggests that there must be parity between authority and responsibility.. They are co-existent and go together, and are two sides of the same coin.

Discipline

Discipline refers to obedience, proper conduct in relation to others, respect of authority, etc. It is essential for the smooth functioning of all organizations. This principle states that every subordinate should receive orders and be accountable to one and only one superior. If an employee receives orders from more than one superior, it is likely to create confusion and conflict. Unity of Command also makes it easier to fix responsibility for mistakes. All those working in the same line of activity must understand and pursue the same objectives. All related activities should be put under one group, there should be one plan of action for them, and they should be under the control of one manager.

It seeks to ensure unity of action, focusing of efforts and coordination of strength. The management must put aside personal considerations and put company objectives first. Therefore the interests of goals of the organization must prevail over the personal interests of individuals. Workers must be paid sufficiently as this is a chief motivation of employees and therefore greatly influences productivity. The quantum and methods of remuneration payable should be fair, reasonable and rewarding of effort. The amount of power wielded with the central management depends on company size. Centralization implies the concentration of decision making authority at the top management. Sharing of authority with lower levels is called decentralization. The organization should strive to achieve a proper balance.

Scalar Chain refers to the chain of superiors ranging from top management to the lowest rank. The principle suggests that there should be a clear line of authority from top to bottom linking all managers at all levels.

It is considered a chain of command. It involves a concept called a “gang plank” using which a subordinate may contact a superior or his superior in case of an emergency, defying the hierarchy of control. However the immediate superiors must be informed about the matter. Social order ensures the fluid operation of a company through authoritative procedure. Material order ensures safety and efficiency in the workplace.

Employees must be treated kindly, and justice must be enacted to ensure a just workplace. Managers should be fair and impartial when dealing with employees.

Stability of Tenure of Personnel

The period of service should not be too short and employees should not be moved from positions frequently. An employee cannot render useful service if he is removed before he becomes accustomed to the work assigned to him. Using the initiative of employees can add strength and new ideas to an organization. Initiative on the part of employees is a source of strength for the organization because it provides new and better ideas. Employees are likely to take greater interest in the functioning of the

organization. This refers to the need of managers to ensure and develop morale in the workplace; individually and communally. Team spirit helps develop an atmosphere of mutual trust and understanding. These can be used to initiate and aid the processes of change, organization, decision making, skill management and the overall view of the management function.

Definition of Business Research

Entrepreneurs are risk takers, who are willing to bet large sums of money and time to build, launch and develop a business. Just because you're a risk taker doesn't mean you don't look before you jump into the lake. Performing smart, thorough business research mitigates your risks, by providing you with the right data to make strategic decisions. Business research helps you learn about the market and your competitors, so that you can find the most cost-effective solutions within your fulfillment chain.

Market Research

Market research is broken down into two main types: quantitative and qualitative. Quantitative market research determines if there is a need and desire in the marketplace for your product or service. It defines a target demographic and then establishes how big that market is. For example, a high-end vacuum salesperson might target married women with a college education and a certain household income level. Others may buy your product but this market would have the money to buy the high-end product and understand the benefits of it. The qualitative analysis considers why your product or service is better than others. It goes beyond discovering the gap in the market that the quantitative research provided and seeks to answer the questions of how and why a person needs or wants your product. By doing this research, you'll develop your "unique selling proposition" which is used for marketing and advertising.

Competitive Analysis

If you've done the qualitative analysis correctly, you'll have a head start on a competitive analysis. The competitive analysis is looking at your products or services and compares them to the market. But it goes one step further. It considers your company's strengths and weaknesses, as well as the opportunities and threats from competitors or outside forces such as regulations. This type of analysis is also called a SWOT Analysis.

Your strengths could be unique or a new product, but this isn't necessarily true. Your strength could be that you do the same thing that other competitors offer but you do it for less. Weaknesses might be low capitalization, meaning that you don't have the same money to launch and market products as do larger or more established competitors.

Most of the opportunities and threats should be derived from the market research section but they need to go further, exploring the number of competitors and how close they are, and to determine if there are any new laws or regulations that could inhibit your sales.

Operations Solutions

Business overhead is costly. Whenever you can save on costs without sacrificing the quality of your products or services, the more profitable your company will be. This is why researching operations solutions are vital and should be done regularly.

Don't assume that the supplier or wholesaler that you've been using for the past two years is still the best or the cheapest. Make calls to his competitors and shop for better deals.

Shipment and fulfillment methods not only get products to customers faster but they can also save you money. Research automation systems such as self-serve kiosks for customers or computerized point of sale systems with inventory management.

This type of research can reduce the cost of getting your products into consumers hands, and can ultimately save you money that can be passed on to consumers or enjoyed as higher net profits.

Product Analysis

Product analysis is the first step of business research. Companies must find a product that meets or exceeds consumer demand, or the product will fail in the economic market place.

One type of analysis is to find an existing product that can be improved through design or features. Another type of product analysis will find emerging markets with high demand and low supply, which allows for companies to sell new products to meet consumer demand.

Market Analysis

Companies will conduct a market analysis to determine how much profit may be earned from current demand. Management will look at which stage of the business cycle the market is currently in, whether emerging, plateau, or declining.

Each stage has its own level of profitability, with the first stage being the highest and the last stage being the lowest profitability. A market analysis will also determine the price points at which products can be sold; for example, high-quality products at a higher price may not tempt consumers to start buying the product based on quality itself.

Financial Analysis

A financial analysis determines the cost of each production item used to produce goods and services. High costs may not allow companies to price goods or services competitively, leading to an unprofitable situation.

Management will examine the costs of raw materials, labor, and manufacturing overhead to find the best raw goods available to produce the most profitable product. Management will also review the best cost application methods, ensuring that all production costs are properly applied to each product or service produced.

Competitor Analysis

Analyzing the current competitors of a market is an important part of business research. Knowing which companies have the best production methods or customer loyalty helps new companies understand how they can create a competitive advantage when entering a new market. Proper business research will also indicate how financially stable companies are and if they can be purchased outright by a company wanting to enter the industry. Buying a competitor may be cheaper than starting new operations for a company.

Growth Analysis

Business research usually includes forecasting the growth and direction of the current industry or market. Knowing which direction the market is headed helps companies determine the stability of new business operations. Entering a slow-growth industry may be unprofitable early on but have better long-term growth potential. High-growth industries will sometimes face a quick downfall, such as the dot.com boom of 2000-2001. Strong growth early in the business cycle quickly gave way to record losses, leading many businesses into bankruptcy from poor growth analysis.

Social Research

Social research is research conducted by social scientists following a systematic plan. Social research methods can be classified along a quantitative/qualitative dimension.

- Quantitative designs approach social phenomena through quantifiable evidence, and often rely on statistical analysis of many cases (or across intentionally designed treatments in an experiment) to create valid and reliable general claims. Related to quantity.
- Qualitative designs emphasize understanding of social phenomena through direct observation, communication with participants, or analysis of texts, and may stress contextual subjective accuracy over generality. Related to quality.

While methods may be classified as quantitative or qualitative, most methods contain elements of both. For example, qualitative data analysis often involves a fairly structured approach to coding the raw data into systematic information, and quantifying intercoder reliability. Thus, there is often a more complex relationship between “qualitative” and “quantitative” approaches than would be suggested by drawing a simple distinction between them.

Social scientists employ a range of methods in order to analyse a vast breadth of social phenomena: from census survey data derived from millions of individuals, to the in-depth analysis of a single agent’s social experiences; from monitoring what is happening on contemporary streets, to the investigation of ancient historical documents. Methods rooted in classical sociology and statistics have formed the basis for research in other disciplines, such as political science, media studies, program evaluation and market research.

Methodology

Social scientists are divided into camps of support for particular research techniques. These disputes relate to the historical core of social theory (positivism and antipositivism; structure and agency). While very different in many aspects, both qualitative and quantitative approaches involve a systematic interaction between theory and data. The choice of method often depends largely on what the researcher intends to investigate. For example, a researcher concerned with drawing a statistical generalization across an entire population may administer a survey questionnaire to a representative sample population. By contrast, a researcher who seeks full contextual understanding of an individuals' social actions may choose ethnographic participant observation or open-ended interviews. Studies will commonly combine, or 'triangulate', quantitative *and* qualitative methods as part of a 'multi-strategy' design.

Sampling

Typically a population is very large, making a census or a complete enumeration of all the values in that population infeasible. A 'sample' thus forms a manageable subset of a population. In positivist research, statistics derived from a sample are analysed in order to draw inferences regarding the population as a whole. The process of collecting information from a sample is referred to as 'sampling'. Sampling methods may be either 'random' (random sampling, systematic sampling, stratified sampling, cluster sampling) or non-random/nonprobability (convenience sampling, purposive sampling, snowball sampling). The most common reason for sampling is to obtain information about a population. Sampling is quicker and cheaper than a complete census of a population.

Methodological Assumptions

Social research is based on logic and empirical observations. Charles C. Ragin writes in his *Constructing Social Research* book that "Social research involved the interaction between ideas and evidence. Ideas help social researchers make sense of evidence, and researchers use evidence to extend, revise and test ideas." Social research thus attempts to create or validate theories through data collection and data analysis, and its goal is exploration, description, explanation, and prediction. It should never lead or be mistaken with philosophy or belief. Social research aims to find social patterns of regularity in social life and usually deals with social groups (aggregates of individuals), not individuals themselves (although science of psychology is an exception here). Research can also be divided into pure research and applied research. Pure research has no application on real life, whereas applied research attempts to influence the real world.

There are no laws in social science that parallel the laws in natural science. A law in social science is a universal generalization about a class of facts. A fact is an observed phenomenon, and observation means it has been seen, heard or otherwise experienced by researcher. A theory is a systematic explanation for the observations that relate to a

particular aspect of social life. Concepts are the basic building blocks of theory and are abstract elements representing classes of phenomena. Axioms or postulates are basic assertions assumed to be true. Propositions are conclusions drawn about the relationships among concepts, based on analysis of axioms. Hypotheses are specified expectations about empirical reality derived from propositions. Social research involves testing these hypotheses to see if they are true.

Social research involves creating a theory, operationalization (measurement of variables) and observation (actual collection of data to test hypothesized relationship). Social theories are written in the language of variables, in other words, theories describe logical relationships between variables. Variables are logical sets of attributes, with people being the “carriers” of those variables (for example, gender can be a variable with two attributes: male and female). Variables are also divided into independent variables (data) that influences the dependent variables (which scientists are trying to explain).

For example, in a study of how different dosages of a drug are related to the severity of symptoms of a disease, a measure of the severity of the symptoms of the disease is a dependent variable and the administration of the drug in specified doses is the independent variable. Researchers will compare the different values of the dependent variable (severity of the symptoms) and attempt to draw conclusions.

Guidelines for “Good Research”

When social scientists speak of “good research” the guidelines refer to how the science is mentioned and understood. It does not refer to how what the results are but how they are figured. Glenn Firebaugh summarizes the principles for good research in his book *Seven Rules for Social Research*. The first rule is that “There should be the possibility of surprise in social research.” As Firebaugh (p. 1) elaborates: “Rule 1 is intended to warn that you don’t want to be blinded by preconceived ideas so that you fail to look for contrary evidence, or you fail to recognize contrary evidence when you do encounter it, or you recognize contrary evidence but suppress it and refuse to accept your findings for what they appear to say.”

In addition, good research will “look for differences that make a difference” and “build in reality checks”. Advises researchers to replicate, that is, “to see if identical analyses yield similar results for different samples of people” (p. 90). The next two rules urge researchers to “compare like with like” and to “study change”; these two rules are especially important when researchers want to estimate the effect of one variable on another (e.g. how much does college education actually matter for wages?). The final rule, “Let method be the servant, not the master,” reminds researchers that methods are the means, not the end, of social research; it is critical from the outset to fit the research design to the research issue, rather than the other way around.

Explanations in social theories can be idiographic or nomothetic. An idiographic approach to an explanation is one where the scientists seek to exhaust the idiosyncratic

causes of a particular condition or event, i.e. by trying to provide all possible explanations of a particular case. Nomothetic explanations tend to be more general with scientists trying to identify a few causal factors that impact a wide class of conditions or events. For example, when dealing with the problem of how people choose a job, idiographic explanation would be to list all possible reasons why a given person (or group) chooses a given job, while nomothetic explanation would try to find factors that determine why job applicants in general choose a given job.

Research in science and in social science is a long, slow and difficult process that sometimes produces false results because of methodological weaknesses and in rare cases because of fraud, so that reliance on any one study is inadvisable.

Ethics

The ethics of social research are shared with those of medical research. In the United States, these are formalized by the Belmont report as:

Respect for Persons

The principle of respect for persons holds that (a) individuals should be respected as autonomous agents capable of making their own decisions, and that (b) subjects with diminished autonomy deserve special considerations. A cornerstone of this principle is the use of informed consent.

Beneficence

The principle of beneficence holds that (a) the subjects of research should be protected from harm, and, (b) the research should bring tangible benefits to society. By this definition, research with no scientific merit is automatically considered unethical.

Justice

The principle of justice states the benefits of research should be distributed fairly. The definition of fairness used is case-dependent, varying between “(1) to each person an equal share, (2) to each person according to individual need, (3) to each person according to individual effort, (4) to each person according to societal contribution, and (5) to each person according to merit.”

Types of Method

The following list of research methods is not exhaustive:

Statistical–quantitative methods:

- Cluster analysis
- Correlation and association
- Multivariate statistics
- Regression analysis
- Social network analysis

- Social sequence analysis
- Surveys and questionnaire
- Structural equation modeling
- Survey research
- Quantitative marketing research

Qualitative methods:

- Analytic induction
- Case study
- Ethnography
- Life history
- Morphological analysis
- Most significant change technique
- Participant observation
- Textual analysis
- Unstructured interview.

Mixed methods:

- Archival research
- Content analysis
- Longitudinal study
- Focus group
- Historical method
- Semi-structured interview
- Structured interview
- Triangulation (social science).

Foundations of Social Research

Sociological Positivism

The origin of the survey can be traced back at least early as the Domesday Book in 1086, while some scholars pinpoint the origin of demography to 1663 with the publication of John Graunt's *Natural and Political Observations upon the Bills of Mortality*. Social research began most intentionally, however, with the positivist philosophy of science in the early 19th century.

Statistical sociological research, and indeed the formal academic discipline of sociology, began with the work of Émile Durkheim (1858–1917). While Durkheim rejected much of the detail of Comte's philosophy, he retained and refined its method, maintaining that the social sciences are a logical continuation of the natural ones into the realm of human activity, and insisting that they may retain the same objectivity, rationalism, and approach to causality. Durkheim set up the first European department of sociology at the University of Bordeaux in 1895, publishing his *Rules of the Sociological Method* (1895). In this text he argued: “[o]ur main goal is to extend scientific

rationalism to human conduct.... What has been called our positivism is but a consequence of this rationalism.”

Durkheim’s seminal monograph, *Suicide* (1897), a case study of suicide rates among Catholic and Protestant populations, distinguished sociological analysis from psychology or philosophy. By carefully examining suicide statistics in different police districts, he attempted to demonstrate that Catholic communities have a lower suicide rate than that of Protestants, something he attributed to social (as opposed to individual or psychological) causes. He developed the notion of objective *sui generis* “social facts” to delineate a unique empirical object for the science of sociology to study. Through such studies he posited that sociology would be able to determine whether any given society is ‘healthy’ or ‘pathological’, and seek social reform to negate organic breakdown or “social anomie”. For Durkheim, sociology could be described as the “science of institutions, their genesis and their functioning”.

Modern Methodologies

In the mid-20th century there was a general—but not universal—trend for U.S. American sociology to be more scientific in nature, due to the prominence at that time of action theory and other system-theoretical approaches. Robert K. Merton released his *Social Theory and Social Structure* (1949). By the turn of the 1960s, sociological research was increasingly employed as a tool by governments and businesses worldwide. Sociologists developed new types of quantitative and qualitative research methods. Paul Lazarsfeld founded Columbia University’s Bureau of Applied Social Research, where he exerted a tremendous influence over the techniques and the organization of social research. His many contributions to sociological method have earned him the title of the “founder of modern empirical sociology”. Lazarsfeld made great strides in statistical survey analysis, panel methods, latent structure analysis, and contextual analysis. Many of his ideas have been so influential as to now be considered self-evident.

ELEMENTS OF RESEARCH

Another step in the research process is to succinctly clarify the research elements. This will require you to define the problem in a way that lets you gather information about the problem. This step often requires you to *conceptualize* and *operationalize*. That is, the problem must be broken down into concepts that allow measurement. In political research, concepts are perceptions used to represent political characteristics. Examples of concepts used in political research are power, influence, political efficacy, political socialization, and political culture.

Next you must translate the concepts into observable events. “Acquiring Scientific Knowledge,” about the party identification of first-time voters and the party choice of their parents. In that example, the concept of political socialization was analyzed by comparing the party identification of first-time voters (observable event number 1) and the party choice of their parents (observable event number 2).

You will use the various elements of research to determine causal explanations in an effort to understand why a certain political phenomenon occurs. We discuss ways to construct causal explanations and other elements of research, such as variables and hypotheses, in Block 9.

MEASUREMENT

Measurement involves assigning numbers to represent some political concept that we want to analyze. We use these numbers to produce statistics to help us determine causal relationships. Understanding measurement is a complex process that requires much effort on the part of the research student. Block 10 provides detailed coverage of the subject.

The Research Design

The next step in the research process is to design the research procedure. This requires you to determine the sources of your data, consider ways to collect your data, and decide how to analyze your data. This subject receives extensive coverage in Block 11.

The Remaining Steps

Many of the remaining steps in the research process will require you to put the research design into action. For example, you will collect and input your data, analyze your data through the application of appropriate statistical techniques, and prepare a report relating the findings to some political theory or important policy issue.

As some have suggested, you may see that research is a cyclical process. It usually starts with a problem and ends in a tentative empirical generalization. The generalization ending one cycle starts the next cycle. This cyclic process continues indefinitely, reflecting the progress of a scientific discipline. The research process is also self-correcting because the sequential process allows the researcher to reevaluate each stage to identify possible errors.

Summarizing the Systematic Research Process

There are eight stages of the systematic research process: problem identification, element clarification, measurement, research design, data collection, data input, data analysis, and generalization of the results through theory and policy implications. In addition, the literature review and theoretical support are important to the process.

Phases of Research Process

If one would like to have truthful information on world around, on nature, on human being, he/she have to look for convincing data. Collecting such data is process complicated, and complex and due to this very difficult. For layman or even for medical student at the beginning of her/his professional career can seem this statement

hypertrophic, overshoot, and not very probable. Their view will change quickly when they are involved in research (even very simple). It is generally accepted that reliable information on human health state are necessary for prevention of disease development and disease treatment. Only way how to obtain such information is research in which are used scientific methods and it is done by qualified and experienced scientists.

Generally, research is understood as systematic, controlled, empiric and critical investigation of hypothetic statements on supposed relationships among phenomenon. It is a process in which thinking is dominant. Characteristic features of research are:

- Systematic solving of scientific problems
- using scientific methods
- System of interconnecting phases and steps.

The following five phases outline a simple and effective strategy for conducting effective research:

- I. The conceptual phase
- II. Phase of construction of research design
- III. Empiric phase
- IV. Analytic phase
- V. Disseminative phase.

Phases of research process are most frequently used in the presented order. But in other guidelines you can find another order of phases and steps. This variability can be ascribed to flexibility of the formal rules of research process which should be adapted to a studied subject and to the special conditions under which the research is done.

Phase of Conception

According Brink at al. (2006) this phase of research involves activities with a strong conceptual element. „Conceptualisation refers to the process of developing refining abstract ideas. During this phase, the researcher categorises and labels his/her impressions.

Thus, the activities include thinking, rethinking, theorising, making decision, and reviewing ideas with colleagues, research partners or mentors/supervisors. The researcher also needs to draw on the skills and abilities of creativity, analysis and insight, as well as on the firm grounding of existing research on the topic of interest”.

Phase of conception is the first phase of original research. In this phase are created content and structure of the planned research. Creation of conception of new research project is structured process. It can be divided into 4 steps as follows:

- 1) Formulation of research problem or research questions, set bounds of them, determine the purpose of the study
- 2) Searching and review the literature relating to the regarding research problem and develop a framework
- 3) Development of the theoretical construction of the future research
- 4) Creation of hypothesis which should be verified/phalsified in future research

Formulation of research problem and set bounds of it

The first step of the research work is to state the scientific problem. It is important to clearly state what your problem is to avoid any confusion later. Formulation of scientific problem is frequently recognized as the most difficult and the most important part of research project.

Precisely it was expressed by A. Einstein (paraphrase): „If I have one hour for solving the problem on which my life depends, than I will devote 40 minutes to study the problem, 15 minutes to analyse the ways how to solve it, and only 5 minutes to solve it”. Scientific problem is frequently stated in form of a question.

If formulation of scientific problem is so difficult and important who than is qualified to do this job? There is consent that it should be a person(s):

- with large and high quality of knowledge in the respective field,
- with high creativity, able to think independently, large knowledge in culture and history, with ability to persist in research despite of serious problems,
- with non-conventional thinking,
- able to doubt on recent valid truth (dogmas),
- able to formulate and publicly present his/her own doubts,
- with appropriate dose of curiosity,
- able to resist to fashionable hypotheses and theories,
- able to preserve independent thinking in the „sphere” of strong scientific and/or political personalities,
- with high grade of perseverance in looking for scientific problem and its definition, able to look at scientific problem from different points of view,
- possessing excellent memory and appropriate dose of emotionality.

How the really important scientific problem can be discovered and defined?

According the definition the problem is something you'd like to know more about, a question you'd like to answer. Such questions can come from many different sources: from lectures or textbooks, from an experiment you have just made (that raised another questions), from articles you've read in scientific journals or even newspapers and magazines.

To identify a scientific problem, than, you can find sources that relate to your topic and look to see what problems are raised in your search.

Write down the problems that you find. Choose one that would be interesting to solve and that is feasible for you to solve (NC State University, 2004). Discovery of good scientific problem can't be planned.

It will emerge or not! It can emerge at any time of day and night, at any situations (sometimes very peculiar) without any identifiable dependence to previous study or research work. Despite of the mentioned uncertainties some situations and activities are more frequently recognized as the source of new ideas, new scientific problems. These are:

- sometimes it can be mere chance or it can be a result of observation an accidentally Recorded phenomenon during research,
- it can be discovered by systematic study of the subject, and the problem emerge as a gap
- it can be identified during study of different sources of technical literature as cotroversies related to some facts, functions, and so on.

At the beginning the scientific problem is usually not well defined. There are not clear limit, its bounds are not set well. So, you may need to narrow it, to identify a more specific topic within the broader one (refinement of research question/problem). This can make it easier to work with. To set up the boundaries of scientific problem is very important step in preparation of future research. To fulfil this aim one have to:

- think about possible causes of observed new, up to now unknown phenomenon,
- create of hypothesis,
- think on whether defined scientific problem is solvable,
- think about methods suitable for solving the defined scientific problem.

How the research problem is formulated?

At the beginning there is e.g. accidental observation of phenomenon which we are not able to explain. Thank to our curiosity we will start with looking for explanation. We formulate questions-why and how the phenomenon originated.

There is a lot of uncertainties and only small amount of certainties at the beginning. The consideration on possible cause(s) of the phenomenon is formulated.

These considerations we discussed with co-workers, we are looking for answers in literature, and we consider personal experience of other researcher. If we are not able to find convincing explanation than we formulate primary research problem. Subsequently, we consider its solubility.

If it seems soluble than what we are thinking on possible kinds of method which can be used for this purpose. If it seem not soluble than it is necessary to go back and start to think about the research problem once again.

Looking for, to define, to set the boundaries of scientific problem are the steps which are essential for quality of created scientific problem.

On the above mentioned steps depends also the importance of the results which will be obtained by solving the problem, and how these results will influence branches of science, the science as whole and whole society. From this point of view we can distinguish so called „serious”research problems solving which brink a very new knowledge in the field.

Their importance is manifested by awarding their authors by Nobel Prize, e.g. in 2008 yr discovery of papiloma virus was done, in 1962 it was discovery of DNA structure, in 2009 it was discovered structure of ribosome and function of telomere and enzyme telomerase. These discoveries are really serious ones. Discovery and solving of „less important” research problem brink, however, very useful results, too.

It can be, e.g. discovery of new important law in scientific branch or introduction of new discoveries to medical practice. Such discoveries are awarded, too but by another type of prizes.

Searching and review the literature relating to the regarding research problem and development a framework

It is regular rule that research projects begins with conducting literature, which means to identify related research, to set the current research project within a conceptual and theoretical context. When conducting the literature review it's important to concentrate on the scientific literature, start with the most valuable research journals in your topical area, use a blind or juried review system on the research journals and do the review early in the research process. In the literature review you might be able to find a study that is quite similar to the one you are thinking of doing and the literature review will help you to find and select appropriate measurement instruments and it will also help you to anticipate common problems in your research context and will help you to avoid common traps.

The aims of literature review are to get an insight and to get a view work of others, and there are some requirements needed to be fulfilled to be an effective review analyser. One of the requirements is to compare different authors views on an issue and in the same time put those with similar conclusions in groups, note the disagreement areas and conclude by summarising what the literature says.

The best way to do a literature review is to use library resources. And to write a good literature review you have to have an ability to show why your research should be carried out, and why you did choose certain theories to work with. And how your work already adds to research carried out. When reading you need to decide which ideas or information are important so you could emphasis them, and you should look at conclusions, theories, arguments that underline the work and look for similarities and differences with closely related work. And when writing you should compare and show relationships between the work already done by for example another researcher and compare it with another research, and see differences and decide who is most convincing.

So, using other words, the aim of this step is to find the current information related to the recognized research problem. Not to be confused with a book review, a literature review surveys scholarly articles, books and other sources (e.g. dissertations, conference proceedings) relevant to a particular issue, area of research, or theory, providing a description, summary, and critical evaluation of each work. The purpose of a literature review is than to offer an overview of significant literature published on a topic.

By study of literature we would like to find the answers to following questions:

- did anybody else formulate the same research problem in the past?
- did anybody else solve the same or similar problem in the past?
- are the results of the previous solving acceptable for us or not?

Result of the searching literature is than more precisely defined research problem or the recent research problem is rejected because it was successfully solved in the past. More details on the topic you will find in the specific chapter devoted to searching for literal information.

Development of the theoretical construction of the future research

Before the real research will start it is necessary to create its theoretical construction (abstract construction, virtual model). The main aim of this step is thinking on the content of presumed research, on its timing (date of beginning, duration of individual stages, duration whole research, date of supposed finish), on its structure (division to stages, phases), on conditions which should be created for successful run of research (persons, money, material, place).

This step give a chance to researcher to think about each detail related to presumed research, to find out very early „weak links of chain” in the planned research, to think on alternative ways in research protocol if some non assumed condition will influence the research process. Creation of good virtual model of the future research will save the time, money and decrease probability of stressful situations during running research.

For creation theoretical construction of presumed research are necessary some conditions:

- the research problem should be clearly defined
- the technical, financial and personal conditions are potentially available
- the social, legal and ethical views are known and are consistent with current state (institutional) law and ethical rules and regulations
- the main aims of research are well defined

The important result of this step is clear framework of the way by which the research problem will be solved.

Formulation of hypothesis

A stabile step in research is formulation of a research question. A research question is a statement that identifies the phenomenon to be studied. A well-thought-out and focused research question leads directly into your hypotheses. What predictions would you make about the phenomenon you are examining? This will be the foundation of your application. Hypotheses are more specific predictions about the nature and direction of the relationship between two variables. For example, “Those researchers who utilize an online grant writing tutorial will have higher priority scores on their next grant application than those who do not.” Such hypothesis can be than verified or phalsified. It is desirable to create strong hypothesis. What are the characteristic features of such hypothesis? It should:

- give insight into a research question,
- be testable and measurable by the proposed experiments,
- spring logically from the experience of the staff

Normally, no more than three primary hypotheses should be proposed for a research study. A proposal that is hypothesis-driven is more likely to be funded than a “fishing expedition” or a primarily descriptive study (The Research Assistant, 2003). Hypothesis is the result of researcher’s creativity. It is a rational assumption on the possible cause(s) of the observed phenomenon. Hypothesis is a source of questions focused to the core of the research problem. Answers to the questions may support (verify) or reject (phalsify) formulated hypothesis. But we have to realise that research process can never completely verify or phalsify the studied hypothesis, it can be done with certain probability, only. Researcher who has not any doubts on his own research results is not real researcher.

Phase of construction of research design

The aim of this phase of research is to prepare general plan of real research. According the UCLA Department of Education (Phil Ender, 1998) this phase is composed of following parts:

- 1) Identification of variables
 - dependent variables (responses, outcome, or criterion variables)
 - independent variables (explanatory or predictor variables)
 - control variables
 - intervening variables
- 2) Construction of operational definitions for variables
- 3) Selection of design for data analysis

In the process of the design creation should be involved following steps:

- a) selection of patients, animals, other objects used for research to solve the defined scientific problem (representative sample should be created)
- b) selection the methods which are planned to be used in proposed research
- c) Selection of technology
- d) protocol of research should be constructed
- e) the control methods should be stated: to control whether the research procedures are correct or not
- f) creation of pilot study
- g) to define the statistical methods used for evaluation of the obtained results
- h) to define the financial, material and personal needs to secure success in the research

Creation of research design can be expressed by other words. E.g. it seems suitable if we are able to define key elements of the respective research procedure. What key elements are?

- description and size of all experimental/research and control groups, as applicable
- a step-by-step list of everything you must do to curry out your research. Think about all the steps that you will need to go through to complete your research, and record exactly what will need to be done in each step

- the research procedure must tell how you will change your one and only independent variable and how you will measure that change
- the research procedure must explain how you will measure the resulting change in the dependent variable(s)
- if applicable, the research procedure should explain how the controlled variables will be maintained at a constant value
- the research procedure should specify how many times you intend to repeat your research (or its parts), so that you can verify that your results are really reproducible
- a good research procedure enables someone else to duplicate your experiment/clinical study/other type of research, exactly!

You have to be aware where will you conduct your experiment. You may need a lot of room for your experiment or you may not be able to move your experiment around from place to place. If you are working with human or animal subjects, you may need a location that is quiet or defined by another way. You will need to think about these limitations before you start your research/experiment so you can find a location in advance that will meet your needs.

Selection of research objects

As a research object can serve genes, chromosomes, individual cells, tissue cultures, animals, persons, populations, and technologic devices (computers, different kinds of models, computer networks and others).

If human being is to be the research object the inclusion and exclusion criteria should be defined first, e.g. health state, type of disease, age - children, adults (age span) size of group(s), different age groups, control group, gender, selected group, patients with special kind of disease, and others. If animals are planned to be a research object than inclusion and exclusion criteria should be defined first, too, e.g. species, age, gender, health state, knock-out or knock-in animals, specific pathogen free (SPF) or not SPF, with modelled disease, anaesthetised, non-anaesthetised and so on.

Research can be rarely realized in whole population. A population - primary sample is any entire collection of people, animals, plants or things from which we may collect data. It is the entire group we are interested in, which we wish to describe or draw conclusions about. In order to make any generalisations about a population, a sample, that is meant to be representative of the population - selected sample, is often studied. For each population there are many possible selected samples. A sample statistic gives information about a corresponding population parameter. For example, the sample mean for a set of data would give information about the overall population mean.

Selection of research methods

Researchers have long debated the relative value of qualitative and quantitative inquiry (Patton, 1990). Phenomenological inquiry, or qualitative research, uses a naturalistic

approach that seeks to understand phenomena in context-specific settings. Logical positivism, or quantitative research, uses experimental methods and quantitative measures to test hypothetical generalizations. Each represents a fundamentally different inquiry paradigm, and researcher actions are based on the underlying assumptions of each paradigm.

There are several considerations when deciding to adopt a qualitative research methodology. Strauss and Corbin (1990) claim that qualitative methods can be used to better understand any phenomenon about which little is yet known. They can also be used to gain new perspectives on things about which much is already known, or to gain more in-depth information that may be difficult to convey quantitatively. Thus, qualitative methods are appropriate in situations where one needs to first identify the variables that might later be tested quantitatively, or where the researcher has determined that quantitative measures cannot adequately describe or interpret a situation. Research problems tend to be framed as open-ended questions that will support discovery of new information.

Quantitative research refers to the systematic empirical investigation of quantitative properties and phenomena and their relationships. The objective of quantitative research is to develop and employ mathematical models, theories and/or hypotheses pertaining to phenomena. The process of measurement is central to quantitative research because it provides the fundamental connection between empirical observation and mathematical expression of quantitative relationships.

From another point of view we can distinguish physiological, biochemical, genetic, mathematical, physical, invasive, and non-invasive and other types of research methods.

Selection of technology

It is necessary to define/to grow serious what kind of devices have to be used, what technical parameters they should have if we would like to be sure that recorded/gathering parameters are really objective and exact. In this process can be helpful if researcher has convincing information on usefulness of the chosen methods in previous similar research done by other researcher. Than he/she can re-use exactly the same methods in own research. Validity of obtained results will be high if research technology in both researches is the same quality. If necessary some adaptation/modification in re-used method can be done but it should be described precisely. More complicated is situation, when researcher has to develop new/original method for solving the research problem. In this case validity of new method should be first proven, and than it can be used in research. This method should be described in full details, its usage clearly justified, and obtained results should undergo to critic analysis.

Protocol of research

What is a protocol of research? It is precise description of all steps done during research. All steps are settled on time axis. The research protocol forms an essential

part of a research. As your research ideas develop into a workable study a written protocol will help to formalise your ideas and gain feedback from others through peer review.

A well written protocol is also necessary for your applications to funding bodies and ethics and research governance committees. The protocol can also act as a manual for members of the research team to ensure they adhere to the methods outlined.

As your study gets underway, the protocol can then be used to monitor the study's progress and evaluate its outcomes. The protocol encourages the reader to think about the study rigorously and provides communication between all of the people involved.

Structure of research protocol can be as it is described in: Writing a research protocol.

Title: The title clearly identifies the study and may contain a brief description of the study design and objectives.

Investigators/Researchers including contact addresses: Everyone who has made a valuable contribution to the study should be named and their contact details given

Abstract/Summary: Summarise the aims or objectives of the study and give a brief outline of the design and methods.

Background/Introduction: The introduction should outline the background to the research, including a critical review of the current knowledge or literature, including published and unpublished work in the area. Any gaps in the evidence should be identified as should the potential value of furthering knowledge in this field, such as theoretical or clinical applications of the study outcomes. If applicable, the research hypothesis should also be included in this section, with an explanation of the reasons for undertaking the work.

Aims and Objectives: Outline concise and precise objectives that should follow on from the hypothesis.

Study design and methods

- **Study design.** What study design is most appropriate to answer your particular research? question
- **Setting.** Where will the research take place? Your study may take place in a number of different sites or you may be visiting patients in their homes. You need to address any practical issues involved, such as safety procedures when doing home visits.
- **Subjects/Patients.** Detailed information regarding your subjects should be given. For example, describe the study population, including a rationale of why they were chosen.
- **Describe the methods by which subjects will be identified and recruited and what inclusion and exclusion criteria will be used.** You will need to justify your sample size and state whether sample size calculations have been used. It may also be necessary to describe the criteria for participation or completion of the study, participant retention strategies and withdrawal criteria.

- Randomisation methods. Some research strategies, such as case control studies or randomised control studies, require a random allocation of patients to the different experimental groups or interventions. You will need to explain what randomisation methods you will use.
- Methods of assessment or measurement. What data will be collected and why. For example how will you measure your participants quality of life, what instruments will you use and are they the most appropriate? If you are using any equipment it should be clearly described.
- Outcome measures/objectives. The measurement outcomes used to support or reject the hypotheses can be stated and separated into primary and secondary outcomes. For example, primary outcomes or endpoints are most important to your hypothesis, there may be only 1 or 2. Secondary outcomes may provide some support to the hypothesis, but without the primary outcomes they could not confirm the hypothesis.
- Interventions (if applicable). Not all studies will involve any interventions, but if yours does a description of the study intervention should be provided. If you are giving a treatment or investigation, the dose, timing, method of providing, administering and receiving the treatment should be detailed. All necessary safeguards and potential risks should be made clear, including the methods by which intervention will be monitored.

Data collection, management and analysis

Explain how the data will be collected and managed and who will have access to it. The method of the data analysis should also be specified and may include the following points:

- method of data entry
- plan of analysis, including assumptions of analysis
- data analysis package
- presentation of demographic and outcome data summaries
- planned presentation of the data, i.e. graphs, tables, figures

Study administration and ethical issues

- outline the methods by which the patient/subject's interests will be safeguarded. For example, the process of risk limitation, how you will maintain confidentiality or anonymise patient's data and how you will monitor any adverse side effects
- state whether there has been user involvement in design of the study.
- if applicable, state whether you have followed the Clinical trials directive, and you have Regulatory authority approval
- you should also state who will provide indemnity in case of harm to your participants through negligence?

- if applicable, the protocol should clearly state who is sponsoring the research study and what interest they have in its outcome. It is also necessary to state whether the sponsors are to provide indemnity in the case of negligent harm to participants.

Resource requirements

The resource implications to the host organisation and any other involved departments should be defined in this section. In addition you must outline the timetable/schedule of the research and all costs.

Study plan

You may wish to include a study plan, showing a brief summary or flow chart of the order, site and timing of all study procedures. It may also be useful to include consent forms and participant information in appendices.

Supervision

Where applicable, the protocol should name the individual(s) who will supervise the research project and the intended arrangements for the supervision.

Dissemination and outcome

How will the study's findings be made available? State whether you intend to publish or present the findings. Any implications for future practice and patient care should also be suggested.

Control mechanisms

Control mechanisms should be stated for each research. The reason for it is that each activity of person, including researcher, can be burden by mistakes. They may be of subjective or/and objective origin. Mistakes can be involved in planning and in running of research. Some mistakes can't be eliminated, so, it will be necessary to take them into account when the results will be interpreted. Other group of mistakes can be identified and excluded during preparation of research, or can be eliminated during pilot study, and some even at the preparation of results for publication. First control mechanism is autocontrol. Author(s) of research project should repeatedly check individual phases and steps of research project. As a second form of control mechanism is recommended peer-review done by co-workers from the same lab, another labs or external expert. Very efficient control form of project quality is carrying out the pilot study. This step can find and eliminate conceptual, formal and technological weak places in the project.

Pilot study

The research is ready for implement the plan. However, where possible, he/she should first carry out a pilot study, which is a small-scale version of „dummy run” of the major

study. Unforeseen problem can arise in the course of a project. By doing a pilot study, the researcher can recognise and address some of the problems by obtaining information for improving the project, making adjustments to the research instruments, to research plan, to protocol, to time schedule, and to other parts of research project, or re-assessing the feasibility of the study.

Select design for data analysis

In this step researcher should consult with specialist on statistics. The aim of this consultation is to choose correct statistic methods for evaluation of results obtained in respective research. Researcher should present to statistician clearly all goals of research. By co-operation both the high probability does exist that appropriate statistic method will be chosen and research result will be evaluated correctly.

Determination of other condition necessary for success of research

Research funding is the most important condition for researcher. Without financial support it is not possible to do valuable research. This is the reason why many senior researchers spend more than a trivial amount of their time applying for grants for research funds. These grants are necessary not only for researchers to carry out their research, but also as a source of merit. Other inevitable conditions for research work are: appropriate material, places for research with appropriate equipment, well prepared and motivated personal, and other, and sometimes very special conditions.

Empiric phase

In this phase, the researcher implements all the plans that he/she made in First phase, step 3, to collect the data. In many studies empirical phase is the most time-consuming part of the investigation.

The amount of time spent, however, varies from study to study. So, we can say that empiric phase is composed of the following activities:

- To do all activities related to gaining scientific results, to sort them, and to evaluate them
- its first step can be pilot study (to tune the research project)

For the gaining of valid results it is necessary to fulfil some essential conditions, e.g.

- well prepared research team – researcher should be skilled in using research methods
- researchers should be manually and mentally skilled, experienced
- specification of environmental condition for carry out the research should be present
- to use relevant and sensitive enough methods, technologies
- precisely fulfil protocol of experiment or clinical study
- minimize possibility of subjective negative influence on production data
- minimize of negative influence of environment to research

The result of this phase is production of data in two main forms:

- qualitative form
- quantitative form: in analogue form in digital form.

The way by which is research conducted and the level of organization used for data collection can importantly influence the amount and quality of obtained results

Examples of qualitative data:

- pain, fatigue, dyspnoea, feeling of cold or heat, paresthesias, dizziness, fear, anger, apathy and others similar

Examples of quantitative data:

- biochemical, electrophysiological, ultrasound, x-ray and other data obtained by measuring and recording different kinds of parameters in clinical or experimental study.

It was mentioned above that for production of objective scientific information is necessary to use methods which will guarantee the highest degree of production unbiased results. This is important mainly:

- when on the same research participate more researchers from the same research team
- when the respective research is conducted at more laboratories or in different hospitals in one or more countries (multicentre studies)
- that researchers are well informed on what parameters should be observed and recorded
- to stress that terminology of recorded parameters have to be well defined and all researchers should understand their meaning.

Key elements of the experimental procedure

If you use experiment as a method in your research than you should be familiar with its key elements. Here they are:

- description and size of all experimental and control groups, as applicable
- a step-by-step list of everything you must do to perform your experiment. Think about all the steps that you will need to go through to complete your experiment, and record exactly what will need to be done in each step
- the experimental procedure must tell how you will change your one and only independent variable and how you will measure that change
- the experimental procedure must explain how you will measure the resulting change in the dependent variable or variables
- if applicable, the experimental procedure should explain how the controlled variables will be maintained at a constant value
- the experimental procedure should specify how many times you intend to repeat your experiment, so that you can verify that your results are reproducible
- a good experimental procedure enables someone else to duplicate your experiment exactly!

Another questions related to experiment should be taken into account, e.g. where will you conduct your experiment? You may need a lot of room for you experiment or you may not be able to move your experiment around from place to place. If you are working with human or animal subjects, you may need a location that is quiet. You will need to think about these limitations before you start your experiment so you can find a location in advance that will meet your needs.

The researcher normally collects the data according to the pre-established plan, and collects actual information, that is data, usage the instrument that has been developed and tested in the pilot study. Rational system for collection of data should involve:

- organisation of the collection, namely:
 - timing of data collection – continuously, at the end of research, day time, part of the year....
 - quantity and quality of persons responsible for data collection – to prepare clear instruction, setting the educational requirements....
 - material requirements – e.g. the method used for archivation of data
- preparation of data for next evaluation
 - control of data completeness and correctness
 - whether the data are classified according their characteristic signs correctly
 - elimination of incorrect data
 - supplement missing data
- control of representativeness of the data
 - elimination of incorrectly chosen patients, animals and their data
- to code of data – give numbers instead of words to each data
 - to transform the data for computer elaboration

The Basics Of Business Forecasting

It is not unusual to hear a company's management speak about forecasts: "Our sales did not meet the forecasted numbers," or "we feel confident in the forecasted economic growth and expect to exceed our targets." In the end, all financial forecasts, whether about the specifics of a business, like sales growth, or predictions about the economy as a whole, are informed guesses.

In this article, we'll look at some of the methods behind financial forecasts, as well as the actual process and some of the risks that crop up when we seek to predict the future.

Financial Forecasting Methods

There are a number of different methods by which a business forecast can be made. All the methods fall into one of two overarching approaches: qualitative and quantitative.

Qualitative Models

Qualitative models have generally been successful with short-term predictions, where the scope of the forecast is limited. Qualitative forecasts can be thought of as expert-driven, in that they depend on market mavens or the market as a whole to weigh in with an informed consensus. Qualitative models can be useful in predicting the short-term success of companies, products and services, but meets limitations due to its reliance on opinion over measurable data. Qualitative models include:

- Market Research Polling a large number of people on a specific product or service to predict how many people will buy or use it once launched.
- Delphi Method: Asking field experts for general opinions and then compiling them into a forecast.

Quantitative Models

Quantitative models discount the expert factor and try to take the human element out of the analysis. These approaches are concerned solely with data and avoid the fickleness of the people underlying the numbers. They also try to predict where variables like sales, gross domestic product, housing prices and so on, will be in the long-term, measured in months or years. Quantitative models include:

- The Indicator Approach: The indicator approach depends on the relationship between certain indicators, for example GDP and unemployment rates, remaining relatively unchanged over time. By following the relationships and then following indicators that are leading, you can estimate the performance of the lagging indicators, by using the leading indicator data.
- Econometric Modeling: This is a more mathematically rigorous version of the indicator approach. Instead of assuming that relationships stay the same, econometric modeling tests the internal consistency of data sets over time and the significance or strength of the relationship between data sets. Econometric modeling is sometimes used to create custom indicators that can be used for a more accurate indicator approach. However, the econometric models are more often used in academic fields to evaluate economic policies.
- Time Series Methods: This refers to a collection of different methodologies that use past data to predict future events. The difference between the time series methodologies is usually in fine details, like giving more recent data more weight or discounting certain outlier points. By tracking what happened in the past, the forecaster hopes to be able to give a better than average prediction about the future. This is the most common type of business forecasting, because it is cheap and really no better or worse than other methods.

How Does Forecasting Work?

There is a lot of variation on a practical level when it comes to business forecasting. However, on a conceptual level, all forecasts follow the same process.

1. A problem or data point is chosen. This can be something like “will people buy a high-end coffee maker?” or “what will our sales be in March next year?”
2. Theoretical variables and an ideal data set are chosen. This is where the forecaster identifies the relevant variables that need to be considered and decides how to collect the data.
3. Assumption time. To cut down the time and data needed to make a forecast, the forecaster makes some explicit assumptions to simplify the process.
4. A model is chosen. The forecaster picks the model that fits the data set, selected variables and assumptions.
5. Analysis. Using the model, the data is analyzed and a forecast made from the analysis.
6. Verification. The forecaster compares the forecast to what actually happens to tweak the process, identify problems or in the rare case of an absolutely accurate forecast, pat himself on the back.

Problems With Forecasting

Business forecasting is very useful for businesses, as it allows them to plan production, financing and so on. However, there are three problems with relying on forecasts:

1. The data is always going to be old. Historical data is all we have to go on and there is no guarantee that the conditions in the past will persist into the future.
2. It is impossible to factor in unique or unexpected events, or externalities. Assumptions are dangerous, such as the assumptions that banks were properly screening borrowers prior to the subprime meltdown, and black swan events have become more common as our dependence on forecasts has grown.
3. Forecasts can't integrate their own impact. By having forecasts, accurate or inaccurate, the actions of businesses are influenced by a factor that can't be included as a variable. This is a conceptual knot. In a worst case scenario, management becomes a slave to historical data and trends rather than worrying about what the business is doing now.

The Bottom Line

Forecasting can be a dangerous art, because the forecasts become a focus for companies and governments, mentally limiting their range of actions, by presenting the short to long-term future as already being determined. Moreover, forecasts can easily breakdown due to random elements that can't be incorporated into a model, or they can be just plain wrong from the beginning.

The negatives aside, business forecasting isn't going anywhere. Used properly, forecasting allows businesses to plan ahead of their needs, raising their chances of keeping healthy through all markets. That's one function of business forecasting that all investors can appreciate.

Forecasting And Its Practical Applications

In an era where forecasts drive entire supply chain networks forecasting is an increasingly critical organizational capability. Forecasting the future may sound like a lofty and theoretical activity when in reality it is a practical business tool like many others. Here is an example. How should a business go about preparing the quarterly sales volume forecasts for their primary product, say, window-glass? The company will certainly want to review the actual sales data for window glass over the last few years. Suppose that the forecaster has access to actual sales data for each quarter over the 15-year period the firm has been in business. Using these historical data, the forecaster can see the general level of sales but more importantly, he or she can also determine what pattern the sales history produces, what trends are visible. A thorough review of the data may reveal some type of seasonal pattern, such as peak sales occurring in the spring as people do spring-cleaning and others prepare to sell their homes during the summer school break. In addition, if the forecaster is able to identify other factors that influence sales, like weather patterns or housing starts, historical data on these factors can also be used in generating forecasts of future sales volumes.

Forecasting Methods

Academics divide forecasting methods into two broad categories: qualitative and quantitative. The division of forecasting methods into qualitative forecasting and quantitative forecasting is based on the availability of historical time series data. If historical data and time series are available, than quantitative methods may be used. If not, qualitative methods are the only option.

Qualitative Forecasting Methods

Qualitative forecasting techniques generally employ the judgment of experts to generate forecasts. A key advantage of these procedures is that they can be applied in situations where historical data are simply not available. Even in situations where such data are available, quantitative forecasting methods are a useful addition to successful forecasting. Three important qualitative forecasting methods are: the Delphi method, scenario writing, and the subject approach.

Delphi Method

In the Delphi method, an attempt is made to develop forecasts through “group consensus.”

A group of experts in a particular field participate. Usually, a panel of these experienced people is asked to respond to a series of questionnaires. The panel members, who should ideally come from a variety of backgrounds (marketing, production, management, finance, purchasing, etc.) are asked to respond to an initial questionnaire. A second questionnaire is then created which incorporates information and opinions gathered in the responses to the first questionnaire. The second questionnaire is then

distributed. Each panelist is asked to reconsider and revise his or her initial response to the questions based on the new information. This process is continued until some degree of consensus among the panelists is reached. It should be noted that the objective of the Delphi method is not to produce a single answer at the end. Instead, it attempts to produce a relatively narrow range of opinions—a range into which most of the panelists' opinions fall.

Scenario Writing Method

Under the scenario writing approach, the forecaster starts with different sets of assumptions. For each set of assumptions, a likely scenario of the business outcome is charted. Thus, the forecaster generates several different future scenarios (corresponding to different sets of assumptions). The decision maker or business person is presented with the different scenarios, and has to decide which scenario is most likely to prevail.

A Subjective Approach Method

The subjective approach allows individuals participating in the forecasting decision to arrive at a forecast based on their feelings, ideas, and personal experiences. Many corporations in the United States have started to increasingly use the subjective approach. Internally, these subjective approaches sometimes take the form of “brainstorming sessions,” in which managers, executives, and employees work together to develop new ideas or to solve complex problems. At other times, the subjective approach may take the form of a survey of the company's sales people. This approach, which is known as the sales force composite or grass roots method, is relied on because salespeople interact directly with purchasers and it is assumed therefore that they have a good feel for which products will or will not sell and in what quantities.

The advantage of using the salespeople's forecasts is that salespeople are highly qualified to explain the demand for products, especially in their own territories. The disadvantage is that salespeople may tend to be optimistic in their estimates since optimism is a characteristic often found in good salespeople. Also, those working in sales may fear that a low sales forecast will lead to layoffs in the sales area. The opinions of salespeople should not be relied on to the exclusion of all else for one additional reason. Salespeople may not be aware of impending changes in other related areas, such as availability of raw materials, national economic developments, or the arrival of a formidable new competitor.

Quantitative Forecasting Methods

Quantitative forecasting methods are used when historical data on variables of interest are available—these methods are based on an analysis of historical data concerning the time series of the specific variable of interest. There are two quantitative forecasting methods. The first uses the past trend of a particular variable in order to make a future forecast of the variable. In recognition of this method's reliance on time series, it is

commonly called the “time series method.” The second quantitative forecasting method also uses historical data. This method is often referred to as the causal method because it relies on the use of several variables and their “cause-and-effect” relationships. Examples of variables that may have this cause-and-effect relationship are: 1) interest rate levels and levels of disposable income; 2) winter weather patterns and demand for heating oil; 3) increasing gas prices and a decline in demand for sports utility vehicles (SUVs). By studying the time series data on two or more variables that have a cause-and-effect relationship with the item for which a forecast is needed, effort is made to incorporate as many relevant factors as possible into the forecast.

In practice, most business people use some combination of these methods and techniques in trying to plan for the future and put together accurate forecasts. With each cycle of forecasting, more is learned about what factor to consider and how to weight their importance in projecting future events.

6

Elements of Scientific Research Methodology

There are many ways of outlining the basic method shared by all fields of scientific inquiry. The following examples are typical classifications of the most important components of the method on which there is wide agreement in the scientific community and among philosophers of science. There are, however, disagreements about some aspects. The following set of methodological elements and organization of procedures tends to be more characteristic of natural sciences than social sciences. In the social sciences mathematical and statistical methods of verification and hypotheses testing may be less stringent.

Nonetheless the cycle of hypothesis, verification and formulation of new hypotheses will resemble the cycle described below.

The essential elements of a scientific method are iterations, recursions, interleavings, and orderings of the following:

- Characterizations (observations, definitions, and measurements of the subject of inquiry)
- Hypotheses (theoretical, hypothetical explanations of observations and measurements of the subject)
- Predictions (reasoning including logical deduction from the hypothesis or theory)
- Experiments (tests of all of the above)

Each element of a scientific method is subject to peer review for possible mistakes. These activities do not describe all that scientists do but apply mostly to experimental sciences (e.g., physics, chemistry). The elements above are often taught in the educational system.

Scientific method is not a recipe: it requires intelligence, imagination, and creativity. It is also an ongoing cycle, constantly developing more useful, accurate and comprehensive models and methods. For example, when Einstein developed the Special and General Theories of Relativity, he did not in any way refute or discount Newton's *Principia*. On the contrary, if the astronomically large, the vanishingly small, and the extremely fast are reduced out from Einstein's theories — all phenomena that Newton could not have observed — Newton's equations remain. Einstein's theories are expansions and refinements of Newton's theories, and observations that increase our confidence in them also increase our confidence in Newton's approximations to them.

A linearized, pragmatic scheme of the four points above is sometimes offered as a guideline for proceeding:

- Define the question
- Gather information and resources (observe)
- Form hypothesis
- Perform experiment and collect data
- Analyse data
- Interpret data and draw conclusions that serve as a starting point for new hypothesis
- Publish results
- Retest (frequently done by other scientists)

The iterative cycle inherent in this step-by-step methodology goes from point 3 to 6 back to 3 again. While this schema outlines a typical hypothesis/testing method, it should also be noted that a number of philosophers, historians and sociologists of science (perhaps most notably Paul Feyerabend) claim that such descriptions of scientific method have little relation to the ways science is actually practiced.

The “operational” paradigm combines the concepts of operational definition, instrumentalism, and utility: The essential elements of a scientific method are operations, observations, models, and a utility function for evaluating models.

- Operation - Some action done to the system being investigated
- Observation - What happens when the operation is done to the system
- Model - A fact, hypothesis, theory, or the phenomenon itself at a certain moment
- Utility Function - A measure of the usefulness of the model to explain, predict, and control, and of the cost of use of it. One of the elements of any scientific utility function is the refutability of the model. Another is its simplicity, on the Principle of Parsimony also known as Occam's Razor.

The Keystones of Science project, sponsored by the journal *Science*, has selected a number of scientific articles from that journal and annotated them, illustrating how different parts of each article embody scientific method. Here is an annotated example of this scientific method example titled *Microbial Genes in the Human Genome: Lateral Transfer or Gene Loss?*

DNA EXAMPLE

Each element of scientific method is illustrated below by an example from the discovery of the structure of DNA:

- *DNA-characterizations*: in this case, although the significance of the gene had been established, the mechanism was unclear to anyone, as of 1950.
- *DNA-hypotheses*: Crick and Watson hypothesized that the gene had a physical basis - it was helical.
- *DNA-predictions*: from earlier work on tobacco mosaic virus, Watson was aware of the significance of Crick's formulation of the transform of a helix. Thus he was primed for the significance of the X-shape in photo 51.
- *DNA-experiments*: Watson sees photo 51.

The examples are continued in "Evaluations and iterations" with *DNA-iterations*.

CHARACTERIZATIONS

Scientific method depends upon increasingly more sophisticated characterizations of the subjects of investigation. (The *subjects* can also be called *unsolved problems* or the *unknowns*.) For example, Benjamin Franklin correctly characterized St. Elmo's fire as electrical in nature, but it has taken a long series of experiments and theory to establish this. While seeking the pertinent properties of the subjects, this careful thought may also entail some definitions and observations; the observations often demand careful measurements and/or counting.

"I am not accustomed to saying anything with certainty after only one or two observations." —Andreas Vesalius (1546).

The systematic, careful collection of measurements or counts of relevant quantities is often the critical difference between pseudo-sciences, such as alchemy, and a science, such as chemistry or biology. Scientific measurements taken are usually tabulated, graphed, or mapped, and statistical manipulations, such as correlation and regression, performed on them. The measurements might be made in a controlled setting, such as a laboratory, or made on more or less inaccessible or unmanipulatable objects such as stars or human populations.

The measurements often require specialized scientific instruments such as thermometers, spectrosopes, or voltmeters, and the progress of a scientific field is usually intimately tied to their invention and development.

Measurements in scientific work are also usually accompanied by estimates of their uncertainty. The uncertainty is often estimated by making repeated measurements of the desired quantity. Uncertainties may also be calculated by consideration of the uncertainties of the individual underlying quantities that are used. Counts of things, such as the number of people in a nation at a particular time, may also have an uncertainty due to limitations of the method used. Counts may only represent a sample of desired quantities, with an uncertainty that depends upon the sampling method used and the number of samples taken.

Measurements demand the use of *operational definitions* of relevant quantities. That is, a scientific quantity is described or defined by how it is measured, as opposed to some more vague, inexact or “idealized” definition. For example, electrical current, measured in amperes, may be operationally defined in terms of the mass of silver deposited in a certain time on an electrode in an electrochemical device that is described in some detail.

The operational definition of a thing often relies on comparisons with standards: the operational definition of “mass” ultimately relies on the use of an artifact, such as a certain kilogram of platinum-iridium kept in a laboratory in France. The scientific definition of a term sometimes differs substantially from its natural language usage.

For example, mass and weight overlap in meaning in common discourse, but have distinct meanings in mechanics. Scientific quantities are often characterized by their units of measure which can later be described in terms of conventional physical units when communicating the work. New theories sometimes arise upon realizing that certain terms had not previously been sufficiently clearly defined. For example, Albert Einstein’s first paper on relativity begins by defining simultaneity and the means for determining length. These ideas were skipped over by Isaac Newton with, “*I do not define time, space, place and motion, as being well known to all.*” Einstein’s paper then demonstrates that they (viz., absolute time and length independent of motion) were approximations. Francis Crick cautions us that when characterizing a subject, however, it can be premature to define something when it remains ill-understood. In Crick’s study of consciousness, he actually found it easier to study awareness in the visual system, rather than to study free will, for example. His cautionary example was the gene; the gene was much more poorly understood before Watson and Crick’s pioneering discovery of the structure of DNA; it would have been counterproductive to spend much time on the definition of the gene, before them.

The Importance of Theory and Methodology in Research

All research contains theory in some form, and social work research is no exception. Much research is clear about the theories being used, and the ways they are applied within the formation of the project. Theory will manifest itself to some degree in:

- the theoretical approach itself (the methodology)
- the arguments about what might happen
- the approach to the fieldwork or data-gathering
- the analysis and synthesis of the findings.

Royse (2008) argues that theory often arises out of a ‘value’ position, and this is apparent within social work research in its attention to anti-oppressive or emancipatory practices.

Sometimes, social work research is carried out pragmatically, without explicitly recognising the theory or methodology behind what is being done. This is not to say that there is not a research design, and an understanding that the research is ‘qualitative’,

‘quantitative’, or ‘mixed-method’, but that the assumptions behind it (what are known as the ontological and epistemological standpoints), have not been clearly stated.

Ontology and Epistemology

Ontology and epistemology are closely linked: ‘commonly regarded as related, sometimes in a linear fashion, with ontological preferences informing epistemological issues’. This is because they are about the ways that we understand the world and the nature of reality, and our understanding of ‘knowledge’ within that.

Ontology is the philosophy of ‘being’, or ‘the nature of the world and what we can know about it’. Different ontological approaches look at how we understand the physical and social world, and its nature. Importantly for social research this considers how we understand ‘reality’ within the social world: whether it is unyielding and there regardless of our thoughts about it (realism), or a product of our knowing about it and constructing it through our individual thoughts (idealism).

Even where not explicitly stated, researchers approach their topic from a particular position and understand the world in a certain way.

This is because we all hold an ‘ontological position’, knowingly or unknowingly, and ‘our values affect our research at every level - theories, hypotheses, choice of methodologies and variables, data analysis strategies - whether we are conscious of our values or not’. Ingleby (1980: 25) takes this understanding of ontological position further, suggesting that ‘they correspond to different mentalities’.

‘Epistemology’ is the philosophy or theory of ‘knowing’. It is ‘concerned with the criteria used to distinguish knowledge claims, or assess their rigour and validity’. McLaughlin (2012: 25) writes that epistemology is about grounding how we know the world: ‘the types of knowledge claims that we can make about the world and how we can then assure the credibility of such claims’.

Scientific Method in Economics

Scientific vs. Non-scientific Methods of Gaining Knowledge

Having finished a brief discussion of the meaning of research and related terms, we are now ready to take up the discussion of the scientific method. Let us start by stating that there are essentially two methods of obtaining knowledge: scientific and non-scientific methods. Let us start with non-scientific method.

Non-Scientific Method: The first method of gaining knowledge is through senses, experience, intuition, and revelation, all of these may be classified as non-scientific methods. Some gain knowledge through physical *senses* - sight, sound, touch, taste, and smell -, and *experience*.

Senses and Experiences. Some knowledge is obtained by senses and experiences. When one gets too close to a fire and gets burned once, he or she gains the knowledge that it is dangerous to be too close to the fire. Other gain knowledge by intuition or revelation.

Intuition. Some rely on intuition as a source of knowledge. *Intuition* is the strong hunch or feeling that what one perceives to be the case is indeed true. If one strongly believes what one perceives is real and true, knowledge thus is obtained. While there is no reason to doubt the truthfulness of the knowledge obtained by intuition, like knowledge obtained from senses and experiences, it is subjective.

Revelation. Some knowledge are obtained by revelation. *Revelation* is the presentation of the truth from a supernatural source, such as deity.

Knowledge acquired via experience, intuition, revelation, and even measurement remain as private knowledge. The validity of knowledge obtained through nonscientific methods cannot be subject to objective testing.

Scientific Method: The second method of obtaining knowledge is the scientific method. This method of gaining knowledge is learning by reasoning. It is considered today to be the most reliable method of gaining knowledge. In contrast with the scientific method, the validity of knowledge obtained by scientific method can be subject to testing.

In all science, research proceeds within the framework of the scientific method. According to Lastrucci, “science may be defined as an objective, logical, and systematic method of analysis of phenomena devised to permit the accumulation of reliable knowledge”. His definition of science contains all essential elements of scientific methods.

First it defines science as the method of analysis. Second, it highlights three major characteristics of the scientific methods as *objective*, *logical* and *systematic*. Let’s explore the meaning of these three characteristics. First, scientific method is *objective*, not in the sense of being value free. But it is objective in the sense that the analysts are not biased or prejudiced or subject to personal whims. Second, scientific method is *logical* in the sense that science follows logical reasoning. Logical reasoning is thinking in reasonable fashion. It is sufficient to point out two types of reasoning process, namely deductive and inductive logic. A full discussion on the logical process will be presented below. Third and finally, the search for truth in science is *systematic*. This means that researchers follow a systematic set of procedures through which knowledge is gained.

Deductive and Inductive Logic

Let us now turn our attention to the logical process in scientific method. First, note that there are two types of reasoning process, namely deductive and inductive logic.

Deductive logic is the process of reasoning from general conditions or premises using assumptions to specific conclusions. Economic theory rests largely on deductive logic. We establish a set of assumptions about conditions and behaviors to arrive at conclusions through logical process.

Examples: utility maximization in the consumer behavior; profit maximization in the producer behavior.

Inductive logic is reasoning from the specific outcomes to a generalized conclusion. This is usually done by observing many individual experiences and cases to formulate a general conclusion. The inductive logic of reasoning is followed in most empirical economic research.

Hypothesis, Theory, and Law

We *deduce* from facts and assumptions to a *hypothesis*. “A *hypothesis* is a tentative assertion of a relationship between factors or events that is subject to verification or rejection.” In short, a hypothesis is a testable proposition of the relationship between or among variables.

How is *hypothesis* related to *theory*? And theory to law? *Hypothesis* is a single statement that attempts to explain a single interesting or puzzling phenomenon. In other words, a hypothesis is a testable proposition on an interesting or puzzling phenomenon. It usually takes the form of an educated guess or conjecture. Usually the hypothesis is based on facts and assumptions.

Theory is a whole system of thought (or systematic explanation) that refers to many phenomena and whose parts are related to one another in deductive, logical form.

Law: A theory that has been subjected to extensive testing over time and across space, and that has won virtually universal acceptance, is called a *law*. For instance, the law of supply and demand refers to the commonly observed phenomena that, in a free market, the forces of supply and demand generally push the price toward its equilibrium level, the price at which the quantity supplied and quantity demanded are equal. Another example is the law of diminishing marginal returns.

Language of Scientific Method

We first learned that the essential element of science lies in its method of analysis, and that the three main characteristics of the scientific method are that it is objective, logical, and systematic. Then we learned about the process of scientific method beginning from, say, deductive logic to hypothesis, from hypothesis to theory, and from theory to law. Now we need to learn the operational aspects of the scientific method by focusing on terms and concepts of research. In particular, we need to learn basic concepts such as assumptions, variables, parameters, and functional forms. Finally, we need to learn something about *ceteris paribus*.

Assumption

In our study of gasoline price hike, we first *assume* that consumers are “rational” in that their decision-making in the purchase of gasoline is consistent with maximization of consumer satisfaction. Likewise, suppliers of gasoline (retail gasoline merchant) are also assumed to be rational in that their decision-making in pricing, inventory, etc. is consistent with profit maximization. We theorize that the retail gasoline price is jointly determined by the forces of demand and supply of gasoline. On the demand side, the

quantity of gasoline demanded depends inversely on the retail price of gasoline, positively consumer income, and a host of other factors. On the supply side, the quantity of gasoline supplied depends positively on the price of gasoline price, cost of production (including price of crude oil and refinery cost), and a host of other factors, including the gasoline tax and environmental cost. Once we introduce the role of assumption in the research, it appropriate to further introduce related basic concepts of research: variables, functional relationship and parameters.

Variables in General

A variable is a quantity of something which varies and you are interested in. Price of gasoline is a variable to an economic analyst studying the recent gasoline price increase in 1996 but not to most motorists and not even to an economic analyst studying the relation between stock and bond prices.

The researcher chooses his or her variables. Choosing variables correctly is one of the first essential step of carrying out research. Therefore, choose variables with extreme care. To choose variable correctly, one has to know the two types of variables: the dependent variable and independent variable.

The Dependent Variable

The dependent variable is that quantity whose change the researcher wants to find out, explain, or predict. In the cause-effect relationship, the *effect* variable is the dependent variable. In the study of the demand for gasoline, the quantity of gasoline demanded is the dependent variable, because the quantity of gasoline demanded changes in response to changes in gasoline price, consumer income, and other demand side factors.

This researcher is ultimately interested in measuring the impacts of these changes on gasoline price. In some cases, however, the dependent variable is not quantity, but represents a qualitative choice. An example of the latter is decision to buy or not; or marry him (or her) or not.

The Independent Variable

The independent variable is a variable whose effect upon the dependent variable one is trying to understand, explain, and predict.

In the cause-effect relationship, it is the *cause* variable. Using the study of the demand for gasoline, the independent variables are the price of gasoline, consumer income, and other variables such as fuel efficiency and population characteristics.

The Parameter

The concept of parameter is a bit more tricky to define. A parameter is a quantity measuring the response of the dependent variable to change in the independent variable, and is usually assumed to remain constant during the period of study. The hypothetical

relationship between the quantity of gasoline demanded and gasoline price over time. On the vertical axis, the quantity of gasoline demanded measured in gallons. On the horizontal axis, the price of gasoline is measured in U.S. dollars per gallon. A scatter plot of cross represents the combination of the quantity of gasoline demanded and corresponding gasoline price in each year over the period. The line is drawn through the scatter plot of the quantity of gasoline demanded and gasoline price as closely as possible. The line represents a best guess of the average relationship between the observed quantity of gasoline demanded and corresponding gasoline price during the period of study.

The Functional Form

The algebraic relationship $Q_g^d = a + bP_g$ is *linear* relationship between the two variables Q^d and P . It is linear both in the parameters and variables. Parameters like a^2 or $1/a$ are non-linear; variables like Q^2 and $1/Q$ are examples of non-linear.

When there is a relationship between two variables, we say that “ y (dependent variable) is a function of x (independent variable)” in the mathematical language. By this we mean to say that the magnitude of y depends on the magnitude of x . But when the dependent variable y is determined by not only by x but also by other independent variables, we say that y is a function of x_1, x_2 , and x_3 . Or in short $y = f(x_1, x_2, x_3)$.

In the gasoline demand example, the quantity of gasoline demanded (Q_g^d) depends upon (or is a linear function of) price of gasoline (P_g) and consumer income (Y). Algebraically, we write: $Q_g^d = f(P_g, Y)$.

Linear functional form of this relationship can be written as $Q^d = a + bP_g + cY$. Since parameters a, b , and c enter the equation are linear, the relationship is linear. But the relationship may be other than linear functional forms. They include the log-linear model, semi-log models, and reciprocal models.

Research Methods and Research Methodology

Research methods provide the specific details of how one accomplishes a research task (procedures and methods); It provides specific and detailed procedures of how to initiate, carry out, and complete a research task by mainly focusing on how to do it. Research methodology deals with general approaches or guidelines to conducting research. It provides the principles for organizing, planning, designing, and conducting research, but it cannot tell you in detail how to conduct a specific, individual research.

In carrying out an applied and quantitative economic research, there are several necessary backgrounds. The researcher should first have solid training in economic theory, quantitative methods (statistics and econometrics), data analysis techniques, and adequate training in micro-computer technology, as well as some training in research methods.

Unfortunately, undergraduate students beginning their research most often do not have these backgrounds. Therefore, they are going to be overwhelmed and intimidated

by the lack of necessary skills. Many often give up in frustration, even if they are willing to persevere and to learn these skills. What is sorely need is a practical guide to initiate, conduct, and complete an applied and quantitative economic research. One proven and effective way of learning these various skills which economists use is *learning by doing by example*.

Types of Research

Basic vs. Applied Research: The distinction between basic and applied research is largely by the focus of its application. This distinction comes from basic science vs. applied science. Example: physics and engineering. Basic research focuses on determining or establishing the basic or fundamental relationships within a discipline without paying attention to any practical applications to the real world. In contrast, applied research is usually conducted to solve a particular and concrete problem.

Descriptive vs. Analytical Research: The distinction between descriptive and analytical research is based on the question it asks. Descriptive research attempts to determine, describe, or identify *what* is, while analytical research attempts to establish *why* it is that way or how it came to be.

The descriptive research uses description, classification, measurement, and comparison to describe what phenomena are. The analytical research usually concerns itself with cause-effect relationships. Examples. Examining the fluctuations of U. S. international trade balance during 1974-1995 is an example of descriptive research; while explaining *why* and *how* U.S. trade balance move in a particular way over time is an example of analytical research.

Another example: Starting from late 1986, the value of U.S. dollar value has steadily increased against the Japanese yen and German Mark. Examining the magnitude of this trend in the value of U.S. dollar is another example of descriptive research; while explaining *how* and *why* this surge in the value of the U.S. dollar is occurring. If one attempts to explain *how* and *why* this surge in the value of U.S. dollar is going to affect the U.S. economy, as well as the economies of Japan and Germany, this is another example of analytical research.

Disciplinary, Subject-Matter, and Problem-Solving Research: By disciplinary research, we mean research “designed to improve a discipline” as Johnson(1986) defines it. It dwells on theories, relationships, and analytical procedures and techniques within the discipline. Examples: Economic research or social research. By subject-matter research, we mean research “on a subject of interest” within a discipline. Example: Research in resource economics or in international economics. By problem-solving research, we mean research “designed to solve a specific problem for a specific decision maker”. It is often multidisciplinary. Example: A multidisciplinary study of on the demand for new mass transit involving economics, sociology, and civil engineering. Or a multidisciplinary study of new medical surgery involving medical doctors, engineers, and an economist.

What is Economics and what do Economists do?

What is Economics ?

Case and Fair define economics as “the study of how individuals and societies choose to use the scarce resources that nature and previous generations have provided” Since it deals with the behavior of human beings and their interactions, it is a social science. Stiglitz also defines it as a social science, but he adds that “it studies the social problem of choice from a scientific viewpoint, which means that it is built on a systematic exploration of the problem of choice”.

This systematic exploration involves both the formulation of *theories* and the examination of *data* to test the validity of theories. This latter definition of economics by Stiglitz is helpful in identifying two major tasks of economists: They are *observing* facts and trends in data and *explaining* and interpreting certain or whole aspect of the economy. From the above definitions of economics, the importance of both theory and measurement is obviously the essential ingredients of meaningful and serious economic research. Theory without measurement is unfruitful and empty; measurement without theory is equally meaningless.

A theory consists of a set of assumptions (or hypotheses), and conclusions derived from those assumptions. Theories are logical exercises: if the assumptions hold, then the results follow. Examples. When theory is formally presented, it is called model. Examples of model car, ship, or airplane to understand how economists use models. Bamoul and Bliner’s definition of theory: “A theory is a deliberate simplification of relationships whose purpose is to explain how those relationships work. It is an explanation of the mechanism behind observed phenomena. The relationship is usually couched in terms of the relationship between or among variables. What is variable? A variable is anything which varies over time and space; then the measurement of variable in terms of numbers or facts expressed in quantitative terms are data.

The data are then quantitative information on various aspects of economics and business. We need to learn how to describe and analyze data. The description of data is one area of statistics: descriptive statistics. Data analysis involves the analysis of statistical data of one variable (univariate distribution). Data analysis also analyzes the relationship between two variables (bivariate distribution). Data analysis may involve the analysis of among more than two variables (multivariate distribution). The presentation of data can either be tabular and graphical. In this part, both the description and computation, illustration, using any easily available computer softwares are to be presented side by side.

What do Economists do?

In a certain sense, economics is really what economists do. Two major tasks on which economists spend their energies are: First, observing facts and trends in data and, second, explaining the relationship(s) between them in a cause-effect fashion.

Observing Facts and Trends in Data

A *fact* is a verifiable observation or phenomenon. It is an observed fact that retail gasoline prices in California are higher than elsewhere in the U.S.

The problem is then to explain why and how this is so. A *fact(s)* is usually presented in terms of *variable(s)*. A *variable* is a quantity of something which varies and the researcher is interested in. There are two types of variables: discrete (or discontinuous) and continuous variables. When a *variable(s)* is continuous, the researcher can identify *atrend* in it. Examples of trend are a linear trend, parabolic or quadratic trend. There are several publications containing fact and trends of economic variables on the overall U.S. economic conditions. The first is the *Statistical Abstract of US and Economic Report of the President*, the annual publication by the Council of Economic Advisors.

Two other well-known publications on more current economic situations are *Economic Indicators*, a monthly publication prepared by the Council of Economic Advisors and *Economic Trends*, another monthly publication by the Federal Reserve Bank of Cleveland. To obtain specific information on some specific aspect of the economy, say health care or energy, one needs to consult specialized publications.

Some examples of interesting facts and trends in the US economy during the last two decades are:

- Health care spending in the U.S. has increased faster than other spending categories than before.
- The Budget deficit and national debt have increased faster than before.
- Foreign exchange rates have fluctuated more widely than before.

When a fact or phenomenon is measured (quantified), organized, and presented in a desirable fashion, we call them *data*. Data provide factual information. They are usually regarded as reliable subject to the implicit understanding that there may be some sampling or measurement errors in the process of collection.

To learn how to work with macroeconomics data on the U.S. economy, consider the following set of major economics data on the U. S. economy for the period from 1960 through 1995. Home Work Assignment.

Spread sheet data & How to Use Excel for Data Analysis [Ed's work]

Measurement of variables in level, first difference, index numbers, percentage changes, per capital measures, converting nominal to real variables, fixed weight vs. chain weight GDP measure, adjustments to data (seasonal adjustments, various moving averages, normalized variables etc. Use of spread sheet (Excel) to manipulate data and graph. When we examine facts and trend in the variable (s), we often present information in a table or in a graph. When we examine facts and trend, we need to be mindful of the distinction between *correlation* vs. *causation* between the two variables.

Explaining and Predicting the Relationship

Even though fact or phenomena provides useful factual knowledge, fact alone is not interesting. Even though facts are interesting, they are not very useful by themselves.

As C.S. Peirce expressed so clearly, “a person can stare stupidly at phenomena; but in the absence of imagination they will not connect themselves together in any rational way”^{*} More interesting and useful are raising analytical questions regarding facts. With the first fact on rapid increase in health-care spending, one can ask *why* and *how* health-care spending increased faster than spending in other sectors? Relating to this question, one can also ask *why* and *how* the price of health-care increased faster than other prices? Observing another fact, say the large fluctuations in foreign exchange rates, one may ask whether or not, and *why* and *how* a change in the international monetary arrangement might affect the pattern of change in exchange rate. To answer this question, we need to have a measure for various exchange rates. They are bilateral versus multilateral, or the effective exchange rate, and the nominal versus real exchange rate.

Given the first fact, the problem is to explain *why and how* retail gasoline prices have increased during this particular period and to explain *why and how* retail gasoline prices in California are substantially higher than in the United States.

Explaining the relationship between the share of health care expenditure out of GDP requires the use of the scientific method. Therefore, it is necessary for us to be familiar with concepts of research, such as hypothesis, theory, and model. We also have to learn operational terms of scientific method, such as variables, assumptions, parameters, and functional form.

Explaining and predicting requires the formulation of economic theories as to specific issue under consideration and testing the validity of theories.

METHODS

TYPES OF RESEARCH DESIGNS

Different types of research designs have different advantages and disadvantages. The design is the structure of any scientific work. It gives direction and systematizes the research.

The method you choose will affect your results and how you conclude the findings. Most scientists are interested in getting reliable observations that can help the understanding of a phenomenon.

There are two main approaches to a research problem:

Quantitative Research

These experiments are sometimes referred to as true science, and use traditional mathematical and statistical means to measure results conclusively.

They are most commonly used by physical scientists, although social sciences, education and economics have been known to use this type of research. It is the opposite of qualitative research.

Quantitative experiments all use a standard format, with a few minor inter-disciplinary differences, of generating a hypothesis to be proved or disproved. This hypothesis must

be provable by mathematical and statistical means, and is the basis around which the whole experiment is designed. Randomization of any study groups is essential, and a control group should be included, wherever possible. A sound quantitative design should only manipulate one variable at a time, or statistical analysis becomes cumbersome and open to question.

Ideally, the research should be constructed in a manner that allows others to repeat the experiment and obtain similar results.

Advantages

Quantitative research design is an excellent way of finalizing results and proving or disproving a hypothesis. The structure has not changed for centuries, so is standard across many scientific fields and disciplines.

After statistical analysis of the results, a comprehensive answer is reached, and the results can be legitimately discussed and published. Quantitative experiments also filter out external factors, if properly designed, and so the results gained can be seen as real and unbiased.

Quantitative experiments are useful for testing the results gained by a series of qualitative experiments, leading to a final answer, and a narrowing down of possible directions for follow up research to take.

Disadvantages

Quantitative experiments can be difficult and expensive and require a lot of time to perform. They must be carefully planned to ensure that there is complete randomization and correct designation of control groups.

Quantitative studies usually require extensive statistical analysis, which can be difficult, due to most scientists not being statisticians. The field of statistical study is a whole scientific discipline and can be difficult for non-mathematicians

In addition, the requirements for the successful statistical confirmation of results are very stringent, with very few experiments comprehensively proving a hypothesis; there is usually some ambiguity, which requires retesting and refinement to the design. This means another investment of time and resources must be committed to fine-tune the results.

Quantitative research design also tends to generate only proved or unproven results, with there being very little room for grey areas and uncertainty. For the social sciences, education, anthropology and psychology, human nature is a lot more complex than just a simple yes or no response.

Qualitative Research

Qualitative research design is a research method used extensively by scientists and researchers studying human behaviour and habits. It is also very useful for product designers who want to make a product that will sell.

For example, a designer generating some ideas for a new product might want to study people's habits and preferences, to make sure that the product is commercially viable. Quantitative research is then used to assess whether the completed design is popular or not.

Qualitative research is often regarded as a precursor to quantitative research, in that it is often used to generate possible leads and ideas which can be used to formulate a realistic and testable hypothesis. This hypothesis can then be comprehensively tested and mathematically analyzed, with standard quantitative research methods.

For these reasons, these qualitative methods are often closely allied with survey design techniques and individual case studies, as a way to reinforce and evaluate findings over a broader scale.

One example of a qualitative research design might be a survey constructed as a precursor to the paper towel experiment. A study completed before the experiment was performed would reveal which of the multitude of brands were the most popular. The quantitative experiment could then be constructed around only these brands, saving a lot of time, money and resources.

Qualitative methods are probably the oldest of all scientific techniques, with Ancient Greek philosophers qualitatively observing the world around them and trying to come up with answers which explained what they saw.

Design

The design of qualitative research is probably the most flexible of the various experimental techniques, encompassing a variety of accepted methods and structures.

From an individual case study to an extensive survey, this type of study still needs to be carefully constructed and designed, but there is no standardized structure. Case studies and survey designs are the most commonly used methods.

Advantages

Qualitative techniques are extremely useful when a subject is too complex to be answered by a simple yes or no hypothesis. These types of designs are much easier to plan and carry out, useful when budgetary decisions have to be taken into account.

The broader scope covered by these designs ensures that some useful data is always generated, whereas an unproved hypothesis in a quantitative experiment can mean that a lot of time has been wasted. Qualitative research methods are not as dependent upon sample sizes as quantitative methods; a case study, for example, can generate meaningful results with a small sample group.

Disadvantages

Whilst not as time or resource consuming as quantitative experiments, qualitative methods still require a lot of careful thought and planning, to ensure that the results obtained are as accurate as possible.

Qualitative data cannot be mathematically analyzed in the same comprehensive way as quantitative results, so can only give a guide to general trends. It is a lot more open to personal opinion and judgment, and so can only ever give observations rather than results. Any qualitative research design is usually unique and cannot be exactly recreated, meaning that they do lack the ability to be peer reviewed.

DIFFERENT RESEARCH METHODS

True Experimental Design

True experimental design is regarded as the most accurate form of experimental research, in that it tries to prove or disprove a hypothesis mathematically, with statistical analysis.

For some of the physical sciences, such as physics, chemistry and geology, they are standard and commonly used. For social sciences, psychology and biology, they can be a little more difficult to set up.

For an experiment to be classed as a true experimental design, it must fit all of the following criteria.

The sample groups must be assigned randomly.

There must be a viable control group.

Only one variable can be manipulated and tested. It is possible to test more than one, but such experiments and their statistical analysis tend to be cumbersome and difficult.

The tested subjects must be randomly assigned to either control or experimental groups.

Advantages

The results of a true experimental design can be statistically analyzed and so there can be little argument about the results.

It is also much easier for other researchers to replicate the experiment and validate the results.

For physical sciences working with mainly numerical data, it is much easier to manipulate one variable, so true experimental design usually gives a yes or no answer.

Disadvantages

Whilst perfect in principle, there are a number of problems with this type of design. Firstly, they can be almost too perfect, with the conditions being under complete control and not being representative of real world conditions. For psychologists and behavioral biologists, for example, there can never be any guarantee that a human or living organism will exhibit 'normal' behaviour under experimental conditions.

True experiments can be too accurate and it is very difficult to obtain a complete rejection or acceptance of a hypothesis because the standards of proof required are so difficult to reach.

True experiments are also difficult and expensive to set up. They can also be very impractical.

While for some fields, like physics, there are not as many variables so the design is easy, for social sciences and biological sciences, where variations are not so clearly defined it is much more difficult to exclude other factors that may be affecting the manipulated variable.

True experimental design is an integral part of science, usually acting as a final test of a hypothesis. Whilst they can be cumbersome and expensive to set up, literature reviews, qualitative research and descriptive research can serve as a good precursor to generate a testable hypothesis, saving time and money.

Whilst they can be a little artificial and restrictive, they are the only type of research that is accepted by all disciplines as statistically provable.

Quasi-experimental Design

Quasi-experimental design is a form of experimental research used extensively in the social sciences and psychology

Whilst regarded as unscientific and unreliable, by physical and biological scientists, the method is, nevertheless, a very useful method for measuring social variables. The inherent weaknesses in the methodology do not undermine the validity of the data, as long as they are recognized and allowed for during the whole experimental process. Quasi experiments resemble quantitative and qualitative experiments, but lack random allocation of groups or proper controls, so firm statistical analysis can be very difficult.

Design

Quasi-experimental design involves selecting groups, upon which a variable is tested, without any random pre-selection processes.

For example, to perform an educational experiment, a class might be arbitrarily divided by alphabetical selection or by seating arrangement. The division is often convenient and, especially in an educational situation, causes as little disruption as possible.

After this selection, the experiment proceeds in a very similar way to any other experiment, with a variable being compared between different groups, or over a period of time.

Advantages

Especially in social sciences, where pre-selection and randomization of groups is often difficult, they can be very useful in generating results for general trends.

E.g. if we study the effect of maternal alcohol use when the mother is pregnant, we know that alcohol does harm embryos. A strict experimental design would include that mothers were randomly assigned to drink alcohol. This would be highly illegal because of the possible harm the study might do to the embryos.

So what researchers does is to ask people how much alcohol they used in their pregnancy and then assign them to groups.

Quasi-experimental design is often integrated with individual case studies; the figures and results generated often reinforce the findings in a case study, and allow some sort of statistical analysis to take place.

In addition, without extensive pre-screening and randomization needing to be undertaken, they do reduce the time and resources needed for experimentation.

Disadvantages

Without proper randomization, statistical tests can be meaningless. For example, these experimental designs do not take into account any pre-existing factors (as for the mothers: what made them drink or not drink alcohol), or recognize that influences outside the experiment may have affected the results. A quasi experiment constructed to analyse the effects of different educational programs on two groups of children, for example, might generate results that show that one programme is more effective than the other.

These results will not stand up to rigorous statistical scrutiny because the researcher also need to control other factors that may have affected the results. This is really hard to do properly. One group of children may have been slightly more intelligent or motivated. Without some form of pre-testing or random selection, it is hard to judge the influence of such factors.

Disadvantages aside, as long as the shortcomings of the quasi-experimental design are recognized, these studies can be a very powerful tool, especially in situations where 'true' experiments are not possible.

They are very good way to obtain a general overview and then follow up with a case study or quantitative experiment, to focus on the underlying reasons for the results generated.

Double Blind Experiment

A double blind experiment is an experimental method used to ensure impartiality, and avoid errors arising from bias. It is very easy for a researcher, even subconsciously, to influence experimental observations, especially in behavioral science, so this method provides an extra check. For example, imagine that a company is asking consumers for opinions about its products, using a survey. There is a distinct danger that the interviewer may subconsciously emphasize the company's products when asking the questions. This is the major reason why market research companies generally prefer to use computers, and double blind experiments, for gathering important data.

The Blind Experiment

The blind experiment is the minimum standard for any test involving subjects and opinions, and failure to adhere to this principle may result in experimental flaws.

The idea is that the groups studied, including the control, should not be aware of in which group they are placed. In medicine, when researchers are testing a new medicine, they ensure that the placebo looks, and tastes, the same as the actual medicine.

There is strong evidence of a placebo effect with medicine, where, if people believe that they are receiving a medicine, they show some signs of improvement in health. A blind experiment reduces the risk of bias from this effect, giving an honest baseline for the research, and allowing a realistic statistical comparison.

The Double Blind E-xperiment

The double blind experiment takes this precaution against bias one-step further, by ensuring that the researcher does not know in which group a patient falls.

Whilst the vast majority of researchers are professionals, there is always a chance that the researcher might subconsciously tip off a patient about the pill they were receiving. They may even favour giving the pill to patients that they thought had the best chance of recovery, skewing the results.

Whilst nobody likes to think of scientists as dishonest, there is often pressure, from billion dollar drug companies and the fight for research grants, to generate positive results.

This always gives a chance that a scientist might manipulate results, and try to show the research in a better light. Proving that the researcher carried out a double blind experiment reduces the chance of criticism.

Other Applications

Whilst better known in medicine, double blind experiments are often used in other fields. Surveys, questionnaires and market research all use this technique to retain credibility.

If you wish to compare two different brands of washing powder, the samples should be in the same packaging. A consumer might have an inbuilt brand identity awareness, and preference, which will lead to favouritism and bias.

An example of the weakness of single blind techniques is in police line-ups, where a witness picks out a suspect from a group. Many legal experts are advocating that these line-ups should be unsupervised, and unprompted.

If the police are fixed on bringing a particular subject to justice, they may consciously, or subconsciously, tip off the witness. Humans are very good at understanding body language and unconscious cues, so the chance of observer's bias should be minimized.

Descriptive Research Design

Descriptive research design is a scientific method which involves observing and describing the behaviour of a subject without influencing it in any way. Many scientific disciplines, especially social science and psychology, use this method to obtain a general overview of the subject.

Some subjects cannot be observed in any other way; for example, a social case study of an individual subject is a descriptive research design and allows observation without affecting normal behaviour. It is also useful where it is not possible to test and measure the large number of samples needed for more quantitative types of experimentation.

These types of experiments are often used by anthropologists, psychologists and social scientists to observe natural behaviors without affecting them in any way. It is also used by market researchers to judge the habits of customers, or by companies wishing to judge the morale of staff. The results from a descriptive research can in no way be used as a definitive answer or to disprove a hypothesis but, if the limitations are understood, they can still be a useful tool in many areas of scientific research.

Advantages

The subject is being observed in a completely natural and unchanged natural environment. A good example of this would be an anthropologist who wanted to study a tribe without affecting their normal behaviour in any way. True experiments, whilst giving analyzable data, often adversely influence the normal behaviour of the subject.

Descriptive research is often used as a pre-cursor to more quantitatively research designs, the general overview giving some valuable pointers as to what variables are worth testing quantitatively. Quantitative experiments are often expensive and time-consuming so it is often good sense to get an idea of what hypotheses are worth testing.

Disadvantages

Because there are no variables manipulated, there is no way to statistically analyse the results. Many scientists regard this type of study as very unreliable and 'unscientific'. In addition, the results of observational studies are not repeatable, and so there can be no replication of the experiment and reviewing of the results.

Descriptive research design is a valid method for researching specific subjects and as a precursor to more quantitative studies. Whilst there are some valid concerns about the statistical validity, as long as the limitations are understood by the researcher, this type of study is an invaluable scientific tool. Whilst the results are always open to question and to different interpretations, there is no doubt that they are preferable to performing no research at all.

Literature Review

Many students are instructed, as part of their research programme, to perform a literature review, without always understanding what a literature review is. Most are aware that it is a process of gathering information from other sources and documenting it, but few have any idea of how to evaluate the information, or how to present it.

A literature review can be a precursor in the introduction of a research paper, or it can be an entire paper in itself, often the first stage of large research projects, allowing the supervisor to ascertain that the student is on the correct path.

A literature review is a critical and in depth evaluation of previous research. It is a summary and synopsis of a particular area of research, allowing anybody reading the paper to establish why you are pursuing this particular research programme. A good literature review expands upon the reasons behind selecting a particular research question.

What is a Literature Review Not

It is not a chronological catalog of all of the sources, but an evaluation, integrating the previous research together, and also explaining how it integrates into the proposed research programme. All sides of an argument must be clearly explained, to avoid bias, and areas of agreement and disagreement should be highlighted.

It is not a collection of quotes and paraphrasing from other sources. A good literature review should also have some evaluation of the quality and findings of the research. A good literature review should avoid the temptation of impressing the importance of a particular research programme.

The fact that a researcher is undertaking the research programme speaks for its importance, and an educated reader may well be insulted that they are not allowed to judge the importance for themselves. They want to be re-assured that it is a serious paper, not a pseudo-scientific sales advertisement.

Whilst some literature reviews can be presented in a chronological order, it is best avoided. For example, a review of Victorian Age Physics, could present J.J. Thomson's famous experiments in a chronological order. Otherwise, this is usually perceived as being a little lazy, and it is better to organize the review around ideas and individual points.

As a general rule, certainly for a longer review, each paragraph should address one point, and present and evaluate all of the evidence, from all of the differing points of view.

Conducting a Literature Review

Evaluating the credibility of sources is one of the most difficult aspects, especially with the ease of finding information on the internet. The only real way to evaluate is through experience, but there are a few tricks for evaluating information quickly, yet accurately.

There is such a thing as 'too much information,' and Google does not distinguish or judge the quality of results, only how search engine friendly a paper is. This is why it is still good practice to begin research in an academic library. Any journals found there can be regarded as safe and credible.

The next stage is to use the internet, and this is where the difficulties start. It is very difficult to judge the credibility of an online paper. The main thing is to structure the internet research as if it were on paper. Bookmark papers, which may be relevant, in one folder and make another subfolder for a 'shortlist.'

The easiest way is to scan the work, using the abstract and introduction as guides. This helps to eliminate the non-relevant work and also some of the lower quality research.

If it sets off alarm bells, there may be something wrong, and the paper is probably of a low quality. Be very careful not to fall into the trap of rejecting research just because it conflicts with your hypothesis. Failure to do this will completely invalidate the literature review and potentially undermine the research project. Any research that may be relevant should be moved to the shortlist folder.

The next stage is to critically evaluate the paper and decide if the research is sufficient quality. Think about it this way: The temptation is to try to include as many sources as possible, because it is easy to fall into the trap of thinking that a long bibliography equates to a good paper. A smaller number of quality sources is far preferable than a long list of irrelevance.

Check into the credentials of any source upon which you rely heavily for the literature review. The reputation of the University or organization is a factor, as is the experience of the researcher.

If their name keeps cropping up, and they have written many papers, the source is usually OK. Look for agreements. Good research should have been replicated by other independent researchers, with similar results, showing that the information is usually fairly safe to use.

If the process is proving to be difficult, and in some fields, like medicine and environmental research, there is a lot of poor science, do not be afraid to ask a supervisor for a few tips. They should know some good and reputable sources to look at. It may be a little extra work for them, but there will be even more work if they have to tear apart a review because it is built upon shaky evidence.

Conducting a good literature review is a matter of experience, and even the best scientists have fallen into the trap of using poor evidence. This is not a problem, and is part of the scientific process; if a research programme is well constructed, it will not affect the results.

Case Study Research Design

The case study research design have evolved over the past few years as a useful tool for investigating trends and specific situations in many scientific disciplines, especially social science, psychology, anthropology and ecology.

This method of study is especially useful for trying to test theoretical models by using them in real world situations. For example, if an anthropologist were to live amongst a remote tribe, whilst their observations might produce no quantitative data, they are still useful to science.

What is a Case Study

Basically, a case study is an in depth study of a particular situation rather than a sweeping statistical survey. It is a method used to narrow down a very broad field of

research into one easily researchable topic. Whilst it will not answer a question completely, it will give some indications and allow further elaboration and hypothesis creation on a subject.

The case study research design is also useful for testing whether scientific theories and models actually work in the real world. You may come out with a great computer model for describing how the ecosystem of a rock pool works but it is only by trying it out on a real life pool that you can see if it is a realistic simulation.

For psychologists, anthropologists and social scientists they have been regarded as a valid method of research for many years. Scientists are sometimes guilty of becoming bogged down in the general picture and it is sometimes important to understand specific cases and ensure a more holistic approach to research.

Case Study Research Design

Some argue that because a case study is such a narrow field that its results cannot be extrapolated to fit an entire question and that they show only one narrow example. On the other hand, it is argued that a case study provides more realistic responses than a purely statistical survey.

The truth probably lies between the two and it is probably best to try and synergize the two approaches. It is valid to conduct case studies but they should be tied in with more general statistical processes.

For example, a statistical survey might show how much time people spend talking on mobile phones, but it is case studies of a narrow group that will determine why this is so.

The other main thing to remember during case studies is their flexibility. Whilst a pure scientist is trying to prove or disprove a hypothesis, a case study might introduce new and unexpected results during its course, and lead to research taking new directions.

The argument between case study and statistical method also appears to be one of scale. Whilst many 'physical' scientists avoid case studies, for psychology, anthropology and ecology they are an essential tool. It is important to ensure that you realise that a case study cannot be generalized to fit a whole population or ecosystem.

Finally, one peripheral point is that, when informing others of your results, case studies make more interesting topics than purely statistical surveys, something that has been realized by teachers and magazine editors for many years. The general public has little interest in pages of statistical calculations but some well placed case studies can have a strong impact.

How to Design and Conduct a Case Study

The advantage of the case study research design is that you can focus on specific and interesting cases. This may be an attempt to test a theory with a typical case or it can be a specific topic that is of interest. Research should be thorough and note taking should be meticulous and systematic.

The first foundation of the case study is the subject and relevance. In a case study, you are deliberately trying to isolate a small study group, one individual case or one particular population.

For example, statistical analysis may have shown that birthrates in African countries are increasing. A case study on one or two specific countries becomes a powerful and focused tool for determining the social and economic pressures driving this. In the design of a case study, it is important to plan and design how you are going to address the study and make sure that all collected data is relevant. Unlike a scientific report, there is no strict set of rules so the most important part is making sure that the study is focused and concise; otherwise you will end up having to wade through a lot of irrelevant information.

It is best if you make yourself a short list of 4 or 5 bullet points that you are going to try and address during the study. If you make sure that all research refers back to these then you will not be far wrong.

With a case study, even more than a questionnaire or survey, it is important to be passive in your research. You are much more of an observer than an experimenter and you must remember that, even in a multi-subject case, each case must be treated individually and then cross case conclusions can be drawn.

How to Analyse the Results

Analyzing results for a case study tends to be more opinion based than statistical methods. The usual idea is to try and collate your data into a manageable form and construct a narrative around it.

Use examples in your narrative whilst keeping things concise and interesting. It is useful to show some numerical data but remember that you are only trying to judge trends and not analyse every last piece of data. Constantly refer back to your bullet points so that you do not lose focus.

It is always a good idea to assume that a person reading your research may not possess a lot of knowledge of the subject so try to write accordingly. In addition, unlike a scientific study which deals with facts, a case study is based on opinion and is very much designed to provoke reasoned debate. There really is no right or wrong answer in a case study.

Survey Research Design

The survey research design is often used because of the low cost and easy accessible information. Conducting accurate and meaningful surveys is one of the most important facets of market research in the consumer driven 21st century.

Businesses, governments and media spend billions of dollars on finding out what people think and feel.

Accurate research can generate vast amounts of revenue; bad or inaccurate research can cost millions, or even bring down governments. The survey research design is a

very valuable tool for assessing opinions and trends. Even on a small scale, such as local government or small businesses, judging opinion with carefully designed surveys can dramatically change strategies.

Television chat-shows and newspapers are usually full of facts and figures gleaned from surveys but often no information is given as to where this information comes from or what kind of people were asked.

A cursory examination of these figures usually shows that the results of these surveys are often manipulated or carefully sifted to try and reflect distort the results to match the whims of the owners.

Businesses are often guilty of carefully selecting certain results to try and portray themselves as the answer to all needs. When you decide to enter this minefield and design a survey, how do you avoid falling into the trap of inaccuracy and bias? How do you ensure that your survey research design reflects the views of a genuine cross-section of the population?

The simple answer is that you cannot; even with unlimited budget, time and resources, there is no way of achieving 100% accuracy. Opinions, on all levels, are very fluid and can change on a daily or even hourly basis.

Despite this, surveys are still a powerful tool and can be an extremely powerful research tool. As long as you design your survey well and are prepared to be self-critical, you can still obtain an accurate representation of opinion.

Establishing the Aims of Your Research

This is the single most important step of your survey research design and can make or break your research; every single element of your survey must refer back to this design or it will be fatally flawed.

If your research is too broad, you will have to ask too many questions; too narrow and you will not be researching the topic thoroughly enough.

Researching And Determining Your Sample Group

This is the next crucial step in determining your survey and depends upon many factors.

The first is accuracy; you want to try and interview as broad a base of people as possible. Quantity is not always the answer; if you were researching a detergent, for example, you would want to target your questions at those who actually use such products.

For a political or ethical survey, about which anybody can have a valid opinion, you want to try and represent a well balanced cross section of society.

It is always worth checking beforehand what quantity and breadth of response you need to provide significant results or your hard work may be in vain.

Before you start the planning, it is important that you consult somebody about the statistical side of your survey research design. This way, you know what number and type of responses you need to make it a valid survey and prevent inaccurate results.

Methodology

How do you make sure that your questionnaire reaches the target group? There are many methods of reaching people but all have advantages and disadvantages. For a college or university study it is unlikely that you will have the facilities to use internet, e-mail or phone surveying so we will concentrate on only the likely methods you will use.

Face to Face

This is probably the most traditional method of the survey research design. It can be very accurate. It allows you to be selective about to whom you ask questions and you can explain anything that they do not understand.

In addition, you can make a judgment about who you think is wasting your time or giving stupid answers.

There are a few things to be careful of with this approach; firstly, people can be reluctant to give up their time without some form of incentive. Another factor to bear in mind is that is difficult to ask personal questions face to face without embarrassing people. It is also very time consuming and difficult to obtain a representative sample.

Finally, if you are going to be asking questions door-to-door, it is essential to ensure that you have some official identification to prove who you are.

Mail

This does not necessarily mean using the postal service; putting in the legwork and delivering questionnaires around a campus or workplace is another method.

This is a good way of targeting a certain section of people and is excellent if you need to ask personal or potentially embarrassing questions. The problems with this method are that you cannot be sure of how many responses you will receive until a long time period has passed. You must also be wary of collecting personal data; most countries have laws about how much information you can keep about people so it is always wise to check with somebody more knowledgeable.

Structuring and Designing the Questionnaire

The design of your questionnaire depends very much upon the type of survey and the target audience. If you are asking questions face to face it is easy to explain if people are unsure of a question. On the other hand, if your questionnaire is going to include many personal questions then mailing methods are preferable (but may violate local legislation). You must keep your questionnaire as short as possible; people will either refuse to fill in a long questionnaire or get bored halfway through.

Cover Note

It is also polite, especially with mailed questionnaires, to send a short cover note explaining what you are doing and how the subject should return the surveys to you.

You should introduce yourself; explain why you are doing the research, what will happen with the results and who to contact if the subject has any queries.

Types Of Question

Multiple choice questions allow many different answers, including don't know, to be assessed. The main strength of this type of question is that the form is easy to fill in and the answers can be checked easily and quantitatively; this is useful for large sample groups.

Rating, on some scale, is a tried and tested form of question structure. This way is very useful when you are seeking to be a little more open-ended than is possible with multiple choice questions. It is a little harder to analyse your responses. It is important to make sure that the scale allows extreme views.

Questions asking for opinions must be open-ended and allow the subject to give their own response; you should avoid entrapment and appear to be as neutral as possible during the procedure. The major problem is that you have to devise a numerical way of analyzing and statistically evaluating the responses which can lead to a biased view, if care is not taken. These types of question should really be reserved for experienced researchers.

The order in which you ask the questions can be important. Try to start off with the most relevant questions first. Also friendly and non-threatening questions put the interviewee at ease. Questions should be simple and straightforward using everyday language rather than perfect grammar.

Try and group questions about similar topics together; this makes it a lot quicker for people to answer questions more quickly and easily.

Some researchers advocate mixing up and randomizing questions for accuracy but this approach tends to be more appropriate for advanced market research. For this type of survey the researcher is trying to disguise the nature of the research and filter out preconceptions.

It is also a good idea to try out a test survey; ask a small group to give genuine and honest feedback so that you can make adjustments. Common mistakes when doing the survey research design.

Analyzing Your Results

This is where the fun starts and it will depend upon the type of questions used.

For multiple choice questions it is a matter of counting up the answers to each question and using statistics to 'crunch the numbers' and test relevance.

Rating type question require a little more work but they follow broadly the same principle. For opinion questions you have to devise some way of judging the responses numerically.

The next step is to devise which statistical test you are going to use and start to enter some numbers to judge the significance of your data.

This is where you have to analyse the results. Be self critical whether your results showed what you expected or not. Any survey has flaws in its method so it is always a good idea to show that you are aware of these.

For example, a university represents only a narrow cross section of society; as long as you are aware of this then your results are valid. If your survey gave unexpected results explain the possible reasons for why this happened and suggestions for refining the techniques and structure of your survey next time.

Identical Twins Study

The identical twins study has been used for a long time, to study the effects of environment and genetics on human development.

Some studies have tried to determine how genetics and environmental factors contribute to intelligence, aggression or substance addictions.

Most of the twin's studies compare identical twins, having 100% genetic similarity, with non identical twins, with about 50% genetic similarity. The researcher compares the occurrence of an individual trait between identical and fraternal twins. If the identical twins show more similarity for this trait than the non-identical twins, then the excess is assumed to be down to genetic factors.

This type of analysis would then allow the researchers to estimate the heritability of specific traits and quantify the effect of genetic factors on the individual trait. Psychologists have long known that a twin study is not a true experimental design, but it has led to some interesting insights into the influence of genes on human behaviour.

For this method, a number of assumptions have to be made; that the identical twins share identical DNA profiles, and that the environmental factors are the same for all participants.

Criticisms

There have been few criticisms of identical twin studies over the years. By their nature, and because of small sample sizes, it is very difficult to quantitatively analyse the results and so all experimentation tends to be observational; the sample groups cannot be random so statistical analysis is impossible.

The experimental methods assume that there is little difference in the environmental factors between fraternal and identical twins, but there is a criticism that the tendency of adults to treat identical twins in exactly the same way makes this assumption invalid. Parents tend to dress identical twins the same way and encourage them to pursue the same interests.

The distinction between environmental factors and genetic influences may not be as black and white as the identical twins study assumes. There is probably an interaction between genes and environment and so the whole picture may be a lot more complex.

In addition, the experiment tends to assume that one gene affects one behavioral trait. Modern genetic research is showing that many different genes can influence behaviour.

The above criticisms all have some validity, but the main point is that twin studies have never claimed to be anything other than observational, identifying and trying to explain trends rather than prove a hypothesis.

Whilst there are some concerns about the validity of the identical twins study, such experiments are certainly better than performing no research at all.

Twins studies are now trying to analyse the environmental factors more. Instead of assuming that the environmental factors are the same, they are now contrasting shared family environment with the individual events suffered by the individual twin.

In addition, identical twins study is constantly evolving into more complex forms, now taking into account whole families and other siblings in addition to the twins.

Research into the human genome is now resurrecting the studies of twins; hereditary trends observed in an identical twins study can now be studied quantitatively in the laboratory. It is now standard practice, when conducting twin's research to analyse DNA from all participants and this is bypassing many of the concerns about the twin study.

Meta Analysis

Meta analysis is a statistical technique developed by social scientists, who are very limited in the type of experiments they can perform.

Social scientists have great difficulty in designing and implementing true experiments, so meta-analysis gives them a quantitative tool to analyse statistically data drawn from a number of studies, performed over a period of time.

Medicine and psychology increasingly use this method, as a way of avoiding time-consuming and intricate studies, largely repeating the work of previous research.

What is meta-analysis

Social studies often use very small sample sizes, so any statistics used generally give results containing large margins of error.

This can be a major problem when interpreting and drawing conclusions, because it can mask any underlying trends or correlations. Such conclusions are only tenuous, at best, and leave the research open for criticism.

Meta analysis is the process of drawing from a larger body of research, and using powerful statistical analyzes on the conglomerated data.

This gives a much larger sample population and is more likely to generate meaningful and usable data.

The Advantages Of Meta Analysis

Meta analysis is an excellent way of reducing the complexity and breadth of research, allowing funds to be diverted elsewhere. For rare medical conditions, it allows researchers to collect data from further afield than would be possible for one research group.

As the method becomes more common, database programs have made the process much easier, with professionals working in parallel able to enter their results and access the data. This allows constant quality assessments and also reducing the chances of unnecessary repeat research, as papers can often take many months to be published, and the computer records ensure that any researcher is aware of the latest directions and results.

The field of meta study is also a lot more rigorous than the traditional literature review, which often relies heavily upon the individual interpretation of the researcher.

When used with the databases, a meta study allows a much wider net to be cast than by the traditional literature review, and is excellent for highlighting correlations and links between studies that may not be readily apparent as well as ensuring that the compiler does not subconsciously infer correlations that do not exist.

The Disadvantages Of Meta Analysis

There are a number of disadvantages to meta analysis, of which a researcher must be aware before relying upon the data and generated statistics.

The main problem is that there is the potential for publication bias and skewed data.

Research generating results not refuting a hypothesis may tend to remain unpublished, or risks not being entered into the database. If the meta study is restricted to the research with positive results, then the validity is compromised.

The researcher compiling the data must make sure that all research is quantitative, rather than qualitative, and that the data is comparable across the various research programs, allowing a genuine statistical analysis.

It is important to pre-select the studies, ensuring that all of the research used is of a sufficient quality to be used.

One erroneous or poorly conducted study can place the results of the entire meta analysis at risk. On the other hand, setting almost unattainable criteria and criteria for inclusion can leave the meta study with too small a sample size to be statistically relevant.

Striking a balance can be a little tricky, but the whole field is in a state of constant development, incorporating protocols similar to the scientific method used for normal quantitative research.

Finding the data is rapidly becoming the real key, with skilled meta analysts developing a skill-set of library based skills, finding information buried in government reports and conference data, developing the knack of assessing the quality of sources quickly and effectively.

Conclusions And The Future

Meta analysis is here to stay, as an invaluable tool for research, and is rapidly gaining momentum as a stand-alone discipline, with practitioners straddling the divide between statisticians and librarians.

The conveniences, as long as the disadvantages are taken into account, are too apparent to ignore, and a meta study can reduce the need for long, expensive and potentially intrusive repeated research studies.

Systematic Reviews

Heavily used by the healthcare sector, systematic reviews are a powerful way of isolating and critically evaluating previous research.

Modern medical research generates so much literature, and fills so many journals, that a traditional literature review could take months, and still be out of date by the time that the research is designed and performed.

In addition, researchers are often guilty of selecting the research best fitting their pre-conceived notions, a weakness of the traditional 'narrative' literature review process.

To help medical professionals, specialist compilers assess and condense the research, entering it into easily accessible research databases. They are an integral part of the research process, and every student of medicine routinely receives a long and extensive training in the best methods for critically evaluating literature.

Addressing the Deficiencies in Narration

The problems with narrative literature came to light a couple of decades ago, when critics realized that reviewers looking at the same body of evidence often generated completely different findings. They drew conclusions based upon their specialty, rather than the compelling evidence contained within the body of research.

It is unclear whether this was a case of conscious or subconscious manipulation (bias), but this particular finding was worrying, especially in a research area where life and death could be at stake.

To address this issue, medical authorities developed a new protocol of systematic reviewing, based upon a structure as strict as the scientific method governing empirical research programs.

The Protocols Underpinning Systematic Reviews

Define a research question, in a similar way to formulating a research question for a standard research design

Locate and select relevant previous research studies, with no attempt at evaluation at this stage. Ideally, research in languages other than English should be used, and the researcher should try to find papers and reports unpublished in journals, such as conference speeches or company reports.

Critically evaluate the studies. The reviewer should assess each study upon criteria based upon quality, strength of the findings and validity. For safety, this process should include at least two independent reviewers, although a greater number is advisable. Combine the results. This is the process of combining all of the findings, sometimes qualitatively, but usually quantitatively, using meta-analysis.

Publish the results. As with any research, the results have to be written and published, usually with a system of independent review. Discussion of the conclusions, as with any research, allows the validity of the findings to be verified.

The Reasoning Behind Systematic Reviews

The principle behind the systematic reviews process is that the researcher critically evaluates previous studies, in a much more comprehensive and systematic way than a standard literature review. In many cases, statistical meta-analysis tools are used to give the review a quantitative foundation, allowing correlations to be documented and conclusions to be drawn. Whilst the techniques are mainly used by medicine and psychology, there is a growing trend towards using systematic reviews in other disciplines. Many branches of science are becoming increasingly fragmented and anarchic, so this layer of analysis aggregates all of the disparate elements.

Systematic reviews, and meta-analysis, are regarded as a cornerstone of healthcare research, essential where it is impractical or unethical to keep repeating old research.

In addition to the potential risks of repeated research upon patients and volunteers, there are now laws in many countries prohibiting excessive research using animals. Systematic reviews are a great way of reducing the amount of suffering caused by vivisection.

Disadvantages Of Systematic Reviews

As with most systems, despite the protocols, systematic reviews do have some inherent weaknesses. The main problem is the rapid advancement of medical research and technology, often meaning that many reviews are out of date before they are even published, forcing researchers to update their findings constantly. The development of specialist organizations for finding and evaluating data minimizes the effects of this particular shortcoming.

As with any subjective review, there is the problem of selection bias, where contradictory research is jettisoned, although most medical researchers are adept at following the proper procedures.

Funding and research grants cause researchers to try to find results that suit their paymasters, a growing problem in many areas of science, not just medicine. The specialist reviewers sidestep this problem, to a certain extent, by producing independent research, uncorrupted by governmental or private healthcare funding, curbing the worst excesses.

Often, a blind system is used, and reviewers are unaware of where the papers they are reviewing came from, or who they are written by. This lessens allegations of favouritism and judging research by the reputation of the researcher rather than on merit. Ultimately, the onus is on the reader to draw their own assessments, using their own experience to judge the quality of the systematic review. Whilst not a perfect system, systematic reviews are far superior to the traditional narrative approach, which often allows a lot of good research to fall through the cracks.

Observational Study

In the fields of social science, psychology and medicine, amongst others, observational study is an essential tool.

This type of research draws a conclusion by comparing subjects against a control group, in cases where the researcher has no control over the experiment.

A research study comparing the risk of developing lung cancer, between smokers and non-smokers, would be a good example of an observational study.

The main reason for performing any observational research is due to ethical concerns.

With the smoking example, a scientist cannot give cigarettes to non-smokers for 20 years and compare them with a control group. This also brings up the other good reason for such studies, in that few researchers can study the long-term effects of certain variables, especially when it runs into decades.

For this study of long-term and subtle effects, they have to use pre-existing conditions and medical records. The researcher may want to study an extremely small sample group, so it is easier to start with known cases and works backwards. The thalidomide cases, for example, are an example of an observational study where researchers had to work backwards, and establish that the drug was the cause of disabilities.

The main problem with observational studies is that the experimenter has no control over the composition of the control groups, and cannot randomize the allocation of subjects. This can create bias, and can also mask cause and effect relationships or, alternatively, suggest correlations where there are none (error in research).

For example, in the smoking example, if the researcher found that there is a correlation between smoking and increased rates of lung cancer, without knowing the full and complete background of the subjects, there is no way of determining whether other factors were involved, such as diet, occupation or genetics. Randomization is assumed to even out external causal effects, but this is impossible in an observational study. There is no independent variable, so it is dangerous to assume cause and effect relationships, a process often misunderstood by the mass media lauding the next wonder food, or sensationalizing a political debate with unfounded results and pseudo-science.

Despite the limitations, an observational study allows a useful insight into a phenomenon, and sidesteps the ethical and practical difficulties of setting up a large and cumbersome medical research project.

Naturalistic Observation

In many scientific disciplines, naturalistic observation is a useful tool for expanding knowledge about a specific phenomenon or species.

In fields such as anthropology, behavioral biology and ecology, watching a person or organism in a natural environment is essential.

Most naturalistic observation is unobtrusive, such as a researcher setting up a camera to film the behaviour of a badger underground. Most nature documentaries are examples of naturalistic observational study, where days, weeks or even years of film are analyzed

and edited, to give an overview of the life cycle of the organism. There is often little attempt at analysis, quantitative or qualitative, but the observational study does uncover unknown phenomena and behaviors.

Obtrusive naturalistic observational study is often used in anthropology, where a researcher lives with a remote tribe for a period of time and records their behaviour. By living there, she is influencing their social interactions and habits, but can still make some excellent observations.

Often, anthropologists will adopt the lifestyle of a particular group of people, in an attempt to understand why they have certain customs and beliefs.

In technical terms, it would be difficult to follow people without discovery, and it would also be unethical to observe without consent, so obtrusive naturalistic observation is the only method that can be used with human subjects.

Many of the producers of the recent glut of reality shows try to claim that their shows are psychological experiments, based around observational study. This is stretching the idea too far, as there are very few people who would not change their behaviour when they are aware that a camera is watching.

In these cases, it is difficult to make any realistic and valid observations about their lifestyle.

Most criticisms of naturalistic observation are based around this principle, and an anthropologist or social scientist has to ensure that they intervene as little as possible.

Field Experiments

For geologists, social scientists and environmental biologists, amongst others, field experiments are an integral part of the discipline.

As the name suggests, a field study is an experiment performed outside the laboratory, in the 'real' world. Unlike case studies and observational studies, a field experiment still follows all of the steps of the scientific process, addressing research problems and generating hypotheses.

The obvious advantage of a field study is that it is practical and also allows experimentation, without artificially introducing confounding variables.

A population biologist examining an ecosystem could not move the entire environment into the laboratory, so field experiments are the only realistic research method in many fields of science.

In addition, they circumvent the accusation leveled at laboratory experiments of lacking external or ecological validity, or adversely affecting the behaviour of the subject.

Social scientists and psychologists often used field experiments to perform blind studies, where the subject was not even aware that they were under scrutiny.

A good example of this is the Piliavin and Piliavin experiment, where the propensity of strangers to help blood covered 'victims' was measured. This is now frowned upon, under the policy of informed consent, and is only used in rare and highly regulated circumstances.

Field experiments can suffer from a lack of a discrete control group and often have many variables to try to eliminate.

For example, if the effects of a medicine are studied, and the subject is instructed not to drink alcohol, there is no guarantee that the subject followed the instructions, so field studies often sacrifice internal validity for external validity.

For fields like biology, geology and environmental science, this is not a problem, and the field experiment can be treated as a sound experimental practice, following the steps of the scientific method.

A major concern shared by all disciplines is the cost of field studies, as they tend to be very expensive.

For example, even a modestly sized research ship costs many thousands of dollars every day, so a long oceanographical research programme can run into the millions of dollars.

Pilot studies are often used to test the feasibility of any long term or extensive research programme before committing vast amounts of funds and resources. The changeable nature of the external environment and the often-prohibitive investment of time and money mean that field experiments are rarely replicable, so any generalization is always tenuous.

Cohort Study

A cohort study is a research programme investigating a particular group with a certain trait, and observes over a period of time.

Some examples of cohorts may be people who have taken a certain medication, or have a medical condition. Outside medicine, it may be a population of animals that has lived near a certain pollutant or a sociological study of poverty.

A cohort study can delve even further and divide a cohort into sub-groups, for example, a cohort of smokers could be sub-divided, with one group suffering from obesity. In this respect, a cohort study is often interchangeable with the term naturalistic observation.

There are two main sub-types of cohort study, the retrospective and the prospective cohort study. The major difference between the two is that the retrospective looks at phenomena that have already happened, whilst the prospective type starts from the present.

Retrospective Cohort Study

The retrospective case study is historical in nature. Whilst still beginning with the division into cohorts, the researcher looks at historical data to judge the effects of the variable.

For example, it might compare the incidence of bowel cancer over time in vegetarians and meat eaters, by comparing the medical histories. It is a lot easier than the prospective, but there is no control, and confounding variables can be a problem, as the researcher cannot easily assess the lifestyle of the subject.

A retrospective study is a very cheap and effective way of studying health risks or the effects of exposure to pollutants and toxins. It gives results quickly, at the cost of validity, because it is impossible to eliminate all of the potentially confounding variables from historical records and interviews alone.

Prospective Cohort Study

In a prospective cohort study, the effects of a certain variable are plotted over time, and the study becomes an ongoing process. To maintain validity, all of the subjects must be initially free of the condition tested for.

For example, an investigation, over time, into the effects of smoking upon lung cancer must ensure that all of the subjects are free of the disease. It is also possible to subgroup and try to control variables, such as weight, occupation type or social status.

They are preferable to a retrospective study, but are expensive and usually require a long period of time to generate useful results, so are very expensive and difficult.

Some studies have been running for decades, but are generating excellent data about underlying trends in a population. The prospective cohort study is a great way to study long-term trends, allowing the researcher to measure any potential confounding variables, but the potential cost of error is high, so pilot studies are often used to ensure that the study runs smoothly.

Ambidirectional Cohort Study

The ambidirectional cohort study is the ultimate method, combining retrospective and prospective aspects. The researcher studies and analyzes the previous history of the cohorts and then continues the research in a prospective manner.

This gives the most accurate results, but is an extremely arduous undertaking, costing time and a great deal of money.

The ambidirectional study shares one major drawback with the prospective study, in that it is impossible to guarantee that any data can be followed up, as participants may decline to participate or die prematurely. These studies need to look at very large samples to ensure that any attributional losses can be absorbed by the statistics.

Longitudinal Study

A longitudinal study is observational research performed over a period of years or even decades, and allows social scientists and economists to study long-term effects in a human population.

A cohort study is a subset of the longitudinal study because it observes the effect on a specific group of people over time. Quite often, a longitudinal study is an extended case study, observing individuals over long periods, and is a purely qualitative undertaking.

The lack of quantitative data means that any observations are speculative, as with many case studies, but they allow a unique and valuable perspective on some aspects of human culture and sociology.

The 'up' Series

The groundbreaking television documentary 'UP' is probably the most famous example of a long-term longitudinal study, a case study of a group of British people from birth.

The original producer, Michael Apted, proposed the hypothesis that children born into a certain social class would remain entrenched in that class throughout their life.

In 1967, he selected children from the rich, poor and middle classes, and proceeded to interview and film them every seven years. The highly acclaimed series is still running, with the next set of interviews to be performed in 2011/2012, and it has provided a unique insight into the development of British culture since the 1960's. Even this series highlights one of the major flaws of a longitudinal study, the problem that there can be no retesting or restart. Apted, with hindsight, wished that he had used more female subjects, showing the importance of the initial planning stage of a longitudinal study. Once a course of action is decided, the clock cannot be turned back, and the results must stand as tested.

The Cross Sectional Study

The cross sectional study looks at a different aspect than the standard longitudinal study.

The longitudinal study uses time as the main variable, and tries to make an in depth study of how a small sample changes and fluctuates over time.

A cross sectional study, on the other hand, takes a snapshot of a population at a certain time, allowing conclusions about phenomena across a wide population to be drawn.

An example of a cross-sectional study would be a medical study looking at the incidence of breast cancer in a population. The researcher can look at a wide range of ages, ethnicities and social backgrounds. If a significant number of women from a certain social background are found to have the disease, then the researcher can investigate further.

This is a relatively easy way to perform a preliminary experiment, allowing the researcher to focus on certain population groups and understand the wider picture.

Of course, researchers often use both methods, using a cross section to take the snapshot and isolate potential areas of interest, and then conducting a longitudinal study to find the reason behind the trend.

This is called panel data, or time series cross-sectional data, but is generally a complicated and expensive type of research, notoriously difficult to analyse.

Such programs are rare, but can give excellent data, allowing a long-term picture of phenomena to be ascertained.

Factorial Design

A factorial design is often used by scientists wishing to understand the effect of two or more independent variables upon a single dependent variable.

Traditional research methods generally study the effect of one variable at a time, because it is statistically easier to manipulate. However, in many cases, two factors may be interdependent, and it is impractical or false to attempt to analyse them in the traditional way.

Social researchers often use factorial designs to assess the effects of educational methods, whilst taking into account the influence of socio-economic factors and background.

Agricultural science, with a need for field-testing, often uses factorial designs to test the effect of variables on crops. In such large-scale studies, it is difficult and impractical to isolate and test each variable individually.

Factorial experiments allow subtle manipulations of a larger number of interdependent variables. Whilst the method has limitations, it is a useful method for streamlining research and letting powerful statistical methods highlight any correlations.

The Basics

Imagine an aquaculture research group attempting to test the effects of food additives upon the growth rate of trout.

A traditional experiment would involve randomly selecting different tanks of fish and feeding them varying levels of the additive contained within the feed, for example none or 10%.

However, as any fish farmer knows, the density of stocking is also crucial to fish growth; if there are not enough fish in a tank, then the wasted capacity costs money. If the density is too high, then the fish grow at a slower rate.

Rather than the traditional experiment, the researchers could use a factorial design and co-ordinate the additive trial with different stocking densities, perhaps choosing four groups. The factorial experiment then needs 4×2 , or eight treatments.

The traditional rules of the scientific method are still in force, so statistics require that every experiment be conducted in triplicate.

This means 24 separate treatment tanks. Of course, the researchers could also test, for example, 4 levels of concentration for the additive, and this would give 4×4 or 16 tanks, meaning 48 tanks in total. Each factor is an independent variable, whilst the level is the subdivision of a factor.

Assuming that we are designing an experiment with two factors, a 2×2 would mean two levels for each, whereas a 2×4 would mean two subdivisions for one factor and four for the other. It is possible to test more than two factors, but this becomes unwieldy very quickly. In the fish farm example, imagine adding another factor, temperature, with four levels into the mix. It would then be $4 \times 4 \times 4$, or 64 runs. In triplicate, this would be 192 tanks, a huge undertaking.

There are a few other methods, such as fractional factorial designs, to reduce this, but they are not always statistically valid. This lies firmly in the realm of advanced statistics and is a long, complicated and arduous undertaking.

The Pros and Cons of Factorial Design

Factorial designs are extremely useful to psychologists and field scientists as a preliminary study, allowing them to judge whether there is a link between variables, whilst reducing the possibility of experimental error and confounding variables.

The factorial design, as well as simplifying the process and making research cheaper, allows many levels of analysis. As well as highlighting the relationships between variables, it also allows the effects of manipulating a single variable to be isolated and analyzed singly.

The main disadvantage is the difficulty of experimenting with more than two factors, or many levels. A factorial design has to be planned meticulously, as an error in one of the levels, or in the general operationalization, will jeopardize a great amount of work. Other than these slight detractions, a factorial design is a mainstay of many scientific disciplines, delivering great results in the field.

Case Control Study

A case control study is a method extensively used by the medical profession, as an easy and quick way of comparing treatments, or investigating the causes of disease. Longitudinal studies are the preferred method, but are often expensive, time consuming and difficult. Whilst this method does suffer from some weaknesses, it is quick, and delivers results quickly.

The case control study uses groups of patients stricken with a disease, and compares them with a control group of patients not suffering symptoms. Medical records and interviews are used to try to build up a historical picture of the patient's life, allowing cross-reference between patients and statistical analysis. Any trends can then be highlighted and action can be taken.

Statistical analysis allows the researcher to draw a conclusion about whether a certain situation or exposure led to the medical condition. For example, a scientist could compare a group of coal miners suffering from lung cancer with those clear of the disease, and try to establish the underlying cause. If the majority of the cases arose in collieries owned by one company, it might indicate that the company's safety equipment and procedures were at fault.

Possibly the most famous case control study using this method was a study into whether bicycle helmets reduce the chance of cyclists receiving bad head injuries in an accident. Obviously, the researcher could not use standard experimentation and compare a control group of non-helmet wearers with helmet wearers, measuring the chances of head injury, as this would be unethical. A case study control was utilized, and the researchers looked at medical records, comparing the number of head injury sufferers wearing helmets against those without. This generated a statistical result, showing that wearing a cycle helmet made it 88% less likely that head injury would be suffered in an accident.

The main weakness of the case control study is that it is very poor at determining cause and effect relationships.

In the cycle helmet example, it could be argued that a cyclist who bothered wearing a helmet may well have been a safer cyclist anyway, and less likely to have accidents. Evidence showed that children wearing helmets were more likely to be from a more affluent class, more used to cycling through parks than city streets. The study also showed that helmets were of little use to adults.

Whilst most agree that cycle helmets are probably a good thing for children, there is not enough evidence to suggest that they should be mandatory for adults outside extreme cycling. These problems serve as a warning that the results of any case control study should not be relied upon, instead acting as a guide, possibly allowing deeper and more rigorous methods to be utilized.

Pilot Study

A pilot study is a standard scientific tool for 'soft' research, allowing scientists to conduct a preliminary analysis before committing to a full-blown study or experiment.

A small chemistry experiment in a college laboratory, for example, costs very little, and mistakes or validity problems easily rectified. At the other end of the scale, a medical experiment taking samples from thousands of people from across the world is expensive, often running into the millions of dollars.

Finding out that there was a problem with the equipment or with the statistics used is unacceptable, and there will be dire consequences.

A field research project in the Amazon Basin costs a lot of time and money, so finding out that the electronics used do not function in the humid and warm conditions is too late.

To test the feasibility, equipment and methods, researchers will often use a pilot study, a small-scale rehearsal of the larger research design. Generally, the pilot study technique specifically refers to a smaller scale version of the experiment, although equipment tests are an increasingly important part of this sub-group of experiments.

For example, the medical researchers may conduct a smaller survey upon a hundred people, to check that the protocols are fine.

The Amazon Researchers may perform an experiment, in similar conditions, sending a small team either to the Amazon to test the procedures, or by using something like the tropical bio-dome at the Eden Project.

Pilot studies are also excellent for training inexperienced researchers, allowing them to make mistakes without fear of losing their job or failing the assignment.

Logistical and financial estimates can be extrapolated from the pilot study, and the research question, and the project can be streamlined to reduce wastage of resources and time.

Pilots can be an important part of attracting grants for research as the results can be placed before the funding body.

Generally, most funding bodies see research as an investment, so are not going to dole out money unless they are certain that there is a chance of a financial return.

Unfortunately, a paper reporting the preliminary pilot study, especially if problems were reported, is often stigmatized and sidelined. This is unfair, and punishes researchers for being methodical, so these attitudes are under a period of re-evaluation.

Discouraging researchers from reporting methodological errors, as found in pilot studies, means that later researchers may make the same mistakes.

The other major problem is deciding whether the results from the pilot study can be included in the final results and analysis, a procedure that varies wildly between disciplines.

Pilots are rapidly becoming an essential pre-cursor to many research projects, especially when universities are constantly striving to reduce costs. Whilst there are weaknesses, they are extremely useful for driving procedures in an age increasingly dominated by technology, much of it untested under field conditions.

7

The Use of Computer in Commerce Research

Computer is use in all Commerce Departments like shopping malls and Big Departmental stores to calculate the Sales and Purchase of the Firm. Computers are very help in the business to Maintain their accounts through the Computer Example: Tally Accounting Software, MARG Software.

Nowadays every Commercial Firms are use Computers for every transaction. Organization:

The ease with which databases, spreadsheets and data can be compiled on a computer has certainly improved the efficiency and management practices of businesses worldwide. Many offices now uses computer programs to handle scheduling, accounting, billing, inventory management, contact management, etc.

It Portable laptop computers, smart phones, wireless internet, air cards and hub spots are the wave of the future when comes to computer uses in business. Today, business can be conducted remotely from almost anywhere.

Retail purchases, credit card transactions, automatic debits... can all be processed over computers, and through the internet today. Banks, credit card companies, and other financial institutions are some of the best examples of computer uses in business. E-commerce is a rapidly growing industry, promising to become one of the top computer related professions in the future. The world may have grown smaller in the internet age, but only because the power of communication has grown by leaps and bounds.

The top computer uses in business include e-mail communication, and networking platforms such as video and web conferencing, social networking, and various other

online chat tools. Even faxing, which has become increasingly obsolete in light of new technologies, is still a relevant business use of the computer.

WONDERS OF COMPUTER IN E-COMMERCE

Electronic mail is the revolutionary service offered by the computers. Video Conferencing is also another major advantage. Electronic shopping through online shopping added favor to purchaser and merchants.

Electronic banking is now at your hand where every bank has online support for transaction of monetary issues. You can easily transfer your money anywhere even from your home.

USES FOR COMPUTERS IN BUSINESS

Almost every business uses computers to complete daily tasks. From making contact with clients to inputting data for reports, computers allow businesses a more efficient way to manage affairs when compared to traditional paper and manila folders. Businesses use a variety of different types of computers such as desktops, laptops, servers, smartphones and tablets, depending on their needs. With computers, employees are able to work anytime, anywhere.

Communication

Communication is key when gaining and maintaining clients and other important contacts. Computers give businesses access to email, instant messaging and custom customer contact systems. Computerized phone systems allow for automated support during off hours and a virtual operator can quickly direct callers to the correct department for faster support.

Marketing

Computers allow businesses to create websites, stunning ads and complete marketing campaigns. Marketing videos can be edited and custom ads created in-house with the use of specialized software.

Businesses can completely develop and manage websites with their own servers or connect remotely to a third-party business to upload their latest content such as articles, product images and blog posts.

Accounting

Accounting without computers presents a high risk for human error. Accounting software allows businesses to simply input their financial data and instantly see gains and losses. All necessary tax reports are available the moment the data is entered. Using computers for invoicing, managing expenses and calculating payroll is vital for ensuring financial data is as accurate as possible.

Storage

Instead of filing cabinets, businesses are able to store millions of files using computers and servers. Data can be stored centrally for easy access from multiple computers or stored locally for individual use. Computerized storage saves space and provides a far more efficient organization strategy. With encryption, passwords and replace keys, data remains secure.

Documents and Reports

Most businesses have some sort of productivity software which typically includes a word processor and spreadsheet application. These two programs allow businesses to create reports, memos, tutorials and even colorful ads for company events. Spreadsheet applications give businesses the chance to organize, manage and calculate both numeric and alphabetic data. With charts and graphs, reporting becomes visual instead of text-based.

Education

Businesses use computers to help educate employees on software, company policy, standard procedures and safety. Instead of hiring teachers, computers can be used to educate employees at their own pace or through an online webinar with live questions and answers. This form of education fits the busy schedules of businesses without sacrificing the quality of the education.

Research

From learning more about the competition to discovering what customers really want, research isn't as difficult as it once was, thanks to computers. Search engines, forums, social networks and industry specific websites provide businesses with a wealth of information and research data.

Computer Applications in Business

Although computing devices are not as recent an invention as one might think, it was not until the latter part of the twentieth century that they reached a level of sophistication that enabled them to become common in the workplace. Computers enhance productivity at work, quickly and accurately performing many of the repetitive and tedious chores associated with document creation and editing, accounting, publishing, and record keeping. In addition, computers are invaluable for enhancing the way that people work together by providing the capabilities to share data, coordinate tasks, and quickly communicate. Computers also are invaluable for managing data and allow organizations to maintain and manipulate databases in ways that would be impossible without technology. The use of computers for business is not without risks and concerns, however, including both issues of security and concerns for the individual's right to privacy.

Information Technology: Computer Applications in Business

As a species, human beings appear to be infinitely inventive and constantly searching for ways to improve their lives. Many important accomplishments and innovations resulting from this quest have changed the course of history along the way. Within the current lifetime, however, arguably none is more important than the progress that has been made in countless disciplines as well as everyday activities as a result of computer technology. The application of computers both at home and at work helps us to better organize our lives, more easily perform repetitive or logical tasks, and access our creativity. Like the Renaissance and the Industrial Revolution before it, the Information Age continues to transform our lives and the ways in which we perform many of our tasks. The trend does not appear to be coming to a quick end, however, and it is reasonable to assume that the proliferation of technological invention and innovation will not continue to change the way that we do things well into the future.

The use of computers to assist in performing tedious tasks, however, is not a recent concept. For centuries, human beings have looked for ways to improve the accuracy of their computations and to better control the world around them. For example, the Antikythera Mechanism — the world's oldest known computing device — dates to the first century B.C. This complex, multi-gearred device is believed to be an early analog computer that was designed to calculate astronomical positions. Despite such intriguing examples, however, computers did not come into their own until the mid-twentieth century. The cumbersome mainframe devices of the time were revolutionized in the 1970s with the invention of the microprocessor. This innovative technology led to the development of the personal computer that soon became commonplace both at home and at work.

Computers have become so much a part of many people's lives that it is difficult to imagine life without them. In the workplace, computers allow workers to write letters, record financial transactions, and send both quick messages and lengthy documents to colleagues across the room or across the world. Computers automatically spell check correspondence and make suggestions on how to improve grammar. Computers allow workers to circumvent the art department and use presentation software to design business presentations in full color, illustrated with high-quality photographs and graphics, with just a few clicks of a mouse. Computers automatically synchronize the calendars in personal digital assistants or smartphones and chime merrily to remind us of appointments, tasks, and deadlines.

In the mid-twentieth century, computers were cumbersome things that belonged to the realm of big business problems and scientific research. For most people, correspondence and technical writing was literally penned by hand and sent to a secretary or a typing pool to be formalized on that modern wonder, the electric typewriter. The phrase "cut and paste" was literal: workers received the crisp white sheets of paper back from the unseen typing pool and proceeded to edit by hand, frequently using scissors and tape to rearrange paragraphs. Then suddenly lives were revolutionized by

the advent of the word processor. Although these devices still were not on every desktop, the secretaries and typists were thrilled because they no longer had to type each draft from scratch; they could rearrange paragraphs, edit, and correct on electronically stored documents and print out a fresh edited copy in a fraction of the time that the work would have taken on a typewriter.

The first word processors were crude, however, when compared to the capabilities that are housed today within personal computers or laptops. They were unitaskers, useful only for typing, editing, and printing documents. Since the introduction of the personal computer to the workplace, the way that we do work has been transformed. Depending on the software applications that are installed on the computers, one can not only type and edit documents but also create layouts and designs for print. Electronic spreadsheets that automatically total columns of figures and quickly paste cells to other pages make the task of double-entry bookkeeping almost fun. Whether they work on independent personal computers or are part of a network, the ability to do carry out work using computers has impacted workers in a wide range of industries. The application of computers to business is virtually as unlimited as the tasks performed in the workplace.

Applications

The examples above illustrate just a few of the ways that computers can enhance individual productivity in the workplace. However, computers in the workplace are useful for much more than enhancing individual productivity. There are numerous applications of computer technology that can enhance the way that workers interact with each other and collaborate to enhance the productivity and effectiveness of the organization as a whole. Computers also are invaluable for storing and manipulating large collections of data in support of numerous tasks across the organization. In addition, the Internet has given businesses an entirely new medium for advertising products and services, gathering data about prospective customers and competitors, and selling products online.

Enterprise-Wide Computer Applications

By definition, an organization is a group of people that is organized to perform work or some other task. The fact that organizations are made up of more than one person gives them the opportunity to achieve synergy and accomplish more together than they could as isolated individuals. Modern computer technology can help in this task. In many businesses, computers are linked together over a local area network that allows multiple computers located near each other (e.g., in the same room or building) to share files and peripheral devices such as printers. This capability is particularly useful if a number of people need to work on the same document (e.g., inputting data, commenting on a document, creating reports from a database). Similarly, computers that are widely dispersed can be linked into a wide area network. Wide area networks typically use high-speed, long-distance communications networks or satellites to connect the

computers within the network. Through them, employees can communicate and cooperate with colleagues in the next office and across the globe.

The linking of computers through networks can enhance the productivity and effectiveness of the entire enterprise. For example, computerized cash registers in a retail store or other points of sale can be linked together so that sales clerks can search across the network to see whether other stores have a particular item in stock. Similarly, the computers at the point of sale can be linked directly to the corporate headquarters so that the store's closing data each evening can be directly and immediately shared with corporate management.

ROLE OF COMPUTERS IN RESEARCH

Computers play a major role today in every field of scientific research from genetic engineering to astrophysics research. This article is a brief overview of the role that computers have played in research and the ways in which they are helping unravel several scientific mysteries. While they are still far away from that supreme ideal of being able to think for themselves as artificially intelligent machines, the brute force of their ability to calculate and process information at phenomenal speeds powers research in every field of human endeavor. In fact, computers have changed the way in which information, or more precisely, knowledge is disseminated around the world through the Internet, via which I am communicating with your right now. In this Buzzle article, I have highlighted the role of computers in research fields, where it works to overcome the horizons of ignorance that still exist.

The first computers developed were mere calculators, with the ability to carry out simple arithmetic operations. Advancement in electronics, the development of the vacuum tube based transistors and devising of digital logic gates led to development of the first digital computer which could perform more advanced logical operations.

These early machines took up whole multistorey buildings and were operated through punch cards. All this changed after the development of semiconductor based transistors which led to the extreme miniaturization of integrated circuits, that made the development of the first personal computers possible.

Since then, computer technology has advanced by leaps and bounds to bring extreme computing power into the hands of the common man. Today computers are ubiquitous, with applications in every field of human endeavor. To no field have computers contributed as much as scientific research. From predicting weather to running astrophysical simulations, computers are the scientist's prime analytical tool in unraveling the mysteries of nature.

ROLE OF COMPUTERS IN SCIENTIFIC RESEARCH

It is truly impossible to capture the whole range of computer applications in scientific research. Here is a bird's-eye view of the role of computers in scientific research, where I shall discuss four of their most important applications.

Data Storage & Analysis

Experimentation is the cornerstone of scientific research. Every experiment in any of the natural sciences generates a lot of data, that needs to be stored and analyzed to derive important conclusions, to validate or disprove hypotheses. Computers attached with experimental apparatuses, directly record data as it's generated and subject it to analysis through specially designed software.

Analyzing tons of statistical data is made possible using specially designed algorithms that are implemented by computers. This makes the extremely time-consuming job of data analysis to be a matter of a few minutes. In genetic engineering, computers have made the sequencing of the entire human genome possible. Data from different sources can be stored and accessed via computer networks set up in research labs, which makes collaboration simpler.

Scientific Simulations

One of the prime uses of computers in pure science and engineering projects is the running of simulations. A simulation is a mathematical modeling of a problem and a virtual study of its possible solutions. Problems which do not yield themselves to experimentation can be studied through simulations carried out on computers.

For example, astrophysicists carry out structure formation simulations, which are aimed at studying how large-scale structures like galaxies are formed. Space missions to the Moon, satellite launches and interplanetary missions are first simulated on computers to determine the best path that can be taken by the launch vehicle and spacecraft to reach its destination safely.

Instrumentation Control

Most advanced scientific instruments come with their own on-board computer, which can be programmed to execute various functions. For example, the Hubble Space Craft has its own on-board computer system which is remotely programmed to probe the deep space. Instrumentation control is one of the most important applications of computers.

Knowledge Sharing Through Internet

Lastly, in the form of Internet, computers have provided an entirely new way to share knowledge. Today, any one can access the latest research papers that are made available for free on websites like Arxiv. Sharing of knowledge and collaboration through the Internet, has made international cooperation on scientific projects possible.

Through various kinds of analytical software programs, computers are contributing to scientific research in every discipline, ranging from biology to astrophysics, discovering new patterns and providing novel insights. When the work in neural network based artificial intelligence advances and computers are granted with the ability to learn and think for themselves, future advances in technology and research will be even

more rapid. If we survive human strife and manage not to self destruct our civilization, future is going to be an exciting time!

Uses of Computers in Scientific Research

Computers have changed the ways in which scientific research is compiled and analyzed. Scientists, engineers and researchers alike are able to compile vast amounts of data and leave it to the computer to work through the data while focusing on another area of the research project. This creates research results with fewer errors and better-engineered products.

Complex Data Analysis

Computers used in scientific research have the ability to analyze data in ways and at speeds not possible with the human eye. They are able to analyze percentages of materials present in a variety of compounds from soil samples to chemicals and even the air you breathe.

Additionally, computers used in this manner can determine trends in data samples. For example, computer analysis of data in research could determine the temperature at which certain chemical compounds break down or the percentages of improvement patients show when administered a certain medication.

Solving Mathematical Equations

Scientific research often requires that complex mathematical equations be solved in order to determine if data is valid or if a certain structure of molecules will remain stable. Computers are integral to this calculation process since scientists can write software programs specifically to provide answers to such questions. This removes the element of human error, which can cost research institutions millions of dollars in fixing a product that was created with even the smallest amount of flawed data.

Prediction Modeling

Scientists and researchers are able to use computer programs to model how data might manifest itself in the future. This ability is useful in predicting climate patterns, simulating how engineered products might perform in the field, predicting the erosion rate of beaches and anticipating the absorption rate of medications in the body. Scientists and engineers are then able to adjust building strategies or chemical compositions of products to ensure safe operation and consumption.

Research Information Networks

Multidisciplinary Research Networks

CIESIN (The Centre for International Earth Science Information Network) is an Information Network research centre within the Earth Institute at Columbia University.

CIESIN works at the intersection of the social, natural, and information sciences, and specializes in on-line data and information management, spatial data integration and training, and interdisciplinary research related to human interactions in the environment.

Information Network can be established with industry and university, and focus on some specific area such as wireless communication research. At New Jersey's Rutgers University industry-university cooperative research centre Wireless Information Networks laboratory focused on wireless technology was founded in 1989. Another example is Centre for Wireless Information Network Studies (CWINS) is a wireless research laboratory with research alliances with other industrial and academic groups. The centre has performed research for government agencies and has close ties with the world-leading organizations in the wireless industry.

Information Network Portals

Information Network may be seen also as a www-portal, blog or library service. These exemplify the human and technical sides of the information networking.

8

Structure of Reports

WRITING RESEARCH

Writing is easy. All you do is stare at a blank sheet of paper until drops of blood form on your forehead. --- Gene Fowler

A major goal of this course is the development of effective technical writing skills. To help you become an accomplished writer, you will prepare several research papers based upon the studies completed in lab. Our research papers are not typical "lab reports." In a teaching lab a lab report might be nothing more than answers to a set of questions. Such an assignment hardly represents the kind of writing you might be doing in your eventual career.

Written and oral communications skills are probably the most universal qualities sought by graduate and professional schools as well as by employers. You alone are responsible for developing such skills to a high level.

RESOURCES FOR LEARNING TECHNICAL WRITING

Before you begin your first writing assignment, Consult all of the following resources, in order to gain the most benefit from the experience.

- General form of a typical research substance.
- Specific guidelines (if any) for the assignment
- McMillan, VE. "Writing Papers in the Biological Sciences, Third Ed." New York: Bedford/St. Martin's, 2001. ISBN 0-312-25857-7 (REQUIRED for Bios 211, 311, recommended for other science courses that include writing)

As you polish up your writing skills please make use of the following resources:

- Instructor feedback on previous assignments
- Common errors in student research papers
- Selected writing rules (somewhat less serious than the other resources)

For Biosciences majors the general guidelines apply to future course work, as can be seen by examining the guidelines for the advanced experimental sciences research paper. Instructions for authors from the Journal of Biological Chemistry editorial board may be helpful as well. Their statement of editorial policies and practices may give you an idea of how material makes its way into the scientific literature.

GENERAL FORM OF A RESEARCH PAPER

An objective of organizing a research paper is to allow people to read your work selectively. When we research a topic, we may be interested in just the methods, a specific result, the interpretation, or perhaps we just want to see a summary of the paper to determine if it is relevant to my study. To this end submitted in the order listed. There are variations of course. Some journals call for a combined results and discussion, for example, or include materials and methods after the body of the paper. The well known journal Science does away with separate parts altogether, except for the abstract.

Your papers are to adhere to the form and style required for the Journal of Biological Chemistry, requirements that are shared by many journals in the life sciences.

GENERAL STYLE

Specific editorial requirements for submission of a manuscript will always supercede instructions in these general guidelines.

To make a paper readable:

- Print or type using a 12 point standard font, such as Times, Geneva, Bookman, Helvetica, etc.
- Text should be double spaced on 8 1/2" x 11" paper with 1 inch margins, single sided
- Number pages consecutively
- Start each new parts on a new page
- Adhere to recommended page limits

Mistakes to avoid:

- Placing a heading at the bottom of a page with the following text on the next page (insert a page break!)
- Dividing a table or figure - confine each figure/table to a single page
- Submitting a paper with pages out of order

In all parts of your paper:

- Use normal prose including substances ("a", "the," etc.)
- Stay focused on the research topic of the paper
- Use paragraphs to separate each important point (except for the abstract)

- Indent the first line of each paragraph
- Present your points in logical order
- Use present tense to report well accepted facts - for example, 'the grass is green'
- Use past tense to describe specific results - for example, 'When weed killer was applied, the grass was brown'
- Avoid informal wording, don't address the reader directly, and don't use jargon, slang terms, or superlatives
- Avoid use of superfluous pictures - include only those figures necessary to presenting results

Title Page

Select an informative title as emphasized in the examples in your writing portfolio example package. Include the name(s) and address(es) of all authors, and date submitted. "Biology lab #1" would not be an informative title, for example.

General intent

An abstract is a concise single paragraph summary of completed work or work in progress. In a minute or less a reader can learn the rationale behind the study, general approach to the problem, pertinent results, and important ceases or new questions.

Writing an abstract

Write your summary after the rest of the paper is completed. After all, how can you summarize something that is not yet written? Economy of words is important throughout any paper, but especially in an abstract. However, use complete sentences and do not sacrifice readability for brevity.

You can keep it concise by wording sentences so that they serve more than one purpose. For example, "In order to learn the role of protein synthesis in early development of the sea urchin, newly fertilized embryos were pulse-labeled with tritiated leucine, to provide a time course of changes in synthetic rate, as measured by total counts per minute (cpm)." This sentence provides the overall question, methods, and type of analysis, all in one sentence.

The writer can now go directly to summarizing the results. Summarize the study, including the following elements in any abstract.

Try to keep the first two items to no more than one sentence each:

- *Purpose of the study:* Hypothesis, overall question, objective
- Model organism or system and brief description of the experiment
- *Results, including specific data:* If the results are quantitative in nature, report quantitative data; results of any statistical analysis should be reported
- Important ceases or questions that follow from the experiment(s)

Style:

- Single paragraph, and concise

- As a summary of work done, it is always written in past tense
- An abstract should stand on its own, and not refer to any other part of the paper such as a figure or table
- Focus on summarizing results - limit background information to a sentence or two, if absolutely necessary
- What you report in an abstract must be consistent with what you reported in the paper
- Correct spelling, clarity of sentences and phrases, and proper reporting of quantities are just as important in an abstract as they are anywhere else

Introduction

Your introductions should not exceed two pages (double spaced, typed).

General Intent

The purpose of an introduction is to acquaint the reader with the rationale behind the work, with the intention of defending it. It places your work in a theoretical context, and enables the reader to understand and appreciate your objectives.

Writing an Introduction

The abstract is the only text in a research paper to be written without using paragraphs in order to separate major points.

Approaches vary widely, however for our studies the following approach can produce an effective introduction.

- *Describe the importance (significance) of the study:* Why was this worth doing in the first place? Provide a broad context.
- *Defend the model:* Why did you use this particular organism or system? What are its advantages? You might comment on its suitability from a theoretical point of view as well as indicate practical reasons for using it.
- *Provide a rationale:* State your specific hypothesis(es) or objective(s), and describe the reasoning that led you to select them.
- Very briefly describe the experimental design and how it accomplished the stated objectives.

Style:

- Use past tense except when referring to established facts. After all, the paper will be submitted after all of the work is completed.
- Organize your ideas, making one major point with each paragraph.
- Present background information only as needed in order support a position. The reader does not want to read everything you know about a subject.
- State the hypothesis/objective precisely - do not oversimplify.
- As always, pay attention to spelling, clarity and appropriateness of sentences and phrases.

Materials and Methods

There is no specific page limit, but a key concept is to keep this part as concise as you possibly can. People will want to read this material selectively. The reader may only be interested in one formula or part of a procedure. Materials and methods may be reported under separate subheadings within this part or can be incorporated together.

General Intent

This should be the easiest part to write, but many students misunderstand the purpose. The objective is to document all specialized materials and general procedures, so that another individual may use some or all of the methods in another study or judge the scientific merit of your work. It is not to be a step by step description of everything you did, nor is a methods part a set of instructions. In particular, it is not supposed to tell a story.

WRITING A MATERIALS AND METHODS PART

Materials:

- Describe materials separately only if the study is so complicated that it saves space this way.
- Include specialized chemicals, biological materials, and any equipment or supplies that are not commonly found in laboratories.
- Do not include commonly found supplies such as test tubes, pipet tips, beakers, etc., or standard lab equipment such as centrifuges, spectrophotometres, pipettors, etc.
- If use of a specific type of equipment, a specific enzyme, or a culture from a particular supplier is critical to the success of the experiment, then it and the source should be singled out, otherwise no.
- Materials may be reported in a separate paragraph or else they may be identified along with your procedures.
- In biosciences we frequently work with solutions - refer to them by name and describe completely, including concentrations of all reagents, and pH of aqueous solutions, solvent if non-aqueous.

Methods:

- Report the methodology
- Describe the methodology completely, including such specifics as temperatures, incubation times, etc.
- To be concise, present methods under headings devoted to specific procedures or groups of procedures
- *Generalize:* Report how procedures were done, not how they were specifically performed on a particular day. For example, report "samples were diluted to a final concentration of 2 mg/ml protein;" don't report that "135 microliters of sample one was diluted with 330 microliters of buffer to make the protein

concentration 2 mg/ml." Always think about what would be relevant to an investigator at another institution, working on his/her own project.

- If well documented procedures were used, report the procedure by name, perhaps with reference, and that's all. For example, the Bradford assay is well known. You need not report the procedure in full - just that you used a Bradford assay to estimate protein concentration, and identify what you used as a standard. The same is true for the SDS-PAGE method, and many other well known procedures in biology and biochemistry.

Style:

- It is awkward or impossible to use active voice when documenting methods without using first person, which would focus the reader's attention on the investigator rather than the work. Therefore when writing up the methods most authors use third person passive voice.
- Use normal prose in this and in every other part of the paper - avoid informal lists, and use complete sentences.

What to avoid:

- Materials and methods are not a set of instructions.
- Omit all explanatory information and background - save it for the discussion.
- Omit information that is irrelevant to a third party, such as what colour ice bucket you used, or which individual logged in the data.

Results:

- The page length of this part is set by the amount and types of data to be reported. Continue to be concise, using figures and tables, if appropriate, to present results most effectively.

General Intent

The purpose of a results part is to present and emphasize your findings. Make this part a completely objective report of the results, and save all interpretation for the discussion.

Writing a Results Part

Important: You must clearly distinguish material that would normally be included in a research from any raw data material that would not be published. In fact, such material should not be submitted at all unless requested by the instructor.

Content:

- Summarize your findings in text and emphasize them, if appropriate, with figures and tables.
- In text, describe each of your results, pointing the reader to observations that are most relevant.
- Provide a context, such as by describing the question that was addressed by making a particular observation.

- Describe results of control experiments and include observations that are not presented in a formal figure or table, if appropriate.
- Analyse your data, then prepare the analysed (converted) data in the form of a figure (graph), table, or in text form.

What to avoid:

- Do not discuss or interpret your results, report background information, or attempt to explain anything.
- Never include raw data or intermediate calculations in a research paper.
- Do not present the same data more than once.
- Text should complement any figures or tables, not repeat the same information.
- Please do not confuse figures with tables - there is a difference.

Style:

- As always, use past tense when you refer to your results, and put everything in a logical order.
- Place figures and tables, properly numbered, in order at the end of the report (clearly distinguish them from any other material such as raw data, standard curves, etc.)
- If you prefer, you may place your figures and tables appropriately within the text of your results part.

Figures and tables:

- Either place figures and tables within the text of the result, or include them in the back of the report (following Literature Cited) - do one or the other
- If you place figures and tables at the end of the report, make sure they are clearly distinguished from any attached materials, such as raw data
- Regardless of placement, each figure must be numbered consecutively and complete with caption
- Regardless of placement, each table must be titled, numbered consecutively and complete with heading
- Each figure and table must be sufficiently complete that it could stand on its own, separate from text

Discussion

Journal guidelines vary. Space is so valuable in the Journal of Biological Chemistry, that authors are asked to restrict discussions to four pages or less, double spaced, typed. That works out to one printed page. While you are learning to write effectively, the limit will be extended to five typed pages. If you practice economy of words, that should be plenty of space within which to say all that you need to say.

General Intent

The objective here is to provide an interpretation of your results and support for all of your ceases, using evidence from your experiment and generally accepted knowledge, if appropriate. The significance of findings should be clearly described.

Writing a Discussion

Interpret your data in the discussion in appropriate depth. This means that when you explain a phenomenon you must describe mechanisms that may account for the observation. If your results differ from your expectations, explain why that may have happened. If your results agree, then describe the theory that the evidence supported. It is never appropriate to simply state that the data agreed with expectations, and let it drop at that.

- Decide if each hypothesis is supported, rejected, or if you cannot make a decision with confidence. Do not simply dismiss a study or part of a study as "inconclusive."
- Research papers are not accepted if the work is incomplete. Draw what you can based upon the results that you have, and treat the study as a finished work
- You may suggest future directions, such as how the experiment might be modified to accomplish another objective.
- Explain all of your observations as much as possible, focusing on mechanisms.
- Decide if the experimental design adequately addressed the hypothesis, and whether or not it was properly controlled.
- Try to offer alternative explanations if reasonable alternatives exist.
- One experiment will not answer an overall question, so keeping the big picture in mind, where do you go next? The best studies open up new avenues of research. What questions remain?
- Recommendations for specific papers will provide additional suggestions.

Style:

- When you refer to information, distinguish data generated by your own studies from published information or from information obtained from other students (verb tense is an important tool for accomplishing that purpose).
- Refer to work done by specific individuals (including yourself) in past tense.
- Refer to generally accepted facts and principles in present tense. For example, "Doofus, in a 1989 survey, found that anemia in basset hounds was correlated with advanced age. Anemia is a condition in which there is insufficient hemoglobin in the blood."

The biggest mistake that students make in discussions is to present a superficial interpretation that more or less re-states the results.

It is necessary to suggest why results came out as they did, focusing on the mechanisms behind the observations.

Literature Cited

You will not be required to properly document sources of all of your information. One reason is that your major source of information is this website, and websites are inappropriate as primary sources. Second, it is problematic to provide a hundred students

with equal access to potential reference materials. List all literature cited in your paper, in alphabetical order, by first author. In a proper research paper, only primary literature is used. Be cautious about using web sites as references - anyone can put just about anything on a web site, and you have no sure way of knowing if it is truth or fiction. If you are citing an on line journal. Some of your papers may not require references, and if that is the case simply state that "no references were consulted."

Writing Research Reports

After you have done your Market Research work, you have to write a report. However, the best research can get put aside without being read. Following are a few tips for writing a good Market Research Report that you can be proud of.

The first thing you have to do is to get your reader's attention with a powerful headline and a good opening summary. If you fail to get your prospects' attention, you will fail to communicate and deliver the benefits of your research.

Always remember that if your report is not read and action taken, your company will receive no benefits from all the research that you have done, and that's a pretty darn shame. But if you do the report right, you are probably about 70 per cent of the way to a promotion. So read on and pay attention.

The trick is writing a research report that will grab the reader's attention without allowing his or her mind to wander even for a second. Or worse: Making a copy mistake that turns her or him off entirely -- and gets your research report instantly tossed into the nearest virtual or literal trash can.

In a very real sense, your body copy is a minefield that must be navigated with the greatest of care: Every word, every sentence and paragraph of body copy represents the chance to either intensify your prospect's focus... or to completely lose him. But take care, in long research reports -- the headline and opening summary represent only about 4 per cent of the total volume of the report copy required. There's a lot of pages where you can lose your reader's attention. Lose him, even for a split second, and you've probably lost him -- and your promotion -- for good!

There are three unforgivable sins that you must be aware of and avoid when writing your report.

Avoid the following three unforgivable sins when crafting your research report:

- Do not confuse your reader...
- Do not bore your reader, and
- Do not set off his BS detector...

By following the seven simple rules, you can avoid all three dangers and produce a top quality Market Research Report.

Rule Number 1: Keep Your Report Logically Organized

Humans are NOT logical animals. But when reading or learning, they generally require that the material be presented in a clear, logical way. That generally means

starting at point"A"... progressing to point"B"... moving on to point"C"... and so on, until you have reached your ultimate cease.

To do that, you must build your case logically and methodically -- much like a mason builds a brick wall. You must lay a solid foundation of research and then build upon each completed analysis argument with the next... brick by brick... in a logical order, proving your research points.

Ask yourself, "What must my reader know first... second... third... and so on, in order to conclude that this research offers the opportunity of a lifetime?"

Rule Number 2: Keep the Report Moving

When a reader's eyes first fall upon your report, a little stopwatch starts ticking in his head. If at any point, he feels you're not moving along quickly enough, you will lose him.

Creating a dynamic flow of information in your research report is absolutely essential for maximum readability. There are three ways to do it...

- Creating and following a "chain of logic" outline helps a lot in this regard -- by ensuring that you make each point once, then move on. If prospects feel like you're going back over stuff you already covered, any sense of momentum you may have established is instantly destroyed.
- Check the momentum of each draft by reading it aloud. Mark the places where you -- as a reader -- begin to become distracted or bored. Once again, highlight any parts that begin to lose you. Each of these parts will kill readership and response if they're still there in the final draft. Edit them or delete them.
- Making each part of copy shorter than the one before is a great way to create momentum. For example -- let's say you have to make ten analysis points in order to complete the report. You could spend 1 1/2 pages making your first analysis... 1 page making your second... 3/4 of a page making your third... 1/2 page making your fourth... and then wrap up the final six points in a series of bullets covering a single page.

Rule Number 3: Keep Your Report Simple

Never ask your prospect to work in order to figure out what you're saying. Two-dollar words, esoteric references and complex sentences are killers in research reports. Subtlety, nuance and complexity are for poets -- NOT market reports!

Try to limit yourself to one complete, clearly presented thought per sentence. When you connect two thoughts in a sentence, make sure they connect directly and clearly with each other. Also be sure to avoid inserting undeveloped or underdeveloped thoughts in sentences or paragraphs. They're like little boobytraps in the report. They stop readers cold.

Rule Number 4: Keep the Report Fat-Free

Readers should feel as though they're getting good value in return for the number of words they're made to read. Your challenge is to never use three words when two will do the job.

Here are four ways to say more with less:

- Use more precise word choices: When you fail to use the word that most precisely and accurately communicates a thought, you wind up using five, six or even ten words instead. When searching for the most precise word, checking synonyms in a thesaurus often gives you the answer.
- Eliminate unnecessary words: Here again, reading copy aloud really helps. Much of the time, for example, the word "that" is totally unnecessary. When in doubt, leave it out!
- Avoid unhelpful repetition: Repetition of key sales points -- a USP or major benefit, for example -- is a beautiful thing. Repeating minor thoughts only slows the copy and bores the reader.
- Figures of speech can help you say more, faster: If a picture is worth a thousand words, metaphors, similes, and other figures of speech are as well.

Rule Number 5: Keep Your Report Believable

Your reader is already skeptical. Making grandiose claims that you can't (or don't) prove beyond the shadow of a doubt will only confirm what he or she already suspects: That you're full of beans. And this will get your promotion trashed in a heartbeat.

Rule Number 6: Keep Your Report Potent

One of the fastest ways to lose your prospect's attention is to fail to focus on his favourite subject: HIM or HER! The word "You" has been called the most powerful word in the English language -- and for good reason. Finding ways to personalize the report -- applying each passage as if had been written for the reader -- is a key to keeping his attention.

Rule Number 7: Avoid Unintended Impressions

Here's where insisting that friends read your report can pay huge dividends. By the time you're ready to stick a fork in your new promotion, you can almost recite it word for word -- frontwards and backwards. That means you're too close to the report to catch things that may be misread... even things that may raise objections or implant an erroneous impression in your reader's mind.

HOW TO WRITE A SCHOLARLY RESEARCH REPORT

Researchers communicate their results and help accumulate knowledge through conference papers, reports, on-line journals and print journals. While there are many

rewards for having research disseminated in a scholarly outlet, the preparation of a good research report is not a trivial task.

The common parts of a research report along with frequently made mistakes. While the emphasis here is on reports prepared for scholarly, peer-reviewed publication, these points are applicable to other forms of research reports. Dissertations and theses, for example, provide more detail than scholarly publications yet they adhere to the same basic scientific writing principles. Since all scientific research involves observation, description and analysis as well as to experimental, research. More detail can be found in the Publication Manual of the American Psychological Association, proposed revisions to the manual.

FIRST STEPS IN WRITING A RESEARCH REPORT

You should constantly think about writing your report at every stage of your research activities. The Publication Manual provides detailed information about the entire process of publication -- from organizing, writing, keying and submitting your manuscript, to seeing the accepted manuscript through production and publication. Of special interest in the fourth edition are the updated parts on reporting statistics, writing without bias, preparing manuscripts with a word processor for electronic production and publishing research in accordance with ethical principles of scientific publishing. You should have a copy.

Plan your report to focus on a single important finding or highly related group of findings. In the process of analyzing your data, you probably uncovered many relationships and gained numerous insights into the problem. Your journal submission, however, should contain only one key point. The point should be so fundamental that you should be able express it in one sentence or, at most, in a paragraph. If you have several key points, consider writing multiple manuscripts.

When writing your manuscript, keep in mind that the purpose is to inform the readers of what you investigated, why and how you conducted your investigation, the results and your ceases. As the investigator and writer, your job is simply to report, not to convince and usually not to advocate. You must provide enough detail so readers can reach their own ceases about the quality of your research and the veracity of your ceases.

Parts of Your Report

Thus, the title should not contain jargon or vernacular. Rather, the title should be short (generally 15 words or less) and clearly indicate what the study is about. If in doubt, try to specify the cause and effect relationship in your key point. Avoid trite and wasteful phrases such as "A study of..." or "An investigation to determine..."

Abstract: The abstract serves two major purposes: it helps a person decide whether to read the paper, and it provides the reader with a framework for understanding the paper if they decide to read it. Thus, your abstract should describe the most important

aspects of the study within the word-limit provided by the journal. As appropriate for your research, try to include a statement of the problem, the people you studied, the dependent and independent variables, the instruments, the design, major findings, and ceases. If pressed for space, concentrate on the problem and, especially, your findings.

Introduction: You will usually start your report with a paragraph or two presenting the investigated problem, the importance of the study, and an overview of your research strategy. You do not need to label this part. Its position within the paper makes that obvious.

The introductory paragraphs are usually followed by a review of the literature. Show how your research builds on prior knowledge by presenting and evaluating what is already known about your research problem. Assume that the readers possess a broad knowledge of the field. Discuss the findings of works that are pertinent to your specific issue. You usually will not need to elaborate on methods.

The goal of the introduction and literature review is to demonstrate "the logical continuity between previous and present work". This does not mean you need to provide an exhaustive historical review. Analyse the relationships among the related studies instead of presenting a series of seemingly unrelated abstracts or annotations. The introduction should motivate the study. The reader should understand why the problem was researched and why the study represents a contribution to existing knowledge. Unless the study is an evaluation of a programme, it is generally inappropriate to attempt to motivate the study based on its social importance.

Method: The method part includes separate descriptions of the sample, the materials, and the procedures. These are subtitled and may be augmented by further parts, if needed.

Describe your sample with sufficient detail so that it is clear what population(s) the sample represents. A discussion of how the sample was formed is needed for replicability and understanding your study." Convenience samples are not unusual in scientific inquiry; their use should not discourage you from seeking a publication outlet for your report.

A description of your instruments, including all surveys, tests, questionnaires, interview forms, and other tools used to provide data, should appear in the materials subsection. Evidence of reliability and validity should be presented. Since reliability is a property of scores from a specific use of a specific instrument for a specific population, you should provide reliability estimates based on your data.

The design of the study, whether it is a case study, a survey, a controlled experiment, a meta-analysis, or some other type of research, is conveyed through the procedures subsection.

It is here that the activities of the researcher are described, such as what was said to the participants, how groups were formed, what control mechanisms were employed, etc. The description is sufficient if enough detail is present for the reader to replicate the essential elements of the study. It is important for the procedures to conform to ethical criteria for researchers.

Results: Present a summary of what you found in the results part. Here you should describe the techniques that you used, each analysis and the results of each analysis.

Start with a description of any complications, such as protocol violations and missing data that may have occurred. Examine your data for anomalies, such as outliers, points of high influence, miscoded data, and illogical responses. Use your common sense to evaluate the quality of your data and make adjustments if need be. Describe the process that you used in order to assure your readers that your editing was appropriate and purified rather than skewed your results.

With today's availability of statistical packages, it is fairly easy to use very sophisticated techniques to analyse your data. Understand the techniques you are using and the statistics that you are reporting. Try to use the simplest, appropriate technique for which you can meet the underlying assumptions.

If you are going to use inferential statistics, you should determine the power a priori based on your anticipated distribution, design, and definition of practical significance. This information must stem from your related literature and not the data that you collected. If you fail to reach statistical significance, then this analysis can be used to show that the finding does not stem from low power.

Where appropriate, compute and report effect sizes or, at a minimum, be sure you provide enough information so effect sizes can be computed. Effect sizes provide a common metric for evaluating results across studies and aid in the design of future studies. They will be needed by anyone who attempts a quantitative synthesis of your study along with the others in your area of research.

For most research reports, the results should provide the summary about what you found rather than an exhaustive listing of every possible analysis and every data point. Use carefully planned tables and graphs. While tables and graphs should be self-explanatory, do not include a table or graph unless it is discussed in the report. Limit them to those that help the reader understand your data as they relate to the investigated problem.

Discussion: At this point, you are the expert on your data set and an authority on the problem you addressed. In this part, discuss and interpret your data for the reader, tell the reader of the implications of your findings and make recommendations. Do not be afraid to state your opinions.

Many authors chose to begin the discussion part by highlighting key results. Return to the specific problem you investigated and tell the reader what you now think and why. Relate your findings to those of previous studies, by explaining relationships and supporting or disagreeing with what others have found. Describe your logic and draw your ceases. Be careful, however, not to over generalize your results. Your ceases should be warranted by your study and your data.

Be sure to recognize the limitations of your study. Try to anticipate the questions a reader will have and suggest what problems should be researched next in order to extend your findings into new areas.

References: There should be a one-to-one match between the references cited in the report and the references listed in the reference part.

PUBLISHING YOUR REPORT

If you intend to publish in a journal, these journals will be the most likely candidates. Review the target audience and publication guidelines for these journals to decide which is best suited to your research. Regardless of scholarly quality, a key question in any editor's mind will be whether your manuscript is suited to the journal's purpose and audience. When considering where to submit, note the style of the journal.

Remember that the review process is conducted by human staff, and so is a fallible process. Peters and Ceci made this point abundantly clear.

Because of high rejection rates and the usual long length of time journals need to make a selection decision, it is tempting to submit a manuscript simultaneously to more than one journal. This, however, is clearly unethical. Most journals appropriately specify that manuscripts under consideration cannot be submitted elsewhere. The editors and reviewers will be taking a considerable amount of time examining your manuscript, usually as volunteers.

You should expect your manuscript to be rejected when it is submitted for the first time. If a manuscript is rejected, you should evaluate the comments and then decide whether to revise, resubmit, or submit it elsewhere. In order to facilitate both your revision and its subsequent evaluation, a resubmission should be accompanied by a description of the issues raised in the review process and your manuscript modifications and other substantive reactions to them.

While very little has been written about ethical standards for authors in the education field, the Uniform Requirements for Manuscripts Submitted to Biomedical Journals, which have been adopted by more than 500 scientific and biomedical journals, address criteria for authorship, acknowledgments, redundant publication, competing manuscripts, and conflict of interest. A concise summary of the Uniform Requirements can be found in Syrett and Rudner.

A key concept in the Uniform Requirements is that individuals identified as authors should have made significant contributions to the conception and design, or analysis and interpretation of data, or both; to drafting of the manuscript or revising it critically for intellectual content; and on final approval of the version of the manuscript to be considered for publication. Being an advisor or head of a research group, does not, in itself, warrant authorship credit.

Bibliography

- Alexander Clarence Flick.: *The Rise of the Mediaeval Church and Its Influence on the Civilisation of Western Europe from the First to the Thirteenth Century*, B. Franklin, 1919.
- Bodenheimer, Danna.: *Real World Clinical Social Work: Find Your Voice and Find Your Way (1st ed.)*, Harrisburg, PA: The New Social Worker Press, 2015.
- Bruce S. Jansson.: *Empowerment Series: Becoming An Effective Policy Advocate*, Cengage Learning, 2017.
- Catheleen Jordan.: *Clinical Assessment for Social Workers: Quantitative and Qualitative Methods*, Oxford University Press, 2015.
- Darley, Gillian, “Hill, Octavia.: *Oxford Dictionary of National Biography*, Oxford University Press, 2004.
- Davies, Martin.: *The Blackwell Companion of Social Work (2nd ed.)*, Oxford, UK; Malden, MA: Blackwell, 2002.
- Fischer, Joel and Kevin J. Corcoran.: *Measures for Clinical Practice and Research: A Sourcebook (4th ed.)*, Oxford, UK; New York, NY: Oxford University Press, 2007.
- Greene, Roberta R.: *Social Work with the Aged and their Families (3rd ed.)*. New Brunswick, NJ: Transaction Publishers, 2008.
- Grinnell, Richard M. and Yvonne A Unrau.: *Social Work Research and Evaluation: Foundations of Evidence-Based Practice (8th ed.)*, Oxford, UK; New York, NY: Oxford University Press, 2008.
- Jerrold R. Brandell.: *Theory and Practice in Clinical Social Work*, Simon and Schuster, 1997.

- Kenneth J. Linfield; Emil J. Posavac.: *Program Evaluation: Methods and Case Studies*, Taylor & Francis, 2018.
- Larry D. Watson; Richard A. Hoefer.: *Developing Nonprofit and Human Service Leaders*, SAGE Publications, 2013.
- Mary Carmel Ruffolo; Brian E Perron; Elizabeth H Voshel.: *Direct Social Work Practice: Theories and Skills for Becoming an Evidence-Based Practitioner (1st ed.)*, Los Angeles: SAGE Publications, 2015.
- Mizrahi, Terry and Larry E. Davis.: *Encyclopedia of Social Work (20th ed.)*, Washington, DC; Oxford, UK; New York, NY: NASW Press and Oxford University Press, 2008.
- Peter J. Pecora; David Cherin; Emily Bruce; Trinidad de Jesus Arguello.: *Strategic Supervision: A Brief Guide for Managing Social Service Organizations*, SAGE Publications, 2010.
- Poppo, Philip R. and Leslie Leighninger.: *The Policy-Based Profession: An Introduction to Social Welfare Policy Analysis for Social Workers (4th ed.)*, Boston, MA: Pearson/ Allyn and Bacon, 2008.
- Ralph Brody; Murali Nair.: *Effectively Managing and Leading Human Service Organizations*, SAGE Publications, 2013.
- Reamer, Frederic G.: *Ethical Standards in Social Work: A Review of the NASW Code of Ethics (2nd ed.)*, Washington, DC: NASW Press, 2006.
- Richard S. Balkin; Gerald A. Juhnke.: *Assessment in Counseling: Practice and Applications*, Oxford University Press, 2018.
- Richardson, Virginia E. and Amanda Smith Barusch.: *Gerontological Practice for the Twenty-First Century: A Social Work Perspective*, New York, NY: Columbia University Press, 2006.
- Sowers, Karen M. and Catherine N. Dulmus.: *Comprehensive Handbook of Social Work and Social Welfare*, Hoboken, NJ: John Wiley & Sons, 2008.
- Thyer, Bruce A. and John S. Wodarski.: *Social Work in Mental Health: An Evidence-Based Approach*, Hoboken, NJ: John Wiley, 2007.
- Turner, Francis J.: *Canadian Encyclopedia of Social Work*, Waterloo, ON: Wilfrid Laurier University Press, 2005.
- Victor J. Danilov.: "Social Activists", *Famous Americans: A Directory of Museums, Historic Sites, and Memorials*, Scarecrow Press, 2013.
- Webb, Stephen.: *Professional Identity and Social Work*, London, UK: Routledge, 2017.
- Wyatt, R.J.: *Octavia Hill and The Crown Estate – a Continuing Legacy?*, The Crown Estate, 2000.

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